



Swiss Centre
for Life Cycle
Inventories

ETH

(PFL)

FEO

EMPA

ART

Documentation of changes implemented in ecoinvent Data v2.1 and v2.2

Data v2.2 (2010)

Roland Hischier (Editor)
ecoinvent Centre

Hans-Jörg Althaus, Christian Bauer, Gabor Doka, Rolf
Frischknecht, Niels Jungbluth, Thomas Nemecek, Andrew
Simons, Matthias Stucki, Jürgen Sutter, Matthias Tuchschnid

ecoinvent report No. 16

St. Gallen, July 2010

Project "ecoinvent data v2.2 and data v2.1"

Commissioners:	Swiss Centre for Life Cycle Inventories, Dübendorf Swiss Federal Office for the Environment (BAFU - FOEN), Bern Swiss Federal Office for Energy (BFE), Bern
ecoinvent Board:	Gérard Gaillard, Agroscope Reckenholz-Tänikon Research Station, ART, Zürich (Chair) Stefanie Hellweg, ETHZ, Zürich Stefan Hirschberg, PSI, Villigen Rainer Zah, Empa, St. Gallen
ecoinvent Advisory Council:	Andreas Ciroth, GreenDeltaTC, Berlin Norbert Egli, BAFU, Bern Mark Goedkoop, PRé Consultants B.V., Amersfoort Lukas Gutzwiler, BfE, Bern Patrick Hofstetter, WWF, Zürich Roland Högger, öbu / Geberit AG, Rapperswil
Institutes of the ecoinvent Centre:	Swiss Federal Institute of Technology Zürich (ETHZ) Paul Scherrer Institute (PSI) Swiss Federal Laboratories for Materials Testing and Research (Empa) Agroscope Reckenholz-Tänikon Research Station (ART)
Participating consultants:	Doka Life Cycle Assessments, Zürich Dr. Frank Werner, Environment & Development, Zürich ESU-services Ltd., Uster
Software Support:	ifu Hamburg GmbH
Management ecoinvent:	Bo Weidema (General Manager) Roland Hirschier (Deputy Manager)

Project "ecoinvent data v2.0"

Commissioners:	Swiss Centre for Life Cycle Inventories, Dübendorf Swiss Federal Office for the Environment (BAFU - FOEN), Bern Swiss Federal Office for Energy (BFE), Bern Swiss Federal Office for Agriculture (BLW), Bern
ecoinvent Board:	Alexander Wokaun (Chair) PSI, Villigen Gérard Gaillard, Agroscope Reckenholz-Tänikon Research Station, ART, Zürich Lorenz Hilty, Empa, St. Gallen Konrad Hungerbühler, ETHZ, Zürich François Maréchal, EPFL, Lausanne
ecoinvent Advisory Council:	Norbert Egli, BAFU, Bern Mark Goedkoop, PRé Consultants B.V. Patrick Hofstetter, WWF, Zürich Roland Högger, öbu / Geberit AG, Rapperswil Christoph Rentsch, BAFU (until January 2006) Mark Zimmermann, BFE (until July 2007)
Institutes of the ecoinvent Centre:	Swiss Federal Institute of Technology Zürich (ETHZ) Swiss Federal Institute of Technology Lausanne (EPFL) Paul Scherrer Institute (PSI) Swiss Federal Laboratories for Materials Testing and Research (Empa) Agroscope Reckenholz-Tänikon Research Station (ART)
Participating consultants:	Basler & Hofmann, Zürich Bau- und Umweltchemie, Zürich Carbotech AG, Basel Chudacoff Oekoscience, Zürich Doka Life Cycle Assessments, Zürich Dr. Werner Environment & Development, Zürich Ecointesys - Life Cycle Systems Sarl. ENERS Energy Concept, Lausanne ESU-services Ltd., Uster Infras AG, Bern Umwelt- und Kompostberatung, Grenchen
Software Support:	ifu Hamburg GmbH
Project leader:	Rolf Frischknecht, ecoinvent Centre, Dübendorf
Marketing and Sales:	Annette Köhler, ecoinvent Centre, Dübendorf Mireille Faiste Emmenegger, ecoinvent Centre, Dübendorf

Project "ecoinvent 2000"

Commissioners:	Swiss Centre for Life Cycle Inventories, Dübendorf Swiss Federal Roads Authority (ASTRA), Bern Swiss Federal Office for Construction and Logistics (BBL), Bern Swiss Federal Office for Energy (BFE), Bern Swiss Federal Office for Agriculture (BLW), Bern Swiss Federal Office for the Environment (BAFU - FOEN), Bern
ecoinvent Board:	Alexander Wokaun (Chair) PSI, Villigen Konrad Hungerbühler, ETHZ, Zürich Lorenz Hilty, Empa, St. Gallen Gérard Gaillard, Agroscope Reckenholz-Tänikon, ART, Zürich François Maréchal, EPFL, Lausanne
ecoinvent Advisory Council:	Norbert Egli, BAFU, Bern Mark Goedkoop, PRé Consultants B.V. Patrick Hofstetter, WWF, Zürich Roland Högger, öbu / Geberit AG, Rapperswil Mark Zimmermann, BFE, Bern
Participating institutes and consultants:	Swiss Federal Institute of Technology Zürich (ETHZ) Swiss Federal Institute of Technology Lausanne (EPFL) Paul Scherrer Institute (PSI) Swiss Federal Laboratories for Materials Testing and Research (Empa) Agroscope Reckenholz-Tänikon Research Station (ART) Chudacoff Oekoscience, Zürich Doka Life Cycle Assessments, Zürich ESU-services Ltd., Uster
Software Development:	ifu Hamburg GmbH
Project leader:	Rolf Frischknecht, ESU-services, Uster

Documentation of changes implemented in ecoinvent Data v2.1 and v2.2

Editor:	Roland Hischier, ecoinvent Centre
Authors:	Hans-Jörg Althaus, Empa Christian Bauer, PSI Gabor Doka, Doka Life Cycle Assessments Rolf Frischknecht, ESU-services Ltd. Roland Hischier, Empa Niels Jungbluth, ESU-services Ltd. Manuele Margni, Ecoinvent - Life Cycle Systems Thomas Nemecek, ART Andrew Simons, PSI Jürgen Sutter, ETHZ-ICB Matthias Stucki, ESU-services Ltd. Matthias Tuchschrnid, SBB
Contact address:	ecoinvent Centre Empa P.O. Box CH-9014 St. Gallen http://www.ecoinvent.org/ support@ecoinvent.org
Responsibility:	This report has been prepared on behalf of one or several Federal Offices listed on the opposite page (see commissioners). The final responsibility for contents and conclusions remains with the authors of this report.
Terms of Use:	Data published in this report are subject to the end user licence agreement (EULA) of ecoinvent, in particular paragraph 7. The valid EULA document can be downloaded via the Internet (www.ecoinvent.org).
Liability:	Information contained herein have been compiled or arrived from sources believed to be reliable. Nevertheless, the authors or their organizations do not accept liability for any loss or damage arising from the use thereof. Using the given information is strictly your own responsibility.

Citation:

Hischier R., Althaus H.-J., Bauer C., Doka G., Frischknecht R., Jungbluth N., Nemecek T., Simons A., Sutter J., Stucki M., Tuchschrnid M. (2010) Documentation of changes implemented in ecoinvent Data v2.1 and v2.2. ecoinvent report No. 16. Swiss Centre for Life Cycle Inventories, St. Gallen.

Part I: Summary of changes implemented in data v2.1 and v2.2

Author: Roland Hischier, ecoinvent Centre, Empa

Citation:

Hischier R., Althaus H.-J., Bauer C., Doka G., Frischknecht R., Jungbluth N., Nemecek T., Simons A., Sutter J., Stucki M., Tuchschnid M. (2010) Documentation of changes implemented in ecoinvent Data v2.1 and v2.2. ecoinvent report No. 16. Swiss Centre for Life Cycle Inventories, St. Gallen.

1 Introduction

This report summarises the changes implemented in the database **ecoinvent data v2.1 and v2.2**. The report is split into three distinguished parts: **Part I** of this report gives an overview of the various changes executed; **Part II** contains the documentation of all corrections in single or in multi-output datasets for version v2.2 and those for version v2.1 that are not reported in an updated version of their regular reports – i.e. report No. 6//VII – Kernenergie (in German), report No. 6/XII – Photovoltaics, report No. 10 – Metals, and report No. 13 – Waste. Last but not least, in **part III**, the annex part of this report, you can find the following documentation parts:

- Annex 1: *Sputtering of ITO layers* (report not available in documentation of version v2.01);
- Annex 2: *Magnesium Dioxide Oxidation* (Potassium Permanganate / new report v2.2);
- Annex 3: *Methylamine* (new report v2.2);
- Annex 4: *Board based on industrial residual wood*
[complements Section 12.4 of Werner et al. 2007 for version v2.1];
- Annex 5: *Insulation Products and Processes*
[updated Part XIV of Kellenberger et al. 2007 for version v2.1];
- Annex 6: *Fibre Cement Products and Processes*
[updated Part IX of Kellenberger et al. 2007 for version v2.1];
- Annex 7: *Life Cycle Inventories for Rail Transport*
[updated chapter 6 of Spielmann et al. 2007 for version v2.2].

2 New Inventory Data

2.1 ecoinvent data v2.1

In the framework of the project ecoinvent data v2.1 a couple of **new datasets** have been integrated into the database. In Tab 2.1 an overview is given of these new datasets as well as the place where they are documented.

Tab 2.1 Overview of new datasets in ecoinvent data v2.1 and the place of their documentation

Name			Documentation
antimony, at refinery	CN	Kg	updated metal report
stibnite ore, 70% stibnite, at mine	CN	Kg	updated metal report
disposal, uranium tailings, non-radioactive emissions	GLO	m3	updated nuclear report
fibreboard soft, latex bonded, at plant (u=7%)	CH	m3	Part III / chapter 4
fibreboard soft, without adhesives, at plant (u=7%)	CH	m3	Part III / chapter 4
treatment, soft fibreboard production effluent, to wastewater treatment, class 3	CH	m3	Part III / chapter 4
polystyrene, extruded (XPS), HFC-134a blown, at plant	RER	Kg	Part III / chapter 5
polystyrene, extruded (XPS), HFC-152a blown, at plant	RER	Kg	Part III / chapter 5
polystyrene, extruded (XPS) CO2 blown, at plant	RER	Kg	Part III / chapter 5
fibre cement facing tile, large format, at plant	CH	Kg	Part III / chapter 6
fibre cement facing tile, small format, at plant	CH	Kg	Part III / chapter 6

2.2 ecoinvent data v2.2

In the framework of the project ecoinvent data v2.2 another more than 150 **new datasets** have been integrated into the database. One part of these new datasets is documented in separate documents – documents that are not part of the ecoinvent website, but can be found on websites from third parties. Tab 2.2 gives an overview of all new datasets as well as the place where their documentation can be found.

Tab 2.2 Overview of new datasets in ecoinvent data v2.2 and the place of their documentation

Name			Documentation
[i] Pesticides			
aclonifen, at regional storage	RER	kg	see separate report (Sutter 2010). Download possibility at www.ecoinvent.org (-> Documentation -> reports)
captan, at regional storage	RER	kg	
chlorothalonil, at regional storage	RER	kg	
chlorotoluron, at regional storage	RER	kg	
diazine-compounds, at regional storehouse	RER	kg	
diazole-compounds, at regional storehouse	RER	kg	
dimethenamide, at regional storage	RER	kg	
folpet, at regional storage	RER	kg	
fosetyl-Al, at regional storage	RER	kg	
fungicides, at regional storehouse	RER	kg	
growth regluators, at regional storehouse	RER	kg	
herbicides, at regional storehouse	RER	kg	
insecticides, at regional storehouse	RER	kg	
isoproturon, at regional storage	RER	kg	
mancozeb, at regional storage	RER	kg	
mecoprop, at regional storage	RER	kg	
metaldehyde, at regional storage	RER	kg	
metamitron, at regional storage	RER	kg	
napropamide, at regional storage	RER	kg	
orbencarb, at regional storage	RER	kg	
pendimethalin, at regional storage	RER	kg	
prosulfocarb, at regional storage	RER	kg	
pyridine-compounds, at regional storehouse	RER	kg	
[ii] Chemicals / Plastics			
concentrated lithium brine (6.7 % Li), at plant	GLO	kg	see separate report (Notter et al. 2010). Download possibility at http://pubs.acs.org/journal/esthag
ethylene carbonate, at plant	CN	kg	
lithium fluoride, at plant	CN	kg	
lithium hexafluorophosphate, at plant	CN	kg	
manganese oxide (Mn2O3), at plant	CN	kg	
phosphorous pentachloride, at plant	CN	kg	
polystyrene foam slab, 100% recycled, at plant	CH	kg	Part II / chapter 1
polystyrene foam slab, 45% recycled, at plant	CH	kg	
polystyrene scrap, old, at plant	CH	kg	
Manganese dioxide oxidation	RER	kg	Part III / chapter 2
potassium carbonate from manganese dioxide oxidation, at plant	RER	kg	
potassium permanganate, at plant	RER	kg	
methylamine, at plant	RER	kg	Part III / chapter 3
[iii] Energy			
electricity, certified eletricity	CH	kWh	see separate report (Leuenberger & Frischknecht 2010b). Download possibility at www.esu-services.ch
electricity, consumer mix	CH	kWh	
electricity, high voltage, certified electricity. at grid	CH	kWh	
electricity, high voltage, consumer mix, at grid	CH	kWh	
electricity, low voltage, certified electricity, at grid	CH	kWh	
electricity, low voltage, consumer mix, at grid	CH	kWh	
electricity, medium voltage, certified electricity, at grid	CH	kWh	
electricity, medium voltage, consumer mix, at grid	CH	kWh	

Name			Documentation
electricity, wood, at distillery	SE	kg	see separate report (Büsser & Frischknecht 2010). Download possibility at www.esu-services.ch
ethanol, 95% in H2O, from wood, at distillery	SE	kg	
ethanol, 99.7% in H2O, from Swedish wood, at service station	CH	kg	
ethanol, 99.7% in H2O, from wood, at distillation	SE	kg	
liquefied petroleum gas, at service station	CH	kg	
petrol, 85% vol. ethanol, from Swedish wood, at service station	CH	kg	
wood, in distillery	SE	kg	
[iv] Electronics			
Anode, lithium-ion battery, graphite, at plant	CN	kg	see separate report (Notter et al. 2010). Download possibility at http://pubs.acs.org/journal/esthag
Cathode, lithium-ion battery, lithium manganese oxide, at plant	CN	kg	
graphite, battery grade, at plant	CN	kg	
separator, lithium-ion battery, at plant	CN	kg	
single cell, lithium-ion battery, lithium manganese oxide/graphite, at plant	CN	kg	
chassis, network main devices	RER	kg	see separate report (Leuenberger & Frischknecht 2010d). Download possibility at www.esu-services.ch
network access devices, internet, at user	CH	unit	
router, IP network, at server	CH	unit	
use, computer, laptop, videoconference	CH	h	
use, computer, laptop, videoconference, certified electricity mix	CH	h	
use, computer, laptop, work at home	CH	h	
use, computer, laptop, work at home, certified electricity mix	CH	h	
use, IP network, videoconference	CH	h	
use, IP network, work at home	CH	h	
use, network access devices	CH	h	
use, network access devices, certified electricity	CH	h	
videoconference, laptop, participant	CH	h	
videoconference, laptop, participant, certified electricity mix	CH	h	
work at home, corporate access	CH	h	
work at home, corporate access, certified electricity mix	CH	h	
[v] Mobility			
bicycle, at regional storage	RER	unit	see separate report (Leuenberger & Frischknecht 2010c). Download possibility at www.esu-services.ch
disposal, bicycle	CH	unit	
disposal, electric bicycle	CH	unit	
disposal, electric scooter	CH	unit	
disposal, scooter	CH	unit	
electric bicycle, at regional storage	RER	unit	
electric scooter, at regional storage	RER	unit	
maintenance, bicycle	CH	unit	
maintenance, electric bicycle	CH	unit	
maintenance, electric scooter	CH	unit	
maintenance, scooter	CH	unit	
operation, electric bicycle	CH	km	
operation, electric bicycle, certified electricity	CH	km	
operation, electric scooter	CH	km	
operation, electric scooter, certified electricity	CH	km	
operation, scooter	CH	km	
scooter, ICE, at regional storage	RER	unit	
transport, bicycle	CH	pkm	
transport, electric bicycle	CH	pkm	
transport, electric bicycle, certified electricity	CH	pkm	
transport, electric scooter	CH	pkm	
transport, electric scooter, certified electricity	CH	pkm	
transport, scooter	CH	pkm	
electric motor, electric vehicle, at plant	RER	kg	see separate report (Leuenberger & Frischknecht 2010a). Download possibility at www.esu-services.ch
disposal, electric vehicle, LiMn2O4	RER	unit	
disposal, passenger car, diesel EURO5, city car	RER	unit	
disposal, passenger car, electric, LiMn2O4, city car	RER	unit	

Name			Documentation	
maintenance, electric vehicle, LiMn2O4	RER	unit		
maintenance, passenger car, diesel, EURO5, city car	RER	unit		
maintenance, passenger car, electric, LiMn2O4, city car	RER	unit		
operation, passenger car, diesel, EURO5, city car	CH	km		
operation, passenger car, electric, LiMn2O4	CH	km		
operation, passenger car, electric, LiMn2O4, certified electricity	CH	km		
operation, passenger car, electric, LiMn2O4, city car	CH	km		
operation, passenger car, electric, LiMn2O4, city car, certified electricity	CH	km		
passenger car, diesel, EURO5, city car, at plant	RER	unit		
passenger car, electric, LiMn2O4, at plant	RER	unit		
passenger car, electric, LiMn2O4, city car, at plant	RER	unit		
transport, passenger car, diesel, EURO5, city car	CH	pkm		
transport, passenger car, electric, LiMn2O4	CH	pkm		
transport, passenger car, electric, LiMn2O4, certified electricity	CH	pkm		
transport, passenger car, electric, LiMn2O4, city car	CH	pkm		
transport, passenger car, electric, LiMn2O4, city car, certified electricity	CH	pkm		
operation, average train	AT	pkm		Part III / chapter 7
operation, average train	IT	pkm		
operation, average train	FR	pkm		
operation, average train	DE	pkm		
operation, average train	BE	pkm		
operation, average train, SBB mix	CH	pkm		
Operation, freight train	AT	tkm		
Operation, freight train	IT	tkm		
Operation, freight train	FR	tkm		
Operation, freight train	DE	tkm		
Operation, freight train	BE	tkm		
Operation, freight train, diesel, with particle filter	CH	tkm		
Operation, high speed train	FR	pkm		
Operation, high speed train	IT	pkm		
Operation, metropolitan train, SBB mix	CH	pkm		
transport, average train	AT	pkm		
transport, average train	IT	pkm		
transport, average train	FR	pkm		
transport, average train	DE	pkm		
transport, average train	BE	pkm		
transport, average train, SBB mix	CH	pkm		
transport, freight, rail	AT	tkm		
transport, freight, rail	IT	tkm		
transport, freight, rail	FR	tkm		
transport, freight, rail	DE	tkm		
transport, freight, rail	BE	tkm		
transport, freight, rail, diesel, with particle filter	CH	tkm		
transport, high speed train	FR	pkm		
transport, high speed train	IT	pkm		
transport, metropolitan train, SBB mix	CH	pkm		
[vi] Waste Treatment				
disposal, spoil from coal mining, in surface landfill	GLO	kg	see separate report (Doka 2009b). Download possibility at www.doka.ch	
disposal, spoil from lignite mining, in surface landfill	GLO	kg		
disposal, tailings from hard coal milling, in impoundment	GLO	kg		

3 Changes/Addings in area of tailings data

According to Doka 2007, tailings are the waste from ore beneficiation (milling). Raw ore extracted from the mine contains *gangue*, which is the part of the ore that does not contain the desired target mineral. In ore beneficiation the respective target mineral is separated as far as possible from gangue, usually using mechanical steps like grinding and floatation. After this beneficiation step gangue becomes a watery suspension sludge called *tailings*. These tailings are stored in artificial pools, so called tailings impoundments, ponds or dams. In these impoundments the solids of the tailings sludge are allowed to settle while the excess water is pumped off and usually recycled back to the beneficiation process.

In ecoinvent version v1 and v2.01, among the datasets of tailings, the following datasets were only covering land-use and land transformation due to a lack of reliable models for the geo-chemical processes in order to include e.g. acid rock drainage (ARD):

- Sulphidic tailings and non-sulphidic tailings from metal mining (3 datasets)
- Uranium tailings (1 dataset)
- Hard coal tailings (1 dataset)

For ecoinvent data v2.1, the data for sulfidic and uranium tailings have been changed – and finally, in the framework of the work on ecoinvent data v2.2 the hard coal and lignite tailing data have been changed resp. added. Non-sulphidic tailings are currently only assessed with a land use exchange. However, they don't undergo ARD reactions and are considered to be less problematic than sulfidic tailings.

The former data gap in the **sulphidic tailings** dataset is filled with emissions data from a specifically designed tailings model, based on the landfill disposal models presented in ecoinvent report No. 13 inventoring emissions to water and groundwater (Doka 2009a). This complemented dataset provides a first informed estimate of the full magnitude of sulfidic tailings burdens for non-ferrous metal products and should encourage further research for more detailed models for future versions of the ecoinvent database. More details about this new datasets can be found in part XIV “Non Ferrous Metal Winning Auxiliary Processes” of the updated metals report (Classen et al. 2009).

The **uranium tailings** model added in 2009 is a similar model providing a first coarse estimate on the typical magnitude of non-radioactive emissions; that is added to the already existing radioactive emissions (in the dataset “tailings, uranium mining”¹). This model also includes a proportionate amount of air emissions from tailings in arid regions. More details about this new datasets can be found in chapter 19 of the updated nuclear energy report (Dones et al. 2009).

Finally, for **hard coal and lignite** an emissions model of **spoil and tailings disposal** is developed again similar to the established landfill models. The developed models are a good estimate for a *global average burden* of the disposal of tailings and spoil. In order to avoid overestimations, the modelling assumptions – within the ecoinvent framework – were chosen to be conservative. The goal is not to assess one particular disposal site or to optimise the disposal site performance, but to obtain a typical global average of burdens from currently practiced disposal processes in order to complement the burdens from other life cycle stages.

¹ In the existing dataset “tailings, uranium mining”, the data about land-use as well as the radioactive emissions are kept, while all emissions to groundwater are replaced by this new dataset “disposal, uranium tailings, non-radioactive emissions”.

4 Changes in LCIA Methods

4.1 New LCIA methods in ecoinvent

The following **new Life Cycle Impact Assessment (LCIA) methods** have been included for the first time (resp. again in case of EPS 2000) into the ecoinvent database:

- EPS 2000,
- Ecological Scarcity, ecofactors 2006 (as reported in Frischknecht et al. 2008),
- Global Warming with the IPCC 2007 factors (as reported in IPCC 2007),
- ReCiPe (on both levels – the Midpoint and the Endpoint level), and
- USEtox.

More information about these methods can be found in the latest version of report No.3 of ecoinvent, the implementation report of the LCIA methods (Hischier et al. 2010).

4.2 Changes in consideration of long-term emissions

Having in mind this new constellation for mining activities (i.e. the inclusion of the tailings effects – as described in chapter 3), the treatment of the long-term emissions was reconsidered. While in a first phase (implemented in version v2.1 of ecoinvent) two different approaches have been used (for more details, see the v2.1 version of report No.3 – i.e. Hischier et al. 2009), in the framework of the work on version v2.2, the whole issue has been revised once more, and a consistent solution for all concerned methods has been found: concerned methods are implemented in two distinct versions – one without characterisation factors for any type of long-term emissions, the other one using the same characterisation factors for short- and long-term emissions. The methods concerned are the CML method, the Eco-Indicator'99 (only the *Egalitarian* and the *Hierarchist* perspectives – as the *Individualist* perspective doesn't have characterisation factors for long-term emissions), the methods EDIP 1997 and 2003, the newly implemented ReCiPe method (both versions, i.e. Midpoint and Endpoint – again only the *Egalitarian* and the *Hierarchist* perspectives each time) and the also newly implemented USEtox model.

For the method “**Ecological scarcity (ecofactors 1997)**” discussions with experts on water protection during the work on the updated version of this method (i.e. the ecofactors 2006) revealed:

- a) in Switzerland there are no legal limits nor target values regarding heavy metals in ground water (statement: heavy metals are no problem in groundwaters in our country);
- b) an extrapolation long-term of surface water factors to groundwater is not appropriate and thus declined by FOEN (Swiss Federal Office for the Environment).

Thus, any extrapolations of eco-factors for heavy metal emissions from surface water to groundwater (be it short or long-term) have been deleted (in comparison to the former implementation of this method in ecoinvent data v2.01).

More information about all methods and their actual implementation can be found in the latest version of report No.3 of ecoinvent, the implementation report of the LCIA methods (Hischier et al. 2010).

5 Further changes

5.1 Corrections in Datasets

Thanks to the user of our data, as well as in the framework of our own work, several errors in dataset of ecoinvent data v2.01 resp. v2.1 have been discovered and corrected. **Part II of this report** shows you in details all these error corrections, together with further changes in a variety of datasets.

5.2 Deleted Datasets

In the framework of the investigations for the upcoming version v3 of ecoinvent, several double entries of substances and/or outdated datasets have been discovered in the database ecoinvent v2.1. The double entries have been examined in details, and the less appropriate datasets have been deleted each time. Further datasets got obsolete due to changes in modelling and/or due to updates with different names. The following list gives an overview of all the datasets deleted for version v2.2, as well as the alternative dataset to be used instead.

Deleted Datasets		Alternative Dataset to be used instead	
DS-ID	Name	DS-ID	Name
61	vinasse, at regional storehouse, CH [kg]	6262	Vinasse, at fermentation plant, CH [kg]
377	butanol, 1-, at plant, RER [kg]	6598	1-butanol, propylene hydroformylation, at plant, RER [kg]
366	acrylonitrile, at plant, RER [kg]	6614	acrylonitrile from Sohio process, at plant, RER [kg]
10455	polylactide, granulate, NatureWorks Nebraska, US [kg]	10456	Polylactice, granulate, at plant, GLO [kg]
334	Imperial smelting furnace, GLO [kg]	-	<i>[MO process – replaced by single output data]</i>
10234	zinc , from Imperial smelting furnace, GLO [kg]	1156	zinc, primary, at regional storage, RER [kg]
6945	helium, gaseous, at plant, RER [kg]	7218	helium, at plant, GLO [kg]
1779	mc-Si wafer, at plant, RER [unit]	6825	single-Si wafer, photovoltaics, at plant, RER [m2]
1781	pc-Si wafer, at plant, RER [unit]	6826	multi-Si wafer, at plant, RER [m2]
1785	photovoltaic cell, mc-Si, at plant, RER [unit]	6827	photovoltaic cell, single-Si, at plant, RER [m2]
1786	photovoltaic cell, pc-Si , at plant, RER [unit]	6828	photovoltaic cell, multi-Si, at plant, RER [m2]
1788	photovoltaic laminate, mc-Si, at plant, RER [unit]	6838	photovoltaic laminate, single-Si, at plant, RER [m2]
1790	photovoltaic laminate, pc-Si, at plant, RER [unit]	6839	photovoltaic laminate, multi-Si, at plant, RER [m2]
1793	photovoltaic panel, mc-Si, at plant, RER [unit]	6836	photovoltaic panel, single-Si, at plant, RER [m2]
1794	photovoltaic panel, pc-Si, at plant, RER [unit]	6837	photovoltaic panel, multi-Si, at plant, RER [m2]

5.3 Changes in Nomenclature

Thanks to a thorough analysis of the nomenclature by our technical partner – IFU Hamburg GmbH – several inconsistencies within the nomenclature of the elementary flows as well as the dataset names have been identified and corrected for ecoinvent data v2.1. The following changes have been made:

- German names of datasets

old name	corrected name
integrierte Papierfabrik, RER, [unit]	Papierfabrik, integriert, RER, [unit]
nicht-integrierte Papierfabrik, RER, [unit]	Papierfabrik, nicht-integriert, RER, [unit]
Proteine aus Gras, ab Vergärung, CH, [kg]	Proteine, aus Gras, ab Vergärung, CH, [kg]
Anlage, Leiterplattenbestückung, Typ SMT, GLO, [unit]	Anlagen Leiterplattenbestückung, Typ SMT, GLO, [unit]
Anlage, Leiterplattenbestückung, Typ THT, GLO, [unit]	Anlagen Leiterplattenbestückung, Typ THT, GLO, [unit]
händischer Verarbeitungsbetrieb, Elektronikschrott, GLO, [unit]	Händischer Verarbeitungsbetrieb, Elektronikschrott, GLO, [unit]
mechanischer Verarbeitungsbetrieb, Elektronikschrott, GLO, [unit]	Mechanischer Verarbeitungsbetrieb, Elektronikschrott, GLO, [unit]
kombinierte Erdgas- und Rohölförderung Offshore, NL, [a]	Kombinierte Erdgas- und Rohölförderung Offshore, NL, [a]
kombinierte Erdgas- und Rohölförderung Offshore, NO, [a]	Kombinierte Erdgas- und Rohölförderung Offshore, NO, [a]
kombinierte Erdgas- und Rohölförderung Onshore, NL, [a]	Kombinierte Erdgas- und Rohölförderung Onshore, NL, [a]
kombinierte Erdgas- und Rohölförderung, NG, [a]	Kombinierte Erdgas- und Rohölförderung, NG, [a]
kombinierte Erdgas- und Rohölförderung Offshore, GB, [a]	Kombinierte Erdgas- und Rohölförderung Offshore, GB, [a]
optischer Aufheller Distyrylbiphenyl-Typ, ab Werk, kg, [RER]	Optischer Aufheller Distyrylbiphenyl-Typ, ab Werk, kg, [RER]
Oriented Strand Board, ab Werk, m3, [RER]	Grobspanplatte (OSB), ab Werk, m3, [RER]

- English names of datasets

old name	corrected name
crude coco nut oil, at plant, PH, [kg]	crude coconut oil, at plant, PH, [kg]
Mini-BHKW, common components for heat+electricity, CH, [unit]	Mini CHP plant, common components for heat+electricity, CH, [unit]
Mini-BHKW, components for electricity only, CH, [unit]	Mini CHP plant, components for electricity only, CH, [unit]
Mini-BHKW, components for heat only, CH, [unit]	Mini CHP plant, components for heat only, CH, [unit]
assembly, generator and motor, Mini-BHKW, CH, [unit]	assembly, generator and motor, Mini CHP plant, CH, [unit]
catalytic converter, three-way, Mini-BHKW, CH, [unit]	catalytic converter, three-way, Mini CHP plant, CH, [unit]
electric parts of Mini-BHKW, CH, [unit]	electric parts of Mini CHP plant, CH, [unit]
electricity, at Mini-BHKW, allocation energy, CH, [kWh]	electricity, at Mini CHP plant, allocation energy, CH, [kWh]
electricity, at Mini-BHKW, allocation exergy, CH, [kWh]	electricity, at Mini CHP plant, allocation exergy, CH, [kWh]
electricity, at Mini-BHKW, allocation heat, CH, [kWh]	electricity, at Mini CHP plant, allocation heat, CH, [kWh]
gas motor Mini-BHKW, CH, [unit]	gas motor Mini CHP plant, CH, [unit]
generator Mini-BHKW, CH, [unit]	generator Mini CHP plant, CH, [unit]
heat exchanger of Mini-BHKW, CH, [unit]	heat exchanger of Mini CHP plant, CH, [unit]
heat, at Mini-BHKW, allocation energy, CH, [MJ]	heat, at Mini CHP plant, allocation energy, CH, [MJ]
heat, at Mini-BHKW, allocation exergy, CH, [MJ]	heat, at Mini CHP plant, allocation exergy, CH, [MJ]
heat, at Mini-BHKW, allocation heat, CH, [MJ]	heat, at Mini CHP plant, allocation heat, CH, [MJ]
heating, sanitary equipment Mini-BHKW, CH, [unit]	heating, sanitary equipment Mini CHP plant, CH, [unit]
maintenance, Mini-BHKW, CH, [unit]	maintenance, Mini CHP plant, CH, [unit]
natural gas, burned in Mini-BHKW, CH, [MJ]	natural gas, burned in Mini CHP plant, CH, [MJ]
natural gas, burned in Mini-BHKW, allocation energy, CH, [MJ]	natural gas, burned in Mini CHP plant, allocation energy, CH, [MJ]
natural gas, burned in Mini-BHKW, allocation exergy, CH, [MJ]	natural gas, burned in Mini CHP plant, allocation exergy, CH, [MJ]
natural gas, burned in Mini-BHKW, allocation heat, CH, [MJ]	natural gas, burned in Mini CHP plant, allocation heat, CH, [MJ]
planning, cogen unit Mini-BHKW, CH, [unit]	planning, cogen unit Mini CHP plant, CH, [unit]
storage 650 l Mini-BHKW, CH, [unit]	storage 650 l Mini CHP plant, CH, [unit]
glass, cullets, sorted, at sorting plant, RER, [kg]	glass cullets, sorted, at sorting plant, RER, [kg]
glass cullet, from fluorescent lamps treatment, at plant, GLO, [kg]	glass cullets, from fluorescent lamps treatment, at plant, GLO, [kg]
Industrial residual wood chopping, stationary electric chopper, at plant, RER, [kg]	industrial residual wood chopping, stationary electric chopper, at plant, RER, [kg]
integrated paper mill, RER, [unit]	paper mill, integrated, RER, [unit]
non-integrated paper mill, RER, [unit]	paper mill, non-integrated, RER, [unit]
proteins from grass, at fermentation, CH, [kg]	proteins, from grass, at fermentation, CH, [kg]
sugar cane, at farm, BR, [kg]	sugarcane, at farm, BR, [kg]
vinylfluoride, at plant, US, [kg]	vinyl fluoride, at plant, US, [kg]
natural gas, at production, RNA, [m3]	natural gas, at production, RNA, [Nm3]

- names of **elementary** flows

old name	corrected name
Aluminum, air, high population density, [kg]	Aluminium, air, high population density, [kg]
Aluminum, air, low population density, [kg]	Aluminium, air, low population density, [kg]
Aluminum, air, low population density, long-term, [kg]	Aluminium, air, low population density, long-term, [kg]
Aluminum, air, lower stratosphere + upper troposphere, [kg]	Aluminium, air, lower stratosphere + upper troposphere, [kg]
Aluminum, air, unspecified, [kg]	Aluminium, air, unspecified, [kg]
Aluminum, soil, agricultural, [kg]	Aluminium, soil, agricultural, [kg]
Aluminum, soil, forestry, [kg]	Aluminium, soil, forestry, [kg]
Aluminum, soil, industrial, [kg]	Aluminium, soil, industrial, [kg]
Aluminum, soil, unspecified, [kg]	Aluminium, soil, unspecified, [kg]
Aluminum, water, fossil-, [kg]	Aluminium, water, fossil-, [kg]
Aluminum, water, ground-, [kg]	Aluminium, water, ground-, [kg]
Aluminum, water, ground-, long-term, [kg]	Aluminium, water, ground-, long-term, [kg]
Aluminum, water, lake, [kg]	Aluminium, water, lake, [kg]
Aluminum, water, ocean, [kg]	Aluminium, water, ocean, [kg]
Aluminum, water, river, [kg]	Aluminium, water, river, [kg]
Aluminum, water, river, long-term, [kg]	Aluminium, water, river, long-term, [kg]
Aluminum, water, unspecified, [kg]	Aluminium, water, unspecified, [kg]
Chlormequat Chloride, soil, agricultural, [kg]	Chlormequat chloride, soil, agricultural, [kg]

6 References

- Büsser & Frischknecht 2010 Büsser S. and Frischknecht R. (2010) Life Cycle Inventory of E85, LPG supply in Switzerland and Biogasmix 2008. ESU-services, Uster (Switzerland)
- Classen et al. 2009 Classen M., Althaus H.-J., Blaser S., Doka G., Jungbluth N., Tuchschnid M. and Faist Emmenegger M. (2009) Life Cycle Inventories of Metals. Final report ecoinvent data v2.1 No.10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, St. Gallen, CH
- Doka 2007 Doka G. (2007) Non-radiological emissions from uranium tailings: A generic, global model for Life Cycle Inventory data. Study on behalf of Paul Scherrer Institute (PSI), Villigen, by Doka Life Cycle Assessments, Zürich (Switzerland)
- Doka 2009a Doka G. (2009a) Life Cycle Inventories of Waste Treatment Services. Final report ecoinvent data v2.1 No. 13. Swiss Centre for Life Cycle Inventories, St. Gallen, CH
- Doka 2009b Doka G. (2009b) Life Cycle Inventory of the disposal of lignite spoil, coal spoil and coal tailings. Study on behalf of ecoinvent Centre, St. Gallen, by Doka Life Cycle Assessments, Zürich (Switzerland), Online-Version under: www.doka.ch.
- Dones et al. 2009 Dones R., Bauer C. and Doka G. (2009) Kernenergie. In: Sachbilanzen von Energiesystemen: Grundlagen für den ökologischen Vergleich von Energiesystemen und den Einbezug von Energiesystemen in Ökobilanzen für die Schweiz (ed. Bauer C.). Final report ecoinvent data v2.1 No. 6, Paul Scherrer Institut Villigen, Swiss Centre for Life Cycle Inventories, St. Gallen, CH, Online-Version under: www.ecoinvent.ch.
- Frischknecht et al. 2008 Frischknecht R., Steiner R. and Jungbluth N. (2008) Ökobilanzen: Methode der ökologischen Knappheit - Ökofaktoren 2006. Öbu, Zürich (Switzerland)
- Hischier et al. 2009 Hischier R., Weidema B., Althaus H.-J., Doka G., Dones R., Frischknecht R., Hellweg S., Humbert S., Jungbluth N., Loerincik Y., Margni M., Nemecek T. and Simons A. (2009) Implementation of Life Cycle Impact Assessment Methods. Final report ecoinvent v2.1 No. 3. Swiss Centre for Life Cycle Inventories, St. Gallen, CH
- Hischier et al. 2010 Hischier R., Weidema B., Althaus H.-J., Doka G., Dones R., Frischknecht R., Hellweg S., Humbert S., Jungbluth N., Loerincik Y., Margni M., Nemecek T. and Simons A. (2010) Implementation of Life Cycle Impact Assessment Methods. Final report ecoinvent v2.2 No. 3. Swiss Centre for Life Cycle Inventories, St. Gallen, CH
- IPCC 2007 IPCC (2007) Climate Change 2007: The Physical Science Basis. In: Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change (ed. Solomon S., Qin D., Manning M., Chen Z., Marquis M., Averyt K. B., Tignor M. and Miller H. L.). pp. 996. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA
- Leuenberger & Frischknecht 2010a Leuenberger M. and Frischknecht R. (2010a) Battery electric vehicles and future concept cars. ESU-services, Uster (Switzerland)
- Leuenberger & Frischknecht 2010b Leuenberger M. and Frischknecht R. (2010b) Electricity mixes: consumer mix, certified electricity and electricity mix. ESU-services, Uster (Switzerland)

Leuenberger & Frischknecht 2010c	Leuenberger M. and Frischknecht R. (2010c) Two wheel vehicles. ESU-services, Uster (Switzerland)
Leuenberger & Frischknecht 2010d	Leuenberger M. and Frischknecht R. (2010d) Virtual Mobility. ESU-services, Uster (Switzerland)
Notter et al. 2010	Notter D. A., Gauch M., Widmer R., Wäger P., Stamp A., Zah R. and Althaus H.-J. (2010) Contribution of Li-ion batteries to the environmental impact of electric vehicles. In: Environmental Science & Technology, accepted for publication, pp.
Spielmann et al. 2007	Spielmann M., Dones R. and Bauer C. (2007) Life Cycle Inventories of Transport Services. Final report ecoinvent v2.0 No. 14. PSI, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch .
Sutter 2010	Sutter J. (2010) Life Cycle Inventories of Pesticides. Final report ecoinvent data v2.2. ETH Zürich - S+U group, Swiss Centre for Life Cycle Inventories, St. Gallen, CH
Werner et al. 2007	Werner F., Althaus H.-J., Künniger T., Richter K. and Jungbluth N. (2007) Life Cycle Inventories of Wood as Fuel and Construction Material. Final report ecoinvent data v2.0 No. 9. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch .

Part II:

Documentation of Changes

Editor: Roland Hischier, ecoinvent Centre, Empa

Authors: Hans-Jörg Althaus, Christian Bauer, Gabor Doka, Rolf Frischknecht, Niels Jungbluth, Thomas Kägi, Thomas Nemecek, Andrew Simons, Jürgen Sutter, Matthias Stucki, Matthias Tuchs Schmid

Citation:

Hischier R., Althaus H.-J., Bauer C., Doka G., Frischknecht R., Jungbluth N., Nemecek T., Simons A., Sutter J., Stucki M. (2010) Documentation of changes implemented in ecoinvent Data v2.1 and v2.2. ecoinvent report No. 16. Swiss Centre for Life Cycle Inventories, St. Gallen.

Content

1	CHANGES IN ECOINVENT DATA V2.2.....	1
1.1	ENERGY SYSTEMS	1
1.1.1	<i>Liquified Gas Transportation.....</i>	<i>1</i>
1.1.2	<i>Electricity mix SBB.....</i>	<i>1</i>
1.2	CONSTRUCTION MATERIALS.....	1
1.2.1	<i>Fibre Cement Facing Tile</i>	<i>1</i>
1.2.2	<i>Gravel mining.....</i>	<i>2</i>
1.2.3	<i>Natural Stone</i>	<i>2</i>
1.2.4	<i>Insulation Materials.....</i>	<i>2</i>
1.2.5	<i>Plastic Windows.....</i>	<i>1</i>
1.3	CHEMICALS	1
1.3.1	<i>Propylene Hydroformation</i>	<i>1</i>
1.3.2	<i>Sohio process</i>	<i>1</i>
1.3.3	<i>Sulphuric acid digestion of spodumene.....</i>	<i>1</i>
1.4	METALS.....	2
1.4.1	<i>Converter Steel, low-alloyed.....</i>	<i>2</i>
1.5	PLASTICS	2
1.5.1	<i>Changes for data of PlasticsEurope due to new coal tailing data.....</i>	<i>2</i>
1.6	AGRICULTURAL PRODUCTS.....	3
1.6.1	<i>Agricultural products from Brazil.....</i>	<i>3</i>
1.6.2	<i>Agricultural products from Germany.....</i>	<i>3</i>
1.6.3	<i>Cotton and Rice seed.....</i>	<i>4</i>
1.6.4	<i>Sheep</i>	<i>4</i>
1.7	WOOD AND WOODEN PRODUCTS	5
1.7.1	<i>Soft Fibreboard.....</i>	<i>5</i>
1.7.2	<i>Sawing of Scandinavian Softwood</i>	<i>6</i>
1.8	TRANSPORT SYSTEMS	6
1.8.1	<i>Operation of lorries.....</i>	<i>6</i>
1.9	MECHANICAL ENGINEERING.....	12
1.9.1	<i>Degreasing of metal parts.....</i>	<i>12</i>
1.9.2	<i>Wire Drawing of Steel.....</i>	<i>12</i>
1.9.3	<i>Sheet Rolling of Copper</i>	<i>12</i>
1.9.4	<i>Wire Drawing of Copper.....</i>	<i>12</i>
1.9.5	<i>Contour Brass and Bronze</i>	<i>12</i>
1.10	ELECTRONIC DATA.....	12
1.10.1	<i>NiMH Batteries</i>	<i>12</i>
1.10.2	<i>LCD Screen.....</i>	<i>13</i>
1.10.3	<i>Laptop Use.....</i>	<i>13</i>
2	CHANGES IN ECOINVENT DATA V2.1.....	14
2.1	ENERGY SYSTEMS	14
2.1.1	<i>Crude oil production.....</i>	<i>14</i>
2.1.2	<i>Hard coal production.....</i>	<i>14</i>
2.1.3	<i>Lignite power plant</i>	<i>14</i>
2.1.4	<i>Hydro power plants.....</i>	<i>14</i>
2.1.5	<i>Inverter data of photovoltaics installations.....</i>	<i>15</i>
2.1.6	<i>Slanted-roof installations for photovoltaics.....</i>	<i>15</i>
2.1.7	<i>Electricity production mix in Denmark and Sweden.....</i>	<i>16</i>

2.1.8	<i>Electricity production and supply mix in Finland</i>	17
2.1.9	<i>Electricity distribution in China</i>	18
2.2	CONSTRUCTION MATERIALS	18
2.2.1	<i>Flat Glass, uncoated</i>	18
2.2.2	<i>Building, multi-storey</i>	18
2.2.3	<i>Glazings</i>	18
2.2.4	<i>Fibre Cement</i>	19
2.2.5	<i>Insulation Materials</i>	19
2.2.6	<i>Expanded Clay / Vermiculite</i>	19
2.3	CHEMICALS	23
2.3.1	<i>Arsine</i>	23
2.3.2	<i>Benzal chloride</i>	23
2.3.3	<i>Benzaldehyde</i>	23
2.3.4	<i>Benzene chlorination</i>	23
2.3.5	<i>Benzyl alcohol</i>	23
2.3.6	<i>Benzyl chloride</i>	24
2.3.7	<i>Chlorinated solvents</i>	24
2.3.8	<i>Chlorosilane amination</i>	24
2.3.9	<i>Ethylene chlorofluorination</i>	24
2.3.10	<i>Ethylene glycol diethyl ether</i>	25
2.3.11	<i>Ethylene glycoles</i>	25
2.3.12	<i>Lithium chloride</i>	25
2.3.13	<i>Lithium chloride electrolysis</i>	26
2.3.14	<i>Maleic Anhydride from direct oxidation of n-butane</i>	26
2.3.15	<i>Modified Solvay Process</i>	26
2.3.16	<i>Phosphor chloride</i>	27
2.4	METALS.....	27
2.4.1	<i>Cadmium substances</i>	28
2.4.2	<i>Mercure</i>	29
2.5	PLASTICS	29
2.5.1	<i>Methological change for integration of data of PlasticsEurope</i>	29
2.5.2	<i>PVC, bulk polymerised, and PVC, at regional storage</i>	30
2.6	AGRICULTURAL PRODUCTS.....	30
2.6.1	<i>Agricultural products from Brazil</i>	30
2.6.2	<i>Agricultural products from Germany</i>	31
2.6.3	<i>Agricultural products from Switzerland</i>	31
2.6.4	<i>Agricultural products from USA</i>	32
2.7	WOOD AND WOODEN PRODUCTS	36
2.7.1	<i>Fibreboard, soft</i>	36
2.7.2	<i>Wood Pellet manufacturing, infrastructure (RER)</i>	36
2.7.3	<i>Tropical wood</i>	36
2.8	BIOENERGY	36
2.8.1	<i>Biowaste to agricultural co-fermentation, covered</i>	36
2.8.2	<i>Biowaste, to anaerobic digestion</i>	37
2.8.3	<i>Anaerobic digestion plant covered, agriculture</i>	39
2.8.4	<i>Methane purification</i>	39
2.9	TRANSPORT SYSTEMS	39
2.9.1	<i>lorry transport</i>	39
2.9.2	<i>Operation of lorries</i>	40
2.9.3	<i>Operation of cars</i>	40
2.9.4	<i>Transport of aircraft</i>	41
2.10	ELECTRONICS DATA.....	42
2.10.1	<i>Printed Wiring board production (Pb-containing finishing)</i>	42
2.10.2	<i>LCD screen</i>	42

2.11	WASTE DISPOSAL.....	42
2.11.1	<i>Hazardous waste incineration direct carbon dioxide emissions.....</i>	42
2.11.2	<i>Energy for digestion of wastewater treatement sludge</i>	43
2.11.3	<i>Wastewater treatement direct carbon dioxide emissions.....</i>	44
2.11.4	<i>Wastewater treatement direct nitrous oxide emissions</i>	45
2.11.5	<i>Changed waste composition</i>	45
3	REPORTING ERRORS IN REPORTS V2.01	46
4	ADDITIONAL LITERATURE.....	47

1 Changes in ecoinvent data v2.2

1.1 Energy systems

1.1.1 Liquified Gas Transportation

DS-ID 1970: transport, liquefied natural gas, freight ship, OCE, [tkm]:

The demand of "operation, maintenance, port" is too high by a factor of about 33. The correct value is 7.23E-13 units/tkm (which corresponds to the demand of the transoceanic tanker).

The input of "port facilities, RER" is missing. The correct demand of "port facilities" is 7.23E-15 units/tkm.

1.1.2 Electricity mix SBB

DS-ID 722: electricity mix, SBB, CH, [kWh]:

The electricity mix was not updated according to the 2005 mix figures as displayed in the electricity mix report (p.85). The shares of nuclear power (Swiss and French production) are significantly higher, see table below:

Name	Location	Infrastructure	Process	Unit	electricity mix, SBB
Location InfrastructureProcess Unit					CH 0 kWh
electricity mix, SBB	CH	0		kWh	1.00E+0
electricity, hydropower, at run-of-river power plant	CH	0		kWh	2.76E-1
electricity, hydropower, at reservoir power plant	CH	0		kWh	2.89E-1
electricity, hydropower, at pumped storage power plant	CH	0		kWh	2.80E-2
electricity, nuclear, at power plant	CH	0		kWh	5.09E-2
electricity, nuclear, at power plant pressure water reactor	FR	0		kWh	2.03E-1
electricity, hydropower, at power plant	FR	0		kWh	1.53E-1

1.2 Construction Materials

1.2.1 Fibre Cement Facing Tile

DS-ID 11180: fibre cement facing tile, large format at plant, CH [kg]:

The consumption of some materials is too high due to an error in averaging data from the primary study. The correct values are given below:

chemicals inorganic, at plant	GLO	1.30E-02 kg
chemicals organic, at plant	GLO	1.88E-03 kg
sulphate pulp, ECF bleached, at plant	RER	5.45E-03 kg
portland cement, strength class Z 42.5, at plant	CH	6.16E-01 kg
portland cement, strength class Z 52.5, at plant	CH	2.04E-01 kg

1.2.2 Gravel mining

DS-ID 26: mining, gravel / sand, CH [kg];

DS-ID 463: gravel, crushed, at mine, CH [kg];

Land occupation and transformation values as well as the value for the recultivation of the mine were 10 times too high. The correct values are:

Occupation, mineral extraction site	2.88E-04	m2a
Transformation, to mineral extraction site	2.88E-05	m2
Transformation, from unknown	3.50E-05	m2
Transformation, to water bodies, artificial	6.27E-06	m2
Occupation, water bodies, artificial	6.27E-04	m2a
recultivation, limestone mine (CH)	8.48E-06	m2

1.2.3 Natural Stone

DS-ID 9238: natural stone plate, cut, at regional storage, CH, [kg]

DS-ID 9239: natural stone plate, grounded, at regional storage, CH, [kg]

DS-ID 9240: natural stone plate, polished, at regional storage, CH, [kg]

Values for transports within mine (represented as “diesel, burned in building machine”) were too high. The correct values are:

DS-ID 9238: natural stone plate, cut, at regional storage	1.071 MJ
DS-ID 9239: natural stone plate, grounded, at regional storage	0.357 MJ
DS-ID 9240: natural stone plate, polished, at regional storage	0.357 MJ

1.2.4 Insulation Materials

DS-ID 1000: Rock wool, at plant, CH [kg];

DS-ID 1001: Rock wool, packed, at plant, CH [kg];

Material and energy flows adapted to plant data of Flumroc from 2008 (was data from 2007).

DS-ID 11791: polystyrene foam slab, 100% recycled, at plant, CH, [kg]

DS-ID 11792: polystyrene foam slab, 45% recycled, at plant, CH, [kg]

The production of polystyrene foam slab made of recycled materials has been added using new data from one Swiss company. There are two new products:

- A. polystyrene foam slab, 100% recycled, at plant
- B. polystyrene foam slab, 45% recycled, at plant

The first one is entirely made of recycled materials (100% of recycled EPS and XPS) and the second one is made of 45% of recycled materials (EPS scrap, secondary, at plant”) and 55% of new EPS (“*polystyrene, expandable, at plant*”).

The data “*foaming, expanding*” from Hischier (2007) used in part 4.5.2.1 is not used for the “*EPS foam slab, 45% recycled, at plant*”. To be consistent, the input data for thermoforming new polystyrene used by the Swiss company was preferred.

A. Polystyrene foam slab, 100% recycled, at plant

Process system: the EPS and XPS are transported from disposal sites to the company by lorry. They are grinded to become grains of about 6mm of diameter. These grains are blown into a tank waiting to be thermoformed. Slabs of 1m² and 40, 50 or 60 mm thick are made by thermoforming. As in report n°11 - part II p.182, the cooling system is closed and the water for the process is the one that stays in the slab after thermoforming and evaporating during drying. There is no pentane emission because secondary materials have no pentane left in them.

These slabs needs more time to be thermoformed then for the 45% recycled slabs, therefore more electricity is used.

B. Polystyrene foam slab, 45% recycled, at plant

Process system: The slab's heart is made of recycled material with the same process as above. The periphery of the slab is made of new polystyrene. The foaming process used is the same as “*foaming, expanding*” from Hischier (2007). But the energy used is rather different, more electricity and less petrol fuels are used. The pentane losses are lower, because of a better recuperation. Slab of 1m² and 140 to 200 mm thick are made by thermoforming. As in report n°11 - part II p.182, the cooling system is closed and the water for the process is the one that stay in the slab after thermoforming and evaporating during drying.

The ecoinvent meta-information for polystyrene foam slab from recycled materials is given in Table 1 and the inputs and outputs flows for “EPS foam slab, 100% recycled, at plant” and “EPS foam slab, 45% recycled, at plant” are shown in Fig. 1 and Fig. 2 respectively.

Table 1 : ecoinvent meta information for polystyrene foam slab production from recycled materials.

Name	polystyrene foam slab, 100 % recycled, at plant	polystyrene foam slab, 45 % recycled, at plant
Location	CH	CH
Infrastructure Process	0	0
Unit	kg	kg
Data Set Version	2.2	2.2
Included Processes	Includes grinding and thermoforming processes	Includes foaming, grinding and thermoforming processes
Amount	1	1
Local Name	Polystyrolplatte expandiert, 100% recycling, ab Werk	Polystyrolplatte expandiert, 45% recycling, ab Werk
Synonyms		
General Comment to reference function	Combination of material and processing module. EPS foam slab has a density of 28 kg/m ³ and a thermal conductivity of 0.036 W/mK	Combination of material and processing module. EPS foam slab has a density of 28 kg/m ³ and a thermal conductivity of 0.036 W/mK
Start Date	2009	2009
End Date	2009	2009

Part II: 1. Changes in ecoinvent data v2.2

Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data from swisspor Romandie SA in Châtel-st-Denis FR	Data from swisspor Romandie SA in Châtel-st-Denis FR
Technology text	Swiss average EPS production; process from 1 factory in Switzerland	Swiss average EPS production; process from 1 factory in Switzerland
Representativeness [%]		
Production volume	Unknown	Unknown
Sampling procedure	None	None
Extrapolations	None	None
Uncertainty Adjustments	None	None

Fig. 1 : Flows for “polystyrene foam slab, 100% recycled, at plant” and their representation in the Ecoinvent database

General Flow information				Representation in ecoinvent							uncertainty			
Input	Precess Name	Output	Remarks	Category	Sub category	Infrastructu	Locatio n	modul name in ecoinvent	mean value	Unité	Source mean value	Type	StDv 95%	General Comment
Transport lorry	polystyrene foam slab, 100% recycled, at plant		Average of 50 km	transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	5.00E-02	tkm	SWISSPOR 2009	1	2.02	(2,4,1,1,1,1); Average of 50 km
EPS recycled			No loss to get 1 kg of polystyrene foam slab, average of 2 different thicknesses	waste management	recycling	No	CH	EPS scrap, secondary, at plant	1.00E+00	kg	SWISSPOR 2009	1	1.12	(2,4,1,1,1,1); No loss to get1 kg of polystyrene foam slab, average of 2 different thicknesses
Electricity processus			Includes crusing, blowing and thermoforming, foam slab manufacturing	electricity	supply mix	No	CH	electricity, low voltage, at grid	3.09E-01	kWh	SWISSPOR 2009	1	1.12	(2,4,1,1,1,1); Includes crusing, induction, foam slab manufacturing
Steam			natural gaz for water vapor heating	natural gas	power plants	No	CH	natural gas, burned in boiler modulating >100kW	8.57E+00	MJ	SWISSPOR 2009	1	1.12	(2,4,1,1,1,1); natural gaz for water vapor heating
Water			For steam process	resource	in water	No		Water, unspecified natural origin	1.36E-04	m3	SWISSPOR 2010	1	1.24	(2,4,1,1,1,5); For steam process
		EPS foam slab recycled		insulation materials	production	No	CH	polystyrene foam slab, 100% recycled, at plant	1.00E+00	kg	SWISSPOR 2009			

Fig. 2 : Flows for “polystyrene foam slab, 45% recycled, at plant” and their representation in the Ecoinvent database

General Flow information				Representation in ecoinvent							uncertainty			
Input	Precess Name	Output	Remarks	Category	Sub category	Infrastructure	Location	modul name in ecoinvent	mean value	Unité	Source mean value	Type	StDv 95%	General Comment
Transport lorry	polystyrene foam slab, 45% recycled, at plant		Average of 50 km	transport systems	road	No	CH	transport, lorry c, fleet average	2.25E-02	tkm	SWISSPOR 2009	1	2.02	(2,4,1,1,1,1); Average of 50 km
Transport lorry			Distance of 430 km	transport systems	road	No	RER	transport, lorry >32t, EURO4	2.48E-01	tkm	SWISSPOR 2009	1	2.02	(2,4,1,1,1,1); 430 km from polystyrene granulate factory to Swisspor factory
EPS recycled			No loss - 45% from 1 kg of polystyrene foam slab, average of 2 different thicknesses	waste management	recycling	No	CH	EPS scrap, secondary, at plant	4.50E-01	kg	SWISSPOR 2009	1	1.12	(2,4,1,1,1,1); No loss - 45% from 1 kg of polystyrene foam slab, no loss, average of 2 different thicknesses
EPS new			55% from 1 kg of polystyrene foam slab + 4.6% pentane weight, average of 6 different thicknesses	plastics	polymers	No	RER	Polystyrene, expandable, at plant	5.77E-01	kg	SWISSPOR 2009	1	1.12	(2,4,1,1,1,1); 55% from 1 kg of polystyrene foam slab + 4.6% pentane weight, no loss, average of 6 different thicknesses

Part II: 1. Changes in ecoinvent data v2.2

Electricity processus			Includes crusing, induction, foam slab manufacturing	electricity	supply mix	No	CH	electricity, low voltage, at grid	1.37E-01	kWh	SWISSPOR 2009	1	1.12	(2,4,1,1,1,1); Includes crusing, induction, foam slab manufacturing
Steam			natural gaz for water vapor heating	natural gas	power plants	No	CH	natural gas, burned in boiler modulating >100kW	8.57E+00	MJ	SWISSPOR 2009	1	1.12	(2,4,1,1,1,1); natural gaz for water vapor heating
Water			For steam process	resource	in water	No		Water, unspecified natural origin	1.36E-04	m3	SWISSPOR 2010	1	1.12	(2,4,1,1,1,1); For steam process
		polystyrene foam slab recycled		insulation materials	production	No	CH	polystyrene foam slab, 45% recycled, at plant	1.00E+00	kg	SWISSPOR 2009			
		Pentane	1.3% pentane in air after foam slab fabrication, when disposed before using	air	Loy population density	-	-	Pentane	7.49E-03	kg	SWISSPOR 2009	1	1.52	(2,4,1,1,1,1); 1.3% pentane in air after foam slab fabrication, when disposed before using

1.2.5 Plastic Windows

DS-ID 7148: window frame, plastic (PVC), U=1.6 W/m²K, at plant, RER, [m²]

The amount of steel needed in the window frame is lowered to 25 kg / m² (visible frame area). This value corresponds to the average of 4 major producers of PVC window frames in Switzerland and 2 major producers of window frame profiles (E-mail communication with N. Helminiak, 22.09.2009)

1.3 Chemicals

1.3.1 Propylene Hydroformation

MODS-ID 168: Propylene hydroformylation, RER, [kg]:

Some values have been corrected.

Tab. 1.1 Inputs and outputs of propylene hydroformation

	kg per kg propylene	Units	Values in v2.01	Correct values
inputs	Water, cooling, unspecified natural origin	m ³	1.5278E-01	1.5278E-02
	transport, lorry >16t, fleet average	tkm	2.11E-01	1.3994E-01
	transport, freight, rail	tkm	1.2554	8.3964E-01
	light fuel oil, burned in industrial furnace 1MW, non-modulating	MJ	12.5	3.4722
	heat, unspecific, in chemical plant	MJ	3.4722	12.5
to air	Carbon dioxide, fossil	kg	2E-03	4.1502E-01
	Propene	kg	1.40E-06	2E-03
	Carbon monoxide, fossil	kg	1.1261E-03	1.40E-06
	Hydrogen	kg	4.1502E-01	1.1261E-03

1.3.2 Sohio process

MODS-ID 169: Sohio process, RER, [kg]:

Some input values have been corrected.

Tab. 1.2 Inputs and outputs of the Sohio process

	kg per kg propylene	Units	Values in v2.01	Correct values
inputs	Water, cooling, unspecified natural origin	[m ³]	0.33896	0.0338961
	water, deionised, at plant	[kg]	42.408	4.24082
	heat, unspecific, in chemical plant	[MJ]	23.2889	2.32889
	chemical plant, organics	[units]	4.37181E-06	4.37181E-10

1.3.3 Sulphuric acid digestion of spodumene

DS-ID 331: sulphuric acid digestion of spodumene, GLO, [kg]

All values for inputs and outputs have been corrected.

Tab. 1.3 AOX emission factor in arsine production

			Value in v2.01	correct value
Raw materials	cooling water	[m3]	0.0179446	0.012541935
	spodumene	[kg]	1	1
	sulphuric acid	[kg]	0.3570850	0.249575552
	sodium carbonate	[kg]	0.3862348	0.269949066
Energy	natural gas	[MJ]	1.2411692	0.867483871
	electricity	[kWh]	0.2097692	0.146612903
waste	silicate residues	[kg]	0.9238462	0.811290323
transports	lkw	[tkm]	0.1743320	0.151952462
	rail	[tkm]	1.0459919	0.911714771
infrastructur	plant	[units]	2.99E-10	2.09032E-10
Emissions to air	heat waste	[MJ]	0.7551692	0.527806452
	sulphate	[kg]	0.0000350	2.44482E-05
Emissions to water	sulphate	[kg]	0.0174549	0.01219966
	sodium	[kg]	0.0083806	0.005857385
	carbonate	[kg]	0.0109312	0.007640068

1.4 Metals

1.4.1 Converter Steel, low-alloyed

3 waste flows were lacking in version 2.1 of the database due to an error with importing the dataset. The documentation of version 2.1 however is correct.

1.5 Plastics

1.5.1 Changes for data of PlasticsEurope due to new coal tailing data

All datasets based on the LCA publications from PlasticsEurope (list see below):

Due to the (new) modelling in the area of coal tailings – distinguishing between the actual tailings and the spoils – the modelling of the following waste types in the datasets from APME / PlasticsEurope had to be adjusted:

Waste category (according to PlasticsEurope)	Modelling in v2.1	Modelling in v2.2
Mineral waste	disposal, hard coal mining waste tailings, in landfill	disposal, spoil from coal mining, in surface landfill
Waste returned to mine	disposal, hard coal mining waste tailings, in landfill	disposal, spoil from coal mining, in surface landfill
Tailings	disposal, hard coal mining waste tailings, in landfill	Disposal, hard coal mining waste tailings, in surface backfill

The following datasets are concerned from these changes:

Part II: 1. Changes in ecoinvent data v2.2

DS-ID	dataset name	DS-ID	dataset name
285	hydrogen, cracking, APME, at plant, RER [kg]	1817	acrylonitrile-butadiene-styrene copolymer, ABS, at plant, RER [kg]
362	acetone cyanohydrin, at plant, RER [kg]	1821	nylon 6, at plant, RER [kg]
363	acetone, liquid, at plant, RER [kg]	1822	nylon 6, glass-filled, at plant, RER [kg]
375	benzene, at plant, RER [kg]	1823	nylon 66, at plant, RER [kg]
412	hydrogen cyanide, at plant, RER [kg]	1824	nylon 66, glass-filled, at plant, RER [kg]
434	pentane, at plant, RER [kg]	1825	polybutadiene, at plant, RER [kg]
451	toluene, liquid, at plant, RER [kg]	1826	polycarbonate, at plant, RER [kg]
457	xylene, at plant, RER [kg]	1829	polyethylene, HDPE, granulate, at plant, RER [kg]
1561	naphtha, APME mix, at refinery, RER [kg]	1830	polyethylene, LDPE, granulate, at plant, RER [kg]
1798	butadiene, at plant, RER [kg]	1831	polyethylene, LLDPE, granulate, at plant, RER [kg]
1799	butene, mixed, at plant, RER [kg]	1832	polymethyl methacrylate, beads, at plant, RER [kg]
1802	epoxy resin, liquid, at plant, RER [kg]	1833	polymethyl methacrylate, sheet, at plant, RER [kg]
1804	ethylene, average, at plant, RER [kg]	1834	polypropylene, granulate, at plant, RER [kg]
1805	ethylene, pipeline system, at plant, RER [kg]	1835	polystyrene, expandable, at plant, RER [kg]
1806	methyl methacrylate, at plant, RER [kg]	1836	polystyrene, general purpose, GPPS, at plant, RER [kg]
1807	methylene diphenyl diisocyanate, at plant, RER [kg]	1837	polystyrene, high impact, HIPS, at plant, RER [kg]
1808	polyols, at plant, RER [kg]	1842	polyvinylchloride, emulsion polymerised, at plant, RER [kg]
1809	propylene, at plant, RER [kg]	1843	polyvinylchloride, suspension polymerised, at plant, RER [kg]
1810	styrene, at plant, RER [kg]	1844	polyvinylidenechloride, granulate, at plant, RER [kg]
1811	toluene diisocyanate, at plant, RER [kg]	1846	styrene-acrylonitrile copolymer, SAN, at plant, RER [kg]
1813	vinyl chloride, at plant, RER [kg]	11122	propylene, pipeline system, at plant, RER [kg]

In case of the following datasets, 100% of the amount of “disposal, hard coal mining waste tailings, in landfill” from ecoinvent v2.1 is changed to “disposal, spoil from coal mining, in surface landfill” due to the fact that all was reported as mineral waste in the original publication:

DS-ID	dataset name	DS-ID	dataset name
379	carbon tetrachloride, at plant, RER [kg]	414	latex, at plant, RER [kg]
391	dichloromethane, at plant, RER [kg]	6016	trichlorethylene, at plant, WEU [kg]
426	methylchloride, at plant, WEU [kg]		

1.6 Agricultural products

1.6.1 Agricultural products from Brazil

DS-ID 6658: sugar cane, at farm, BR, [kg]:

The values for heavy metal emissions to soil were wrong.

Tab. 1.4 Use of fertilizers in the cultivation of soybean

Product	values in v2.01	correct values
	kg kg-1 sugar cane	kg kg-1 sugar cane
Cadmium	2.80329E-08	2.8088E-08
Chromium	-2.55265E-09	2.10087E-07
Copper	-3.98189E-10	1.30091E-07
Nickel	-5.13146E-09	1.04905E-07
Lead	-2.56573E-09	2.15266E-07
Tin	-6.41433E-08	2.160456E-06

1.6.2 Agricultural products from Germany

DS-ID 6576: rape seed conventional, at farm/DE, [kg]:

In the dataset the emission of fossil CO₂ from the use of urea was absent.

Tab. 1.5 Emissions to air from soybean cultivation

Emission	values in v2.01	correct values
CO ₂ , fossil [kg/kg rape seed]	-	1.07459E-02

1.6.3 Cotton and Rice seed

The datasets “cotton seed, at regional storehouse, US” and “rice seed, at regional storehouse, US”, which were modelled with “electricity, low voltage, at grid, GR”, were corrected to “electricity, low voltage, at grid US”.

1.6.4 Sheep

Multi output dataset “sheep husbandry, per head and year, US” with associated single output datasets “wool, sheep, at farm, US” and “sheep for slaughtering, live weight, at farm, US”: the allocation has been corrected according to Tab. 1.6.

Tab. 1.6 Corrected allocation factors for “sheep husbandry, per head and year, US”.

Exchanges	Allocation rule	Allocation to wool, sheep, at farm, US	Allocation to sheep for slaughtering, live weight, at farm, US
Heavy metal emissions Energy, gross calorific value, in biomass Heat, waste Carbon dioxide, in air	Mass allocation	13.3%	86.7%
All other exchanges	Economic allocation	22.8%	77.2%

The prices for wool and sheep for the economic allocation were taken from USDA (2007).

The outputs of heavy metals to water (ground and surface) and to soil were too high. The correct values are given in Tab. 1.7.

Tab. 1.7 Corrected heavy metal emissions for “wool, sheep, at farm, US” and “sheep for slaughtering, live weight, at farm, US”.

	wool, sheep, at farm	sheep for slaughtering, live weight, at farm
Heavy metals to surface water		
Cadmium (Cd, kg/kg product)	1.18E-09	5.08E-10
Chromium (Cr, kg/kg product)	2.52E-06	1.09E-06
Copper (Cu, kg/kg product)	2.38E-06	1.03E-06
Lead (Pb, kg/kg product)	7.18E-09	3.10E-09
Mercury (Hg, kg/kg product)	0.00E+00	0.00E+00
Nickel (Ni, kg/kg product)	5.10E-07	2.20E-07
Zinc (Zn, kg/kg product)	7.03E-06	3.04E-06
Heavy metals to groundwater		
Cadmium (Cd, kg/kg product)	5.07E-11	2.19E-11
Chromium (Cr, kg/kg product)	4.58E-07	1.98E-07
Copper (Cu, kg/kg product)	8.80E-08	3.80E-08
Lead (Pb, kg/kg product)	4.57E-11	1.97E-11
Mercury (Hg, kg/kg product)	0.00E+00	0.00E+00
Nickel (Ni, kg/kg product)	0.00E+00	0.00E+00
Zinc (Zn, kg/kg product)	9.67E-07	4.18E-07
Heavy metals to soil		
Cadmium (Cd, kg/kg product)	-5.18E-10	-2.24E-10
Chromium (Cr, kg/kg product)	-2.90E-06	-1.25E-06
Copper (Cu, kg/kg product)	-2.41E-06	-1.04E-06
Lead (Pb, kg/kg product)	-5.80E-09	-2.51E-09
Mercury (Hg, kg/kg product)	0.00E+00	0.00E+00
Nickel (Ni, kg/kg product)	-4.30E-07	-1.86E-07
Zinc (Zn, kg/kg product)	-5.37E-06	-2.32E-06

1.7 Wood and Wooden Products

1.7.1 Soft Fibreboard

Input of wood chips and the related transports are too low in version 2.1. Correct values are:

DS-ID 2446: fibreboard soft, at plant (u=7%), CH, [m3]:

wood chips, softwood, from industry, u=40%, at plant: 3.09E-01 m3

transport, lorry >32t, EURO3: 1.22E+01 tkm

DS-ID 11166: fibreboard soft, latex bonded, at plant (u=7%), CH, [m3]:

wood chips, softwood, from industry, u=40%, at plant: 5.38E-01 m3

transport, lorry >32t, EURO3: 3.43E+01 tkm

DS-ID 11167: fibreboard soft, without adhesives, at plant (u=7%), CH, [m3]:

wood chips, softwood, from industry, u=40%, at plant: 4.84E-01 m3
 transport, lorry >32t, EURO3: 1.85E+01 tkm

1.7.2 Sawing of Scandinavian Softwood

MODS-ID 104: sawing / debarking, Scandinavian softwood, NORDEL, [m3] with

DS-ID 2505: sawn timber, Scandinavian softwood, raw, plant-debarked, u=70%, at plant, NORDEL, [m3];

DS-ID 2441: chips, Scandinavian softwood (plant-debarked), u=70%, at plant, NORDEL, [m3];

DS-ID 2497: sawdust, Scandinavian softwood (plant-debarked), u=70%, at plant, NORDEL, [m3];

The input amounts of the allocation correction factors have been calculated in a wrong way. The correct values are the following:

Name	Wrong value in v2.1	Corrected value in v2.2
softwood, allocation correction, 1	-0.545	-1.0554
softwood, allocation correction, 2	0.402	0.777
softwood, allocation correction, 3	0.143	0.279

1.8 Transport Systems

1.8.1 Operation of lorries

DS-ID 10755: operation, lorry 3.5-7.5t, EURO3, RER, [vkm];

DS-ID 10756: operation, lorry 3.5-7.5t, EURO4, RER, [vkm];

DS-ID 10757: operation, lorry 3.5-7.5t, EURO5, RER, [vkm];

DS-ID 7291: operation, lorry 7.5-16t, EURO3, RER, [vkm];

DS-ID 7292: operation, lorry 7.5-16t, EURO4, RER, [vkm];

DS-ID 7293: operation, lorry 7.5-16t, EURO5, RER, [vkm];

DS-ID 1915: operation, lorry 3.5-16t, fleet average, RER, [vkm];

DS-ID 7294: operation, lorry 16-32t, EURO3, RER, [vkm];

DS-ID 7295: operation, lorry 16-32t, EURO4, RER, [vkm];

DS-ID 7296: operation, lorry 16-32t, EURO5, RER, [vkm];

DS-ID 7297: operation, lorry >32t, EURO3, RER, [vkm];

DS-ID 7298: operation, lorry >32t, EURO4, RER, [vkm];

DS-ID 7299: operation, lorry >32t, EURO5, RER, [vkm];

DS-ID 1926: operation, lorry >16t, fleet average, RER, [vkm];

Corrections made to the European Commission's transport model database (TREMOVE), from which the ecoinvent data is partially sourced, meant that it was necessary to correct the above datasets with regard to some of the fuel dependant and non-fuel dependant emissions. Progression through the datasets highlighted further corrections necessary due to minor data processing errors.

Only the corrected unit processes are given below. All quantities in kg/vkm.

Part II: 1. Changes in ecoinvent data v2.2

Name	Category	Operation, lorry, 3.5-7.5t, EURO3 [RER]	Operation, lorry, 3.5-7.5t, EURO4 [RER]	Operation, lorry, 3.5-7.5t, EURO5 [RER]
Diesel, low sulphur, at regional storage, CH		9.73E-2	9.32E-2	9.45E-2
Carbon dioxide, fossil	air	3.08E-1	2.96E-1	3.00E-1
Sulphur dioxide	air	9.73E-6	9.32E-6	9.45E-6
Cadmium	air	1.51E-9	1.47E-9	1.48E-9
Copper	air	1.81E-6	1.80E-6	1.80E-6
Chromium	air	2.78E-8	2.76E-8	2.76E-8
Nickel	air	2.31E-8	2.29E-8	2.29E-8
Zinc	air	7.05E-7	7.01E-7	7.03E-7
Lead	air	1.05E-7	1.05E-7	1.05E-7
Selenium	air	9.73E-10	9.32E-10	9.45E-10
Mercury	air	1.95E-12	1.86E-12	1.89E-12
Chromium VI	air	9.73E-12	9.32E-12	9.45E-12
Carbon monoxide, fossil	air	5.73E-4	2.99E-5	2.94E-5
Nitrogen oxides	air	2.41E-3	1.53E-3	8.96E-4
Particulates, <2.5 um	air	8.93E-5	4.40E-5	4.39E-5
NMVOC, unspecified origin	air	1.12E-4	4.45E-6	4.25E-6
Methane, fossil	air	2.52E-5	1.96E-6	2.29E-6
Benzene	air	7.84E-11	3.11E-12	2.97E-12
Toluene	air	1.12E-11	4.45E-13	4.25E-13
Xylene	air	9.85E-10	3.91E-11	3.74E-11
Formaldehyde	air	9.41E-9	3.73E-10	3.57E-10
Acetaldehyde	air	5.12E-9	2.03E-10	1.94E-10
Heat, waste	air	4.42E+0	4.23E+0	4.29E+0
Zinc, ion	water	2.23E-6	2.23E-6	2.23E-6
Copper, ion	water	5.28E-8	5.28E-8	5.28E-8
Cadmium, ion	water	7.89E-10	7.89E-10	7.89E-10
Chromium, ion	water	3.76E-9	3.76E-9	3.76E-9
Nickel, ion	water	1.02E-8	1.02E-8	1.02E-8
Lead	water	3.25E-8	3.25E-8	3.25E-8
Zinc	soil	2.23E-6	2.23E-6	2.23E-6
Copper	soil	5.28E-8	5.28E-8	5.28E-8
Cadmium	soil	7.89E-10	7.89E-10	7.89E-10
Chromium	soil	3.76E-9	3.76E-9	3.76E-9
Nickel	soil	1.02E-8	1.02E-8	1.02E-8
Lead	soil	3.25E-8	3.25E-8	3.25E-8

Part II: 1. Changes in ecoinvent data v2.2

Name	Category	Operation, lorry, 7.5-16t, EURO3 [RER]	Operation, lorry, 7.5-16t, EURO4 [RER]	Operation, lorry, 7.5-16t, EURO5 [RER]
Diesel, low sulphur, at regional storage, CH		1.72E-1	1.57E-1	1.60E-1
Carbon dioxide, fossil	air	5.44E-1	4.99E-1	5.07E-1
Sulphur dioxide	air	1.72E-5	1.57E-5	1.60E-5
Cadmium	air	2.26E-9	2.11E-9	2.14E-9
Copper	air	1.94E-6	1.91E-6	1.92E-6
Chromium	air	3.15E-8	3.08E-8	3.09E-8
Nickel	air	2.84E-8	2.73E-8	2.75E-8
Zinc	air	7.80E-7	7.65E-7	7.68E-7
Lead	air	1.05E-7	1.05E-7	1.05E-7
Selenium	air	1.72E-9	1.57E-9	1.60E-9
Mercury	air	3.44E-12	3.15E-12	3.20E-12
Chromium VI	air	1.72E-11	1.57E-11	1.60E-11
Carbon monoxide, fossil	air	1.19E-3	4.06E-5	3.85E-5
Nitrogen oxides	air	4.71E-3	2.78E-3	1.65E-3
Particulates, <2.5 um	air	8.43E-5	5.09E-5	5.05E-5
NM VOC, unspecified origin	air	7.81E-5	1.98E-6	2.29E-6
Methane, fossil	air	2.53E-5	1.98E-6	1.90E-6
Benzene	air	5.46E-11	1.39E-12	1.60E-12
Toluene	air	7.81E-12	1.98E-13	2.29E-13
Xylene	air	6.87E-10	1.74E-11	2.01E-11
Formaldehyde	air	6.56E-9	1.67E-10	1.92E-10
Acetaldehyde	air	3.57E-9	9.06E-11	1.04E-10
Heat, waste	air	7.80E+0	7.14E+0	7.26E+0
Zinc, ion	water	2.23E-6	2.23E-6	2.23E-6
Copper, ion	water	5.28E-8	5.28E-8	5.28E-8
Cadmium, ion	water	7.89E-10	7.89E-10	7.89E-10
Chromium, ion	water	3.76E-9	3.76E-9	3.76E-9
Nickel, ion	water	1.02E-8	1.02E-8	1.02E-8
Lead	water	3.25E-8	3.25E-8	3.25E-8
Zinc	soil	2.23E-6	2.23E-6	2.23E-6
Copper	soil	5.28E-8	5.28E-8	5.28E-8
Cadmium	soil	7.89E-10	7.89E-10	7.89E-10
Chromium	soil	3.76E-9	3.76E-9	3.76E-9
Nickel	soil	1.02E-8	1.02E-8	1.02E-8
Lead	soil	3.25E-8	3.25E-8	3.25E-8

Part II: 1. Changes in ecoinvent data v2.2

Name	Category	Operation, lorry, 16-32t, EURO3 [RER]	Operation, lorry, 16-32t, EURO4 [RER]	Operation, lorry, 16-32t, EURO5 [RER]
Diesel, low sulphur, at regional storage, CH		2.35E-1	2.05E-1	2.09E-1
Carbon dioxide, fossil	air	7.44E-1	6.51E-1	6.62E-1
Sulphur dioxide	air	2.35E-5	2.05E-5	2.09E-5
Cadmium	air	2.89E-9	2.59E-9	2.63E-9
Copper	air	2.04E-6	1.99E-6	2.00E-6
Chromium	air	3.47E-8	3.32E-8	3.34E-8
Nickel	air	3.28E-8	3.07E-8	3.09E-8
Zinc	air	8.43E-7	8.13E-7	8.17E-7
Lead	air	1.05E-7	1.05E-7	1.05E-7
Selenium	air	2.35E-9	2.05E-9	2.09E-9
Mercury	air	4.69E-12	4.11E-12	4.18E-12
Chromium VI	air	2.35E-11	2.05E-11	2.09E-11
Carbon monoxide, fossil	air	1.36E-3	5.75E-5	5.75E-5
Nitrogen oxides	air	6.44E-3	3.75E-3	2.21E-3
Particulates, <2.5 um	air	1.11E-4	5.82E-5	5.85E-5
NMVOC, unspecified origin	air	8.15E-5	3.59E-6	3.71E-6
Methane, fossil	air	7.85E-5	5.02E-6	5.02E-6
Benzene	air	5.70E-11	2.51E-12	2.60E-12
Toluene	air	8.15E-12	3.59E-13	3.71E-13
Xylene	air	7.17E-10	3.16E-11	3.26E-11
Formaldehyde	air	6.85E-9	3.01E-10	3.12E-10
Acetaldehyde	air	3.72E-9	1.64E-10	1.69E-10
Heat, waste	air	1.07E+1	9.32E+0	9.48E+0
Zinc, ion	water	5.02E-6	5.02E-6	5.02E-6
Copper, ion	water	1.19E-7	1.19E-7	1.19E-7
Cadmium, ion	water	1.78E-9	1.78E-9	1.78E-9
Chromium, ion	water	8.48E-9	8.48E-9	8.48E-9
Nickel, ion	water	2.30E-8	2.30E-8	2.30E-8
Lead	water	7.31E-8	7.31E-8	7.31E-8
Zinc	soil	5.02E-6	5.02E-6	5.02E-6
Copper	soil	1.19E-7	1.19E-7	1.19E-7
Cadmium	soil	1.78E-9	1.78E-9	1.78E-9
Chromium	soil	8.48E-9	8.48E-9	8.48E-9
Nickel	soil	2.30E-8	2.30E-8	2.30E-8
Lead	soil	7.31E-8	7.31E-8	7.31E-8

Part II: 1. Changes in ecoinvent data v2.2

Name	Category	Operation, lorry, >32t, EURO3 [RER]	Operation, lorry, >32t, EURO4 [RER]	Operation, lorry, >32t, EURO5 [RER]
Diesel, low sulphur, at regional storage, CH		2.99E-1	2.53E-1	2.57E-1
Carbon dioxide, fossil	air	9.49E-1	8.02E-1	8.15E-1
Sulphur dioxide	air	2.99E-5	2.53E-5	2.57E-5
Cadmium	air	3.53E-9	3.07E-9	3.11E-9
Copper	air	2.15E-6	2.07E-6	2.08E-6
Chromium	air	3.79E-8	3.56E-8	3.58E-8
Nickel	air	3.73E-8	3.40E-8	3.43E-8
Zinc	air	9.07E-7	8.61E-7	8.65E-7
Lead	air	1.05E-7	1.05E-7	1.05E-7
Selenium	air	2.99E-9	2.53E-9	2.57E-9
Mercury	air	5.99E-12	5.06E-12	5.14E-12
Chromium VI	air	2.99E-11	2.53E-11	2.57E-11
Carbon monoxide, fossil	air	1.64E-3	7.86E-5	6.96E-5
Nitrogen oxides	air	8.28E-3	4.78E-3	2.81E-3
Particulates, <2.5 um	air	1.96E-4	6.09E-5	6.13E-5
NMVOC, unspecified origin	air	1.69E-4	3.27E-7	4.09E-7
Methane, fossil	air	7.56E-5	4.60E-6	4.58E-6
Benzene	air	1.18E-10	2.29E-13	2.86E-13
Toluene	air	1.69E-11	3.27E-14	4.09E-14
Xylene	air	1.49E-9	2.88E-12	3.60E-12
Formaldehyde	air	1.42E-8	2.75E-11	3.44E-11
Acetaldehyde	air	7.71E-9	1.50E-11	1.87E-11
Heat, waste	air	1.36E+1	1.15E+1	1.17E+1
Zinc, ion	water	5.95E-6	5.95E-6	5.95E-6
Copper, ion	water	1.41E-7	1.41E-7	1.41E-7
Cadmium, ion	water	2.11E-9	2.11E-9	2.11E-9
Chromium, ion	water	1.00E-8	1.00E-8	1.00E-8
Nickel, ion	water	2.72E-8	2.72E-8	2.72E-8
Lead	water	8.67E-8	8.67E-8	8.67E-8
Zinc	soil	5.95E-6	5.95E-6	5.95E-6
Copper	soil	1.41E-7	1.41E-7	1.41E-7
Cadmium	soil	2.11E-9	2.11E-9	2.11E-9
Chromium	soil	1.00E-8	1.00E-8	1.00E-8
Nickel	soil	2.72E-8	2.72E-8	2.72E-8
Lead	soil	8.67E-8	8.67E-8	8.67E-8

Name	Category	Operation, lorry, 3.5-16t, fleet average [RER]	Operation, lorry, >16t, fleet average [RER]
Diesel, low sulphur, at regional storage, CH		1.20E-1	2.68E-1
Carbon dioxide, fossil	air	3.79E-1	8.48E-1
Sulphur dioxide	air	1.20E-5	2.68E-5
Cadmium	air	1.74E-9	3.22E-9
Copper	air	1.85E-6	2.10E-6
Chromium	air	2.89E-8	3.63E-8
Nickel	air	2.47E-8	3.51E-8
Zinc	air	7.28E-7	8.76E-7
Lead	air	1.05E-7	1.05E-7
Selenium	air	1.20E-9	2.68E-9
Mercury	air	2.39E-12	5.35E-12
Chromium VI	air	1.20E-11	2.68E-11
Carbon monoxide, fossil	air	8.99E-4	1.73E-3
Nitrogen oxides	air	3.84E-3	8.87E-3
Particulates, <2.5 um	air	1.54E-4	2.57E-4
NM VOC, unspecified origin	air	2.86E-4	2.19E-4
Methane, fossil	air	2.50E-5	7.67E-5
Benzene	air	2.00E-10	1.53E-10
Toluene	air	2.86E-11	2.19E-11
Xylene	air	2.52E-9	1.93E-9
Formaldehyde	air	2.40E-8	1.84E-8
Acetaldehyde	air	1.31E-8	1.00E-8
Heat, waste	air	5.43E+0	1.22E+1
Zinc, ion	water	2.23E-6	5.48E-6
Copper, ion	water	5.28E-8	1.30E-7
Cadmium, ion	water	7.89E-10	1.94E-9
Chromium, ion	water	3.76E-9	9.26E-9
Nickel, ion	water	1.02E-8	2.51E-8
Lead	water	3.25E-8	7.99E-8
Zinc	soil	2.23E-6	5.48E-6
Copper	soil	5.28E-8	1.30E-7
Cadmium	soil	7.89E-10	1.94E-9
Chromium	soil	3.76E-9	9.26E-9
Nickel	soil	1.02E-8	2.51E-8
Lead	soil	3.25E-8	7.99E-8

1.9 Mechanical Engineering

1.9.1 Degreasing of metal parts

DS-ID 10123: degreasing, metal part in alkaline bath/RER, [m²]:

The demand of "treatment, sewage, to wastewater treatment, class 3" is too high by a factor of 1'000. The correct value is 0.013m³/m².

1.9.2 Wire Drawing of Steel

DS-ID 1179: Wire drawing, steel, RER [kg]:

Chloride to water emission was changed from 1.14 kg/kg to 1.15E-3 kg/kg

1.9.3 Sheet Rolling of Copper

DS-ID 1173: Sheet rolling, copper, RER [kg]:

Electricity consumption was calculated based on consumption of Al and densities of Cu and Al. The assumption was that higher density leads to higher electricity demand per kg material. This assumption is not justified and was dismissed. The new assumption is that copper sheet rolling demands the same amount of electricity as sheet rolling of steel (0.223 kWh/kg instead of 1.81 kWh/kg).

1.9.4 Wire Drawing of Copper

DS-ID 1178: Wire drawing, copper, RER [kg]:

Electricity consumption is assumed to be twice the value of sheet rolling of copper. Thus, this amount is reduced with the reduction of "sheet rolling, copper" (from 3.61 to 0.446 kWh/kg).

1.9.5 Contour Brass and Bronze

DS-ID 1161: Contour brass, RER [kg];

DS-ID 1162: Contour bronze, RER [kg]:

Electricity consumption is assumed to be the same as for sheet rolling of copper. Thus, this amount is reduced with the reduction of "sheet rolling, copper" (from 1.81 to 0.223 kWh/kg).

1.10 Electronic Data

1.10.1 NiMH Batteries

DS-ID 7066: electrode, negative, Ni, at plant, GLO, [kg]:

The dataset "glazing, double (2-IV), U<1.1 W/m²K, laminated safety glass, at plant" is a wrong proxy for the sintering process of the nickel material.

According to Elliot & Munir 1968), the energy consumption in the nickel sinter process is about 33 kcal per mole of sintered nickel. Due to a lack of further sources, this value, that equals to 0.66 kWh/kg nickel is used here. According to Raichenko et al. 1976), 100% of the energy consumed is electrical energy. Hence, a split of 60:40 between the Chinese and the Japanese electricity mixes is used, similar to the final battery assembly.

DS-ID 7001: battery, NiMH, rechargeable, prismatic, at plant, GLO, [kg]:

The energy consumption during the assembly of the NiMH battery, is too high due to an error in the assumptions used for the calculation of the respective values.

According to Rantik 1999) a NiMH battery of type AB₅ has the following energy consumption – per kg of battery – in the final assembly step: 0.544 kWh electricity and 8.35 MJ of heat energy. Due to a lack of further sources, these values are used here. For the heat, 100% natural gas is assumed – while for the electricity input, still a split of 60:40 to the Chinese and the Japanese electricity mix is used, representing the actual production situation. The corrected values are thus the following:

	battery, NiMH, rechargeable, prismatic, at plant	
	value in v2.1	corrected value
electricity, medium voltage, at grid / CN	12.888	0.326
electricity, medium voltage, at grid / JP	8.592	0.218
heat, light fuel oil, at industrial furnace 1MW / RER	8.413	-
heat, natural gas, at industrial furnace low-NOx >100kW / RER	59.099	8.35

1.10.2 LCD Screen

DS-ID 7105: assembly, LCD module, GLO, [kg];

DS-ID 10169: assembly, LCD screen, GLO, [kg]:

The amount of NF₃ and SF₆ emissions to air, emitted during the assembly of the LCD module and the LCD screen, is due to an error in the original data source too high in the two datasets “assembly, LCD module” resp. “assembly, LCD screen”.

According to the IPCC guidelines (Bartos et al. 2006), the LCD manufacturing industry has the following tier 2b emission factors for the two mentioned fluorinated components: NF₃ 0.3 g/m², SF₆ 1.2 g/m² (m² of screen surface). Due to a lack of further sources, these values are used for the LCD production chain in ecoinvent. The corrected values in the datasets are the following:

	assembly, LCD module		assembly, LCD screen	
	Value in v2.1	Corrected value	Value in v2.1	Corrected value
Sulphur hexafluoride, to air	1.90E-3	2.79E-5	4.87E-4	7.15E-6
Nitrogen fluoride, to air	6.38E-2	6.98E-6	1.63E-2	1.79E-6

1.10.3 Laptop Use

DS-ID 6982: use, computer, laptop, office use, CH [h]:

The dataset was erroneously linked to the use modes of a desktop computer (for active, off and sleep/standby mode). This has been corrected – and now, this dataset is linked to the respective data for the use of a laptop computer.

2 Changes in ecoinvent data v2.1

2.1 Energy systems

2.1.1 Crude oil production

DS-ID 1644: crude oil, production NL, at long distance transport, RER, [kg]:

The v2.01 dataset only includes crude oil from onshore production. In fact, 33% of crude oil in the Netherlands is produced offshore and 67% is produced onshore (according to the ecoinvent report on natural gas). The v2.1 dataset is corrected accordingly. Environmental impacts of offshore production are slightly higher as compared to onshore production.

2.1.2 Hard coal production

DS-ID 835/909/910/911/912/913/914/915: hard coal, at regional storage, WEU/AU/EEU/RLA/RNA/ZA/RU/CPA, [kg]:

The uncertainty factor of the input “hard coal, at mine” in the datasets “hard coal, at regional storage” was changed from 10 to 1.01.

2.1.3 Lignite power plant

DS-ID 1036: lignite, burned in power plant, PL, [MJ]:

Mercury emissions from Polish lignite plants were changed from 9.21E-8 kg/MJ to 1.27E-8 kg/MJ. The figure, although still relatively high compared to other countries, was obtained from the Polish Geological Institute.

2.1.4 Hydro power plants

DS-ID 986: reservoir hydropower plant, CH, [unit];

DS-ID 987: reservoir hydropower plant, alpine region, RER, [unit];

DS-ID 988: reservoir hydropower plant, non alpine regions, RER, [unit];

DS-ID 989: run-of-river hydropower plant, CH, [unit];

DS-ID 990: run-of-river hydropower plant, RER, [unit];

Quantities were corrected according to the lifetime of the components and the total net output. The disposal of steel was removed from the DS assuming this material to be recycled in order to be consistent with other similar datasets.

For the reservoir power plant the total volume of concrete was also recalculated, which affected quantities of materials, disposal and particle emissions from concrete mixing.

Tab. 2.1 Corrected input data (kg/unit) for reservoir hydropower plants (DS-ID 986, 987 and 988).

Unit process	v2.01	v2.1
Chromium steel 18/8, at plant, RER	9.02E7	1.92E8
Reinforcing steel, at plant, RER	1.61E8	1.71E8
Steel, low-alloyed, at plant, RER	4.10E8	7.68E8
Portland cement, strength class Z42.5, at plant, CH	7.55E9	8.97E9
Tap water, at user, RER	3.77E9	4.95E9
Gravel, round, at mine, CH	6.41E10	7.80E10
Disposal, building, reinforced concrete, to final disposal, RER	7.18E10	9.21E10
Disposal, steel, 0% water, to inert material landfill, CH	5.00E8	Removed
Particulates, <2.5 µm	2.10E6	2.47E6
Particulates, >10 µm	2.43E7	2.57E7
Particulates, 2.5< µm and <10 µm	9.22E6	1.01E7

Tab. 2.2 Corrected input data (kg) for run-of-river hydropower plants (DS-ID 989 and 990).

Unit process	V2.0	V2.1
Chromium steel 18/8, at plant, RER	1.85E7	3.72E7
Reinforcing steel, at plant, RER	1.85E8	1.90E8
Steel, low-alloyed, at plant, RER	2.53E8	5.11E8
Disposal, building, reinforced concrete, to final disposal, RER	2.67E10	2.34E10
Disposal, steel, 0% water, to inert material landfill, CH	1.93E7	Removed

2.1.5 Inverter data of photovoltaics installations

DS-ID 1778: inverter, 2500W, at plant, RER, [unit];

DS-ID 6851: inverter, 500W, at plant, RER, [unit];

DS-ID 6852: inverter, 500kW, at plant, RER, [unit]

The steel input in the inverter datasets must be substituted by another steel input process. Quantities remain the same.

Tab. 2.3 Corrected inverter data input of the photovoltaics

Dataset ID	Name	Input in ecoinvent process v2.01	Correct input process	Amount
DS-ID 1778	inverter, 2500W, at plant, RER, [unit]	steel, electric, un- and low-alloyed, at plant	steel, low-alloyed, at plant, RER, [kg] (#1154)	9.8 Kg/unit
DS-ID 6851	inverter, 500W, at plant, RER, [unit];	steel, electric, un- and low-alloyed, at plant	steel, low-alloyed, at plant, RER, [kg] (#1154)	0.078 Kg/unit
DS-ID 6852	inverter, 500kW, at plant, RER, [unit]	steel, electric, un- and low-alloyed, at plant	steel, low-alloyed, at plant, RER, [kg] (#1154)	1438 Kg/unit

2.1.6 Slanted-roof installations for photovoltaics

DS-ID 6847: 3kWp slanted-roof installation, a-Si, laminated, integrated, on roof, CH, [unit]

DS-ID 6848: 3kWp slanted-roof installation, a-Si, panel, mounted, on roof, CH, [unit]

The mounting systems are not correct in the two datasets.

Tab. 2.4 Corrected mounting system input of the photovoltaics

Dataset ID	Name	Input in ecoinvent Data v2.01	Correct input process	Amount
DS-ID 6847	3kWp slanted-roof installation, a-Si, laminated, integrated, on roof, CH, [unit]	slanted-roof construction, mounted, on roof	slanted-roof construction, integrated, on roof	46.512 m2/unit
DS-ID 6848	3kWp slanted-roof installation, a-Si, panel, mounted, on roof, CH, [unit]	slanted-roof construction, integrated, on roof	slanted-roof construction, mounted, on roof	46.512 m2/unit

2.1.7 Electricity production mix in Denmark and Sweden

DS-ID 673: electricity, production mix DK, DK, [kWh];

DS-ID 690: electricity, production mix SE, SE, [kWh];

The electricity production mix of Sweden and Denmark was based on the values of the year 2000. Tab. 2.5 and Tab. 2.6 indicate the correct values, as already shown in the ecoinvent report v2.0.

Tab. 2.5 electricity production mix, Denmark, 2004, at busbar of the power plant

product	Name	Location	Unit	electricity, production mix DK	Uncertainty	Standard deviation	GeneralComment
	Location Infrastructure Process Unit			DK 0 kWh		%	
technosphere	electricity, production mix DK	DK	0 kWh	1.00E+0			(1,1,1,1,1,1); SDG for output: 1;
	electricity, hard coal, at power plant	NORDEL	0 kWh	4.57E-1	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, oil, at power plant	DK	0 kWh	3.99E-2	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, natural gas, at power plant	NORDEL	0 kWh	2.45E-1	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, hydropower, at power plant	DK	0 kWh	6.71E-4	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, at wind power plant	RER	0 kWh	1.72E-1	1	1.21	(1,1,1,3,3,1); national and international statistics
	electricity, at cogen ORC 1400kWth, wood, allocation exergy	CH	0 kWh	4.48E-2	1	1.21	(1,1,1,2,3,1); national and international statistics
	electricity, at cogen with biogas engine, allocation exergy	CH	0 kWh	6.48E-3	1	1.21	(1,1,1,2,3,1); national and international statistics

Tab. 2.6: electricity production mix, Sweden, 2004, at busbar of the power plant

product	Name	Location	Unit	electricity, production mix SE	Uncertainty	Standard deviation	GeneralComment
	Location Infrastructure Process Unit			SE 0 kWh		%	
technosphere	electricity, production mix SE	SE	0 kWh	1.00E+0			(1,1,1,1,1,1); SDG for output: 1;
	electricity, hard coal, at power plant	NORDEL	0 kWh	6.65E-3	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, peat, at power plant	NORDEL	0 kWh	4.54E-3	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, oil, at power plant	SE	0 kWh	1.29E-2	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, natural gas, at power plant	NORDEL	0 kWh	4.93E-3	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, industrial gas, at power plant	NORDEL	0 kWh	5.34E-3	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, hydropower, at power plant	SE	0 kWh	4.01E-1	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, hydropower, at pumped storage power plant	SE	0 kWh	6.67E-4	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, nuclear, at power plant	UCTE	0 kWh	5.05E-1	1	1.05	(1,1,1,3,1,1); national and international statistics
	electricity, at wind power plant	RER	0 kWh	6.06E-3	1	1.21	(1,1,1,3,3,1); national and international statistics
	electricity, at cogen ORC 1400kWth, wood, allocation exergy	CH	0 kWh	4.36E-2	1	1.21	(1,1,1,2,3,1); national and international statistics
	electricity, at cogen with biogas engine, allocation exergy	CH	0 kWh	6.26E-4	1	1.21	(1,1,1,2,3,1); national and international statistics

2.1.8 Electricity production and supply mix in Finland

DS-ID708: electricity mix, FI, [kWh]:

The electricity supply mix of Finland was modelled based on the values of the year 2000. Tab. 2.7 indicates the correct values, as already shown in the ecoinvent report v2.0.

Tab. 2.7: electricity supply mix, Finland, 2004, at busbar of the power plant

product	Name Location InfrastructureProcess Unit	Location	Infrastructure reProcess	Unit	electricity mix			GeneralComment
					FI 0 kWh	Uncertainty Standard Deviation	%	
technosphere	electricity mix	FI	0	kWh	1.00E+0			(1,1,1,1,1,1); SDG for output: 1;
	electricity, hard coal, at power plant	NORDEL	0	kWh	1.66E-1	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, peat, at power plant	NORDEL	0	kWh	6.58E-2	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, oil, at power plant	FI	0	kWh	6.18E-3	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, natural gas, at power plant	NORDEL	0	kWh	1.29E-1	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, industrial gas, at power plant	NORDEL	0	kWh	5.37E-3	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, hydropower, at power plant	FI	0	kWh	1.55E-1	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, nuclear, at power plant	UCTE	0	kWh	2.32E-1	1	1.05	(1,1,1,3,1,1); national and international statistics
	electricity, at wind power plant	RER	0	kWh	1.28E-3	1	1.21	(1,1,1,3,3,1); national and international statistics
	electricity, at cogen ORC 1400kWh, wood, allocation exergy	CH	0	kWh	1.03E-1	1	1.21	(1,1,1,2,3,1); national and international statistics
	electricity, at cogen with biogas engine, allocation exergy	CH	0	kWh	2.22E-4	1	1.21	(1,1,1,2,3,1); national and international statistics
	electricity, production mix NO	NO	0	kWh	1.02E-3	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, production mix SE	SE	0	kWh	1.08E-2	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, production mix CENTREL	CENTREL	0	kWh	1.19E-1	1	1.05	(1,1,1,3,1,1); national and international statistics

DS-ID 675: electricity, production mix FI, FI, [kWh]:

The production mix was modelled with the values of the Finnish supply mix. Tab. 2.8 shows the correct values.

Tab. 2.8 electricity production mix, Finland, 2004, at busbar of the power plant

product	Name Location InfrastructureProcess Unit	Location	Infrastructure reProcess	Unit	electricity, production mix FI			GeneralComment
					FI 0 kWh	Uncertainty Standard Deviation	%	
technosphere	electricity, production mix FI	FI	0	kWh	1.00E+0			(1,1,1,1,1,1); SDG for output: 1;
	electricity, hard coal, at power plant	NORDEL	0	kWh	1.91E-1	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, peat, at power plant	NORDEL	0	kWh	7.57E-2	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, oil, at power plant	FI	0	kWh	7.11E-3	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, natural gas, at power plant	NORDEL	0	kWh	1.48E-1	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, industrial gas, at power plant	NORDEL	0	kWh	6.18E-3	1	1.05	(1,1,1,2,1,1); national and international statistics
	electricity, hydropower, at power plant	FI	0	kWh	1.79E-1	1	1.05	(1,1,1,1,1,1); national and international statistics
	electricity, nuclear, at power plant	UCTE	0	kWh	2.67E-1	1	1.05	(1,1,1,3,1,1); national and international statistics
	electricity, at wind power plant	RER	0	kWh	1.47E-3	1	1.21	(1,1,1,3,3,1); national and international statistics
	electricity, at cogen ORC 1400kWh, wood, allocation exergy	CH	0	kWh	1.18E-1	1	1.21	(1,1,1,2,3,1); national and international statistics
	electricity, at cogen with biogas engine, allocation exergy	CH	0	kWh	2.55E-4	1	1.21	(1,1,1,2,3,1); national and international statistics

2.1.9 Electricity distribution in China

DS-ID 6679: electricity, high voltage, at grid, CN, [kWh];

DS-ID 6681: electricity, medium voltage, at grid, CN, [kWh];

DS-ID 6680: electricity, low voltage, at grid, CN, [kWh];

There is a change of the loss rates of high voltage, medium voltage and low voltage. The correct values are: high voltage = 0.94 % (including 0.5 % due to transport over large distances), medium voltage = 1.03 %, a low voltage = 27 %.

Tab. 2.9 Correct values regarding the electricity distribution in China

Voltage level:	Losses CH	Consumption CN, assumed	Losses at Voltage level	Ecoinvent model					
				Final consumption at Voltage	Calculated electricity available at	Calculated losses	additional (long distance transmission)	Heat to air	Heat to soil
	%	%	TWh/a	TWh/a	TWh/a	%	%	%	%
Low voltage	80%	20%	111	412	412	27.0		25	75
Medium Voltage	13%	60%	18	1236	1759.2	1.03		55	45
High voltage	7%	20%	10	412	2189.27	0.44	0.5	95	5
Total	100%	100%	139	2'060					

2.2 Construction materials

2.2.1 Flat Glass, uncoated

DS-ID 806: flat glass, uncoated, at plant, RER, [kg]:

The CO₂ Emission (fossil) is too low by a factor of 2.7. The correct value is 0.693 kg/kg. The value shown in the ecoinvent report v2.0 is correct.

2.2.2 Building, multi-storey

DS-ID 549: building, multi-storey, RER, [m³]:

The inputs of reinforcing steel and concrete are confused. The correct values are:

Concrete: 9.27E-2 m³/m³

Steel: 7.17 kg/m³

The values shown in the ecoinvent report v2.0 are correct.

2.2.3 Glazings

DS-ID 7140: glazing, double (2-IV), U<1.1 W/m²K, at plant, RER, [m²];

DS-ID 7142: glazing, triple (3-IV), U<0.5 W/m²K, at plant, RER, [m²];

DS-ID 7137: disposal, building, glazing 2-IV, U<1.1 W/m²K, to final disposal, CH, [m²];

DS-ID 7138: disposal, building, glazing 2-IV, U<1.1 W/m²K, LSG, to final disposal, CH, [m²];

DS-ID 7139: disposal, building, glazing 3-IV, U<0.5 W/m²K, to final disposal, CH, [m²];

The density on which the calculation of the amount of Polybutadiene is based was too low by a factor of 1000.

The assumed loss of Argon and especially Krypton was too high by a factor of 12. According to Glass Trösch AG, the loss of Krypton is below 5%. A value of 5% should be assumed.

The correct inputs are shown in Tab. 2.10.

Tab. 2.10 Correct values of industrial gases and rubber of glazing datasets

		glazing, double (2- IV), U<1.1 W/m2K m2	glazing, triple (3- IV), U<0.5 W/m2K m2	disposal, building, glazing 2- IV, U<1.1W/ m2K m2	disposal, building, glazing 2- IV, U<1.1W/ m2K, LSG m2	disposal, building, glazing 3- IV, U<0.5W/ m2K m2
argon, liquid, at plant	kg	2.44E-02				
krypton, gaseous, at regional storage	kg		7.70E-02			
polybutadiene, at plant	kg	9.37E-03	1.87E-02			
disposal, rubber, unspecified, 0% water, to municipal incineration	kg			4.48E-01	1.50E+00	6.76E-01

2.2.4 Fibre Cement

DS-ID 514: fibre cement facing tile, at plant, CH, [kg];

DS-ID 11180: fibre cement facing tile, large format, at plant, CH, [kg]

DS-ID 11181: fibre cement facing tile, small format, at plant, CH, [kg]:

Data for the mentioned three types of fibre cement tiles were changed. These changes are reported in part III of this report here (Annex 4, part III: replaces part IX of report No. 7, Life Cycle Inventories of Building Products).

2.2.5 Insulation Materials

All datasets of the below mentioned types of insulation materials:

Data for some insulation materials were changed. These changes are reported in part III of this report here (Annex 3, part III: replaces part XIV of report No. 7, Life Cycle Inventories of Building Products). The datasets of the following insulation materials were changed:

- Rock wool
- Expanded polystyrene

2.2.6 Expanded Clay / Vermiculite

DS-ID 462: expanded vermiculite, at plant, CH, [kg];

DS-ID 480: expanded clay, at plant, DE, [kg]:

Expanded vermiculite, at plant: Dataset for ship transport is corrected

Expanded vermiculite, at plant and Expanded clay, at plant: Modelling of incineration of oil is changed. The amount of oil incinerated and the specific emissions from this process are replaced by the generic dataset for incineration of oil in industrial furnace in ecoinvent.

Fig. 2.1 and Fig. 2.2 show the correct in – and outputs for the two datasets.

Part II: 2. Changes in ecoinvent data v2.1

General Flow information					Representation in ecoinvent							Uncertainty Information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
vermiculite, at mine	→	swelling of vermiculite					construction materials	additives	No	ZA	vermiculite, at mine	1.14E+00 kg		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
transport of total amount of raw vermiculite with freight rail	→			transport distance: 400 km		transport systems	train	No	RER	transport, freight, rail	4.55E-01 tkm		Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
transport of total amount of raw vermiculite with freight ship	→			transport distance: 13'500 km		transport systems	ship	No	OCE	transport, transoceanic freight ship	1.53E+01 tkm		Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
transport of total amount of raw vermiculite with lorry	→			transport distance: 42 km		transport systems	road	No	RER	transport, lorry >16t, fleet average	4.77E-02 tkm		Richter et al. (1995)	1	2.15	(4,5,3,1,1,5);	
infrastructure (incl. Land-use and machines)	→			clay manufacturing plant used as proxy; amount copied from module "brick, at plant"		construction materials	additives	Yes	CH	mine, clay	2.00E-10 unit		estimation	1	5.00	estimation	
water	→			assumed that tap water has been used used for high-lift truck, module "diesel, burnded in building machine" used as proxy, amount: 0.000267 l/kg		water supply	production	No	RER	tap water, at user	1.34E-02 kg		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
diesel	→			UCTE supply mix used as proxy.		construction processes	machinery	No	GLO	diesel, burned in building machine	9.61E-03 MJ		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
electricity	→			the amount of light fuel oil (0.055kg/kg expanded vermiculite) is equivalent to about 2.4 MJ/kg		electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.71E-02 kWh		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);	
light fuel oil	→			One paper bag for 25 kilo product. Assumed weight of paper bag 0.05 kg		oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	2.40E+00 MJ		Richter et al. (1995)	1	1.21	(4,1,1,1,1,1);	
packing material (paper)	→					paper & cardboard	cardboard & corrugated board	No	RER	corrugated board base paper, kraftliner, at plant	2.00E-03 kg		Richter et al. (1995)	1	1.41	(4,5,3,5,1,5);	
packing material (plastics)	→					plastics	processing	No	RER	packaging film, LDPE, at plant	4.81E-04 kg		Richter et al. (1995)	1	1.41	(4,5,3,5,1,5);	
	→		waste heat			from electricity uses	air	unspecified			Heat, waste	9.77E-02 MJ		calculated after the electricity use	1	1.40	(4,5,3,1,1,5);
	→		vermiculite waste			losses during production process						5.08E-02 kg		Richter et al. (1995)			
	→		disposal of dust from filter			total amount: 1kg/t product	waste management	sanitary landfill	No	CH	disposal, inert material, 0% water, to sanitary landfill	1.00E-03 kg		EPA (1998), chapter 11.28	1	5.00	estimation
	→		expanded vermiculite				construction materials	additives	No	CH	expanded vermiculite, at plant	1.00E+00 kg					

Fig. 2.1 Representation of "expanded Vermiculite, at plant" in ecoinvent v2.1

Part II: 2. Changes in ecoinvent data v2.1

General Flow information					Representation in ecoinvent							Uncertainty Information			
Input		Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
clay, at mine	→	swelling of clay			construction materials	additives	No	CH	clay, at mine	1.10E+00 kg		Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
transport with lorry within production in Germany	→			50 tkm for 1,1 t of raw material (from mine beside production plant; about 5 km)	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.55E-02 tkm		Hutter et al. (1999)	1	2.34	(3,5,5,3,1,5);
infrastructure (incl. Land-use and machines)	→			clay manufacturing plant used as proxy; amount copied from module "brick, at plant"	construction materials	additives	Yes	CH	mine, clay	2.00E-10 unit		estimation	1	5.00	estimation
water	→			copied from vermiculite production	water supply	production	No	RER	tap water, at user	1.34E-02 kg		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
heavy fuel oil as additive	→			about 5 w-%	oil	fuels	No	RER	heavy fuel oil, at regional storage	5.50E-02 kg		http://www.agoef.de/baustoffe_daemmung/geblaehete.html	1	1.68	(4,5,5,3,1,5);
diesel	→			copied from "expanded vermiculite", used for high-lift truck, module "diesel, burned in building machine" used as proxy, amount: 0.000267 l/kg	construction processes	machinery	No	GLO	diesel, burned in building machine	9.61E-03 MJ		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
electricity	→				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.50E-02 kWh		Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
heavy fuel oil for burning	→			the used amount of heavy fuel oil (0.075 l/kg expanded clay) is equivalent to about 3.09 MJ/kg	oil	heating systems	No	RER	heat, heavy fuel oil, at industrial furnace 1MW	3.09E+00 MJ		Hutter et al. (1999)	1	1.64	(3,5,5,3,1,5);
packing material (paper)	→			copied from "expanded vermiculite", 0.04 paper bags for one kilo product. Assumed weight of paper bag 0.05 kg	paper & cardboard	cardboard & corrugated board	No	RER	corrugated board base paper, kraftliner, at plant	2.00E-03 kg		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
packing material (plastics)	→			copied from "expanded vermiculite"	plastics	processing	No	RER	packaging film, LDPE, at plant	4.81E-04 kg		Richter et al. (1995)	1	1.40	(4,5,3,1,1,5);
			→ waste heat	from electricity uses	air	unspecified			Heat, waste	1.26E-01 MJ		calculated after the electricity use	1	1.64	(3,5,5,3,1,5);
			→ vermiculite waste	losses during production process						1.00E-01 kg					
			→ expanded clay		construction materials	additives	No	DE	expanded clay, at plant	1.00E+00 kg					

Fig. 2.2 Representation of "expanded clay, at plant" in ecoinvent v2.1

2.3 Chemicals

2.3.1 Arsine

DS-ID 7239: arsine, at plant, GLO, [kg]:

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.11 AOX emission factor in arsine production

	Value in v2.01	correct value
AOX (kg)	1.41	0

2.3.2 Benzal chloride

DS-ID 6615: benzal chloride, at plant, RER, [kg]:

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.12 AOX emission factor in benzal chloride production

	Value in v2.01	correct value
AOX (kg)	4.821E-01	0

2.3.3 Benzaldehyde

DS-ID 6616: benzaldehyde, at plant, RER, [kg]:

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.13 AOX emission factor in benzaldehyde production

	Value in v2.01	correct value
AOX (kg)	6.78E-01	2.16E-03

2.3.4 Benzene chlorination

MODS-ID 165: benzene chlorination, RER, [kg]:

All values are too low by a factor of 10, except the benzene input, which is correct.

The derived unit process datasets:

DS-ID 6650: monochlorobenzene, at plant, RER, [kg];

DS-ID 6653: o-dichlorobenzene, at plant, RER, [kg];

DS-ID 6654: p-dichlorobenzene, at plant, RER, [kg];

DS-ID 6639: hydrochloric acid from benzene chlorination, at plant, RER, [kg]:

The benzene input is too low by a factor of ten. All other values (inputs and emissions) are correct.

2.3.5 Benzyl alcohol

DS-ID 6617: benzyl alcohol, at plant, RER, [kg]:

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.14 AOX emission factor in benzyl alcohol production

	Value in v2.01	correct value
AOX (kg)	4.14E-01	0

2.3.6 Benzyl chloride

DS-ID 6618: benzyl chloride, at plant, RER, [kg]:

The calculation of AOX and emission of chlorine and chloride are not correct.

Tab. 2.15 AOX, chlorine and chloride emission factor in benzyl chloride production

	Values in v2.01	correct values
AOX (kg)	3.01E-01	0
Chlorine, to water (kg)	3.01E-01	0
Chloride, to water (kg)	-	3.01E-01

2.3.7 Chlorinated solvents

DS-ID 452: trichloromethane, at plant, RER, [kg]:

A small error in the reaction formula was corrected. The input of chlorine is about five percent higher than in v1.3. The correct value is 0.848 kg/kg.

The ecoinvent report shows the correct data as implemented in ecoinvent Data v2.01.

2.3.8 Chlorosilane amination

MODS-ID 205: chlorosilane amination, GLO, [kg]:

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.16 AOX emission factor in chlorosilane amination

	Value in v2.01	correct value
AOX (kg)	1.56E-02	0

The derived unit process datasets:

DS-ID 7243: ammonium chloride from chlorosilane, at plant, GLO, [kg];

DS-ID 7220: hexamethyldisilazane, at plant, GLO, [kg]:

The AOX emission factor changes according to the changes in the multi-output dataset.

2.3.9 Ethylene chlorofluorination

MODS-ID 203: ethylene chlorofluorination, GLO, [kg]:

The calculation of AOX and emission of chlorine and chloride are not correct.

Tab. 2.17 AOX, chlorine and chloride emission factor in ethylene chlorofluorination

	Values in v2.01	correct values
AOX (kg)	1.73	0
Chlorine, to water (kg)	6E-01	0
Chloride, to water (kg)	1.128	1.728

The derived unit process datasets:

DS-ID 7227: monochloropentafluoroethane, at plant, GLO, [kg];

DS-ID 7219: hexafluoroethane, at plant, GLO, [kg];

The inputs and outputs change according to the changes in the multi-output dataset.

2.3.10 Ethylene glycol diethyl ether

DS-ID 6630: ethylene glycol diethyl ether, at plant, RER, [kg];

The AOX calculation is not correct. The correct emission factor is '1.17E-03'.

Tab. 2.18 AOX emission factor in ethylene glycol diethyl ether production

	Value in v2.01	correct value
AOX (kg)	3.07E-01	1.17E-03

2.3.11 Ethylene glycoles

MODS-ID 110: ethylene oxide, in ethylene glycol plant, RER, [kg];

The COD value in the multioutput process (and thus in the three subsequent single output processes of the three products) was due to a calculation error a factor 1'000 too high. The correct value is 2.876E-04 kg.

The derived unit process datasets:

DS-ID 393: diethylene glycol, at plant, RER, [kg];

DS-ID 402: ethylene glycol, at plant, RER, [kg];

DS-ID 454: triethylene glycol, at plant, RER, [kg];

The COD value in the output changes according to the changes in the multi-output dataset.

2.3.12 Lithium chloride

DS-ID 7221: lithium chloride, at plant, GLO, [kg];

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.19 AOX emission factor in lithium chloride production

	Value in v2.01	correct value
AOX (kg)	4.21E-02	0

2.3.13 Lithium chloride electrolysis

MODS-ID 210: lithium chloride electrolysis, GLO, [kg]:

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.20 AOX emission factor in lithium chloride electrolysis

	Value in v2.01	correct value
AOX (kg)	1E-01	0

The derived unit process datasets:

DS-ID 7224: lithium, at plant, GLO, [kg];

DS-ID 7249: chlorine, gaseous, lithium chloride electrolysis, at plant, GLO, [kg]:

The AOX emission factor changes according to the changes in the multi-output dataset.

2.3.14 Maleic Anhydride from direct oxidation of n-butane

MODS-ID 22: direct oxidation of n-butane, RER, [kg]:

The CO₂ emission to air are too high. Instead of the former data source, the following stoichiometric calculation is used for the calculation of the total amount of CO₂ emissions to air:

- efficiency of process: 63%
- input amount of n-butane: 1.105 kg
- hereof not transformed 37%, i.e. 408.9 g or 7.03 mol of n-butane
- assuming a 100% transformation into CO₂ this leads to 1.238 kg of CO₂

The derived unit process datasets:

DS-ID 447: steam from direct oxidation of n-butane, at plant, RER, [kg];

DS-ID 418: maleic anhydride from the direct oxidation of n-butane, at plant, RER, [kg]:

The output emissions of CO₂ change according to the changes in the multi-output dataset.

2.3.15 Modified Solvay Process

MODS-ID 206: modified Solvay process, GLO, [kg]:

The values for the input of raw materials are calculated with a wrong stoichiometry. That is why the values of transport service requirements and of some emissions change too.

Tab. 2.21 Consumption of raw materials and auxiliaries for the modified Solvay process

Input per kg ammonium	Values in v2.01	Correct values
Sodium chloride (kg)	1.71	3.412
Carbon dioxide (kg)	2.588	1.294

Tab. 2.22 Total transport amounts for the modified Solvay process

tkm kg-1 ammonium	Values in v2.01	Correct values
Lorry	5.294-01	5.706E-01
Train	3.176	3.424

Tab. 2.23 Process emissions to air from the modified Solvay process

Output	Value in v2.01	Correct value
CO ₂ , fossil, to air (kg)	1.29E02	6.47E-02

Tab. 2.24 Process emissions to water from the modified Solvay process

Output	Values in v2.01	Correct values
sodium (kg)	3.38E-02	6.76E-02
chloride (kg)	5.15E-02	1.03E-01
AOX (kg)	5.15E-02	1.03E-01

Tab. 2.25 Product output of ammonium chloride and sodium carbonate from the modified Solvay process

kg per kg ammonium	Values in v2.01	Correct values
Ammonium chloride, at plant, GLO	3.055	2.962
Sodium carbonate, from modified Solvay process, at plant, GLO	3.055	2.962

The derived unit process datasets:

DS-ID 7238: ammonium chloride, at plant, GLO, [kg];

DS-ID 7246: sodium carbonate from ammonium chloride production, at plant, GLO, [kg];

The inputs and outputs change according to the changes in the multi-output dataset.

2.3.16 Phosphor chloride

DS-ID 7212: phosphoryl chloride, at plant, RER, [kg];

The AOX calculation is not correct. The correct emission factor is '0'.

Tab. 2.26 AOX emission factor in lithium chloride electrolysis

	Value in v2.01	correct value
AOX (kg)	3.64E-2	0

2.4 Metals

The following datasets were changed or added. All these changes are reported directly in the updated metal report (v2.1)

- steel, converter, low-alloyed, at plant
- stibnite ore, 70% stibnite, at mine
- antimony, at refinery
- disposal, sulfidic tailings, off-site

- disposal, lead smelter slag, 0% water, to residual material landfill
- aluminium, primary, liquid, at plant
- ferronickel, 25% Ni, at plant
- zinc coating, pieces, adjustment per um
- molybdenum, at regional storage

2.4.1 Cadmium substances

DS-ID 6902: cadmium, semiconductor-grade, at plant, US, [kg];

DS-ID 6905: cadmium chloride, semiconductor-grade, at plant, US, [kg];

DS-ID 6906: cadmium telluride, semiconductor-grade, at plant, US, [kg];

The cadmium emissions are too high due to a mistake regarding the filtration of airborne emissions in the cadmium purification process. The HEPA-filter efficiency (1-99.97 %) must not be multiplied with the amount of (liquid) wastes, but should be multiplied with the amount of airborne process emissions in order to receive the cadmium emissions to nature. That way calculated emissions of 6.06 E-06 kg/kg in the purification process of “cadmium, semiconductor-grade” are consistent with table 12 in Fthenakis (2004).

cadmium, semiconductor-grade, at plant:

$$\underbrace{(1 - 99.97 \%) *}_{\text{Filter...efficiency}} \underbrace{2 \%}_{\text{Emissions}} * 1 \text{ kgCd} = \underbrace{6 * 10^{-6} \text{ kgCd}}_{\text{emissions...to...air}}$$

cadmium chloride, semiconductor-grade:

$$\underbrace{(1 - 99.97 \%) *}_{\text{Filter...efficiency}} \underbrace{(2 \%) * 0.645 \text{ kgCd}}_{\text{Emissions}} = \underbrace{3.876 * 10^{-6} \text{ kgCd}}_{\text{emissions...to...air}}$$

cadmium telluride, semiconductor-grade, at plant:

$$\underbrace{(1 - 99.97 \%) *}_{\text{Filter...efficiency}} \underbrace{(2 \%) * 1 \text{ kgCdTe}}_{\text{Emissions}} = \underbrace{6 * 10^{-6} \text{ kgCdTe}}_{\text{emissions...to...air}}$$

Tab. 2.27 Corrected values of cadmium emissions (to air/high population density) of the “cadmium, semiconductor grade” datasets

Dataset ID	Name	Value in ecoinvent Data v2.01	Correct value	Unit
DS-ID 6902	cadmium, semiconductor-grade, at plant, US, [kg];	1.62E-04	6.0E-06	Kg/kg
DS-ID 6905	cadmium chloride, semiconductor-grade, at plant, US, [kg]	4.4572E-5	3.876E-06	Kg(kg)
DS-ID 6906	cadmium telluride, semiconductor-grade, at plant, US, [kg]	1.5E-5	6.0E-06	Kg/kg

Additional Reference:

Bartos et al. 2006 Bartos S., Beu L. S., Burton C. S., Fraust C. L., Illuzzi F., Mocella M. T. and Raoux S. (2006) Electronics Industry Emissions. In: *2006 IPCC Guidelines for National*

Greenhouse Gas Inventories (ed. IPCC). The Intergovernmental Panel on Climate Change (IPCC), Hayama, Kanagawa (Japan)

Elliot & Munir 1968 Elliot A. G. and Munir Z. A. (1968) The sintering of nickel/aluminium spheres to nickel plates. *In: Journal of Materials Science*, 3(2), pp. 150-157.

Raichenko et al. 1976 Raichenko A. I., Kolchinskii M. Z. and Levina D. A. (1976) Theory and Technology of Sintering, Thermal and Chemicothermal Treatment Processes. *In: Poroshkovaya Metallurgiya*, 10(166), pp. 19-26.

Rantik 1999 Rantik M. (1999) Life Cycle Assessment of Five Batteries for Electric Vehicles under Different Charging Regimes. Kommunikationsforskningsberedningens (KFB) / Chalmers University of Technology, Stockholm (Sweden)

DS-ID 6904: cadmium sulphide, semiconductor-grade, at plant, US, [k];

The waterborne cadmium emissions in this dataset are mismatched as “from nature”. However, they are emissions “to nature”.

Tab. 2.28 Corrected allocation of the waterborne cadmium emissions of the “cadmium sulphide, semiconductor-grade” dataset

Dataset ID	Name	Value in ecoinvent Data v2.01	Correct value
DS-ID 6904	cadmium sulphide, semiconductor-grade, at plant, US, [k]	From nature; water/unspecified	To nature; water/unspecified

2.4.2 Mercure

DS-ID 1111: mercury, liquid, at plant, GLO, [kg];

The input of heat out of heavy fuel oil („heat, heavy fuel oil, at industrial furnace 1MW”) was due to a calculation error a factor of 10 too high – the correct value is 124 MJ/kg mercury (instead of 1'240 MJ/kg).

2.5 Plastics

2.5.1 Methodological change for integration of data of PlasticsEurope

All datasets based on the LCA publications from PlasticsEurope (list see below):

The treatment of two types of waste – regulated chemical waste, unregulated chemical waste – reported in the PlasticsEurope data has been changed due to the high influence of the former choice (i.e. the disposal in an underground deposit) in the updated Swiss LCIA method “ecological scarcity 2006”, especially in case of the ABS dataset (DS ID 1817) where the amount of “regulated chemical waste” had grown for a factor more than 20 compared to the former data used.

According to PlasticsEurope, this increase is attributable to the process data (and not the background energy chain) due to the inclusion of new production sites. A more detailed analysis concerning the origin of these waste amounts was however not available from the author. According to other experts from the chemical industry, today all burnable waste is incinerated in adequate incineration plants.

Hence, the disposal in the underground deposit was replaced for the data in ecoinvent v2.1 by a disposal as hazardous waste in a hazardous waste incineration (HWI).

Tab. 2.29 Changes in the transformation from PlasticsEurope to Ecoinvent for the waste data

PlasticsEurope	ecoinvent v2.01	Corrected values for v2.1
1 mg regulated chemical waste	10 ⁻⁶ kg disposal, hazardous waste, 0% water, to underground deposit	10 ⁻⁶ kg disposal, hazardous waste, 25% water, to hazardous waste incineration
1 mg unregulated chemical waste	10 ⁻⁶ kg disposal, hazardous waste, 0% water, to underground deposit	10 ⁻⁶ kg disposal, hazardous waste, 25% water, to hazardous waste incineration

These changes have been done for all datasets based on the LCA publications from PlasticsEurope – i.e. the following datasets:

DS-ID	dataset name	DS-ID	dataset name
1561	naphtha, APME mix, at refinery, RER [kg]	1829	polyethylene, HDPE, granulate, at plant, RER [kg]
1805	ethylene, pipeline system, at plant, RER [kg]	1830	polyethylene, LDPE, granulate, at plant, RER [kg]
1804	ethylene, average, at plant, RER [kg]	1831	polyethylene, LLDPE, granulate, at plant, RER [kg]
1809	propylene, at plant, RER [kg]	1834	polypropylene, granulate, at plant, RER [kg]
11122	propylene, pipeline system, at plant, RER [kg]	1836	polystyrene, general purpose, GPPS, at plant, RER [kg]
1810	styrene, at plant, RER [kg]	1835	polystyrene, expandable, at plant, RER [kg]
1799	butene, mixed, at plant, RER [kg]	1837	polystyrene, high impact, HIPS, at plant, RER [kg]
375	benzene, at plant, RER [kg]	1832	polymethyl methacrylate, beads, at plant, RER [kg]
1807	methylene diphenyl diisocyanate, at plant, RER [kg]	1833	polymethyl methacrylate, sheet, at plant, RER [kg]
1811	toluene diisocyanate, at plant, RER [kg]	1817	acrylonitrile-butadiene-styrene copolymer, ABS, at plant, RER [kg]
1798	butadiene, at plant, RER [kg]	1846	styrene-acrylonitrile copolymer, SAN, at plant, RER [kg]
1806	methyl methacrylate, at plant, RER [kg]	1825	polybutadiene, at plant, RER [kg]
362	acetone cyanohydrin, at plant, RER [kg]	1826	polycarbonate, at plant, RER [kg]
366	acrylonitrile, at plant, RER [kg]	1821	nylon 6, at plant, RER [kg]
434	pentane, at plant, RER [kg]	1822	nylon 6, glass-filled, at plant, RER [kg]
1808	polyols, at plant, RER [kg]	1823	nylon 66, at plant, RER [kg]
363	acetone, liquid, at plant, RER [kg]	1824	nylon 66, glass-filled, at plant, RER [kg]
457	xylene, at plant, RER [kg]	1844	polyvinylidenechloride, granulate, at plant, RER [kg]
285	hydrogen, cracking, APME, at plant, RER [kg]	1813	vinyl chloride, at plant, RER [kg]
286	hydrogen, liquid, at plant, RER [kg]	1842	polyvinylchloride, emulsion polymerised, at plant, RER [kg]
412	hydrogen cyanide, at plant, RER [kg]	1843	polyvinylchloride, suspension polymerised, at plant, RER [kg]
451	toluene, liquid, at plant, RER [kg]		

2.5.2 PVC, bulk polymerised, and PVC, at regional storage

DS-ID 1841: polyvinylchloride, bulk polymerised, at plant, RER, [kg];

DS-ID 1840: polyvinylchloride, at regional storage, RER, [kg];

Due to the fact that the last update of PlasticsEurope only resulted in new, updated dataset for the two other types of polyvinylchloride (emulsion polymerised, suspension polymerised). As in addition, bulk polymerised polyvinylchloride has a very small market share below 5% (personal communication PlasticsEurope), no proper production data for bulk PVC are reported – instead, a mix of the two other PVC types according to their market volumes is used.

Similar, for the dataset “polyvinylchloride, at regional storage” only the market shares of emulsion polymerised and suspension polymerised PVC are taken into account – and no bulk polymerised PVC is taken into account anymore.

2.6 Agricultural products

2.6.1 Agricultural products from Brazil

DS-ID 6658: soybeans, at farm, BR, [kg];

The table showing the use of fertilizers is not correct in the ecoinvent report.

In the dataset the values are correct except for the input for ammonium nitrate phosphate that is calculated as “ammonium nitrate phosphate as N” instead of “ammonium nitrate phosphate as P2O5”.

Tab. 2.30 Use of fertilizers in the cultivation of soybean

Product	values in v2.01 kg kg ⁻¹ soybean	correct values kg kg ⁻¹ soybean
ammonium sulphate as N	5.6604E-04	-
urea as N	1.6981E-03	-
ammonium nitrate as N	5.6604E-04	-
diammonium phosphate as N	4.9528E-04	-
potassium nitrate as N	2.1226E-04	-
diammonium phosphate as P ₂ O ₅	1.1675E-02	5.307E-03
single superphosphate as P ₂ O ₅	7.5236E-03	3.420E-03
triple superphosphate as P ₂ O ₅	4.1509E-03	1.887E-03
phosphate rock as P ₂ O ₅	1.2972E-03	5.896E-04
ammonium nitrate phosphate as P ₂ O ₅	1.2972E-03	5.896E-04
potassium chloride as K ₂ O	2.4371E-02	1.179E-02
Total fertilizers	5.385E-02	2.358E-02

The values of the emissions of N₂O and NO_x were too low.

Tab. 2.31 Emissions to air from soybean cultivation

Emission	values in v2.01	correct values
N ₂ O [kg/kg soybean]	9,8E-04	1.258E-03
NO _x [kg/kg soybean]	2.06E-04	2.642E-04

2.6.2 Agricultural products from Germany

DS-ID 6957: rape seed conventional, Saxony-Anhalt, at farm, DE, [kg]:

The input and output of the pesticide "Clomazone" is too high by a factor of 1'000. The correct value for the input "cyclic N-compounds, at regional storehouse" is 9.70E-5 kg/kg rape seed and for the output "Clomazone" to "soil/agricultural" it is 2.38E-5 kg/kg rape seed.

2.6.3 Agricultural products from Switzerland

DS-ID 149: sugar beet seed IP, at regional storehouse, CH, [kg]:

The values for "Energy, gross calorific value, in biomass" and "Carbon dioxide, in air" of sugar beet seed have been confounded. The correct values are -84.60 MJ/kg sugar beet seed and -7.21 kg CO₂/kg sugar beet seed.

DS-ID 190: barley grains extensive, at farm, CH, [kg]:

DS-ID 191: barley grains IP, at farm, CH, [kg]:

DS-ID 192: barley grains organic, at farm, CH, [kg]:

DS-ID 193: barley straw extensive, at farm, CH, [kg]:

DS-ID 194: barley straw IP, at farm, CH, [kg]:

DS-ID 195: barley straw organic, at farm, CH, [kg]:

DS-ID 196: fava beans IP, at farm, CH, [kg]:

DS-ID 197: fava beans organic, at farm, CH, [kg]:

DS-ID 198: fodder beets IP, at farm, CH, [kg]:
 DS-ID 200: grain maize IP, at farm, CH, [kg]:
 DS-ID 201: grain maize organic, at farm, CH, [kg]:
 DS-ID 202: green manure IP, until April, CH, [ha]:
 DS-ID 203: green manure IP, until February, CH, [ha]:
 DS-ID 204: green manure IP, until January, CH, [ha]:
 DS-ID 205: green manure IP, until march, CH, [ha]:
 DS-ID 216: potatoes IP, at farm, CH, [kg]:
 DS-ID 217: potatoes organic, at farm, CH, [kg]:
 DS-ID 218: protein peas, IP, at farm, CH, [kg]:
 DS-ID 219: protein peas, organic, at farm, CH, [kg]:
 DS-ID 220: rape seed extensive, at farm, CH, [kg]:
 DS-ID 221: rape seed IP, at farm, CH, [kg]:
 DS-ID 222: rye grains extensive, at farm, CH, [kg]:
 DS-ID 223: rye grains IP, at farm, CH, [kg]:
 DS-ID 224: rye grains organic, at farm, CH, [kg]:
 DS-ID 225: rye straw extensive, at farm, CH, [kg]:
 DS-ID 226: rye straw IP, at farm, CH, [kg]:
 DS-ID 227: rye straw organic, at farm, CH, [kg]:
 DS-ID 228: silage maize IP, at farm, CH, [kg]:
 DS-ID 229: silage maize organic, at farm, CH, [kg]:
 DS-ID 230: soy beans IP, at farm, CH, [kg]:
 DS-ID 231: soy beans organic, at farm, CH, [kg]:
 DS-ID 234: sugar beets IP, at farm, CH, [kg]:
 DS-ID 235: sunflower IP, at farm, CH, [kg]:
 DS-ID 236: wheat grains extensive, at farm, CH, [kg]:
 DS-ID 237: wheat grains IP, at farm, CH, [kg]:
 DS-ID 238: wheat grains organic, at farm, CH, [kg]:
 DS-ID 239: wheat straw extensive, at farm, CH, [kg]:
 DS-ID 240: wheat straw IP, at farm, CH, [kg]:
 DS-ID 241: wheat straw organic, at farm, CH, [kg]:

The fossil CO₂ emission from urea are underestimated by a factor 2.

Swiss arable crops: the emissions of N₂O from (liquid and solid manure) is overestimated. The lower the rate of application of manure, the higher the overestimation. The error occurred only, where liquid or solid manure was applied.

2.6.4 Agricultural products from USA

DS-ID 6970: rice, at farm, US, [kg]:

The emission “Methane, biogenic” to “air, low population density” from rice production are missing. The correct value is 0.0407 kg CH₄/kg rice, which corresponds to an increase of the global warming potential (IPCC 2001, 100 years) by 0.9361 kg CO₂-eq./kg rice (increase by a factor 3 from 0.46634 kg CO₂-eq./kg rice to 1.40244 kg CO₂-eq./kg rice). For the dataset “rice seed, at regional storehouse, US” the GWP 100a will increase from 0.53156 to 1.46766 kg CO₂-eq./kg rice seed (factor 2.76).

The irrigation (m³ water) is too low by a factor 1000. The correct value is 1.08 m³ water/kg rice.

The input of "diammonium phosphate, as N, at regional storehouse" is missing. The correct value is 1.71E-03 kg N/kg rice.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Tab. 2.32.

DS-ID 6968: potatoes, at farm, US, [kg]:

The output of sulfur to "soil/forestry" is corrected to "soil/agriculture".

The irrigation (m³ water) is too low by a factor 1000. The correct value is 0.11 m³ water/kg potato.

The input of "diammonium phosphate, as N, at regional storehouse" is missing. The correct value is 1.69E-03 kg N/kg potato.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Tab. 2.32.

DS-ID 6976: sheep for slaughtering, live weight, at farm, US, [kg]:

The "occupation and transformation, pasture and meadow" is too low. The correct value is 4.52 m²/kg live weight.

The value for "shed" input is too high by a factor of 50 due to the missing allocation of the shed lifespan. The correct value is 2.22E-4 m²/kg live weight.

DS-ID 6973: wool, sheep, at farm, US, [kg]:

The "occupation and transformation, pasture and meadow" is too low. The correct value is 126.35 m²/kg wool.

The value for "shed" input is too high by a factor of 50 due to the missing allocation of the shed lifespan. The correct value is 6.24E-3 m²/kg wool.

The irrigation (m³ water) is too low by a factor 1000. The correct value is 0.54 m³ water/kg wool.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Table A

DS-ID 6976: sheep for slaughtering, live weight, at farm, US, [kg]:

The irrigation (m³ water) is too low by a factor 1000. The correct value is 1.92E-02 m³ water/kg live weight.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Tab. 2.32.

DS ID 6977: cotton fibres, at farm, US, [kg]:

The energy demand of the ginning process is approximated by the process "grain drying, high temperature". The amount of kg "grain drying, high temperature" is too low due to a calculation error. The correct value is 0.198 kg "grain drying, high temperature"/kg cotton fibre

The irrigation (m³ water) is too low by a factor 1000. The correct value is 1.47 m³ water/kg cotton fibres.

The input of "diammonium phosphate, as N, at regional storehouse" is missing. The correct value is 2.46E-02 kg N/kg cotton fibres.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Tab. 2.32.

DS ID 6974: cotton seed, at farm, US, [kg]:

The energy demand of the ginning process is approximated by the process "grain drying, high temperature". The amount of kg "grain drying, high temperature" is too low due to a calculation error. The correct value is 1.97E-2 kg "grain drying, high temperature"/kg cotton seed

The irrigation (m³ water) is too low by a factor 1000. The correct value is 0.146 m³ water/kg cotton seed.

The input of "diammonium phosphate, as N, at regional storehouse" is missing. The correct value is 2.45E-03 kg N/kg cotton seed.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Tab. 2.32.

DS-ID 6969: rape seed, at farm, US, [kg]:

The input of "diammonium phosphate, as N, at regional storehouse" is missing. The correct value is 7.44E-03 kg N/kg rape seed.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Table A

DS-ID 6972: wheat, at farm, US, [kg]:

The irrigation (m³ water) is too low by a factor 1000. The correct value is 7.25E-02 m³ water/kg wheat.

The input of "diammonium phosphate, as N, at regional storehouse" is missing. The correct value is 7.24E-03 kg N/kg wheat.

The output of heavy metals to water (ground and surface) and to soil is too low by a factor 1000. The correct values are given in Tab. 2.32.

Tab. 2.32 Heavy metal emissions from agricultural production in the US.

	cotton fibres, at farm	cotton seed, at farm	potatoes, at farm	rape seed, at farm	rice, at farm	wheat grains, at farm	wool, sheep, at farm	sheep for slaughtering, live weight, at farm
Heavy metals to surface water								
Cadmium (Cd, kg/kg product)	5.15E-07	5.15E-07	2.68E-08	6.87E-07	1.28E-07	4.18E-07	5.29E-05	1.10E-07
Chromium (Cr, kg/kg product)	5.64E-05	5.63E-05	2.77E-06	8.30E-05	1.49E-05	4.68E-05	9.19E-02	1.90E-04
Copper (Cu, kg/kg product)	4.21E-05	4.20E-05	2.32E-06	5.81E-05	1.08E-05	3.39E-05	1.01E+00	2.09E-03
Lead (Pb, kg/kg product)	1.06E-05	1.06E-05	1.02E-06	8.87E-06	1.83E-06	6.89E-06	4.77E-05	9.88E-08
Mercury (Hg, kg/kg product)	0.00E+00	0.00E+00	3.95E-09	2.91E-09	1.23E-09	2.52E-09	0.00E+00	0.00E+00
Nickel (Ni, kg/kg product)	3.88E-05	3.88E-05	2.30E-06	4.98E-05	9.47E-06	2.93E-05	4.79E-01	9.91E-04
Zinc (Zn, kg/kg product)	5.79E-05	5.79E-05	4.46E-06	6.12E-05	1.26E-05	4.16E-05	0.00E+00	0.00E+00
Heavy metals to groundwater								
Cadmium (Cd, kg/kg product)	2.22E-08	2.22E-08	1.16E-09	2.96E-08	5.51E-09	1.80E-08	1.23E-03	2.54E-06
Chromium (Cr, kg/kg product)	1.03E-05	1.02E-05	5.05E-07	1.51E-05	2.71E-06	8.52E-06	2.48E+00	5.14E-03
Copper (Cu, kg/kg product)	1.56E-06	1.56E-06	8.59E-08	2.15E-06	4.00E-07	1.25E-06	7.34E+00	1.52E-02
Lead (Pb, kg/kg product)	6.73E-08	6.72E-08	6.48E-09	5.64E-08	1.16E-08	4.38E-08	7.50E-03	1.55E-05
Mercury (Hg, kg/kg product)	0.00E+00	0.00E+00	1.26E-10	9.33E-11	3.95E-11	8.07E-11	5.32E-01	1.10E-03
Nickel (Ni, kg/kg product)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.63E+00	5.45E-03
Zinc (Zn, kg/kg product)	7.97E-06	7.96E-06	6.13E-07	8.43E-06	1.74E-06	5.72E-06	0.00E+00	0.00E+00
Heavy metals to soil								
Cadmium (Cd, kg/kg product)	1.56E-06	1.55E-06	2.75E-07	-3.22E-07	9.57E-08	8.03E-07	-5.41E-04	-1.12E-06
Chromium (Cr, kg/kg product)	-4.21E-05	-4.20E-05	2.21E-07	-8.20E-05	-1.40E-05	-4.00E-05	-3.03E+00	-6.27E-03
Copper (Cu, kg/kg product)	-3.75E-05	-3.75E-05	1.75E-06	0.00E+00	2.20E-05	0.00E+00	-2.51E+00	-5.21E-03
Lead (Pb, kg/kg product)	-7.97E-06	-7.96E-06	-7.11E-07	-7.46E-06	-1.45E-06	-5.33E-06	-6.06E-03	-1.25E-05
Mercury (Hg, kg/kg product)	0.00E+00	0.00E+00	-1.13E-08	-3.45E-09	-1.37E-09	-2.42E-09	0.00E+00	0.00E+00
Nickel (Ni, kg/kg product)	-3.30E-05	-3.30E-05	-1.61E-06	-4.58E-05	-8.48E-06	-2.59E-05	-4.49E-01	-9.30E-04
Zinc (Zn, kg/kg product)	-2.52E-05	-2.52E-05	-3.79E-07	-4.94E-05	-1.57E-05	-3.09E-05	-5.61E+00	-1.16E-02

DS-ID 6659: soybeans, at farm, US, [kg]:

The values of the emissions of N₂O and NO_x were too low.

Tab. 2.33 Emissions to air from soybean cultivation

Emission	values in v2.01	correct values
N ₂ O [kg/kg soybean]	3.55E-04	1.298E-03
NO _x [kg/kg soybean]	7.45E-05	2.726E-04

2.7 Wood and wooden products

2.7.1 Fibreboard, soft

DS-ID 2446: fibreboard soft, at plant (u=7%), CH, [m3];

DS-ID 11166: fibreboard soft, latex bonded, at plant (u=7%), CH, [m3];

DS-ID 11167: fibreboard soft, without adhesives, at plant (u=7%), CH, [m3];

Data for softboard production were completely updated and boards with different binders were introduced. The text in Annex chapter 2 (part III of this report here) complements section 12.4 in the original report.

2.7.2 Wood Pellet manufacturing, infrastructure (RER)

DS-ID 2357: wood pellet manufacturing, infrastructure, RER, [unit];

The general comment to the reference function in the meta information contained an error. The life time output was specified as 1E8 kg instead of 1E8 m3. This error is corrected in v2.1.

2.7.3 Tropical wood

DS-ID 10208: roundwood, parana pine (SFM), under bark, u=50%, at forest road, BR, [m3];

The negative wood resource demand was inventoried as “wood, hard, standing” instead of “wood, soft, standing”. This error is corrected in v2.1.

2.8 Bioenergy

2.8.1 Biowaste to agricultural co-fermentation, covered

MODS-ID 411: biowaste, to agricultural co-fermentation, covered, CH, [kg];

DS-ID 10761: disposal, biowaste, to agricultural co-fermentation, covered, CH, [kg];

DS-ID 10762: biogas, from biowaste, at agricultural co-fermentation, covered, CH, [kg];

DS-ID 10772: application, digested matter from biowaste in agricultural co-digestion, covered, CH, [kg];

The heavy metal emissions to soil were 20 to 40 % too low. Iodide, Nitrogen, and Sulfur emissions to agricultural soil were missing. This influences the results of the specific datasets with regard to LCIA methods, which consider heavy metals. The ecoinvent report shows the correct data.

Tab. 2.34 Emission factors to soil, agricultural and applied allocation factors of multi-output dataset "biowaste, to agricultural co-fermentation, covered"

		Multi-output dataset			
		allocation factors [%]			
		biowaste, to agricultural co-fermentation, covered	biogas, from biowaste, at agricultural co-fermentation, covered	disposal, biowaste, to agricultural co-fermentation, covered	application, digested matter from biowaste in agricultural co-digestion, covered
Name		CH	CH	CH	CH
Location		0	0	0	0
InfrastructureProcess		0	0	0	0
Unit	Unit	kg	Nm3	kg	Kg
emissions to soil, agricultural					
Boron	kg	1.02E-05	0	50	50
Chloride	kg	4.00E-03	0	50	50
Bromine	kg	6.00E-06	0	50	50
Fluoride	kg	2.00E-04	0	50	50
Arsenic	kg	2.00E-06	0	50	50
Cadmium	kg	1.38E-07	0	50	50
Cobalt	kg	5.00E-06	0	50	50
Chromium	kg	8.00E-06	0	50	50
Copper	kg	1.80E-05	0	50	50
Iodide	kg	5.50E-08	0	50	50
Mercury	kg	7.00E-08	0	50	50
Manganese	kg	4.30E-06	0	50	50
Molybdenum	kg	4.00E-07	0	50	50
Nickel	kg	5.42E-06	0	50	50
Nitrogen	kg	3.68E-03	0	50	50
Lead	kg	1.86E-05	0	50	50
Selenium	kg	5.00E-07	0	50	50
Sulfur	kg	1.50E-03	0	50	50
Tin	kg	8.00E-06	0	50	50
Vanadium	kg	3.00E-06	0	50	50
Zinc	kg	5.82E-05	0	50	50
Silicon	kg	4.00E-02	0	50	50
Iron	kg	6.00E-04	0	50	50
Calcium	kg	2.18E-02	0	50	50
Aluminum	kg	1.00E-02	0	50	50
Potassium	kg	3.50E-03	0	50	50
Magnesium	kg	2.82E-03	0	50	50
Sodium	kg	1.50E-03	0	50	50

2.8.2 Biowaste, to anaerobic digestion

MODS-ID 143: biowaste, to anaerobic digestion, CH, [kg];:

DS-ID 6247: disposal, biowaste, to anaerobic digestion, CH, [kg];

DS-ID 6164: biogas, from biowaste, at storage, CH, [kg];

DS-ID 6248: digested matter, application in agriculture, CH, [kg];

The heavy metal emissions to soil were 20 to 40 % too low. Iodide, Nitrogen, and Sulfur emissions to agricultural soil were missing. This influences the results of the specific datasets with regard to LCIA methods, which consider heavy metals. The ecoinvent report shows the correct data.

Tab. 2.35 Emission factors to soil, agricultural and applied allocation factors of multi-output dataset "biowaste, to anaerobic digestion"

Name	Unit	Multi-output dataset	Allocation factors [%]		
		biowaste, to aerobic digestion	biogas, from biowaste, at storage	disposal, biowaste, to anaerobic digestion	digested matter, application in agriculture
Location		CH	CH	CH	CH
InfrastructureProcess		0	0	0	0
Unit		kg	Nm3	kg	Kg
emissions to soil, agricultural					
Boron	kg	1.02E-05	0	50	50
Chloride	kg	4.00E-03	0	50	50
Bromine	kg	6.00E-06	0	50	50
Fluoride	kg	2.00E-04	0	50	50
Arsenic	kg	2.00E-06	0	50	50
Cadmium	kg	1.38E-07	0	50	50
Cobalt	kg	5.00E-06	0	50	50
Chromium	kg	8.00E-06	0	50	50
Copper	kg	1.80E-05	0	50	50
Iodide	kg	5.50E-08	0	50	50
Mercury	kg	7.00E-08	0	50	50
Manganese	kg	4.30E-06	0	50	50
Molybdenum	kg	4.00E-07	0	50	50
Nickel	kg	5.42E-06	0	50	50
Nitrogen	kg	3.68E-03	0	50	50
Lead	kg	1.86E-05	0	50	50
Selenium	kg	5.00E-07	0	50	50
Sulfur	kg	1.50E-03	0	50	50
Tin	kg	8.00E-06	0	50	50
Vanadium	kg	3.00E-06	0	50	50
Zinc	kg	5.82E-05	0	50	50
Silicon	kg	4.00E-02	0	50	50
Iron	kg	6.00E-04	0	50	50
Calcium	kg	2.18E-02	0	50	50
Aluminum	kg	1.00E-02	0	50	50
Potassium	kg	3.50E-03	0	50	50
Magnesium	kg	2.82E-03	0	50	50
Sodium	kg	1.50E-03	0	50	50

2.8.3 Anaerobic digestion plant covered, agriculture

DS-ID 6171: anaerobic digestion plant, agriculture, CH, [unit]:

The material inputs of polyethylene, synthetic rubber and glued laminated timber were too low by a factor of about 300. Those inputs and the related transport service and waste management requirements were corrected. The ecoinvent report shows the correct data of v2.1 (Jungbluth et al. 2007). The data are displayed in Tab. 2.36.

Tab. 2.36 Unit process raw data of the anaerobic digestion plant, agricultural bio waste, v2.1

Explanations	Name	Location	Infrastructure	Process	Unit	anaerobic	uncertainty	StandardDeviation	GeneralComment
						digestion	type	95%	
						plant,			
	Location InfrastructureProcess Unit					CH 1 unit			
Technosphere	concrete, normal, at plant	CH	0	m3	7.85E+1	1	1.26	(3,4,2,1,1,5); literature studies	
	reinforcing steel, at plant	RER	0	kg	6.48E+3	1	1.26	(3,4,2,1,1,5); literature studies	
	chromium steel 18/8, at plant	RER	0	kg	7.80E+2	1	1.26	(3,4,2,1,1,5); literature studies	
	polystyrene, high impact, HIPS, at plant	RER	0	kg	3.42E+2	1	1.26	(3,4,2,1,1,5); literature studies	
	polyvinylchloride, at regional storage	RER	0	kg	4.95E+1	1	1.26	(3,4,2,1,1,5); literature studies	
	polyethylene, HDPE, granulate, at plant	RER	0	kg	2.55E+1	1	1.26	(3,4,2,1,1,5); literature studies	
	synthetic rubber, at plant	RER	0	kg	1.80E+2	1	1.26	(3,4,2,1,1,5); literature studies	
	glued laminated timber, outdoor use, at plant	RER	0	m3	5.54E+0	1	1.26	(3,4,2,1,1,5); literature studies	
	copper, at regional storage	RER	0	kg	7.50E+1	1	1.26	(3,4,2,1,1,5); literature studies	
	transport, lorry 20-28t, fleet average	CH	0	tkm	6.73E+3	1	2.10	(4,5,na,na,na,na); standard distances for used materials	
	transport, freight, rail	CH	0	tkm	2.23E+4	1	2.10	(4,5,na,na,na,na); standard distances for used materials	
	disposal, building, reinforced concrete, to final disposal	CH	0	kg	1.79E+5	1	1.26	(3,4,2,1,1,5); literature studies	
	disposal, polystyrene, 0.2% water, to municipal incineration	CH	0	kg	3.42E+2	1	1.26	(3,4,2,1,1,5); literature studies	
	disposal, polyvinylchloride, 0.2% water, to municipal incineration	CH	0	kg	4.95E+1	1	1.26	(3,4,2,1,1,5); literature studies	
	disposal, building, polyethylene/polypropylene products, to final disposal	CH	0	kg	2.55E+1	1	1.26	(3,4,2,1,1,5); literature studies	
	disposal, rubber, unspecified, 0% water, to municipal incineration	CH	0	kg	1.80E+2	1	1.26	(3,4,2,1,1,5); literature studies	
	disposal, wood untreated, 20% water, to municipal incineration	CH	0	kg	2.74E+3	1	1.26	(3,4,2,1,1,5); literature studies	
resource, land	Occupation, industrial area, built up			m2a	2.22E+3	1	1.58	(2,4,2,1,1,5); literature studies	
	Transformation, from heterogeneous, agricultural			m2	1.11E+2	1	2.07	(2,4,2,1,1,5); literature studies	
	Transformation, to industrial area, built up			m2	1.11E+2	1	2.07	(2,4,2,1,1,5); literature studies	
Outputs	anaerobic digestion plant, agriculture	CH	1	unit	1.00E+0				

2.8.4 Methane purification

DS-ID 6176: methane, 96 vol-%, from biogas, at purification, CH, [Nm3]:

Tab. 2.37 Change of unit process and input value within the dataset: methane, 96 vol-%, from biogas, at purification, CH (DS-ID 6176).

v2.01 name	v2.01 input	v2.1 name	v2.1 input
Facilities, chemical production, RER	4.0E-11	Chemical plant, organics, RER	4.0E-10

2.9 Transport Systems

2.9.1 lorry transport

DS-ID 7307: transport, lorry >32t, EURO4, RER, [tkm]:

The dataset incorrectly included «lorry 28t», «maintenance lorry 28t», and «disposal lorry 28t» as inputs. These inputs are deleted.

2.9.2 Operation of lorries

DS-ID 1915: Operation, lorry, 3.5-16t, fleet average, RER, [vkm];
 DS-ID 1916: Operation, lorry, 3.5-20t, fleet average, CH, [vkm];
 DS-ID 1918: Operation, lorry, 3.5-20t, empty, fleet average, CH, [vkm];
 DS-ID 1921: Operation, lorry, 20-28t, fleet average, CH, [vkm];
 DS-ID 1923: Operation, lorry, 20-28t, empty, fleet average, CH, [vkm];
 DS-ID 1925: Operation, lorry, 20-28t, full, fleet average, CH, [vkm];
 DS-ID 1927: Operation, lorry, >28t, fleet average, CH, [vkm];
 DS-ID 1931: Operation, lorry, >28t, full, fleet average, CH, [vkm];
 DS-ID 6113: Operation, lorry, 28t, rape methyl ester 100%, CH, [km];
 DS-ID 7294: Operation, lorry, 16-32t, EURO3, RER, [vkm];
 DS-ID 7295: Operation, lorry, 16-32t, EURO4, RER, [vkm];
 DS-ID 7296: Operation, lorry, 16-32t, EURO5, RER, [vkm];
 DS-ID 7297: Operation, lorry, >32t, EURO3, RER, [vkm];
 DS-ID 7298: Operation, lorry, >32t, EURO4, RER, [vkm];
 DS-ID 7299: Operation, lorry, >32t, EURO5, RER, [vkm];

The datasets of v2.01 were missing the dataset entry for ‘Ethane, 1,1,1,2-tetrafluoro-, HFC-134a’, an emission to air, unspecified, caused by the use of the air conditioning unit. The same quantity had been applied to all vehicle types in v1.3 and the same value was copied into all v2.1 datasets. The input value is 2.29E-6kg/vkm.

2.9.3 Operation of cars

DS-ID 6555: Operation, passenger car, diesel, EURO3, CH, [km];
 DS-ID 6556: Operation, passenger car, diesel, EURO4, CH, [km];
 DS-ID 6557: Operation, passenger car, diesel, EURO5, CH, [km];
 DS-ID 6559: Operation, passenger car, petrol, EURO3, CH, [km];
 DS-ID 6560: Operation, passenger car, petrol, EURO4, CH, [km];
 DS-ID 6561: Operation, passenger car, petrol, EURO5, CH, [km];
 DS-ID 6558: Operation, passenger car, petrol, 15% ETBE with ethanol from biomass, EURO4, CH, [km];
 DS-ID 6672: Operation, passenger car, petrol, 4% ETBE with ethanol from biomass, EURO4, CH, [km];

The emissions to water, unspecified, and soil, unspecified, of lead, cadmium, copper, chromium, nickel and zinc due to tyre and brake abrasion had not been updated in v2.01 for all these datasets. The emissions and values from v2.01 and v2.1 are shown in the following table.

Tab. 2.38 Updated emissions to water and soil due to passenger vehicle tyre and brake ware (kg/km).

Emissions category	Emission	v2.01	v2.1 (kg/km)
Water, unspecified	Lead	1.44E-8	3.93E-9
	Cadmium, ion	2.89E-9	9.55E-11
	Copper, ion	4.33E-8	6.39E-9
	Chromium, ion	2.60E-8	4.55E-10
	Nickel, ion	2.31E-8	1.23E-9
	Zinc, ion	2.32E-6	2.70E-7
Soil, unspecified	Lead	1.44E-8	3.93E-9
	Cadmium	2.89E-9	9.55E-11
	Copper	4.33E-8	6.39E-9
	Chromium	2.60E-8	4.55E-10
	Nickel	2.31E-8	1.23E-9
	Zinc	2.32E-6	2.70E-7

Emissions of this type should be the same for all passenger car datasets.

2.9.4 Transport of aircraft

DS-ID 1892: Transport, aircraft, freight, RER, [tkm];

DS-ID 1893: Transport, aircraft, freight, Europe, RER, [tkm];

DS-ID 1894: Transport, aircraft, freight, Intercontinental RER, [tkm];

DS-ID 1895: Transport, aircraft, passenger RER, [pkm];

DS-ID 1896: Transport, aircraft, passenger, Europe RER, [pkm];

DS-ID 1897: Transport, aircraft, passenger, Intercontinental RER, [pkm];

For the transport of passengers and freight on intra-European, intercontinental and average flights the demand factors for the unit process 'Operation, maintenance, airport, RER' were recalculated. The demand factor for the unit process 'Airport, RER' is then a division of this based on a lifetime of 100 years and so this was necessarily altered. Demand factors for aircraft were also recalculated based on allocation by weight transported. The following table outlines the changes.

Tab. 2.39 Corrected inputs to aircraft transport datasets in units/tkm or units/pkm. v2.01 data in brackets.

DS-ID	1892	1893	1894	1895	1896	1897
DS Name and unit	Transport, aircraft, freight, RER (tkm)	Transport, aircraft, freight, Europe, RER (tkm)	Transport, aircraft, freight, Intercontinental RER (tkm)	Transport, aircraft, passenger RER (pkm)	Transport, aircraft, passenger, Europe RER (pkm)	Transport, aircraft, passenger, Intercontinental RER (pkm)
Unit process						
Airport, RER	8.02E-13 (3.23E-12)	2.35E-14 (2.97E-11)	8.24E-13 (2.48E-12)	1.50E-13 (6.87E-13)	3.32E-14 (2.97E-12)	1.73E-13 (2.48E-13)
Operation, maintenance, airport, RER	8.02E-11 (5.89E-10)	2.35E-12 (2.97E-9)	8.24E-11 (2.48E-10)	1.50E-11 (3.28E-10)	3.32E-14 (2.97E-10)	1.73E-11 (2.48E-11)
Disposal, airport, RER	8.02E-13 (3.23E-12)	2.35E-14 (2.97E-11)	8.24E-13 (2.48E-12)	1.50E-13 (6.87E-13)	3.32E-14 (2.97E-12)	1.73E-13 (2.48E-13)
Aircraft, medium haul, RER	-	2.39E-9 (1.08E-9)	-	-	2.38E-10 (2.59E-10)	-
Aircraft, long haul, RER	-	-	3.16E-10 (1.76E-10)	-	-	3.13E-11 (4.22E-11)
Aircraft, freight, RER	4.28E-10 (2.10E-10)	-	-	-	-	-
Aircraft passenger, RER	-	-	-	9.89E-11 (5.03E-11)	-	-

2.10 Electronics data

2.10.1 Printed Wiring board production (Pb-containing finishing)

DS-ID 10996: printed wiring board, surface mount, lead-containing surface, at plant, GLO, [m2]:

The input of lead (Pb) was erroneously reported as Nickel (Ni) in v2.01 – the input has been corrected to “lead, at regional storage”. The value is correct.

2.10.2 LCD screen

DS-ID 6993 : LCD flat screen, 17 inches, at plant, GLO, [unit]:

The amount of LCD module per 17-Inch LCD screen is too low – the correct value is 4.006 kg (per screen) instead of 2.304 kg. The error was due to the fact that the backlight, being part of the dataset “LCD module”, has been left out in the original calculations.

2.11 Waste disposal

2.11.1 Hazardous waste incineration direct carbon dioxide emissions

DS-ID 2055: disposal, antifreezer liquid, 51.8% water, to hazardous waste incineration, CH, [kg];

DS-ID 2056: disposal, bilge oil, 90% water, to hazardous waste incineration, CH, [kg];

DS-ID 2058: disposal, emulsion paint remains, 0% water, to hazardous waste incineration, CH, [kg];

DS-ID 2059: disposal, hazardous waste, 25% water, to hazardous waste incineration, CH, [kg];

DS-ID 2060: disposal, paint remains, 0% water, to hazardous waste incineration, CH, [kg];
 DS-ID 2061: disposal, refinery sludge, 89.5% water, to hazardous waste incineration, CH, [kg];
 DS-ID 2062: disposal, separator sludge, 90% water, to hazardous waste incineration, CH, [kg];
 DS-ID 2063: disposal, solvents mixture, 16.5% water, to hazardous waste incineration, CH, [kg];
 DS-ID 2064: disposal, used mineral oil, 10% water, to hazardous waste incineration, CH, [kg];

Due to a mistake in the HWI calculation tool, HWI disposal datasets of carbon-bearing wastes have direct CO₂ emissions which are systematically too low by a factor of 1.57. Thanks to Ivan Muñoz Ortiz, Autonomous University of Barcelona, for reporting this error. This mistake only affects the direct CO₂ emissions from the waste material itself; CO₂ emissions from added fuel oil incineration in HWI are correct. The correction increases the GWP of HWI datasets. The largest GWP increases occur in high-carbon wastes like used mineral oil (26%), solvents mixture (12%) and paint remains (5%).

The mistake is corrected in the affected 9 HWI datasets. The culpable cell in the Excel calculation tool is also corrected (the corrected version is called 13_HWIv2.xls). Users of the waste calculation tools should be alerted to use the corrected version¹.

2.11.2 Energy for digestion of wastewater treatment sludge

Affected are *all* datasets for wastewater treatment "treatment," [m³] *except* the unpolluted wastewaters (DS-IDs 2280 + 2281), i.e. DS-IDs: 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282:

The energy expenditures to treat raw digester sludge in the plant are calculated on *dry mass* basis. In the calculation tool the dry mass of raw digester sludge was calculated without the hydrogen and oxygen in organic compounds, only carbon mass from organic compounds. Thanks to Ivan Muñoz Ortiz, Autonomous University of Barcelona, for reporting this error. Hydrogen and oxygen in organic compounds make up 27.6% of the total dry mass of raw digester sludge from average municipal wastewater. The energy expenditure was thus underestimated. Sludge digestion causes on average 20% of the electricity demand and 90% of the heat demand of a wastewater treatment plant. The impacts from direct energy use are however only small for average wastewater (2% of total impact with Eco-indicator⁹⁹HA), thus this error is not serious.

The erroneous calculation in the Excel tool is corrected (the corrected version is called 13_WWTv2.xls). Users of the waste calculation tools should be alerted to use the corrected version².

¹ In case you cannot simply replace the old HWI calculation tool – e.g. you have customised your HWI tool or incorporated it into a larger file network –, you can correct the mistake yourself:

1. In the HWI tool run the macro 'AllVisibleSheet'
2. Select the sheet 'air emissions'
3. Locate cell D97. The caption left of this cell should read 'CO₂ from remaining carbon'
4. Replace the formula with the expression '=D95/12*44'. D95 is the cell two rows up for 'carbon in flue gas without CO-carbon'.

This increases the direct CO₂ emissions from waste by a factor of 1.57. Btw. there is a similar calculation procedure in the MSWI tool (CO₂ from C in flue gas without CO) which is correct in its original form and does not have to be fixed.

² In case you cannot simply replace the old WWTP calculation tool – e.g. you have customised your WWTP tool or incorporated it into a larger file network –, you can correct the mistake yourself:

1. In the WWTP tool run the macro 'AllVisibleSheet'
2. Select the sheet 'digester'
3. Locate cells B88 and B89. The caption left of those cells should read 'O/C ratio' and 'H/C ratio' and originally contain the

2.11.3 Wastewater treatment direct carbon dioxide emissions

DS-ID 2260: treatment, black chrome coating effluent, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2262: treatment, concrete production effluent, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2263: treatment, condensate from light oil boiler, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2264: treatment, fibre board production effluent, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2265: treatment, glass production effluent, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2266: treatment, heat carrier liquid, 40% C₃H₈O₂, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2268: treatment, maize starch production effluent, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2269: treatment, particle board production effluent, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2270: treatment, pig iron production effluent, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2271: treatment, plywood production effluent, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2272: treatment, potato starch production effluent, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2273: treatment, rainwater mineral oil storage, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2274: treatment, sewage, from residence, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2275: treatment, sewage, to wastewater treatment, class 1, CH, [m3];
 DS-ID 2276: treatment, sewage, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2277: treatment, sewage, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2278: treatment, sewage, to wastewater treatment, class 4, CH, [m3];
 DS-ID 2279: treatment, sewage, to wastewater treatment, class 5, CH, [m3];

Due to a stoichiometry error, the direct carbon dioxide emissions to air are calculated 16.6% too small. Thanks to Ivan Muñoz Ortiz, Autonomous University of Barcelona, for reporting this error. This concerns only *carbon-containing* wastewaters. For most wastewaters the effect is small or, in case biogenic CO₂ is not characterised in LCIA, zero. For most wastewaters the effect on global warming impacts are only a few percent and negligible for fully aggregated LCIA impacts. The exception is the treatment of heat carrier liquid (from solar collector plants) which is a highly concentrated liquid with 379 g carbon per litre wastewater, where the underestimation is moderate (13% for global warming, 10% for Eco-indicator'99HA)

The pertaining datasets were recalculated. The erroneous calculation in the Excel tool is corrected (the corrected version is called 13_WWTv2.xls). Users of the waste calculation tools should be alerted to use the corrected version³.

values 0.55 and 0.125. Check this.

4. Then go to the cell B12 and enter the formula '=B11*B88'. In cell C12 enter the formula '=B11*B89'.

This takes care of the correct calculation of dry mass of raw digester sludge. For completeness you should also complement the formulas in range C12:E13, headed 'GSD', 'arithmetic mean' and 'arithmetic variance', which are helper cells to calculate uncertainty. For the GSD (column C) just refer to the value for carbon ('=C11' in C12 and C13). Copy the formulas from cells D11:E11 to both rows D12:E13.

³ In case you cannot simply replace the old WWTP calculation tool – e.g. you have customised your WWTP tool or incorporated it into a larger file network –, you can correct the mistake yourself:

1. In the WWTP tool run the macro 'AllVisibleSheet'

2.11.4 Wastewater treatment direct nitrous oxide emissions

DS-ID 2262: treatment, concrete production effluent, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2263: treatment, condensate from light oil boiler, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2265: treatment, glass production effluent, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2268: treatment, maize starch production effluent, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2272: treatment, potato starch production effluent, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2273: treatment, rainwater mineral oil storage, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2274: treatment, sewage, from residence, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2275: treatment, sewage, to wastewater treatment, class 1, CH, [m3];
 DS-ID 2276: treatment, sewage, to wastewater treatment, class 2, CH, [m3];
 DS-ID 2277: treatment, sewage, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2278: treatment, sewage, to wastewater treatment, class 4, CH, [m3];
 DS-ID 2279: treatment, sewage, to wastewater treatment, class 5, CH, [m3];

Due to a stoichiometry error, the direct nitrous oxide emissions to air (N₂O) are calculated a factor of two too large. Thanks to Ivan Muñoz Ortiz, Autonomous University of Barcelona, for reporting this error. This concerns only *nitrogen-containing* wastewaters. Even for high-nitrogen wastewaters (such as maize starch production effluent with 1.5 g N₂O per m³ wastewater) the overestimation was only moderate regarding global warming impacts (up to 14%) and small for fully aggregated LCIA impacts (up to 1.5% for Eco-indicator'99HA).

The pertaining datasets were recalculated. The erroneous calculation in the Excel tool is corrected (the corrected version is called 13_WWTV2.xls). Users of the waste calculation tools should be alerted to use the corrected version⁴.

2.11.5 Changed waste composition

DS-ID 2264: treatment, fibre board production effluent, to wastewater treatment, class 3, CH, [m3];
 DS-ID 2271: treatment, plywood production effluent, to wastewater treatment, class 3, CH, [m3];

The zinc content in the wastewater from fibre board and plywood production was deemed to high and lowered. The changes are described in report 13, part I, chapter 4.18.1 "Wastewater from processed wood boards production".

2. Select the sheet 'treatment'

3. Locate cell AD12. Enter the formula '=AB8/12*44'.

⁴ In case you cannot simply replace the old WWTP calculation tool – e.g. you have customised your WWTP tool or incorporated it into a larger file network –, you can correct the mistake yourself:

1. In the WWTP tool run the macro 'AllVisibleSheet'

2. Select the sheet 'digester'

3. Locate cell O19. Enter the formula '=(14*2+16)/28' or simply the value 1.571429.

3 Reporting errors in Reports v2.01

ecoinvent report no. 15, chapter 7.2.7, page 64, the correct text is: The content of heavy metals in the synthetic rubber are 1.6 % (or 16 g/kg_{rubber}) for zinc, 0.0026 % (or **0.026** g/kg_{rubber}) for lead and 0.0006 % (or **0.006** g/kg_{rubber}) for cadmium (Frischknecht *et al.* 1996).

ecoinvent report no. 15, chapter 4.4.3, page 35: equation for P_{er} should correctly be $P_{er} = S_{er} * P_{cs} * F_r * F_{erw}$ (without the factor 10'000).

4 Additional Literature

- Bartos et al. 2006 Bartos S., Beu L. S., Burton C. S., Fraust C. L., Illuzzi F., Mocella M. T. and Raoux S. (2006) Electronics Industry Emissions. In: *2006 IPCC Guidelines for National Greenhouse Gas Inventories* (ed. IPCC). The Intergovernmental Panel on Climate Change (IPCC), Hayama, Kanagawa (Japan).
- Elliot & Munir 1968 Elliot A. G. and Munir Z. A. (1968) The sintering of nickel/aluminium spheres to nickel plates. In: *Journal of Materials Science*, **3**(2), pp. 150-157.
- Raichenko et al. 1976 Raichenko A. I., Kolchinskii M. Z. and Levina D. A. (1976) Theory and Technology of Sintering, Thermal and Chemicothermal Treatment Processes. In: *Poroshkovaya Metallurgiya*, **10**(166), pp. 19-26.
- Rantik 1999 Rantik M. (1999) Life Cycle Assessment of Five Batteries for Electric Vehicles under Different Charging Regimes. Kommunikationsforskningsberedningens (KFB) / Chalmers University of Technology, Stockholm (Sweden).

Part III – ANNEX

Citation:

Hischier R., Althaus H.-J., Bauer C., Doka G., Frischknecht R., Jungbluth N., Nemecek T., Simons A., Sutter J., Stucki M., Tuchschnid M. (2010) Documentation of changes implemented in ecoinvent Data v2.1 and v2.2. ecoinvent report No. 16. Swiss Centre for Life Cycle Inventories, St. Gallen.

Content

1	Sputtering of ITO layers	5
2	Magnesium Dioxide Oxidation (Potassium Permanganate)	27
3	Methylamine	39
4	Board based on industrial residual wood	49
5	Insulation Products and Processes [updated Part XIV of Kellenberger et al. 2007]	65
6	Fibre Cement Products and Processes [updated Part IX of Kellenberger et al. 2007]	152
7	Life Cycle Inventories for Rail Transport	187

As already mentioned in part I, not part of this report are the following documentations, as they are documented in separate documents, available as indicated below:

Topic	Title of publication	Access
Pesticides	Sutter, J. (2010) Life Cycle Inventories of Pesticides; Final report ecoinvent data v2.2; ETH Zürich - S+U group, Swiss Centre for Life Cycle Inventories: St. Gallen (Switzerland).	http://www.ecoinvent.org (-> Documents -> reports)
Coal tailings	Doka, G. (2009) Life Cycle Inventory of the disposal of lignite spoil, coal spoil and coal tailings; Study on behalf of ecoinvent Centre, St. Gallen, by Doka Life Cycle Assessments: Zürich (Switzerland).	http://www.doka.ch (-> publications)
Lilon Batteries	Notter, D., Gauch, M., Widmer, R., Wäger, P., Stamp, A., Zah, R. and Althaus, H-J. (2010) Contribution of Li-ion batteries to the environmental impact of electric vehicles, in: <i>Environmental Science & Technology</i> , accepted for publication. [-> Article + Supporting Materials]	http://pubs.acs.org/journal/esthag
Transport by two wheel vehicles	Leuenberger, M., and Frischknecht, R. (2010) Two wheel vehicles, Uster: ESU-services.	http://www.esu-services.ch
Transport by electric cars	Leuenberger, M., and Frischknecht, R. (2010) Battery electric vehicles and future concept cars, Uster: ESU-services.	http://www.esu-services.ch
Virtual mobility	Leuenberger, M., and Frischknecht, R. (2010) Virtual Mobility, Uster: ESU-services.	http://www.esu-services.ch
Electricity mixes, Switzerland	Leuenberger, M., and Frischknecht, R. (2010) Electricity mixes: consumer mix, certified electricity and electricity mix, Uster: ESU-services.	http://www.esu-services.ch
Biofuels	Büsser, S., and Frischknecht, R. (2010) Life Cycle Inventory of E85, LPG supply in Switzerland and Biogasmix 2008, Uster: ESU-services.	http://www.esu-services.ch

1 Report "Sputtering of ITO layers"

Autor: Mischa Classen, Empa

Review: Roberto Dones, PSI

Last Changes: 2007

1.1 Introduction

Transparent and conductive oxide (TCO) layers are widely used for many microelectronics applications, such as transparent electrical contacts electrodes in displays, touch screens, thin film solar cells, etc. Indium-Tin-Oxide (ITO) is one of the most investigated and used transparent conductive oxides due to its high electrical conductivity and high transparency in the visible light wavelength range. There are several different methods for ITO deposition (sputtering, evaporation, chemical vapour deposition, sol-gel method, spray pyrolysis, etc.). DC and RF magnetron sputtering are the most attractive techniques for industrial development (Kurdesau et al. (2006)).

1.2 Material Characterisation

The ITO layers described in this chapter are formed by means of sputtering. In this process a so called sputtering target consisting of a base plate packed with ITO is bombarded in an evacuated chamber with charged particles (electrons) that eject surface atoms out of the target. These liberated atoms travel across chamber to the substrate.

1.2.1 ITO Targets

ITO planar sputtering targets are used for the deposition of transparent conductive films and can be employed in a variety of applications including Touch Panels and different types of Displays such as TFT-LCDs, OLEDs and PDPs. Technical data for a typical ITO target (Fig. 1.1) with an $\text{In}_2\text{O}_3/\text{SnO}_2$ in a 90/10 wt% ratio are given in Tab. 1.1. The targets consist of a sintered and pressed ITO body and a backing plate. Thin film adhesion and diffusion barrier layers are applied to the back of each target body. The ITO body is bonded by a temperature controlled metallic solder seal made of pure indium to a backing plate from Cu or Ti.

Tab. 1.1 Physical Properties of a typical commercially available ITO target with a composition of $\text{In}_2\text{O}_3/\text{SnO}_2$ in a 90/10 wt% ratio (Umicore (2006)).

Property	Value
Maximum theoretical density	7.16 g/cm ³
Ultra high density grade	> 99.5 %
Flexural strength	> 150 MPa
Electrical resistivity at 20 °C	< 170 $\mu\Omega\text{cm}$
Thermal conductivity	at 20 °C 11 – 12 W/mK at 250 °C 10 – 11 W/mK
Purity	The total of all of the following impurities does not exceed 100 ppm: Al, K, Sb, Bi, Na, Ti, Cr, Ni, Zn, Cu, Mg, Fe, Pb



Fig. 1.1 Indium Tin Oxide (ITO) High Density Sputtering Target. Maximal dimensions possible are 610x350x9mm (Umicore (2006)).

1.2.2 ITO Layers

ITO layers are transparent conductive films from tin-doped indium oxide (ITO). ITO is an n-type semiconductor with a band gap between 3.5 eV and 4.3 eV and a maximum charge carrier concentration in the order of 10^{21} cm^{-3} . Consequently, ITO is transparent to visible and near infrared light, which has a low resistivity in the range of $1\text{--}3 \times 10\text{--}4 \text{ } \Omega \text{ cm}$. With high luminous transmittance larger than 90%, high infrared reflectance, good electrical conductivity, excellent substrate adherence, hardness and chemical inertness, ITO thin films are widely used in anti-static coatings, heat mirrors, solar cells, flat panel displays, liquid crystal displays, electroluminescent devices, sensors and organic light-emitting diodes.

In LCD technology the back light has to travel through two transparent electrodes, which operate as electric directors for the liquid crystal. ITO is capable of acting as such a transparent electrode that has to maintain transparency and specific resistivity and is therefore widely used in the manufacturing of TFT screens (Gan et al. (2006), Betz et al. (2005)). Lee & Ok Park (2006) report different film thickness depending on the application, ranging from 40 nm to 300 nm.

Fig. 1.2 shows (i) the commercial color-STN LCD with 150-nm-thick top-ITO and 330-nm-thick bottom-ITO and (ii) the TFT LCD with 150-nm-thick top-ITO (common ITO) and 50-nm-thick bottom-ITO (pixel ITO)

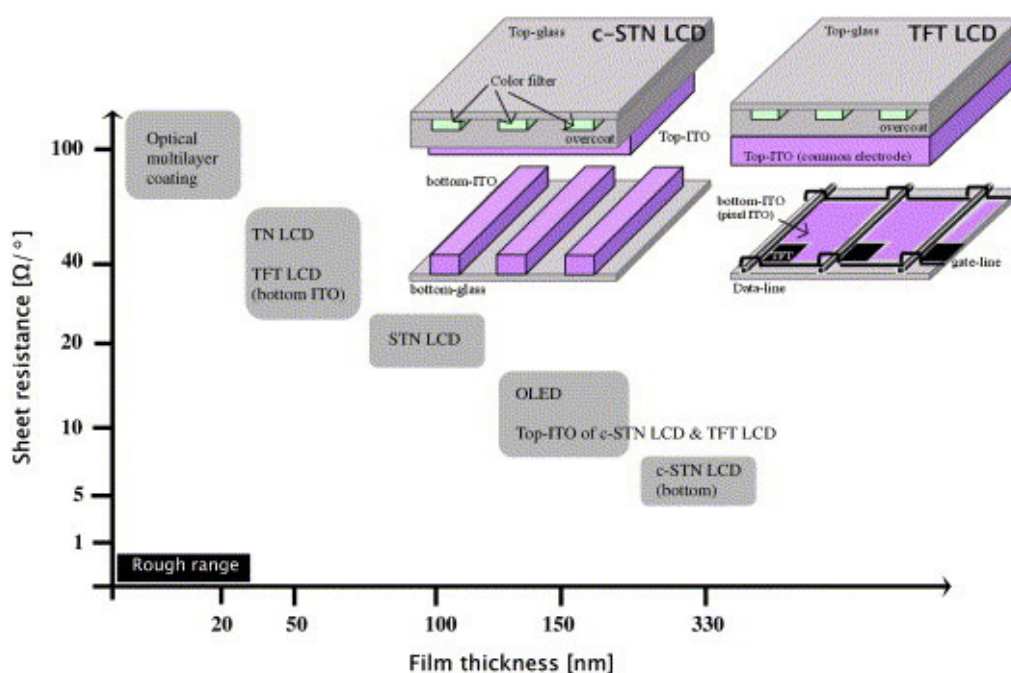


Fig. 1.2 Overview of the rough ranges for the thickness and sheet resistance of ITO electrode in various displays.

1.3 Production and Use

Today virtually all of the LCD produced use ITO as conducting layer. ITO is the transparent conductor of choice in terms of high transmissivity of visible light and low electric resistivity compared to other transparent conductors, such as tin oxide (SnO_2) and zinc oxide (ZnO) (Granqvist & Hultaker (2002)). The sputtering of the substrate is done by display manufacturers. The number of major panel vendors is small. Several of the most well-known are Sharp Corporation, Samsung Electronics, LG.Philips LCD and AU Optronics. As of April 2006, the three biggest manufacturers were LG.Philips and Samsung with 22% of the market, followed by AU Optronics with a 19% share (Wikipedia Contributors (2006)).

Jansseune (2003) estimated that in 2002 45% of the indium supplied went as ITO into coating, whereas in 2004 approximately 72% of the indium was used for ITO coatings (Jorgenson & George (2004)). At the end of 2006 Hewett (2006) estimated the share of indium into ITO as 80% of the total indium supply. In 2004 perhaps 65% of the indium used in coating went into display devices, and the remainder (env. 7%) into transparent heat reflection. ITO coatings are also used in architectural glass and solar panel applications (high infrared reflectance), cathode-ray tubes (CRTs), low-pressure sodium lamps, and windshield glass. (Jorgenson & George (2004)).

In 2005 the USGS estimated that 70% of the Indium was used in coatings; electrical components and semiconductors, 12%; solders and alloys, 12%; and, research and other 6% (USGS (2006)).

Only small amounts of ITO scrap are recycled worldwide. The reason for the low recycling rate is the lack of infrastructure for collecting indium-containing products, although recycling is supposed to expand significantly if the current price of indium is sustained. ITO use is highly inefficient, as only about 15% of ITO ends up making LCDs, whereas the rest is scrap. The major problem in recycling ITO scrap is the high cost associated with the process. The process takes about 12 weeks from collection of scrap to fabrication of secondary indium products. A large increase in ITO scrap could be difficult for the recycling industry to handle because of large capital costs, environmental restrictions, and storage space (USGS (2006)).

Taking an average concentration of 2 ppm¹ of ITO in collected backlights², 90% recovery rate and 95% reprocessing efficiency, the required amount of scrap backlights to gain one kg indium would amount to 585 tonnes. Even with prices as high as 1'000 \$ per kg ITO a global collection system is not probable to evolve, and therefore no post consumer scrap recycling may be considered for the time being.

1.4 System Characterisation

The sputtering of ITO films is reflected within ecoinvent as a series of three related datasets:

- ITO powder, for target production, at plant (GLO) [kg]
- ITO, sintered target, at plant (GLO) [kg]
- Sputtering, ITO, for LCD (GLO) [kg]

These datasets describe the production of nanoscaled ITO powder from indium and tin metal and scrap ITO (prompt scrap), the manufacturing by coprecipitation, moulding and pressing of the target body, sintering, bonding to the backplate, sputtering and the arising and recycling of ITO scrap (Fig. 1.3).

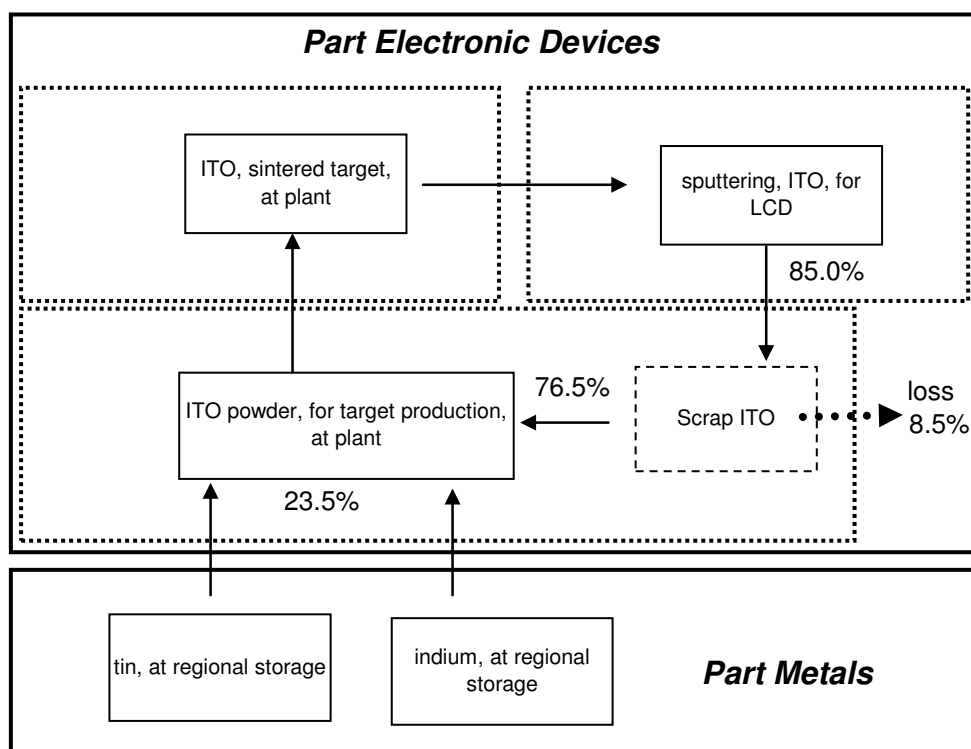


Fig. 1.3 System representation of the datasets treated in this chapter.

¹ Content in 17 inch LCD module (1.8 kg) 1.76 ppm, in 15 inch LCD module (1.1 kg) 2.05 ppm for film thickness of 500 nm.

² A backlight is the form of illumination used in an LCD display.

Tab. 1.2 Ecoinvent meta information for the processes for the production of the target and the required auxiliaries.

Name	ITO powder, for target production, at plant	ITO, sintered target, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	Electricity and heat demand for precipitation; recovery of spent ITO targets,	Production of nanoscaled powder, sintering, bonding, and production of backplate.
Amount	1	1
Local Name	ITO pulver, für target produktion, ab Werk	ITO, gesintertes target, ab Werk
Synonyms	Indium Tin Oxide // Indium Zinn Oxid	Indium Tin Oxide // Indium Zinn Oxid // sputtering target
General Comment	This dataset describes the production of nanoscaled ITO powder by coprecipitation for the use in sputtering targets with a density close to the theoretical one. No direct process emissions are considered due to a lack of information. The input raw materials are prompt scrap from the sputtering process, amounting to 77% of the raw material, and raw materials won from primary metals, making 23%. 8.5% of the total ITO is lost during scrap processing.	This module relates to 1 kg ITO net bound to a backplate. It comprises the forming of a high-density sintered body from ITO and its attachment to a copper backplate. Dimensions assumed: 610mm x 350mm. No direct process emissions are considered due to a lack of respective information.
Start Date	2000	2003
End Date	2005	2005
Data valid for entire period	1	1
Geography Text	Water provision and disposal approximated with Swiss conditions	
Technology Text	Relates to coprecipitation of ITO from a raw aqueous solution and subsequent calcination. The raw solution can be prepared from both pure metals as well as from spent targets.	
Representativeness		
Production Volume		
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see General Comment
Uncertainty Adjustments	none	none

Tab. 1.3 Ecoinvent meta information for the sputtering process.

Name	Sputtering, ITO, for LCD
Location	GLO
Infrastructure Process	0
Unit	m3
Dataset Version	2.0
Included Processes	Sputtering process and infrastructure, as well as the transportation of the targets. No pre and post treatment of the surfaces are considered.
Amount	1
Local Name	Sputtering, ITO, für LCD
Synonyms	Indium Tin Oxide // Indium Zinn Oxid // TFT // Transparent conductive Oxide Layer // TCO // physical vapour deposition // PVD
General Comment	This dataset is established for the manufacturing of LCD screens. Due to a lack of structured data the inventory bases on assumptions and lab-scale technology. Direct process emissions are not known and disregarded. The dataset therefore must not be used for the direct comparison of different film deposition technologies. The functional unit of this dataset is cubic meter resulting from the sputtered area times the film thickness. Film thickness depends on display technology: STN LCD: 150nm top; 330nm bottom; 480nm total. TFT LCD: 150nm top; 50nm bottom; 200nm total. Default value: 340nm total, resulting in typical values for a 17" monitor (893.55 cm ²) of 3.04E-10 m ³ . 85% of the target ends up as scrap.
Start Date	2005
End Date	2005
Data valid for entire period	1
Geography Text	
Technology Text	Relates to RF magnetron sputtering technique with commercially available sputtering targets. Pre-sputtering phase and subsequent main sputtering.
Representativeness	
Production Volume	
Sampling Procedure	Literature
Extrapolations	see General Comment
Uncertainty Adjustments	none

1.5 Production of indium oxide nanoscale powder

ITO powder is mainly used for production of an ITO sintered body (target for sputtering), and contains tin oxide in an amount of usually from 2 to 20 wt %.

The higher the density of the sintered body used, the higher the electric conductivity of the ITO thin film that can be obtained. Therefore, it is known that a high density sintered body having a relative density of 99% or more is suitable as an ITO sintered body used for a sputtering target. The density that is achievable depends on the type of powder used. Depending on the methods of producing such powders, the quality and density of the resulting films may differ. Increasing demands for the film properties require fine, homogenous, and crystalline powders.

The actual manufacturing process is not disclosed to the public. Teckcominco (2006) indicates that metallic indium ingots are processed into an oxide and then mixed with tin-oxide to form indium-tin-oxide (ITO). However, this process probably is outdated today.

Hattori et al. (2006) (Sumitomo Corp.) mention the co-precipitation of ITO (In and Sn oxides) as common process, where ITO is precipitated simultaneously from indium(III) and tin(IV) chlorides in aqueous solution with ammonia water and subsequent calcination of the resulting filtrate, as claimed by Sasaki et al. (1995) and Nishihara et al. (1995) (patents held by Mitsubishi).

In Hattori et al. (2006) a similar process by means of a mixture of tetravalent and bivalent tin ions (SnCl_2) is proposed. Because there are not so much differences between the two precipitation methods, from the inventory point of view the illustrative description of Example 1 in the more recent patent of Hattori et al. (2006) is taken to reflect the process used. It consists of the following steps:

- Preparation of aqueous solution
- Precipitation
- Filtration
- Calcination
- Washing
- Pulverising (wet milling)

1.5.1 Life Cycle Inventory for ITO powder, for target production, at plant

This dataset describes the production of nanoscaled ITO powder by coprecipitation for the use in sputtering targets with a density close to the theoretical one. The raw materials are mainly prompt scrap from the sputtering process, amounting to approximately 77% of the total input of raw material. During scrap processing, 10% of the ITO is lost. Of the raw materials, approximately 23.5% are won from primary metals. The metadata of the established inventory is given in (Tab. 1.2); its exchanges are shown in Fig. 1.4).

Preparation of the raw aqueous solution. The preparation of the raw aqueous solution is effected by dissolution of tin and indium metal in hydrochloric acid with subsequent regulation of the acidity with sodium hydroxide (Tab. 1.5). In a similar manner a solution can be formed with prompt scrap from sputtering and target manufacturing (Tab. 1.6). In the latter case an ion exchange resin column is used to purify the solution namely from zirconium. This purification step has been disregarded, because it can be assumed that the operation of the column has only little demand in energy and causes virtually no emissions. Based on the arising of 85% of ITO as scrap from sputtering and the assumption that of this returned scrap 10% is lost during its recovery, a share of 76.5% of ITO raw material originates from secondary sources and 23.5% from primary sources (see also Fig. 1.3).

Precipitation. For the precipitation the combined solutions are heated up to 55°C and the pH adjusted to 8.0 with sodium hydroxide after 2.5 h. The total amount of $1\text{E-}2$ mol per litre is roughly estimated. Further, a batch process is taken with heating cycles where the reaction mass is heated from room temperature to 55°C, assuming a system efficiency of 75% without heat recovery (Tab. 1.7).

Filtration. An amount of 3%wt of ammonia in the washing water was assumed. The energy consumption for drying the filtrate was assessed with a comparable drying process in the ceramics industry (spray dryer) (Tab. 1.7).

Calcination. The energy consumption has been assessed with the values for calcining bauxite documented in Mori & Adelhardt (1998), assuming that a gas fired furnace is used (Tab. 1.7).

Washing and pulverising. The calcined ITO is washed and afterwards dried, assuming moisture content of 20%. The energy for drying was assessed with the theoretical evaporation energy required and a thermal efficiency of 80%. Pulverising is done in a ball mill for eight hours. A mill was considered with an average power consumption of 1.5 kW considering the range of available values between 0.1 kW for a 6 kg batch mill and 22 kW for a small continuous 1.5 t/h system, based on a non-representative search of devices for sale. Of the ethanol employed, 2% is considered to evaporate, the rest being recovered. The replenishment of the evaporated ethanol is approximated with the dataset for methanol (Tab. 1.7).

Infrastructure, disposal and transport. The infrastructure of the equipment is estimated assuming an output of 25 t per year. The resulting demand of infrastructure is reflected with the dataset "facilities, chemical production" (Tab. 1.4). The buildings are approximated with the respective

dataset for a chemical plant (50'000 t/a output and 25 years lifetime per unit) scaled to 1 kg ITO output. The wastewater is assumed to be treated. This process is approximated with the treatment of effluents from ceramics industry. A total of 2.37 kg auxiliary chemicals (acids, bases, chemicals and raw materials) have to be transported 100 km by lorry and 600 km by rail (Frischknecht et al. (2004), p. 12).

1.5.2 Data Quality Considerations

This dataset describing the production of nanoscaled ITO powder is based on rather theoretical considerations that reflect the recent process to manufacture powder for target production. Due to large data gaps no direct emissions but the ethanol evaporation is included in this inventory. Additionally, only a part of the environmental burdens associated with the infrastructure are probably considered, since clean working conditions require a more extensive infrastructure and additional water, energy etc. consumption than assumed in this inventory. Although some of the assumptions exhibit a high uncertainty, the overall quality of the dataset is sufficient to reflect the production process in the context of sputtering.

This process is distinctly different from other manufacturing processes for ITO powder, namely the production from the metal oxides as mentioned in Teckcominco (2006).

Tab. 1.4 Infrastructure requirements for equipment. Account of considered apparatus to manufacture ITO nanosized powder.

	kg	output tpa	lifetime in a	total output kg	kg / kg output
Reaction vessel	250	25	5	126595	2.0E-03
Filtration apparatus	250	25	10	253189	9.9E-04
Furnace	4000	25	10	253189	1.6E-02
Mill	100	25	5	126595	7.9E-04
Total					1.96E-2

Tab. 1.5 Raw aqueous solution from primary materials. Production of 1 kg ITO slurry. 23.5% of the ITO powder is made from this (primary) source,

Preparation Aqueous Solution	Unit	Value	
Indium-Chloride solution	g	1782	Hattori et al. (2006)
Indium concentration	%wt	23.3%	Hattori et al. (2006)
Indium metal (5N)	kg	0.415	Hattori et al. (2006)
Hydrochloric acid for dissolution	%wt	35%	Hattori et al. (2006)
Hydrochloric acid (per 100%)	kg	0.478	
Tin chloride solution	g	182	Hattori et al. (2006)
Tin Concentration	%wt	23.30%	Hattori et al. (2006)
Tin Metal (4N5)	kg	0.0424	Hattori et al. (2006)
Hydrochloric acid for dissolution	%wt	35%	Hattori et al. (2006)
Hydrochloric acid (per 100%)	kg	0.0489	
Combined solutions	kg	1.96	Hattori et al. (2006)
Regulation of pH with NaOH	kg	1.96	assumption: same order as combined solutions
Sodium hydroxide (to adjust pH)	%wt	25%	Hattori et al. (2006)
Sodium hydroxide (per 100%)	kg	0.491	
Amount of ITO from Metals in slurry	kg	1	

Tab. 1.6 Raw aqueous solution from prompt scrap (spent targets, articles, etc.). Production of 1 kg ITO solution. 76.5% of the ITO powder is made from this source,

Preparation Aqueous Solution	Unit	Value	Remarks
Ground ITO target	kg	1.01	Hattori et al. (2006). Prompt scrap, no burdens attributed.
Hydrochloric acid 35%	kg	0.886	
Hydrochloric acid, for spent target	kg	0.310	Hattori et al. (2006). value reports to 100% HCl
ITO losses	%	1%	Assumption, due to incomplete dissolution, curd, etc.
Energy for dissolution (spent targets)	kWh	5.0	Assumption: 2 l with 808 g ITO @ 80 °C for 40 hours (100 W to keep temperature)
Exchange resin		???	According to Hattori et al. (2006), disregarded.
Tin metal		not accounted	Only secondary sources regarded.
Indium Chloride		not accounted	Only secondary sources regarded.
Sodium Hydroxide		0.491	to obtain pH 0.02 - same as for primary process
ITO in solution	kg	1	

Tab. 1.7 Co-precipitation of ITO from raw aqueous solution. Production of 1 kg ITO nanoscaled powder.

Preparation Aqueous Solution	Unit	Value	Remarks / sources
Precipitation			
Deionised water	kg	2.08	Hattori et al. (2006)
water temperature	°C	55	Hattori et al. (2006)
Reaction mass	kg	3.46	combined solutions, sodium hydroxide, water
Energy for heating up	kWh	0.209	Assumption: batch with heating cycles, efficiency 75%
Sodium hydroxide (as 100%)	kg	0.240	Assumption: 1 e-2 mol per litre, adjust pH after 2.5 h to 8.0
Filtration			
alkali water (NH ₃ , pH 10)	kg	2.0	Hattori et al. (2006)
Wash stages	-	5	Hattori et al. (2006)
total amount for filtration	kg	10.0	multiplication of the two items above.
... of which water	kg	9.73	Deionised water for filtration
... Ammonia (filtration)	kg	0.30	Assumption: 3% _{wt} of total amount
drying the filtrate	kWh	0.152	Spray dryer 38 kW @ 250 kg / h (Timokhova (2002))
Calcination			
Energy consumption for calcination (natural gas)	MJ / kg	1.33	Bauxite calcination 150 Nm ³ Gas / t Al, 4.1 t Bauxit (dry) / t Al for fluidised bed furnace (Mori & Adelhardt (1998))
Washing			
Batch size	kg	1	Hattori et al. (2006)
Deionised Water	kg / kg	3	
Ammonia in wash solution)	kg	0.003	regulate pH to 8-9: 1e-5 mol CH ₃
Electricity for stirring	kWh	0.075	30 minutes @ 0.1 kW and 75% efficiency
Deionised Water for washing	kg	3	Assumption: same relation as for Filtration
Moisture content	%	20%	Assumption
Electricity for drying cake	kWh	0.157	Evaporation of water (41.65 J/mol) with 80% thermal efficiency
Pulverising (wet milling)			
Batch charge	kg	10	Assumption. Patent states 100 g ITO 100 g ethanol, 1000 g zirconia balls 5mm dia
Ethanol	kg / kg	1	Hattori et al. (2006)
Evaporation losses	%	2%	Assumption
Milling duration	h	8	100 rpm
Installed Power	kW	1.5	Assumption: range between 0.1 kW (6 kg batch charge) and 22 kW (small continuous 1.5t/h), 1.48 kW chosen.
Electricity consumption milling	kWh	1.19	Assumption: range between 0.1 kW (6 kg batch charge) and 22 kW (small continuous 1.5t/h), 1.48 kW chosen.
Ethanol evaporated	kg	0.02	Assumption. Patent states 100 g ITO 100 g ethanol, 1000 g zirconia balls 5mm dia
ITO losses	%	0%	
ITO nanopowder product	kg	1	End product

1. Report “Sputtering of ITO layers”

General Flow information				Representation in ecoinvent								Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment

Prep. of solution form spent targets & primary metals (23.5%) Preparation of raw solution form prompt scrap (76.5%) Precipitation Filtration Calci-nation	Indium metal (5N)	ML	ITO powder, for target production, at plant, part I															
	Tin Metal (4N5)	ML																
	Total hydrochloric acid for metal dissolution	ML																
	Sodium hydroxide (per 100%)	ML																
	ground ITO target	ML																
	Hydrochloric acid, for spent target	ML																
	Energy for dissolution (spent targets)	ML																
	Energy for heating up	ML																
	Sodium hydroxide (per 100%)	ML																
	deionised water for filtration	ML																
	Ammonia (filtration)	ML																
	drying the filtrate	ML																
	Energy consumption calcination (gas)	ML																

	metals	extraction	No	RER	indium, at regional storage	9.76E-02	kg	Hattori et al, 2006	1	2.07	(3,3,1,3,5,5,3)
	metals	extraction	No	RER	tin, at regional storage	9.96E-03	kg	Hattori et al, 2006	1	2.07	(3,3,1,3,5,5,3)
	chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	1.24E-01	kg		1	2.07	(3,3,1,3,5,5,3)
	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	1.15E-01	kg		1	2.10	(4,3,1,3,5,5,3)
Prompt scrap, no burdens attributed.						7.73E-01	kg	Hattori et al, 2006	1		
value reports to 100% HCl	chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	2.37E-01	kg	Hattori et al, 2006	1	2.07	(3,3,1,3,5,5,3)
2 l containing 808 g ITO @ 80 °C for 40 hours (100 W to keep temperature)	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.79E+00	kWh	Assumption	1	2.10	(4,3,1,3,5,5,2)
Assumption: batch with heating cycles, efficiency 75%	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.09E-01	kWh		1	2.10	(4,3,1,3,5,5,2)
	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	1.15E-01	kg		1	2.10	(4,3,1,3,5,5,3)
	water supply	production	No	CH	water, deionised, at plant	9.73E+00	kg		1	2.07	(3,3,1,3,5,5,3)
	chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	3.01E-01	kg	Assumption: 3%wt content	1	2.29	(5,3,1,3,5,5,3)
Spray dryer 38 kW @ 250 kg / h	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.52E-01	kWh	Timokhova 2002	1	2.07	(3,3,1,3,5,5,2)
Bauxite calcination 150 Nm3 Gas / t Al, 4.1 t Bauxit (dry) / t Al for fluidised bed furnace (Mori 1998)	natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	1.33E+00	MJ		1	1.58	(3,1,1,3,4,5,1)

Fig. 1.4 Flows for "ITO powder, for target production, at plant" and their representation in the ecoinvent database.

1. Report “Sputtering of ITO layers”

General Flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
washing	Ammonia in wash solution	ITO powder, for target production, at plant, part II	regulate pH to 8-9: 1e-5 mol CH3	chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	3.00E-02	kg		1	2.29	(5,3,1,3,5,5,3)	
	Electricity for stirring		30 minutes @ 0.1 kW and 75% efficiency	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	7.50E-01	kWh		1	2.10	(4,3,1,3,5,5,2)	
	Deionised Water for washing		Assumption: same relation as for Filtration	water supply	production	No	CH	water, deionised, at plant	3.00E+01	kg		1	2.07	(3,3,1,3,5,5,3)	
	Electricity for drying cake		Evaporation of water (41.65 J/mol) with 80% thermal efficiency	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.57E+00	kWh		1	2.10	(4,3,1,3,5,5,2)	
	Electricity consumption milling		Assumption: range between 0.1 kW (6 kg batch charge) and 22 kW (small continous 1.5t/h), 1.48 kW chosen. 8 h milling.	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.19E+00	kWh		1	2.10	(4,3,1,3,5,5,2)	
	Ethanol evaporated		Assumption: patent states 100g ethanol per 100g ITO, losses estimated with 2%	chemicals	organics	No	GLO	methanol, at plant	2.00E-02	kg		1	2.10	(4,3,1,3,5,5,3)	
	Transport services manufacturing, lorry		100 km distance	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.32E-01	tkm		1	2.76	(4,3,1,3,5,5,5)	
	Transport services manufacturing, rail		600 km distance	transport systems	train	No	RER	transport, freight, rail	1.39E+00	tkm		1	2.76	(4,3,1,3,5,5,5)	
	Water to be cleaned		to treatment, aprrox with ceramics production	waste management	wastewater treatment	No	CH	treatment, ceramic production effluent, to wastewater treatment, class 3	4.41E-02	m3		1	2.29	(5,3,1,3,5,5,6)	
	Infrastructure Plant ITO Powder		Proxy: Chemical plant with 50000 tpa output and 25a lifetime	chemicals	organics	Yes	RER	chemical plant, organics	8.00E-10	unit	Assumption	1	3.95	(5,3,1,3,5,5,9)	
Pulverising (wet milling)	Infrastructure supplementary to chemical plant (apparatus for ITO powder production)	200 tpa In into ITO produced at 20 facilities worldwide -> 25 tpa ITO per facility	construction processes	buildings	Yes	RER	facilities, chemical production	1.96E-02	kg	Assumptions	1	3.95	(5,3,1,3,5,5,9)		
	Ethanol emissions		air	high population density			Ethanol	2.00E-02	kg		1	2.33	(4,3,1,3,5,5,18)		
	waste heat		air	high population density			Heat, waste	2.76E+01	MJ		1	2.10	from electricity consumption		
	ITO powder, for target production, at plant		electronics	module	No	GLO	ITO powder, for target production, at plant	1.00E+00	kg						

Fig. 1.4 Flows for "ITO powder, for target production, at plant" and their representation in the ecoinvent database. (cont.)

1.6 Manufacturing of Targets

The targets consist of the sintered ITO body that is bonded to a Cu or Ti backing plate with pure indium based on a proprietary solder technique. The production process is held disclosed by the manufacturers (e.g. Umicore): “Umicore Thin Film Products uses its own proprietary bonding method, based on a flux-free solder technique. Thin film adhesion and diffusion barrier layers are applied to the back of each target, followed by a temperature controlled metallic solder seal between target and backing plate. The bonding is compliant to accommodate mechanical and thermal stress. Bonding integrity is verified by ultrasonic scanning.” (Umicore (2006))

1.6.1 Life Cycle Inventory of “ITO, sintered target, at plant”

This module comprises the forming of a high-density sintered body from ITO and its attachment to a copper backplate. Dimensions of the target are chosen according to Umicore (2006) to 610 mm × 350 mm × 9 mm. These dimensions require 13.73 kg ITO, assuming a maximum density of 7.16 g/cm³. The metadata of the established inventory is given in (Tab. 1.2), their exchanges are shown in Fig. 1.5).

According to Hattori et al. (2006) the body is pressed in two stages, first uniaxially at 10 MPa, then subjected to cold isostatic pressing (CIP) under a moulding pressure of 250 MPa. This moulded body is sintered for 3 hours in oxygen atmosphere at normal pressure and 1500°C.

No data was available about the moulding of the body. The energy requirement for building up pressure depends mainly on the velocity of the action. Since this is not a high throughput process, the pressing may be done with little energy expenditure. It is therefore disregarded in this inventory.

In order to estimate the energy requirements for sintering, three different approaches are chosen: two furnaces of different scales (i-ii) (Graule (2007)), and (iii) on the heat flow through a furnace volume with an insulation with heat transfer coefficient of 2.5 W/m²K (Tab. 1.8).

The materials requirement for the backplate is chosen based on the dimensions and the assumption of 5 mm thick copperplate (at 105% of the targets length and width) and 200 µm indium film (soldering). Electricity demand for soldering is based on Fujino & Suga (2003) who studied PCB soldering and stated a value of 0.328 Wh/cm² for lead-free reflux soldering with a working temperature of 265 °C. Given a much higher heat capacity of the backplate (copper) at the one hand but a lower melting point of indium (156.6 °C) this value is kept as approximation.

The ITO is considered to be transported over 5'000 km by plane and 250 km by lorry (copperplate also 250 km). The dataset “chemical plant, organics” representing a plant with 50'000 tpa output and 25a lifetime per unit is scaled to 1 kg output and taken as approximation to reflect the infrastructure and the land requirements.

Tab. 1.8 Sintering. Energy requirements estimated with three approaches and a sintering time of 3 hours.

		(i) Ceramics oven	(ii) Laboratory scale oven	(iii) heat flow
Volume	m ³	10	0.05	
Surface	m ²			0.53
Power	kW	100	7.50	
Insulation	W/m ² K			2.5
Number of Targets	-	200	1.00	
Power per Target	kW / target	0.5	7.50	1.96
Geo mean (this study)	kW / target		1.95	
Energy requirements	kWh / target		5.84	

1.6.2 Data Quality Considerations

The target manufacturing is only a rough estimate of the processes involved. It probably does underestimate the total environmental burden, because it does not account for the extensive fabrication processes under clean working conditions. The high uncertainties are reflected in the dataset with respective entries into the pedigree matrix (Frischknecht et al. (2004)). No direct process emissions are considered in this dataset.

1. Report "Sputtering of ITO layers"

General Flow information					Representation in ecoinvent							Uncertainty information					
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Weight ITO sintered body	ML	ITO, sintered target, at plant					electronics	module	No	GLO	ITO powder, for target production, at plant	1.00E+00	kg		1	2.10	(4,2,1,3,5,5,3)
Energy requirements for pressing	ML			negligible		electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	0.00E+00	kWh	Assumptions	1	2.28	(5,2,1,3,5,5,2)	
Energy requirements for sintering	ML					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.24E-01	kWh		1	2.10	(4,2,1,3,5,5,2)	
Copper	ML			5 mm copperplate (641mmx368mm)		metals	extraction	No	GLO	copper, primary, at refinery	7.66E-01	kg		1	2.06	(2,2,1,3,5,5,3)	
Processing Copper plate	ML					metals	processing	No	RER	sheet rolling, copper	7.66E-01	kg		1	2.06	(2,2,1,3,5,5,3)	
Indium for soldering	ML			200 um indium film for soldering		metals	extraction	No	RER	indium, at regional storage	2.27E-02	kg		1	2.10	(4,2,1,3,5,5,3)	
Electricity for soldering	ML			Industry value for Pb-free Reflow-soldering assumed as proxy		electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	5.09E-02	kWh	Fujino & Suga 2003	1	2.07	(3,2,1,3,5,5,2)	
Transportation by plane: ITO	ML			5'000 km assumed		transport systems	airplane	No	RER	transport, aircraft, freight	5.00E+00	tkm		1	2.72	(3,2,1,3,5,5,5)	
Transportation by lorry	ML			250 km assumed for copperplate and ITO		transport systems	road	No	RER	transport, lorry >16t, fleet average	4.47E-01	tkm		1	2.72	(3,2,1,3,5,5,5)	
Infrastructure Plant target manufacturer	ML			Proxy: Chemical plant with 50'000 tpa output and 25a lifetime		chemicals	organics	Yes	RER	chemical plant, organics	8.00E-10	unit	Assumption	1	3.76	(4,2,1,3,5,5,9)	
	ML		waste heat			air	high population density				Heat, waste	1.71E+00	MJ		1	2.10	from electricity consumption
	ML		ITO, sintered target, at plant			electronics	module	No	GLO	ITO, sintered target, at plant	1.00E+00	kg					

Fig. 1.5 Flows for "ITO, sintered target, at plant" and their representation in the ecoinvent database.

1.7 Sputtering

Among several methods used for ITO deposition (sputtering, evaporation, chemical vapour deposition, sol-gel method, spray pyrolysis, etc.), direct current (DC) and radio frequency (RF) magnetron sputtering are the most attractive techniques for industrial development because of high deposition rate, good reproducibility, and possibility of using commercially available large area sputtering systems (Kurdesau et al. 2006). The sputtering process is also referred to as physical vapour deposition (PVD). An example for a sputtering production line is the vertical in-line system "Pegasus" of Leybold Optics (Fig. 1.6).



Fig. 1.6 Vertical in-line sputtering system A1500V (Leybold Optics³)

As little as 15% of a given target is actually deposited on the substrate being coated. The rest of the ITO resides as scrap in the form of spent targets, ITO films on the coating chamber walls and ITO bearing solutions generated by the etch processes at LCD factories (O'Neill (2004)).

Although polycrystalline ITO films deposited at elevated substrate temperature can provide the required properties, the industrial coatings are typically produced under low substrate temperature to generate highly amorphous structures for the advantage of a much more rapid and hence cost-efficient patterning process step. A post-deposition heat treatment is then applied in order to transform amorphous ITO into polycrystalline ITO with superior functionality. Moreover, the orientation of anisotropic molecules of nematic liquid crystal between the transparent electrodes has to be determined by a separate aligning film (Betz et al. (2005)).

1.7.1 Life Cycle Inventory for Sputtering, ITO, for LCD

This dataset is established for the manufacturing of LCD screens. The functional unit of this dataset is cubic meter resulting from the sputtered area times the film thickness. Film thickness depends on display technology (s. 1.2.2 ITO Layers): STN LCD: 150 nm top; 330 nm bottom; 480 nm total. TFT LCD: 150 nm top; 50 nm bottom; 200 nm total. LCA practitioner using this dataset without specific information on film thickness should use the average value of 340 nm total ITO thickness, resulting in typical values for a 17" monitor (893.55 cm²) of 3.04E-10 m³ of "sputtering, ITO, for LCD".

The unit process is displayed in Fig. 1.7, respective meta-information is compiled in Tab. 1.3.

The sputtering energy requirements have been assessed based on Lee & Ok Park (2006) and Kurdesau et al. (2006) who report on RF magnetron sputtering experiments. A pre sputtering sequence of

³ <http://www.leyboldoptics.com/> accessed in april 2007

10 min at 4.3 W/cm² sputtering power and a main sputtering sequence of 5 min at 2.15 W/cm² are assumed (Tab. 1.9).

The consumption of argon and oxygen (3%vol) are disregarded because it is assumed to have no significance and likewise the rinsing in methanol and subsequent UV / ozone treatment are disregarded due to a lack of data. The prompt scrap from ITO sputtering is assumed to be recollected to form new ITO powder. This scrap is accounted for in the dataset "ITO powder, for target production, at plant" (s. Fig. 1.3).

The targets are assumed to be transported mainly by air freight, as infrastructure a vertical in-line system from Leybold⁴ (Pegasus, 2 million sqm output per year, lifetime 5 years) was taken to assess the infrastructure.

Tab. 1.9 Unit process ITO Sputtering per m². Assumptions and sources, in ecoinvent the process is scaled by (1/3.4 E-7 m) to give the reference unit m³.

Sputtering	Unit	per m²	Remarks
Density sputtered ITO	kg/m ³	7160	
Film thickness	m	3.40E-07	340 nm film thickness assumed
ITO requirements per m ²	kg	2.43E-03	
Efficiency of transfer to surface	%	15%	USGS (2006)
ITO on target required (net)	kg	1.62E-02	Only mass ITO
Scrap ITO	kg	1.38E-02	Only 15% of the target transferred to film, the rest is scrap to be reused in ITO powder production. USGS (2006)
Energy consumption Pre sputtering	kWh	7.96	10 min @ 4.3 W/cm ² USGS (2006) , 90% energy efficiency assumed
Energy consumption Sputtering	kWh	1.99	5 min @ 2.15 W/cm ² Lee & Ok Park (2006) , 90% energy efficiency assumed
Infrastructure Plant Sputtering	m3	5E-5	Hall 500 sqm, output 5E7 m ² in 25 years lifetime
Infrastructure additional facilities Sputtering	kg	2.00E-04	2 t infra with 2 mill m ² output per year, 5 year lifetime (e.g. Leybold Pegasus 1,500/1,400)
Transport services sputtering, lorry	tkm	2.43E-04	100 km distance
Transport services sputtering, plane	tkm	1.22E-02	5000 km distance

⁴ <http://www.leyboldoptics.com/en/pegasus.html>, accessed in april 2007.

1. Report "Sputtering of ITO layers"

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Net requirements target	sputtering, ITO, for LCD		Only mass of net ITO bound to backplate	electronics	module	No	GLO	ITO, sintered target, at plant	4.77E+04	kg	0	1	2.11	(4,4,1,3,5,5,3)
Energy consumption Pre sputtering			10 min @ 4.3 W/cm2, 90% energy efficiency assumed	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.34E+07	kWh	Lee et al 2006	1	2.11	(4,4,1,3,5,5,2)
Energy consumption Sputtering			5 min @ 2.15 W/cm2, 90% energy efficiency assumed	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	5.86E+06	kWh	Lee et al 2006	1	2.11	(4,4,1,3,5,5,2)
Infrastructure Plant Sputtering			Hall 500 sqm, output 5e7 m2 in 25 years lifetime	construction processes	buildings	Yes	RER	building, multi-storey	1.47E+02	m3	Assumptions	1	3.77	(4,4,1,3,5,5,9)
Infrastructure additional facilities Sputtering			2 t infra with 2 mill m2 output p year, 5 year lifetime (e.g. Leybold Pegasus 1,500/1,400)	construction processes	buildings	Yes	RER	facilities, chemical production	5.88E+02	kg	Assumptions	1	3.77	(4,4,1,3,5,5,9)
Transport services sputtering, lorry			100 km distance	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.16E+02	tkm	Assumptions	1	2.77	(4,4,1,3,5,5,5)
Transport services sputtering, plane			5000 km distance	transport systems	airplane	No	RER	transport, aircraft, freight	3.58E+04	tkm	Assumptions	1	2.77	(4,4,1,3,5,5,5)
		waste heat		air	high population density			Heat, waste	1.05E+08	MJ		1	2.11	from electricity consumption
		Scrap ITO		Only 15% of the target transferred to film, the rest is scrap to be reused in ITO powder production					4.06E+04	kg				
		sputtering, ITO, for LCD			metals	extraction	No	GLO	sputtering, ITO, for LCD	1.00E+00	m3			

Fig. 1.7 Flows for "Sputtering, ITO, for LCD" and their representation in the ecoinvent database.

1.7.2 Data Quality Considerations

The production process is rather new and not much is known from public sources about quantitative process measures. Therefore the data has been established based on reasoning and assumptions from up-to-date lab-scale technology. No direct emissions have been considered. Infrastructure intensity has been approximated with infrastructure from chemical industry. The uncertainties have been calculated with the pedigree matrix approach. Parts of the inventory which are missing due to lack of data or inappropriate assumptions are not sufficient reflected by this method. The uncertainty measure of this dataset therefore may be underestimated.

Note that this process only comprises the pure sputtering and no pre or post treatment. Such preceding or consecutive preparation steps have to be modelled within the dataset referring to this exchange flow.

The data quality is sufficient to reflect the sputtering of conductive films in TFT screen technology in the context of a complete screen. However, this dataset must be used when different manufacturing techniques for ITO films – e.g. chemical vapour deposition, DC magnetron sputtering, solgel technique, etc – are to be compared with each others.

1.8 References

- | | |
|-----------------------------|--|
| Betz et al. (2005) | Betz U., Kharrazi Olsson M., Marthy J., Escolá M. F. and Atamny F. (2005) Thin films engineering of indium tin oxide: Large area flat panel displays application. In: Surface & Coating Technology, 200(20-21), pp. 5751-5759, Online-Version under: http://dx.doi.org/10.1016/j.surfcoat.2005.08.144 . |
| Frischknecht et al. (2004) | Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Hirschier R., Hellweg S., Nemecek T., Rebitzer G. and Spielmann M. (2004) Overview and Methodology. Final report ecoinvent 2000 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch . |
| Fujino & Suga (2003) | Fujino M. and Suga T. (2003) Evaluation of energy consumption of lead-free soldering process. In proceedings from: Environmentally Conscious Design and Inverse Manufacturing, 2003. EcoDesign '03. 2003 3rd International Symposium on, 8-11 Dec. 2003. |
| Gan et al. (2006) | Gan Y., Liu J. and Zeng S. (2006) Transparent conductive indium tin oxide film fabricated by dip-coating technique from colloid precursor. In: Surface and Coatings Technology, 201(1-2), pp. 25-29, Online-Version under: http://dx.doi.org/10.1016/j.surfcoat.2005.10.027 . |
| Granqvist & Hultaker (2002) | Granqvist C. G. and Hultaker A. (2002) Transparent and conducting ITO films: new developments and applications. In: Thin Solid Films, 411(1), pp. 1-5, Online-Version under: http://dx.doi.org/10.1016/S0040-6090(02)00163-3 . |
| Graule (2007) | Graule T. (2007) Dübendorf. Email to Classen M. on: "Infos zu Offenleistung". |
| Hattori et al. (2006) | Hattori T., Fujiwara S. and Saegusa K. (2006) Method of Producing Indium Tin Oxide Powder. Patent Number U.S. 7,115,219 B2, held by Sumitomo Chemical Company, Limited, Osaka (JP) |
| Hewett (2006) | Hewett C. (2006) A volatile year. In: Metal Bulletin, Online-Version under: http://www1.metalbulletin.com/story_2003.asp?storycode=1332427 . |
| Jansseune (2003) | Jansseune T. (2003) The indium market and compound semiconductors. In: Compound Semiconductor Week. Number September. |
| Jorgenson & George (2004) | Jorgenson J. D. and George M. W. (2004) Indium. In: USGS Open File Reports 2004-1300. U.S. Geological Survey, Online-Version under: http://minerals.usgs.gov/minerals/pubs/commodity/indium . |

- Kurdesau et al. (2006) Kurdesau F., Khripunov G., da Cunha A. F., Kaelin M. and Tiwari A. N. (2006) Comparative study of ITO layers deposited by DC and RF magnetron sputtering at room temperature. In: Journal of Non-Crystalline Solids, 352(9-20), pp. 1466-1470, Online-Version under: <http://dx.doi.org/10.1016/j.jnoncrysol.2005.11.088>
- Lee & Ok Park (2006) Lee H.-C. and Ok Park O. (2006) The evolution of the structural, electrical and optical properties in indium-tin-oxide thin film on glass substrate by DC reactive magnetron sputtering. In: Vacuum, 80(8), pp. 880-887, Online-Version under: <http://dx.doi.org/10.1016/j.vacuum.2005.11.069>
- Mori & Adelhardt (1998) Mori G. and Adelhardt W. (1998) Stoffmengenflüsse und Energiebedarf bei der Gewinnung ausgewählter mineralischer Rohstoffe; Teilstudie Aluminium. In: Geologisches Jahrbuch, Vol. Sonderhefte SH 2. Bundesanstalt für Geowissenschaften und Rohstoffe, Hannover. ISBN 3-510-95824-1.
- Nishihara et al. (1995) Nishihara A., Hayashi T. and Sekiguchi M. (1995) MANUFACTURE OF CONDUCTIVE OXIDE POWDER. Patent Number Japan 07-021831, held by MITSUBISHI MATERIALS CORP
- O'Neill (2004) O'Neill B. (2004) Indium: Supply, Demand & Flat Panel Displays. In proceedings from: Presented at Minor Metals 2004, London, June 2004.
- Sasaki et al. (1995) Sasaki T., Sasaki H. and Senna T. (1995) PRODUCTION OF ITO SINTERED BODY. Patent Number Japan 07-247162, held by MITSUBISHI MATERIALS CORP
- Teckcominco (2006) Teckcominco (2006) Indium Applications at December from <http://products.teckcominco.com/Products/IndiumApplications.html>
- Timokhova (2002) Timokhova M. I. (2002) Certain specifics of quasi-isostatic molding (a review). In: Glass and Ceramics, 59(1-2), pp. 22-26, Online-Version under: <Go to ISI>://000177012300007.
- Umicore (2006) Umicore (2006) Sputtering Targets and PVD Materials for Displays. . Online-Version under: <http://www.thinfilmpproducts.umicore.com>.
- USGS (2006) USGS (2006) Indium. In: Mineral Commodity Summaries. U.S. Geological Survey.
- Wikipedia Contributors (2006) Wikipedia Contributors (2006) TFT LCD. Wikipedia, The Free Encyclopedia. Retrieved at 13 December 2006 from http://en.wikipedia.org/w/index.php?title=TFT_LCD&oldid=93781736. Date of last revision: 12 December 2006.

2 Manganese dioxide oxidation

Author: Jürgen Sutter, ETH S+U, Zurich
Last changes: 2010

2.1 Introduction

The manganese dioxide oxidation process is the most important production processes for potassium permanganate.

Potassium manganate(VII) (CAS-No. 7722-64-7, KMnO_4 , Mr 158.03, δ 2.703 g/cm³, decomp. > 200 °C) gives dark purple to bronze-coloured rhombic crystals. Its solutions are faint pink to deep violet, depending on concentration. The solids content of a saturated aqueous solution at 20 °C is ca. 6 wt % KMnO_4 (11 wt % at 40 °C and 20 wt % at 65 °C). Permanganate solutions are most stable in the neutral or near-neutral pH region; acidic or alkaline solutions decompose with loss of oxygen. Potassium permanganate is also soluble in several organic solvents, including acetone, glacial acetic acid, methanol, and sulfolane (tetrahydrothiophene-1,1-dioxide). Such solutions usually have limited stability because the permanganate ion slowly attacks the solvent.

When dry potassium permanganate is heated to 200 – 300 °C, an exothermic, autocatalytic decomposition takes place with the evolution of oxygen. Contact with combustibles, especially water-soluble organic substances (such as polyhydroxy compounds), can lead to spontaneous ignition and potentially violent combustion.

The usefulness of permanganate as an oxidant is a result of several unique properties. One is the compound's ability to function as an oxidant at all pHs. Also interesting is the fact that permanganate oxidations can be carried out in both aqueous and several nonaqueous solvents. Phase-transfer catalysis greatly extends the range of usable solvents.

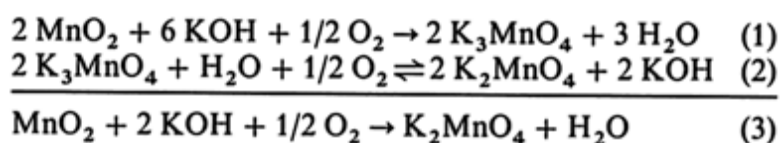
Permanganate oxidizes a wide variety of inorganic and organic compounds. It is often regarded as the oxidant of choice for olefins and aromatic side chains even though it can also interact with other functional groups. Permanganate oxidations can be used not only for specific modification of organic molecules but also to destroy such materials as in pollution abatement applications.

For this inventory the functional unit is 1 kg of manganese ore input.

2.2 Production Technologies (Ullmann's 2002)

Most permanganate is made from MnO_2 -containing ore by fusion followed by electrolysis.

The industrial preparation of potassium permanganate and its precursors (K_3MnO_4 and K_2MnO_4) begins with a finely ground MnO_2 ore which is subjected to alkaline oxidative fusion:



In a separate step, potassium manganate(VI) is subsequently converted to potassium permanganate, usually by anodic oxidation.

Several continuous or batch-type fusion processes are in current use, but they can be categorized in two groups:

1. Roasting processes that employ a MnO_2 : KOH molar ratio of 1 : 2 to 1 : 3 (the reaction mixture is a solid).
2. Liquid-phase processes with a MnO_2 : KOH molar ratio $\geq 1 : 5$.

Roasting processes usually involve two steps, with reactions (1) and (2) carried out sequentially. In the primary step, finely ground manganese dioxide is mixed with 50 % KOH (frequently in a highly dispersed form) and treated with air or oxygen at 300 – 400 °C. The KOH/MnO_2 reaction mixture is usually introduced as a fine spray into internally or externally heated rotary kilns (see Fig. 1). Stationary spray drier-like reactors are also in use. Most of the water quickly evaporates, and the highly concentrated KOH reacts within a few minutes with MnO_2 and oxygen to form K_3MnO_4 (Eq. 1). The dry, blue-green reaction product is ground to increase its surface area and then subjected to secondary oxidation (Eq. 2), in which it is exposed to a moisture-laden air or oxygen stream at 190 – 210 °C. Several hours are required to effect nearly complete conversion to the black-green K_2MnO_4 .

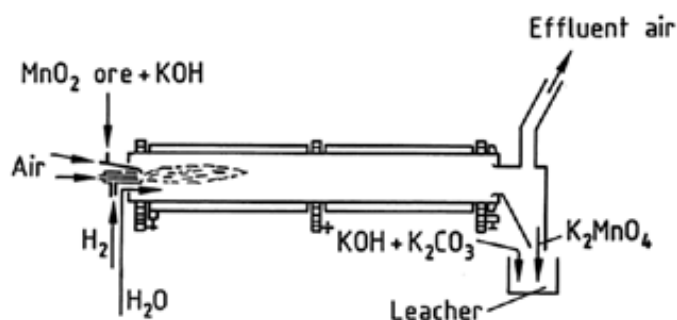


Fig. 2.1 Rotary kiln with internal heating for the production of potassium manganate(VI)

Rotary kilns are commonly employed for this step, but other types of solid – gas contactor (e.g., specially designed “plate” reactors) are also utilized. The removal of hard crusts that form on the inner walls of the reactor can cause substantial downtime.

In **liquid-phase processes** (see Fig. 5) finely ground MnO_2 ore is added to an excess of concentrated, molten potassium hydroxide (70 – 90 wt %) at 200–350 °C. The reaction mixture is agitated vigorously and a stream of finely dispersed air or oxygen is passed through the molten mass. The MnO_2 is converted to K_2MnO_4 in a single step (Eq. 3), although such an oxidation can also be effected in two stages. Potassium manganate(VI) crystallizes from the melt, from which it must then be separated by suitable means (e.g., thickening, decantation, filtration, and centrifugation).

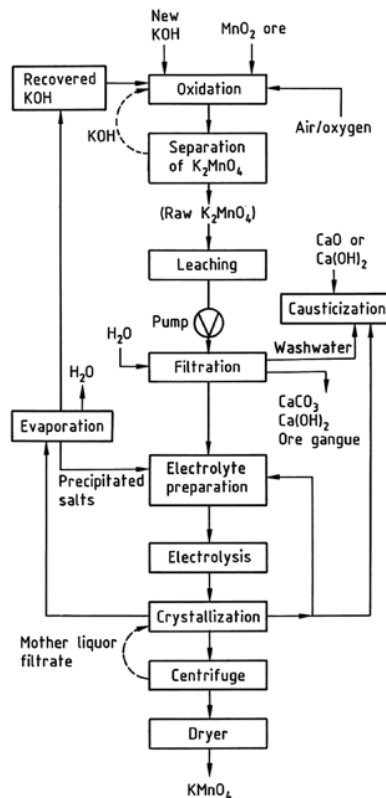


Fig. 2.2 Production of potassium permanganate: liquid-phase oxidation process (overall flow schematic)

Oxidation of the manganate(VI) to permanganate is always accomplished by electrolysis:



Crude manganate from the fusion process is first leached in dilute potassium hydroxide (90 – 250 g KOH per litre, depending upon the nature of the subsequent electrolysis). The resulting leach solution is usually filtered to separate insolubles, originating from the gangue portion of the ore (see Fig. 2).

Effective control of key parameters is decisive for the overall success of the electrolysis, including the concentrations of K₂MnO₄, KMnO₄, and KOH, as well as the temperature (usually ca. 60 °C).

Cell Design

Typical electrolysis cells employ voltages of 2.3 – 3.8 V and anodic current densities of 50 – 1500 A/m². Current yield depends on cell design and mode of operation, and ranges from 40 – 90 %. Nickel or Monel Metal are the preferred materials for the anode, whereas iron is usually the metal of choice for the cathode.

Cells may be either monopolar or bipolar, rectangular or circular, and may be designed for either flow-through or batch operation.

Crystallization

Crystallization of potassium permanganate is effected either directly in a specially designed electrolytic cell or in separate crystallizers. Modern installations use single- or multistage vacuum

crystallizers, which produce a crystalline product directly from the mother liquor in such purity that only centrifugation and drying are required before the KMnO_4 is packaged and sold (see Fig. 8).

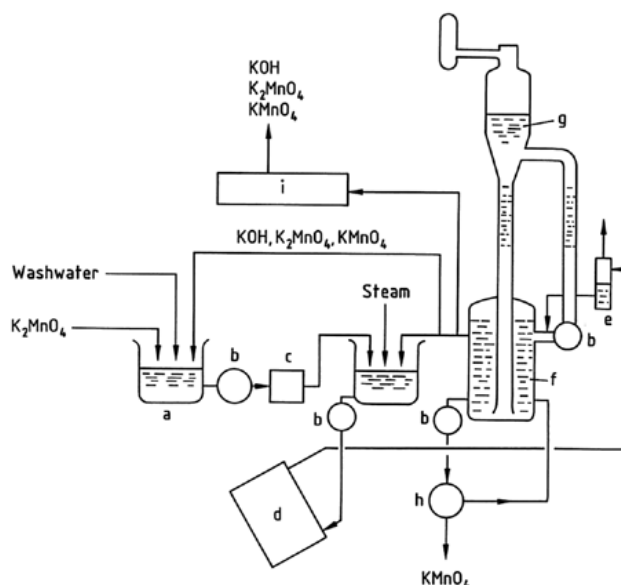


Fig. 2.3 Production of potassium permanganate: continuous electrolyzing – crystallizing system

a) Leach tank; b) Pump; c) Filter; d) Electrolysis cell; e) Gas separator; f) Crystallizer; g) Vacuum evaporator; h) Centrifuge; i) Evaporator

Removal of Impurities and Recycling

When the crude manganate is leached with dilute KOH a solid residue remains. This is composed of the insoluble portion of the ore gangue (e.g., iron oxide hydrates, precipitated aluminium silicate), unreacted MnO_2 ore, and some precipitated MnO_2 (from disproportionation of K_2MnO_4). These insoluble constituents are removed by the use of thickeners, vacuum filters, or filter presses. The Bitterfeld process operates with unfiltered electrolyte, but here the electrolyzers have been specially designed to cope with suspended matter.

In order to recycle potassium hydroxide generated in the electrolysis step, the spent mother liquor is first concentrated by evaporation to a relative density of ca. 1.575 – 1.585. This “recovered” caustic potash is then recycled to the front end of the process and added to either the roaster or the liquid-phase oxidizer. Alternatively, after removal of dissolved impurities, the recovered KOH can be treated with carbon dioxide to produce potassium carbonate. This method has the drawback that there is a limited market for the substantial quantities of potassium carbonate coproduct.

Dissolved impurities (in particular silicate and aluminate from the gangue portion of the ore) must be removed from the process liquors in order to avoid formation of rock-like deposits on pipeline and equipment walls. Equally undesirable is the build-up of carbonate in the process liquors due to reaction of atmospheric CO_2 with KOH . Excessive carbonate concentrations in the recycle system reduce the rate of conversion of ore into manganates and can also cause low current efficiencies in the electrolytic cell.

All three major impurities (Si , Al , and carbonate) can be removed together with such minor impurities as Cu , Pb , Ni , and Co by treatment with calcium hydroxide.

Waste Disposal

The treatment and disposal of solid wastes from permanganate production is an ecological as well as an economic problem. The waste product contains not only unreacted ore but also some precipitated MnO_2 . The main constituents are derived from the insoluble portion of the ore gangue (Fe, Si, Al compounds) but these are accompanied by significant quantities of potassium and calcium salts. One permanganate producer has proposed recovering most of the adsorbed and chemically-bound potassium by treatment with calcium hydroxide. After washing, the treated waste could then be deposited in landfills. The patent literature describes a number of potential uses for the waste from permanganate production, including extraction of residual manganese, or use of the waste as an absorbent for air pollutants, as a micronutrient additive for fertilizers, or in building materials.

2.3 Use (Ullmann's 2002)

The United States is the leading consumer of permanganate followed by Western Europe and Japan intended for environmental applications, such as drinking water treatment (removal of tastes, odours, iron, and manganese; control of trihalomethanes); wastewater treatment (destruction of hydrogen sulphide and other toxic and/or corrosive compounds, including phenols and plating wastes); and air purification (degradation of malodorous or toxic constituents in industrial off-gases).

In other parts of the world (particularly Eastern Europe and India), the principal use of potassium permanganate is in chemical processing, especially the manufacture of synthetic organics (e.g., saccharine, chloroamphenicol, ascorbic acid, isonicotinic acid, pyrazinoic acid). Even in the Western world, chemical processing applications are significant. Thus, numerous intermediates for pharmaceuticals and pesticides are made via permanganate oxidation, as are many fine organics, including flavouring. Significant tonnages are also consumed for the oxidative destruction or precipitation of undesirable impurities, often ones that impart unwanted colours or odours. KMnO_4 is used in the purification of such chemicals as methanol, ethanol, acetic acid, caprolactam, adiponitrile, plasticizers, carbon dioxide for dry ice manufacture, zinc chloride, and hydrofluoric acid. In recent years, KMnO_4 has played a major role in the bleaching of indigo and other dyes for cotton twill fashion garments. This involves processes known as “frosting”, “icing”, or “acid washing”. Permanganate is also used for bleaching beeswax, natural sponges, jute fibres, and certain clays.

Other important uses include surface treatment of carbon steels and stainless steels (descaling and desmutting, especially in wire manufacture), etching of rubber and plastics, and decontamination of nuclear reactors. Fish farmers utilize KMnO_4 to prevent oxygen depletion and to control fish parasites.

Small quantities of methyl formate are used as solvents and insect control agents. With the anticipated extension of C1 chemistry, methyl formate may be used as an intermediate for a number of products.

2.4 System Characterisation

In this study the manganese dioxide oxidation process is described. The functional unit is 1 kg manganese ore. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. Within the manganese dioxide oxidation process, potassium permanganate, potassium carbonate and hydrogen are derived as co-product. Therefore an allocation between the three products was made.

2.5 Raw materials and auxiliaries

Tab. 2.1 shows values for the use of raw materials for the manganese dioxide oxidation process. Pletcher et al. (1990) reports an average yield of 90% for the whole process. The values for the consumption of manganese ore and potassium hydroxide are calculated with this yield. The consumption of oxygen was calculated with an assumed yield of 100%. The amount of carbon dioxide is calculated with an assumed loss of 0.2%.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption.

Tab. 2.1 Consumption of raw materials and auxiliaries for the manganese dioxide oxidation process per kg manganese ore

Input	Values	Remarks
Manganese ore [kg]	1	Functional unit
Potassium hydroxide [kg]	6.44E-1	Stoichiometric calculation with a yield of 90%
Oxygen [kg]	1.66E-1	Stoichiometric calculation with a yield of 100%
Carbon dioxide [kg]	2.09E-2	Stoichiometric calculation with an estimated emission of 0.2%
Cooling water [m ³]	3.92E-2	Estimated from Gendorf 2000

2.6 Energy

Pletcher et al. (1990) reports an electricity consumption of 600 kWh per ton product for a batch process and of 500 kWh per ton product for a continuous process. For this study an average value is used.

Tab. 2.2 Energy consumption for the manganese dioxide oxidation process

Input	per kg manganese ore
Electricity (kWh)	8.99E-1

2.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 2.3 summarises the total transport amounts for the production of 1 kg potassium permanganate.

Tab. 2.3 Total transport amounts for the manganese dioxide oxidation process

tkm kg ⁻¹ manganese ore	lorry	train
Total	1.81E-01	1.09

2.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of manganese dioxide oxidation plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

2.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the manganese dioxide oxidation process. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the carbon dioxide input. The assumed values for the emissions to air are given in Tab. 2.4.

Tab. 2.4 Process emissions to air from the manganese dioxide oxidation process

Output	Value	Remarks
Carbon dioxide, fossil, to air (kg)	4.10E-4	Estimated as 0.2 % of input
Waste heat (MJ)	3.24	calculated from electricity input

2.10 Waste

The unreacted manganese ore (10% of the input) is assumed to be disposed as inert waste.

2.11 Allocation of potassium permanganate, potassium carbonate, and hydrogen

Within the manganese dioxide oxidation process potassium permanganate, potassium carbonate, and hydrogen are obtained. While the first two are considered as co-products, the produced hydrogen amount – due to the small amount – is considered as a by-product and thus no burdens are allocated to the hydrogen output.

The input of manganese ore and oxygen and the disposal of waste ore are allocated to potassium permanganate. For the other inputs and outputs a stoichiometric allocation of the three products is made. The used allocation factors are given in Tab. 2.5.

Tab. 2.5 Allocation of manganese ore and oxygen to the outputs of the manganese dioxide oxidation

Process	Manganese dioxide oxidation	
	kg per kg manganese ore	Allocation factor (stoichiometric)
Potassium permanganate, at plant, RER	1.634	97.47
Potassium carbonate from manganese dioxide oxidation, at plant, RER	0.049	2.53

2.12 Data Quality Considerations

Tab. 2.6 shows the life cycle inventory and the data quality indicators for the manganese dioxide oxidation process. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

2. Manganese dioxide oxidation

Tab. 2.6 Unit process raw data for the manganese dioxide oxidation process

401	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	Manganese dioxide oxidation	UncertaintyType	StandardDeviation95%	GeneralComment	potassium permanganate, at plant	potassium carbonate from manganese dioxide oxidation, at plant
							RER				RER	RER
	662		Location				0				0	0
	493		InfrastructureProcess				kg				kg	kg
	403		Unit									
allocated products resource, in water technosphere	-	2	potassium permanganate, at plant	RER	0	kg	1.63E+0				100	0
	-	2	potassium carbonate from manganese dioxide oxidation, a	RER	0	kg	4.85E-2				0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	3.92E-2	1	1.30	(4,5,na,na,na,na); Estimation	97.470	2.530
	5	-	manganese concentrate, at beneficiation	GLO	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	100.000	-
	5	-	potassium hydroxide, at regional storage	RER	0	kg	6.44E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	97.470	2.530
	5	-	oxygen, liquid, at plant	RER	0	kg	1.66E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	100.000	-
	5	-	carbon dioxide liquid, at plant	RER	0	kg	2.09E-2	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	97.470	2.530
	5	-	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	1.00E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	100.000	-
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	8.99E-1	1	1.51	(1,1,5,3,1,3); Pletcher/Walsh 1990	97.470	2.530
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.81E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	97.470	2.530
	5	-	transport, freight, rail	RER	0	tkm	1.09E+0	1	2.09	(4,5,na,na,na,na); Standard distance 600km	97.470	2.530
	5	-	chemical plant, organics	RER	1	unit	6.54E-10	1	3.09	(4,5,na,na,na,na); Estimation	97.470	2.530
emission air, high population density	-	4	Heat, waste	-	-	MJ	3.24E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input	97.470	2.530
	-	4	Carbon dioxide, fossil	-	-	kg	4.10E-4	1	1.30	(4,5,na,na,na,na); Estimation	97.470	2.530

2.13 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hischer R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under:
<http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Pletcher et al. 1990 Pletcher, Derek/Walsh, Frank C.: Industrial electrochemistry. 2nd ed. Chapman and Hall, London 1990
- Ullmann's 2002 Reidies, Arno: Manganese compounds. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2005 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7th Electronic Release Edition. Wiley InterScience, New York, Online-Version under:
http://mrw.interscience.wiley.com/emrw/9783527306732/ueic/article/a16_123/current/html?hd=All,potassium&hd=All,permanganate#a16_123-sec1-0004

3 Methylamine

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2010

3.1 Introduction

Methylamine (CH_3NH_2 ; CAS 74-89-5, methanamine, aminomethane) is a colourless gas at room temperature.

For this inventory the functional unit is 1 kg of pure methylamine. The most important chemical and physical properties of methylamine used in this inventory are given here.

Tab. 3.1 Chemical and physical properties of methylamine (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	31.06	g mol^{-1}	
Density	662	kg m^{-3}	liquide
Boiling point	-6.3	$^{\circ}\text{C}$	
Melting point	-93.46	$^{\circ}\text{C}$	

3.2 Use

Methylamine is an intermediate in the synthesis of pharmaceuticals (e.g., theophylline), pesticides (carbaryl, sodium metham, carbofuran), surfactants, photographic developers, explosives, and solvents such as N-methyl-2-pyrrolidone.

3.3 Production Technologies (Ullmanns 2004)

Methylamines are prepared commercially by the exothermic reaction of methanol with ammonia over an amorphous silica – alumina catalyst at 390 – 430 $^{\circ}\text{C}$. All three possible methylamines are produced.

The equilibrium composition favors the formation of trimethylamine, whereas market demand is greater for mono- and especially dimethylamine. Excess mono- and trimethylamines are recycled over the same catalyst, leading to additional dimethylamine by dismutation or disproportionation.

The kinetics of the overall reaction have been studied by using several amorphous solid acid catalysts. Amine synthesis can be modeled by six bimolecular equilibrium reactions, three involving amine synthesis and three describing dismutation:

Amination reactions



Dismutation reactions

All six equilibrium constants are known. A parallel reaction leads to the formation of dimethylether, which may also be an intermediate in the synthesis of the amines:



Research has been directed toward improving the selectivity for dimethylamine through the use of shape-selective zeolite catalysts.

Reactant selectivity is a consequence of one of the reactants being too large to pass through the zeolite channels, whereas product selectivity is observed when only certain products are of the proper size and shape to diffuse out of the channels. The activity of zeolites with respect to the conversion of methanol is much higher than that of amorphous silica – alumina, but coke formation within the catalyst pores limits catalyst life even at 300 °C (instead of 400 – 430 °C used with amorphous silica – alumina catalysts).

Purification

The crude reaction mixture consists essentially of excess ammonia; mono-, di-, and trimethylamines; reaction water; and unconverted methanol. Purification is generally effected in a train of four to five distillation columns.

In one technique (Fig. 2) the operating pressures of the columns are chosen in such a way that the top condensers can be cooled by standard 30 °C cooling water, and that the bottom reboilers are fed with saturated steam of about 195 °C. The first column (a) (known as the ammonia column), which operates at 1.7 MPa, removes excess ammonia. The latter is recycled to the synthesis section. In the second column (b), pure trimethylamine is recovered overhead by using extractive distillation with water and a pressure of 1.0 MPa. Extraction water, reaction water, and unconverted methanol are removed from the bottom of the third column (c); the mixture of mono- and dimethylamines is distilled overhead. This mixture is separated in the fourth distillation column (d) at 0.7 MPa, MMA being the overhead and DMA the bottom product.

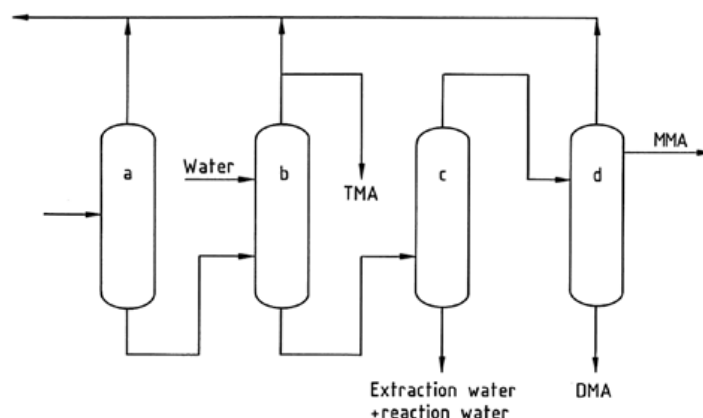


Fig. 3.1 Schematic of methylamine purification (Legend: a) Ammonia column; b) Extractive distillation column; c) Column for water removal; d) MMA and DMA product column)

In another technique, (Fig. 3), a supplementary column (b) operating at 0.7 – 0.9 MPa is inserted after the ammonia column (a) for bottoms removal of reaction water, certain impurities, and unconverted methanol. Most of the extraction water may be recycled; the small remaining quantity of bottom water, containing some impurities and methanol, is disposed of easily either by incineration or biological treatment.

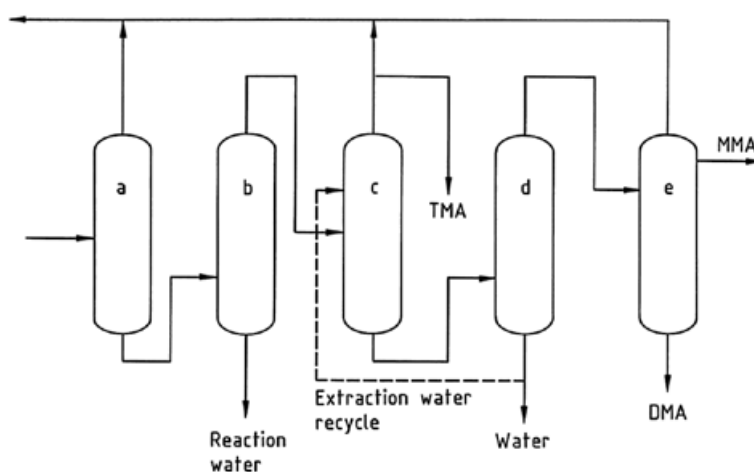


Fig. 3.2 Methylamine purification with supplementary column (Legend: a) Ammonia column; b) Supplementary column; c) Extractive distillation column; d) Column for water removal; e) MMA and DMA product column)

3.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg methylamine, at plant, global**. For this study the production of methylamine by an amination of methanol is considered. All data in the present report are referred to 1 kg methylamine 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final

product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

3.5 Raw materials and auxiliaries

As there was no data available for the production of methylamine, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the methylamine production.

Tab. 3.2 Consumption of raw materials and auxiliaries for the methylamine production

Input	Values	Remarks
Methanol (kg)	1.15	Calculated with a yield of 95 %
Ammonia (kg)	0.577	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

3.6 Energy

Energy data for the amination of methanol are taken from Kim 2003.

Tab. 3.3 Energy consumption for the methylamine production

Input	per kg methylamine
Heat, natural gas burned in industrial furnace > 100 kW	1.76
Electricity (kWh)	0.358

3.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 3.4 summarises the total transport amounts for the methylamine production.

Tab. 3.4 Total transport amounts for the methylamine production

tkm kg ⁻¹ methylamine	lorry	train
Total transports	1.72E-01	1.03

3.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of methylamine production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

3.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the methylamine production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the input of methanol and ammonia.

Tab. 3.5 Process emissions to air from the methylamine production

Output	Value	Remarks
Waste heat (MJ)	1.29	calculated from electricity input
Methanol, to air (kg)	2.29E-03	estimated as 0.2 % of the input
Ammonia, to air (kg)	1.15E-02	estimated as 0.2 % of the input
CO ₂ , to air (kg)	1.39E-01	From wastewater treatment

3.10 Emissions to water

The remaining amount of unreacted methanol and ammonia is assumed to leave the production process with the waste water. Further it was assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for methanol is assumed with 90%. The carbon contained in the removed isopropanol is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC, and DOC used in this inventory are calculated from the amount of methanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC is used.

For the ammonia within the waste water a removal efficiency of 50% was assumed. The nitrogen emitted to air (N₂) and remaining in the sewage sludge was neglected.

Tab. 3.6 Process emissions to water from the methylamine production

Output	Value	Remarks
Methanol (kg)	1.11E-02	calculated from mass balances
Ammonia (kg)	2.77E-03	calculated from mass balance
BOD, COD (kg)	1.62E-02	calculated from water emissions
TOC, DOC (kg)	4.21E-03	calculated from water emissions

3.11 Life cycle inventory of methylamine production and data quality considerations

Tab. 3.7 shows the life cycle inventory and the data quality indicators for the methylamine production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

3. Methylamine

Tab. 3.7 Unit process raw data for the methylamine production

401	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	methylamine, at plant	UncertaintyType	StandardDeviation95%	GeneralComment
	662		Location				RER			
	493		InfrastructureProcess				0			
	403		Unit				kg			
resource, in water technosphere	-	0	methylamine, at plant	RER	0	kg	1,00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2,40E-2	1	1,38	(4,5,1,1,1,5); Estimation
	5	-	methanol, at plant	GLO	0	kg	1,15E+0	1	1,38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	ammonia, liquid, at regional storehouse	RER	0	kg	5,77E-1	1	1,38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	1,76E+0	1	1,13	(1,2,3,1,1,3); Kim 2003
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3,58E-1	1	1,13	(1,2,3,1,1,3); Kim 2003
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1,72E-1	1	2,14	(4,5,1,1,1,5); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	1,03E+0	1	2,14	(4,5,1,1,1,5); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4,00E-10	1	3,09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	1,29E+0	1	1,38	(4,5,1,1,1,5); Calculated from the electricity input
emission air, high population density	-	4	Methanol	-	-	kg	2,29E-3	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	Ammonia	-	-	kg	1,15E-3	1	1,37	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1,39E-1	1	1,30	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	1,12E-2	1	3,09	(4,5,na,na,na,na); Estimation
	-	4	Ammonium, ion	-	-	kg	2,77E-3	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1,62E-2	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1,62E-2	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	4,21E-3	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	TOC, Total Organic Carbon	-	-	kg	4,21E-3	1	1,62	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Methanol	-	-	kg	1,12E-2	1	3,09	(4,5,na,na,na,na); Estimation
	-	4	Ammonium, ion	-	-	kg	2,77E-3	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1,62E-2	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1,62E-2	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	4,21E-3	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	TOC, Total Organic Carbon	-	-	kg	4,21E-3	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	1,12E-2	1	3,09	(4,5,na,na,na,na); Estimation
	-	4	Ammonium, ion	-	-	kg	2,77E-3	1	1,62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1,62E-2	1	1,62	(4,5,na,na,na,na); Estimation

3.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	methylamine, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	production of methylamine including materials, energy uses, infrastructure, and emissions
	404	Amount	1
	490	LocalName	Methylamin, ab Werk
	491	Synonyms	CH5N
	492	GeneralComment	The process "methylamine, at plant, RER" is modelled for the production of methylamine from methanol in Europe. Raw materials are modelled with a stoichiometric calculation. Emissions are estimated. Energy consumptions, infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organika
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2010
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	amination of methanol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	8
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	chemicals

3.13 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschler R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under:
<http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Kim 2003 Kim S., Overcash M.: Energy in chemical manufacturing processes: gate-to-gate information for life cycle assessment. In: Journal of Chemical Technology & Biotechnology vol. 78, no. 9: 995-1005(11). 2003
- Ullmanns 2004 August B. van Gysel/Willy Musin: Methylamines. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: DOI: 10.1002/14356007.a16_535

4 Report “Board based on industrial residual wood”

Autor: Frank Werner, Werner Umwelt & Entwicklung

Review: Hans-Jörg Althaus, Empa

Last Changes: 2009 [complements Section 12.4 of ecoinvent 2.0 report no. 9]

4.1 Fibreboard, MDF (updated 2008)

Fibreboard consists, like paper, of lignocellulose-containing fibres. Both materials are related with regard to production and construction, and the history of fibreboard is closely associated with that of pulp and paper.

The first fibreboard was softboard, i.e. an insulating board. It was produced in the United Kingdom around 1900 - and somewhat later in the United States - from waste fibres that were unsuitable for paper production because they were too short.

Hardboard was developed in the United States in the 1920s to utilise sawmill waste. The first hardboard was produced, like paper, in a wet process from a thin aqueous fibre suspension.

4.1.1 Types of fibreboard

The classification and nomenclature of the different types of fibreboard on the market seem confusing because the names either have a historic origin or represent current types of boards or their properties. Two important criteria used in classification are the production process and the board density (Tab. 4.1).

Tab. 4.1 Classification of fibreboards

Name	low density fibreboard	medium density fibreboard	high density fibreboard
	Softboard	MDF	Hardboard
Density [kg/m ³]	< 250	350 - 800	>800

All three fibreboard types can be produced in wet or dry processes. Because of the sieve that is needed for draining the water, fibreboard from the wet process usually shows the imprints of the sieve on the underside. These boards are called S1S (smooth one side). The products of the dry process, showing a smooth texture on both sides are called S2S (smooth two sides). It is also possible to produce S2S hardboard in the wet process by first producing a softboard and pressing it between two smooth plates.

Most commonly, softboard and hardboard are produced in a wet process whereas MDF is usually produced in a dry process.

Besides the classical wood-based fibreboard, a fibreboard bonded with gypsum is becoming increasingly popular for use in interior furnishing for partitions and panels. Plasterboard can be produced from waste paper and gypsum from the desulfurisation of exhaust gases. The production process is similar to that used for wood particle board using mineral binders (see below).

4.1.2 Production of fibreboard

Raw Material

Fibreboard can be produced from all lignocellulose-containing materials with industrially utilisable fibres. However, for reasons of availability, wood is almost always used. To minimise variations in the process, only one type or a very limited number of types of wood in pre-defined ratios are used.

Deciduous and coniferous woods differ in their anatomical construction. Coniferous wood fibres are generally two to three times longer than those of deciduous wood. Unlike paper, in fibreboard the fibre length is relatively unimportant. However, to a certain extent it determines the orientation of fibres in the board. Short fibres tend to arrange vertically and longer ones horizontally. Fibres with thin walls nestle together better in the press than those with thick walls. The density of the wood used determines the thickness of the fibre mat. The lower the density, the thicker the mat, and consequently the higher are the degree of compression, the contact between fibres, and the resulting strength of the board. Dry processes are generally less sensitive to the properties of wood than wet ones. In the wet process, deciduous woods leads to better hardboard whereas coniferous woods are better suited for softboard. Coniferous wood require longer steaming times and, therefore, more energy for fibre production than deciduous woods. Mixtures of deciduous woods are mostly available in groups classified according to their density. The lower the density of the deciduous wood, the longer is the digestion time.

For cost reasons, efforts are made to use residual wood such as sidings instead of roundwood to the extent possible. Debarked chips from sawmills give good fibreboard. Also wood chips from forestry are used as a raw material.

Sawdust is less suitable because of the irregular particle sizes and short fibres, which requires expensive preparation. Bark is not suitable because of its mineral components and dark colour. Of the lignocellulose-containing annual crops, only sugarcane bagasse is important for softboard production.

Raw Material Processing

Raw material processing and fibre production are very similar for the wet and dry processes, frequently even identical.

In a first step, roundwood, sidings etc. are chipped. Chips are most frequently produced by using disk chippers or cylinder chopping machines. The cutting disk of the disk chipper, arranged vertically or diagonally, is fitted with 4-16 blades, depending on the size. The positions of the blades determine the dimensions of the chips.

Fine chips (<4 mm sieve mesh) and coarse chips (>40 mm sieve mesh) are separated from normal chips by sieving on flat or drum sifters. The fine chips are incinerated, and the coarse ones are cut up further in hammer or wing beater mills. The optimum chip size is 20-25 × 20 × 5 mm.

To remove unwanted material and mineral components, chips are washed before further processing. This is particularly important for chips from forestry. Washing is carried out in specially designed plants with pre-washers for separating coarse components and a screw drainer for separating fine mineral particles. The chips are then stored in reinforced concrete or metal silos. Storage acts as a buffer between chip and fibre production and at the same time allows the moisture to distribute more uniformly.

Fibre Production

The aim of defibration is to break up the previously chopped raw material as carefully as possible into individual fibres and bundles of fibres. This is carried out in a thermo-mechanical process. The lignin-rich middle lamellae that bind the fibres are first softened by heat and steam. The fibres are then separated by mechanical abrasion. They are often broken in the process, giving fibre fragments. The anatomical nature of the fibres and their state after defibration are important for forming the fibre mat

and for the properties of the board. This applies particularly to the wet process in which the freeness (draining behaviour) and the type of matting are very important.

Thermal treatment of the chips before and during defibration causes part of the hemicelluloses to go into solution. Increasing the temperature and the length of the treatment period leads to increased softening of the interfibre bonding. In the wet process, this improves the natural adhesion between fibres during board production. At the same time, the sugar content of the wastewater and the energy consumption increase, so that a compromise between environmentally friendly and economic process design and board properties is usually aimed at. Fibre production is energy intensive; it consumes more than half of the total process energy.

An important discovery was made by H. Mason, who developed the Mason cannon in 1926. In his process, the chips are first softened in a boiler under high pressure. This is followed by a sudden pressure release accompanied by a loud cracking sound, which indicates breakdown of the wood structure into fibres and bundles of fibres. These are pressed in a hot press to form hardboard without the addition of binders. This technique is used exclusively in the United States by the Masonite company (wet process).

In all other processes defibration is carried out by using grinding disks. Some defibrators operate at atmospheric, others at elevated pressure. Also, machines exist with only one rotating grinding disk (Sunds) or with two counterrotatory disks (Sprout-Bauer). Refiners operating at atmospheric pressure have the disadvantages of high energy consumption and moderate quality of the fibres produced.

The refiner now predominantly used, which operates under pressure, gives better results. The Asplund defibrator (Sunds) is a prototype. This makes use of the fact that from ca. 170 °C, separation of fibres in the middle lamella is considerably easier and is associated with a significant lowering of the energy consumption and an improvement in fibre quality.

In the defibration process, washed chips are fed into the pre-heater through a conical opening using a screw feeder. The pre-heater is usually a vertical, slightly conical pressure container, which is continuously charged with saturated steam at 150-180 °C. The steam heats the chips by condensation. Blowing steam out through the opening is hindered by the plug formed by the chips.

In 1-3 min the chips reach a conveyor screw (at the base of the pre-heater) through which they are transferred to the grinding disks of the refiner, where defibration takes place under steam pressure.

The important parameters affecting the quality of the fibre material and the energy consumption are the residence time and vapour pressure in the pre-heater, throughput, defibrating pressure, distance between grinding disks (0.1-0.4 mm), number of revolutions per minute, state of the grinding disk, nature of the raw material, and moisture level.

Defibration is tested by determining the degree of grinding or by fractionating the fibre material. The degree of grinding is a measure of the freeness of the fibre material and determined by means of the dewatering speed of a fibre suspension (standard freeness tester, defibrator-seconds, DS). In the wet process, higher degrees of grinding usually result in an improvement in board properties.

After defibration fibres are blown out through a pressure-regulated exit valve. From this stage onward the wet and dry processes differ and are described separately below.

Wet Process

The fibre material is fed into silos for mixing or is further defibrated in disk refiners. For hard and semi-hard fibreboard a degree of grinding of 18-30 DS is aimed at, and for porous boards 42-70 DS.

A system of tanks and conveyors is used for mixing, storing, and adding adhesives and additives. A control panel connected to the system regulates concentration, temperature, and pH.

The fibre mat is formed on a sieve by drainage of water from the fibre suspension and subsequent sedimentation of the fibres. Mat formation was previously carried out as a batch process in tanks, but cylinder wet machines and conveyor sieve de-watering machines are predominantly used now. The

main component of the latter is a continuously moving sieve. Fibre material is added through a headbox at a concentration of 1-2% and is subsequently drained and compressed by means of a forming table, suction box, and wet pressing section.

To improve the surface properties of the smooth side of the fibreboard, fibre material of higher quality can be applied to the mat on the forming table through a second headbox. After the wet press the thickness of the mat is 20-25 mm, which gives a final thickness of 3-5 mm. The mat is then separated into individual sections corresponding to the pressed board format of the hot press.

In the production of softboard or S2S-hardboard by the wet process, the drained fibre mats are dried in single- or multi-deck dryers at 150-170 °C from moisture levels of 100-120% to about 4%.

To prevent dried boards from catching fire during subsequent storage, they are cooled immediately after drying by spraying with water and aspiration of ambient air.

S1S board is further de-watered in hydraulic multi-daylight presses by hot pressing using an established pressure–time program, compressed, and strengthened by curing the binders that are naturally present or have been added. For S1S, the use of a draining sieve is necessary to remove the large quantities of water. For hardboard the pressing time is generally 2.0 – 3.5 min/mm at a pressing temperature of 180 – 200 °C. A typical pressing diagram consists of three phases, the compression and first high-pressure stage, drying, and the second high-pressure stage.

An improvement in the properties of fibreboard can be obtained by thermal after-treatment in chambers or continuously in tunnels or by impregnating with drying oils. Treatment results in a more uniform moisture distribution in the board as well as better dimensional stability.

Wet processes require large quantities of water. Pollutants, such as extractives, hemicelluloses, and lignin, are present in the wastewater, which has to be treated. For such environmental reasons the dry process has been developed..

Dry Process

After defibration the fibres pass down a blow line in which glue is added and then reach the fibre dryer. Alternatively, they can be blown directly into the fibre dryer by making use of the vapour pressure present in the refiner. In this case, they are subsequently glue-blended in high-speed centrifugal machines, like in particle board production. In almost all plants, glue blending occurs before drying, thus avoiding spots of adhesive on the board surface that impair its quality. Paraffin added to improve hydrophobic properties is generally charged in the molten state to the lower conveyor screw of the refiner.

Tube dryers are now used exclusively for drying the fibres. They are mostly two-stage dryers with tube lengths of 80-100 m and diameters of 1.3-1.6 m. The air temperature at the entry is 140-170 °C. Because of the high air speed (ca. 30 m/s) and the small dimensions of the fibres, the drying time is only a few seconds. The moisture level of the fibres is adjusted to 9-11% by regulating the exit temperature of the dryer.

The fibres reach the forming station via cyclones and fibre hoppers. The fibre mat is formed either mechanically by using a snowfall felter or pneumatically by using a pendistor on a vacuum screen conveyor.

The uniformity of distribution of the fibre material over the surface of the screen significantly affects fibreboard properties.

The mat coming out of the forming station has a very low bulk density and contains a large quantity of air, which must be removed before hot pressing. To decrease the size of the press opening, shorten the closing times of the hot press, and improve the transportability of the mat and the surface quality of the finished board, the mat is precompressed in continuously operating belt prepresses. Mat density is thus doubled or tripled.

Hot pressing of the fibre mat occurs either batchwise on multi-daylight presses with preliminary separation into mat sheets or, as is now increasingly common, continuously on double-belt presses or calenders for thin boards (up to 8 mm thickness). The presses are the same as for particle board production. To heat the fibre mat more rapidly and to shorten pressing time, the fibre mat is often preheated by microwaves before entering the hot press.

The bulk density profile (perpendicular to the board thickness) and thus the properties of the board can be widely varied through the type of compression and the temperature control. High compression in the outer zones and as uniform a density as possible in the inner zones are pursued.

4.1.3 Binders and additives in fibreboard

In the production of fibreboard by the wet process, no binders are required in principle because of the binding forces already present in the wood. However, adhesives are necessary for the dry process.

The main purpose of additives is to improve the mechanical properties and water resistance of fibreboard. In softboard production the fibres are bonded predominantly by hydrogen bonds. For hydrophobing, molten paraffin (ca. 1% slack wax based on dry fibre) or asphalt emulsion is added. For hardboard, lignin is the most commonly used binder, which is activated in the hot press.

Wet process softboard is produced mainly with the addition of highly condensed, alkaline phenol formaldehyde (PF) resins (0.5-2.0% based on dry fibre), which are fixed on the fibre by the addition of $\text{Al}_2(\text{SO}_4)_3$. Paraffin (0.1-0.5% based on dry fibre) and drying oils, such as linseed oil, are also added.

For S2S board (i.e. hardboard) produced by the wet process, phenolic resins cannot be used because the severe drying conditions would result in premature curing. Resins such as rosin are used instead. They are fixed on the fibre by lowering the pH with $\text{Al}_2(\text{SO}_4)_3$.

Dry process hardboard is generally produced by using PF resins (ca. 6% based on dry fibre). For MDF produced by the dry process, urea formaldehyde (UF) resins, generally modified with melamine, are the most important binders (10-14% solid resin based on dry fibre). To a small extent, mainly for special boards, melamine urea formaldehyde (MUF), PF, and polymeric methylene diphenyl diisocyanate (PMDI) are used. As with S1S boards, paraffin is used for hydrophobing.

4.1.4 Use of fibreboard

The main areas of use for fibreboard are:

Insulation board (softboard) is used in interior and exterior applications, mostly as parts of walls, and for ceiling panelling, frequently in the form of acoustic panels.

Hardboard produced by the wet process generally consists of thin boards. It is employed mainly in buildings for inner and outer wall panelling. It is also used in the production of furniture or doors or moulded for use in motor vehicle interiors.

Because of its excellent edge and surface properties and its good workability, MDF is used mainly in furniture production.

4.1.5 Fibreboard inecoinvent database

The background information to the processes are given in Tab. 4.2, flows are shown in:

- fibreboard soft, at plant (updated 2009) Fig. 4.1/Fig. 4.2
- fibreboard soft, latex bonded, at plant (updated 2009) Fig. 4.3/Fig. 2.4
- fibreboard soft, without adhesives, at plant (updated 2009) Fig. 4.5/Fig. 4.6

- medium density fibreboard, at plant Fig. 4.7
- fibreboard hard, at plant Fig. 4.8
- treatment, soft fibre board production effluent, to WWT, class 3 Fig. 4.9

4.1.6 Data quality

Data quality is assessed using the pedigree matrix as described in Frischknecht & Jungbluth (2003).

The datasets for fibreboard soft have been updated based on specific production data for the wet process from the two plants in Switzerland from the year 2009. They represent both the current production and the current consumption mix for Switzerland.

The available input data for MDF and for fibreboard hard are supposed to be updated with the next release of ecoinvent. These datasets should not be compared with the updated datasets on soft fibreboard.

4. Report "Board based on industrial residual wood"

Tab. 4.2 Ecoinvent meta information for the soft fibre board production processes (new)

Name	fibreboard soft, at plant (u=7%)	fibreboard soft, latex bonded, at plant (u=7%)	fibreboard soft, without adhesives, at plant (u=7%)	treatment, soft fibre board production effluent, to WWT, class 3
Location	CH	CH	CH	CH
Infrastructure Process	0	0	0	0
Unit	m3	m3	m3	m3
Data Set Version	2.1	2.1	2.1	2.1
Included Processes	includes all inputs to the production processes, transport, wastes and emissions to air and water; does not include any packaging material	includes all inputs to the production processes, transport, wastes and emissions to air and water; does not include any packaging material; 100% allocation as a default	includes all inputs to the production processes, transport, wastes and emissions to air and water; does not include any packaging material	infrastructure materials for municipal wastewater treatment plant, transports, dismantling. Land use burdens.
Amount	1	1	1	1
Local Name	Faserplatte weich, ab Werk (u=7%)	Faserplatte weich, mit Latex, ab Werk (u=7%)	Faserplatte weich, ohne Klebstoff, ab Werk (u=7%)	Behandlung, Abwasser Weichfaserplattenproduktion, in ARA, Gr.Kl. 3
Synonyms				0
General Comment to reference function				wastewater purified in a medium size municipal wastewater treatment plant (capacity class 3), with an average capacity size of 24900 per-capita-equivalents PCE. Average data from two production sites
	140 kg/m3	240 kg/m3	220 kg/m3	
Start Date	2008	2008	2008	2007
End Date	2009	2009	2009	2008
Data Valid For Entire Period	1	1	1	1
Other Period Text				0
Geography text	representative for Swiss production and Swiss consumption	representative for Swiss production and Swiss consumption	representative for Swiss production and Swiss consumption	specific to the technology mix encountered in Switzerland in 2000. Well applicable to modern treatment practices in Europe, North America or Japan.
Technology text	wet process, medium enterprise technology (2008)	wet process, medium enterprise technology (2008)	wet process, medium enterprise technology (2008)	three stage wastewater treatment (mechanical, biological, chemical) including sludge digestion (fermentation) according to the average technology in Switzerland
Representativeness [%]	1	1	1	1
Production Volume	unknown	unknown	unknown	unknown
Sampling Procedure	plant data	plant data	plant data	plant data
Extrapolations	see Geography and Technology	see Geography and Technology	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	highly representative data with generally low uncertainty. Uncertainty related to emissions into water and related to two proxy processes related to heat production exist.	highly representative data with generally low uncertainty. Uncertainty related to emissions into water and related to two proxy processes related to heat production exist.	highly representative data with generally low uncertainty. Uncertainty related to emissions into water and related to two proxy processes related to heat production exist.	estimated from plant data on two sewage flows with the respective ecoinvent tool

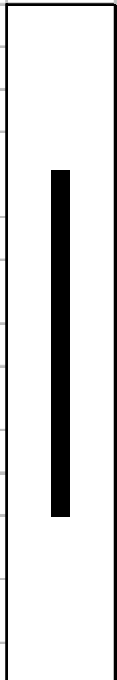
4. Report "Board based on industrial residual wood"

Fig. 4.1 Flows for "fibreboard soft, at plant" and its representation in the ecoinvent database, part 1 (updated 2009)

General Flow information				Representation in Ecoinvent 2000								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Industrial residual wood, sidings, softwood	→		at 0.45t /m3 (u=0%)	wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, air dried, u=20% at plant	3.31E-02	m3	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
Industrial residual wood, chips, softwood	→		at 0.45t /m3 (u=0%)	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	2.20E-01	m3	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
Industrial wood from forests, softwood	→		at 0.45t /m3 (u=0%)	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140% at forest road	2.42E-03	m3	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
Latex emulsion (50%)	→		Adhesive	chemicals	organics	No	RER	latex, at plant	0.00E+00	kg	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
PVAc (60%)	→		Adhesive	plastics	monomers	No	RER	vinyl acetate, at plant	2.10E+00	kg	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
Paraffin emulsion (60%)	→		hydrophobation agent	chemicals	organics	No	RER	paraffin, at plant	6.72E-01	kg	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
Aluminiumsulphate (17%)	→		flocculant;referring to solid Al2(SO4)3	chemicals	inorganics	No	RER	aluminium sulphate, powder, at plant	4.76E-03	kg	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
Electricity medium voltage - at grid CH	→			electricity	supply mix	No	CH	electricity, medium voltage, at grid	7.98E+01	kWh	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,2)
Wood residues into industrial furnace	→		this ecoinvent process contains also wood input of 0.000136 m3 wood/MJ that has to be excluded (see below)	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 1000kW	3.71E+02	MJ	calculated based on ecoinvent process	100	105	(1,1,1,1,1,1)
Correction for wood input	→		0.000136 m3/MJ	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	-5.04E-02	m3	calculated based on ecoinvent process	100	105	(1,1,1,1,1,1)
Animal fat into industrial furnace	→		animal fat considered a by-product and thus no input inventoried; combustion of fuel oil extra light as proxy; corrections for fuel input see below	oil	heating systems	No	CH	heat, light fuel oil, at industrial furnace 1MW	1.47E+02	MJ	Pavatex SA (Werner 2009)	100	107	(1,1,1,1,3,1)
Fuel correction	→		correction: exclusion of fuel input, as combustion of light fuel oil is used as a proxy for the combustion of animal fat	oil	fuels	No	CH	light fuel oil, at regional storage	-3.60E+00	kg	0.02342 kg fuel oil x 105 per MJ heat	100	105	(1,1,1,1,1,1)
CO2 correction (minus fossil)	→		correction: exclusion of CO2 emissions from combustion of fuel oil incineration process used as proxy for the combustion of animal fat	air	high population density			Carbon dioxide, fossil	-1.14E+01	kg	0.074 kg CO2 x 105 per MJ heat	100	105	(1,1,1,1,1,1)
CO2 correction (plus biogenic)	→		correction: inclusion of biogenic CO2 emissions from combustion of fuel oil incineration process used as proxy for the combustion of animal fat	air	low population density			Carbon dioxide, biogenic	1.14E+01	kg	0.074 kg CO2 x 105 per MJ heat	100	105	(1,1,1,1,1,1)
Heavy fuel oil, into industrial furnace	→			oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	7.49E+01	MJ	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,1)
Heat from natural gas, industrial furnace	→			natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace low-NOx >100kW	3.56E+02	MJ	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,1)
Drinking water	→			water supply	production	No	CH	tap water, at user	3.07E+02	kg	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
Process and cooling water	→			resource	in water			Water, cooling, unspecified natural origin	8.07E-01	m3	Pavatex SA (Werner 2009)	100	105	(1,1,1,1,1,4)
plant	→		calculated based on half the output with 1/10 of plant size capred to inventoried plant	wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	6.67E-09	unit		100	3.00	(1,1,1,1,1,9)

4. Report "Board based on industrial residual wood"

Fig. 4.2 Flows for "fibreboard soft, at plant" and its representation in the ecoinvent database, part 2 (updated 2009)

General Flow information					Representation in Ecoinvent 2000									Uncertainty information			
Input		Process Name	Output		Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in Ecoinvent 2000	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Lorry 40 t	→				chemicals: 100 - 1000 km, wood: 50 km, fuels: 50 km	transport systems	road	No	RER	transport, lorry >32t, EURO3	9.38E+00	tkm	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,5)	
Lorry 28 t	→				waste processes according to ecoinvent	transport systems	road	No	RER	transport, lorry 16-32t, EURO3	9.86E-02	tkm	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,5)	
			→	Fibreboard soft, at plant		density ca. 140 kg/m3	wooden materials	extraction	No	CH	fibreboard soft, at plant (u=7%)	1.00E+00	m3	Pavatex SA (Werner 2009)	1.00	1.00	
			→	Concentrate to anaerobic digestion in WWTP			biomass	fuels	No	CH	biowaste, to anaerobic digestion	0.00E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→	Concentrate to MWI		Raw sewage sludge as proxy process	waste management	municipal incineration	No	CH	disposal, raw sewage sludge, to municipal incineration	5.18E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→	Waste water/condensate to WWTP			waste management	wastewater treatment	No	CH	treatment, soft fibre board production effluent, to WWTP, class 3	2.15E-01	m3	Pavatex SA (Werner 2009)	1.00	1.21	(1,1,1,1,1,5,6)
			→	Process and cooling water		not inventoried	resource	in water			Water, cooling, unspecified natural origin	-8.07E-01	m3	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→	NM VOC		value from 1 plant; extrapolated	air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	4.93E-03	kg	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,23)
			→	Particles from wood drying		value from 1 plant; extrapolated	air	high population density			Particulates, > 10 um	5.42E-03	kg	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,26)
			→	Waste heat into air			air	high population density			Heat, waste	2.87E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,2)
			→	Municipal waste to MWI			waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	2.26E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→	Ashes/dust to residual material landfill			waste management	sanitary landfill	No	CH	disposal, wood ash mixture, pure, 0%water, to sanitary landfill	4.72E-01	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→	Waste into inert landfill		value from 1 plant; extrapolated	waste management	inert material landfill	No	CH	disposal, inert waste, 5%water, to inert material landfill	4.26E-02	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)

4. Report "Board based on industrial residual wood"

Fig. 4.3 Flows for "fibreboard soft, latex bonded, at plant" and its representation in the ecoinvent database, part 1 (updated 2009)

General Flow information					Representation in Ecoinvent 2000								Uncertainty information		
Input		Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in Ecoinvent 2000	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Industrial residual wood, sidings, softwood	→			at 0.45t /m3 (u=0%)	wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, air dried, u=20% at plant	4.76E-02	m3	at 0.45t /m3 (u=0%)	1.00	1.05	(1,1,1,1,1,4)
Industrial residual wood, chips, softwood	→			at 0.45t /m3 (u=0%)	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	3.84E-01	m3	at 0.45t /m3 (u=0%)	1.00	1.05	(1,1,1,1,1,4)
Industrial wood from forests, softwood	→			at 0.45t /m3 (u=0%)	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=40% at forest road	2.16E-03	m3	at 0.45t /m3 (u=0%)	1.00	1.05	(1,1,1,1,1,4)
Latex emulsion (50%)	→			adhesive	chemicals	organics	No	RER	latex, at plant	5.40E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
PVAc (60%)	→			adhesive	plastics	monomers	No	RER	vinyl acetate, at plant	3.60E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Paraffin emulsion (60%)	→			hydrophobation agent	chemicals	organics	No	RER	paraffin, at plant	1.87E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Aluminiumsulphate (17%)	→			flocculant; referring to solid Al2(SO4)3	chemicals	inorganics	No	RER	aluminium sulphate, powder, at plant	2.86E-01	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Electricity medium voltage - at grid CH	→				electricity	supply mix	No	CH	electricity, medium voltage, at grid	1.30E+02	kWh	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,2)
Wood residues into industrial furnace	→			this ecoinvent process contains also wood input of 0.000136 m3 wood/MJ that has to be excluded (see below)	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 1000kW	4.82E+02	MJ	calculated based on ecoinvent process	1.00	1.05	(1,1,1,1,1,1)
Correction for wood input	→			correction for wood input: 0.000136 m3/MJ	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	-6.56E-02	m3	calculated based on ecoinvent process	1.00	1.05	(1,1,1,1,1,1)
Animal fat into industrial furnace	→			animal fat considered a by-product and thus no input inventoried; combustion of fuel oil extra light as proxy; corrections for fuel input see below	oil	heating systems	No	CH	heat, light fuel oil, at industrial furnace 1MW	5.33E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.07	(1,1,1,1,1,3,1)
Fuel correction	→			correction: exclusion of fuel input, as combustion of light fuel oil is used as a proxy for the combustion of animal fat	oil	fuels	No	CH	light fuel oil, at regional storage	-1.31E+01	kg	0.02342 kg fuel oil x 1.05 per MJ heat	1.00	1.05	(1,1,1,1,1,1)
CO2 correction (minus fossil)	→			correction: exclusion of CO2 emissions from combustion of fuel oil incineration process used as proxy for the combustion of animal fat	air	high population density			Carbon dioxide, fossil	-4.14E+01	kg	0.074 kg CO2 x 1.05 per MJ heat	1.00	1.05	(1,1,1,1,1,1)
CO2 correction (plus biogenic)	→			correction: inclusion of biogenic CO2 emissions from combustion of fuel oil incineration process used as proxy for the combustion of animal fat	air	low population density			Carbon dioxide, biogenic	4.14E+01	kg	0.074 kg CO2 x 1.05 per MJ heat	1.00	1.05	(1,1,1,1,1,1)
Heavy fuel oil, into industrial furnace	→				oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	2.72E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,1)
Heat from natural gas, industrial furnace	→				natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace low-NOx >100kW	3.17E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,1)
Drinking water	→				water supply	production	No	CH	tap water, at user	6.40E+02	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Process and cooling water	→				resource	in water			Water, cooling, unspecified natural origin	7.20E-01	m3	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
plant	→			wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	6.67E-09	unit	calculated based on half the output with 1/10 of plant size capped to inventoried plant	1.00	3.00	(1,1,1,1,1,9)	

4. Report "Board based on industrial residual wood"

Fig. 4.4 Flows for "fibreboard soft, latex bonded, at plant" and its representation in the ecoinvent database, part 2 (updated 2009)

General Flow information					Representation in Ecoinvent 2000								Uncertainty information		
Input		Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in Ecoinvent 2000	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Lorry 40 t	→			chemicals: 100 - 1000 km, wood: 50 km, fuels: 50 km	transport systems	road	No	RER	transport, lorry >32t, EURO3	2.95E+01	tkm	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,5)
Lorry 28 t	→			Waste processes according to ecoinvent	transport systems	road	No	RER	transport, lorry 16-32t, EURO3	9.80E-02	tkm	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,5)
			→ Fibreboard soft, at plant	Density ca. 240 kg/m3	wooden materials	extraction	No	CH	fibreboard soft, latex bonded, at plant (u=7%)	1.00E+00	m3	Pavatex SA (Werner 2009)	1.00	1.00	
			→ Concentrate to anaerobic digestion in WWTP		biomass	fuels	No	CH	biowaste, to anaerobic digestion	0.00E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→ Concentrate to MWI	Raw sewage sludge as proxy process	waste management	municipal incineration	No	CH	disposal, raw sewage sludge, to municipal incineration	4.62E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→ Waste water/condensate to WWTP		waste management	wastewater treatment	No	CH	treatment, soft fibre board production effluent, to WWT, class 3	4.11E-01	m3	Pavatex SA (Werner 2009)	1.00	1.21	(1,1,1,1,1,6)
			→ Process and cooling water	not inventoried	resource	in water			Water, cooling, unspecified natural origin	-7.20E-01	m3	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→ NM VOC	value from 1 plant; extrapolated	air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	8.45E-03	kg	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,23)
			→ Particles from wood drying	value from 1 plant; extrapolated	air	high population density			Particulates, > 10 um	9.29E-03	kg	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,26)
			→ Waste heat into air		air	high population density			Heat, waste	4.68E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,2)
			→ Municipal waste to MWI		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	2.29E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→ Ashes/dust to residual material landfill		waste management	sanitary landfill	No	CH	disposal, wood ash mixture, pure, 0%water, to sanitary landfill	5.30E-01	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			→ Waste into inert landfill	value from 1 plant; extrapolated	waste management	inert material landfill	No	CH	disposal, inert waste, 5%water, to inert material landfill	1.55E-01	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)

4. Report "Board based on industrial residual wood"

Fig. 4.5 Flows for "fibreboard soft, without adhesives, at plant" and its representation in the ecoinvent database, part 1 (updated 2009)

General Flow information				Representation in Ecoinvent 2000								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in Ecoinvent 2000	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Industrial residual wood, sidings, softwood	→		at 0.45 t /m3 (u=0%)	wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, air dried, u=20% at plant	5.29E-02	m3	at 0.45 t /m3 (u=0%)	1.00	1.05	(1,1,1,1,1,4)
Industrial residual wood, chips, softwood	→		at 0.45 t /m3 (u=0%)	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	3.44E-01	m3	at 0.45 t /m3 (u=0%)	1.00	1.05	(1,1,1,1,1,4)
Industrial wood from forests, softwood	→		at 0.45 t /m3 (u=0%)	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140% at forest road	3.98E-03	m3	at 0.45 t /m3 (u=0%)	1.00	1.05	(1,1,1,1,1,4)
Latex emulsion (50%)	→		adhesive	chemicals	organics	No	RER	latex, at plant	0.00E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
PVAc (60%)	→		adhesive	plastics	monomers	No	RER	vinyl acetate, at plant	0.00E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Paraffin emulsion (60%)	→		hydrophobation agent	chemicals	organics	No	RER	paraffin, at plant	1.06E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Aluminiumsulphate (17%)	→		flocculant; referring to solid Al2(SO4)3	chemicals	inorganics	No	RER	aluminium sulphate, powder, at plant	7.48E-03	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Electricity medium voltage - at grid CH	→			electricity	supply mix	No	CH	electricity, medium voltage, at grid	1.26E+02	kWh	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,2)
Wood residues into industrial furnace	→		this ecoinvent process contains also wood input of 0.000136 m3 wood/MJ that has to be excluded (see below)	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 1000kW	5.94E+02	MJ	calculated based on ecoinvent process	1.00	1.05	(1,1,1,1,1,1)
Correction for wood input	→		correction for wood input: 0.000136 m3/MJ	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40% at plant	-8.08E-02	m3	calculated based on ecoinvent process	1.00	1.05	(1,1,1,1,1,1)
Animal fat into industrial furnace	→		animal fat considered a by-product and thus no input inventoried; combustion of fuel oil extra light as proxy; corrections for fuel input see below	oil	heating systems	No	CH	heat, light fuel oil, at industrial furnace 1MW	2.06E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.07	(1,1,1,1,3,1)
Fuel correction	→		correction: exclusion of fuel input, as combustion of light fuel oil is used as a proxy for the combustion of animal fat	oil	fuels	No	CH	light fuel oil, at regional storage	-5.05E+00	kg	0.02342 kg fuel oil x 1.05 per MJ heat	1.00	1.05	(1,1,1,1,1,1)
CO2 correction (minus fossil)	→		correction: exclusion of CO2 emissions from combustion of fuel oil incineration process used as proxy for the combustion of animal fat	air	high population density			Carbon dioxide, fossil	-1.60E+01	kg	0.074 kg CO2 x 1.05 per MJ heat	1.00	1.05	(1,1,1,1,1,1)
CO2 correction (plus biogenic)	→		correction: inclusion of biogenic CO2 emissions from combustion of fuel oil incineration process used as proxy for the combustion of animal fat	air	low population density			Carbon dioxide, biogenic	1.60E+01	kg	0.074 kg CO2 x 1.05 per MJ heat	1.00	1.05	(1,1,1,1,1,1)
Heavy fuel oil, into industrial furnace	→			oil	heating systems	No	CH	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	1.05E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,1)
Heat from natural gas, industrial furnace	→			natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace low-NOx >100kW	5.84E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,1)
Drinking water	→			water supply	production	No	CH	tap water, at user	4.72E+02	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
Process and cooling water	→			resource	in water			Water, cooling, unspecified natural origin	1.33E+00	m3	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,4)
plant	→			wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	6.67E-09	unit	calculated based on half the output with 1/10 of plant size capped to inventoried plant	1.00	3.00	(1,1,1,1,1,9)

4. Report "Board based on industrial residual wood"

Fig. 4.6 Flows for "fibreboard soft, without adhesives, at plant" and its representation in the ecoinvent database, part 2 (updated 2009)

General Flow information					Representation in Ecoinvent 2000								Uncertainty information		
Input		Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Lorry 40 t	→			Chemicals: 100 - 1000 km, wood: 50 km, fuels: 50 km	transport systems	road	No	RER	transport, lorry >32t, EURO3	1.42E+01	tkm	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,5)
Lorry 28 t	→			Waste processes according to ecoinvent	transport systems	road	No	RER	transport, lorry 16-32t, EURO3	1.61E-01	tkm	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,5)
			Fibreboard soft, at plant	Density ca. 220 kg/m3	wooden materials	extraction	No	CH	fibreboard soft, without adhesives, at plant (u=7%)	1.00E+00	m3	Pavatex SA (Werner 2009)	1.00	1.00	
			Concentrate to anaerobic digestion in WWTP		biomass	fuels	No	CH	biowaste, to anaerobic digestion	0.00E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			Concentrate to MWI	Raw sewage sludge as proxy process	waste management	municipal incineration	No	CH	disposal, raw sewage sludge, to municipal incineration	8.52E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			Waste water/condensate to WWTP		waste management	wastewater treatment	No	CH	treatment, soft fibre board production effluent, to WWTP, class 3	3.34E-01	m3	Pavatex SA (Werner 2009)	1.00	1.21	(1,1,1,1,5,6)
			Process and cooling water	not inventoried	resource	in water			Water, cooling, unspecified natural origin	-1.33E+00	m3	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			NM VOC	value from 1 plant; extrapolated	air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	7.75E-03	kg	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,23)
			Particles from wood drying	value from 1 plant; extrapolated	air	high population density			Particulates, > 10 um	8.52E-03	kg	Pavatex SA (Werner 2009)	1.00	2.00	(1,1,1,1,1,26)
			Waste heat into air		air	high population density			Heat, waste	4.54E+02	MJ	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,2)
			Municipal waste to MWI		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9%water, to municipal incineration	3.69E+00	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			Ashes/dust to residual material landfill		waste management	sanitary landfill	No	CH	disposal, wood ash mixture, pure, 0%water, to sanitary landfill	7.65E-01	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)
			Waste into inert landfill	value from 1 plant; extrapolated	waste management	inert material landfill	No	CH	disposal, inert waste, 5%water, to inert material landfill	5.97E-02	kg	Pavatex SA (Werner 2009)	1.00	1.05	(1,1,1,1,1,6)

Fig. 4.7 Flows for "medium density fibreboard, at plant" and its representation in the ecoinvent database

See ecoinvent 2.0 report no. 9 (Werner et al., 2006), Fig. 12.6

Fig. 4.8 Flows for "fibreboard hard, at plant" and its representation in the ecoinvent database

See ecoinvent 2.0 report no. 9 (Werner et al., 2006), Fig. 12.7

Fig. 4.9 Flows for "treatment, soft fibre board production effluent, to WWT, class 3" and its representation in the ecoinvent database

Name	Location	Category	SubCategory	Infrastructure Process	Unit	treatment, soft fibre board production effluent, to WWT, class 3	Uncertainty Type	Standard Deviation 95%
Location						CH		
InfrastructureProcess						0		
Unit						m3		
Ammonium ion		water	river		kg	7.71E-04	1	134.2%
Nitrite		water	river		kg	4.50E-05	1	150.4%
Nitrogen		water	river		kg	3.43E-05	1	150.5%
NM VOC, non-methane volatile organic compounds, unspecified origin		air	high population density		kg	6.83E-05	1	252.5%
Carbon monoxide, biogenic		air	high population density		kg	4.90E-03	1	196.8%
Carbon dioxide, biogenic		air	high population density		kg	6.12E+00	1	134.0%
Methane, biogenic		air	high population density		kg	1.50E-02	1	187.2%
Nitrogen oxides		air	high population density		kg	5.09E-05	1	149.8%
Ammonia		air	high population density		kg	5.21E-06	1	164.1%
Dinitrogen monoxide		air	high population density		kg	9.73E-06	1	149.0%
Cyanide		air	high population density		kg	1.47E-07	1	243.0%
Phosphorus		air	high population density		kg	3.51E-08	1	169.7%
Iron		air	high population density		kg	6.85E-09	1	491.5%
Aluminum		air	high population density		kg	1.63E-08	1	516.6%
BOD5, Biological Oxygen Demand		water	river		kg	2.90E-01	1	145.4%
COD, Chemical Oxygen Demand		water	river		kg	8.83E-01	1	147.4%
TOC, Total Organic Carbon		water	river		kg	2.37E-01	1	145.9%
DOC, Dissolved Organic Carbon		water	river		kg	2.21E-01	1	145.8%
Sulfate		water	river		kg	2.10E-04	1	149.1%
Nitrate		water	river		kg	3.38E-03	1	131.7%
Phosphate		water	river		kg	5.57E-05	1	149.8%
Chloride		water	river		kg	2.19E-04	1	483.9%
Iron, ion		water	river		kg	7.02E-08	1	485.8%
Aluminum		water	river		kg	8.69E-10	1	416.3%
BOD5, Biological Oxygen Demand		water	ground-, long-term		kg	4.18E-03	1	170.9%
COD, Chemical Oxygen Demand		water	ground-, long-term		kg	1.28E-02	1	170.9%
TOC, Total Organic Carbon		water	ground-, long-term		kg	5.05E-03	1	170.9%
DOC, Dissolved Organic Carbon		water	ground-, long-term		kg	5.05E-03	1	170.9%
Nitrate		water	ground-, long-term		kg	5.85E-06	1	150.8%
Phosphate		water	ground-, long-term		kg	4.11E-06	1	6105.0%
Iron, ion		water	ground-, long-term		kg	9.62E-05	1	1088.0%
Aluminum		water	ground-, long-term		kg	7.73E-06	1	512.7%
municipal waste incineration plant	CH	waste management	municipal incineration	1	unit	5.86E-10	1	118.5%
process-specific burdens, municipal waste incineration	CH	waste management	municipal incineration	0	kg	2.34E+00	1	118.5%
slag compartment	CH	waste management	municipal incineration	1	unit	7.51E-12	1	144.4%
process-specific burdens, slag compartment	CH	waste management	municipal incineration	0	kg	4.22E-03	1	144.4%
residual material landfill facility	CH	waste management	residual material landfill	1	unit	3.81E-12	1	145.0%
process-specific burdens, residual material landfill	CH	waste management	residual material landfill	0	kg	1.83E-03	1	145.0%
Heat, waste		air	high population density		MJ	2.03E+01	1	128.4%
Heat, waste		water	river		MJ	3.17E+01	1	138.3%
electricity from waste, at municipal waste incineration plant	CH	waste management	municipal incineration	0	kWh	3.37E-01	1	382.1%
heat from waste, at municipal waste incineration plant	CH	waste management	municipal incineration	0	MJ	1.97E+00	1	119.3%
sodium hydroxide, 50% in H2O, production mix, at plant	RER	chemicals	inorganics	0	kg	4.96E-07	1	153.7%
quicklime, milled, packed, at plant	CH	construction materials	additives	0	kg	9.00E-08	1	158.3%
hydrochloric acid, 30% in H2O, at plant	RER	chemicals	inorganics	0	kg	2.31E-08	1	156.5%
iron (III) chloride, 40% in H2O, at plant	CH	chemicals	inorganics	0	kg	3.35E-04	1	160.6%
chemicals inorganic, at plant	GLO	chemicals	inorganics	0	kg	3.85E-08	1	156.5%
cement, unspecified, at plant	CH	construction materials	binder	0	kg	7.31E-04	1	151.8%
disposal, cement, hydrated, 0% water, to residual material landfill	CH	waste management	residual material landfill	0	kg	1.83E-03	1	151.8%
transport, freight, rail	RER	transport system	rail	0	tkm	4.81E-04	1	129.5%
transport, lorry 28t	CH	transport system	road	0	tkm	4.75E-02	1	111.7%
ammonia, liquid, at regional storehouse	CH	chemicals	inorganics	0	kg	8.22E-06	1	230.7%
natural gas, burned in industrial furnace low-NOx >100kW	RER	natural gas	heating systems	0	MJ	8.01E-04	1	230.7%
titanium dioxide, production mix, at plant	RER	chemicals	inorganics	0	kg	2.35E-07	1	410.0%
chromium oxide, flakes, at plant	RER	chemicals	inorganics	0	kg	4.80E-09	1	410.0%
electricity, low voltage, at grid	CH	electricity	supply mix	0	kWh	8.32E-01	1	126.8%
iron sulphate, at plant	RER	metals	extraction	0	kg	2.45E-04	1	161.7%
aluminium sulphate, powder, at plant	RER	chemicals	inorganics	0	kg	6.62E-05	1	161.7%
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	waste management	municipal incineration	0	kg	1.55E-02	1	107.0%
disposal, paper, 11.2% water, to municipal incineration	CH	waste management	municipal incineration	0	kg	1.55E-02	1	107.0%
sewer grid, class 3	CH	waste management	wastewater treatment	1	km	2.18E-07	1	100.0%
residential sewer grid	CH	waste management	wastewater treatment	1	km	2.87E-07	1	100.0%
wastewater treatment plant, class 3	CH	waste management	wastewater treatment	1	unit	5.69E-09	1	100.0%
treatment, soft fibre board production effluent, to WWT, class 3	CH			0	m3	1.00E+00		

New References:

Werner F (2009): Ökobilanzinventare für Weichfaserplatten der Firma Pavatex SA, Hintergrundbericht für die Kritische Prüfung gemäss Richtlinien von ecoinvent 2.1, Werner Umwelt & Entwicklung, Zürich (unveröffentlicht).

Annex 5

[updated Part XIV of Kellenberger et al. 2007]

Insulation Products and Processes

Data v2.1 (2009)

Authors: Hans-Jörg Althaus, Empa
Dominic Notter
Daniel Kellenberger, Empa
Mischa Classen, Empa
Philipp Thalmann, Bau- und Umweltchemie

Reviewer: Roberto Dones, Paul Scherrer Institut
Rolf Frischknecht, ESU-services
Hans-Jörg Althaus, Empa

Information to this report version 2.1

The following changes of insulation datasets were made in version 2.1 of ecoinvent compared to version 2.0:

Rock wool, at plant: New flow data from Flumroc are implemented

Expanded polystyrene, at plant: Dataset for specific blowing agents are established.

Acknowledgement

The cooperation of Mrs. M. Mebold from Flumroc AG is very much appreciated.

Also we would like to thank Pittsburgh Corning (Schweiz) AG for commissioning the new LCI study on foamglas.

Summary

This part deals with the production of insulation materials used in the building sector. These are glass wool, foam glass, rock wool (and basalt from mine as raw material), foamglas, polystyrene foam, cellulose fibre and urea formaldehyde foam.

Since the application of rock wool and polystyrene foam slab is very much different from the application of cellulose fibres, the blowing in of the fibres is included in the inventory. The inventories of the insulation materials in ecoinvent represent products of comparable function. Nevertheless the data are not meant for direct comparison of the different materials, rather in a broader context and if the importance of these products for an inventory is not too high.

Table of Contents

INFORMATION TO THIS REPORT VERSION 2.1	I
ACKNOWLEDGEMENT	I
SUMMARY	I
TABLE OF CONTENTS	III
1 INTRODUCTION	1
1.1 Glass wool	1
1.2 Rock wool	1
1.3 Foamglas	1
1.4 Polystyrene Foam	2
2 RESERVES, RESOURCES AND RAW MATERIALS	2
2.1 Glass Wool	2
2.2 Rock Wool / Basalt	2
2.3 Foamglas	3
2.4 Polystyrene Foam	3
2.5 Cellulose Fibre	3
2.6 Urea Formaldehyde Foam	4
3 CHARACTERISATION AND USE	4
3.1 Glass Wool	4
3.1.1 Physical Properties	4
3.1.2 Chemical Properties	4
3.1.3 Use	4
3.2 Rock wool / Basalt	4
3.2.1 Characterisation	4
3.2.2 Use	6
3.3 Foamglas	7
3.4 Polystyrene Foam Slab	7
3.5 Cellulose Fibre	9
3.6 Urea Formaldehyde Foam	9
4 SYSTEM CHARACTERISATION	9
4.1 Glass Wool	9
4.1.1 General	9
4.1.2 Melting	10
4.1.3 Fibre Production	10
4.1.4 Product Forming	10
4.1.5 Life Cycle Inventories	11
4.1.6 Data Quality	11
4.2 Basalt	17
4.2.1 General	17

4.2.2	Process Description	17
4.2.3	Energy Consumption	17
4.2.4	Transports.....	17
4.2.5	Emissions	17
4.2.6	Infrastructure	18
4.2.7	Assignment to ecoinvent datasets.....	18
4.2.8	Life Cycle Inventory.....	20
4.2.9	Data Quality	20
4.3	Rock Wool.....	24
4.3.1	Process Description	24
4.3.2	Life Cycle Inventories	26
4.3.3	Comparison to other Data.....	39
4.3.4	Data Quality	39
4.4	Foamglas	40
4.4.1	System characterisation	40
4.4.2	Foamglas in ecoinvent.....	47
4.4.3	Data quality	58
4.5	Polystyrene Foam Slab (EPS).....	58
4.5.1	Process Description	58
4.5.2	Polystyrene Foam Slab (EPS) in ecoinvent	58
4.6	Polystyrene extruded (XPS).....	61
4.6.1	Process Description	61
4.6.2	Polystyrene, extruded (XPS) in ecoinvent.....	62
4.7	Cellulose Fibre (inclusive blowing in).....	71
4.7.1	Process Description	71
4.7.2	Cellulose Fibre Production and Application in ecoinvent	71
4.8	Urea Formaldehyde Foam.....	74
4.8.1	Process description.....	74
4.8.2	Infrastructure	74
4.8.3	Emissions	74
4.8.4	Waste.....	74
4.8.5	Urea Formaldehyde Foam Production in ecoinvent	74
5	CUMULATIVE RESULTS AND INTERPRETATION	79
6	REFERENCES	79

1 Introduction

This part deals with the production of insulation materials used in the building sector and basalt as resource for rock wool. The following materials are analysed:

- Glass wool
- Rock wool and basalt
- Foamglas
- Polystyrene foam (EPS / XPS)
- Cellulose fibre (inclusive blowing in)
- Urea formaldehyde foam

Since data for polyurethane foams (hard and soft) are adopted from APME data, they are included in the report on plastics (Hischier (2007))

1.1 Glass wool

“Glass fibres for insulation are fibres of vitreous structure produced from silicate melts by mechanical or pneumatic processes. For insulation, non-textile fibres are used exclusively. Such fibres are also called glass wool, slag wool, rock wool, or mineral wool.” (Fitzer et al. (1997))

1.2 Rock wool

With adaptations to the Swiss situation the following introduction for mineral wool and specially rock wool is based on (EPA (1998)): Mineral wool often is defined as any fibrous glassy substance made from minerals (typically natural rock materials such as basalt or diabase) or mineral products such as slag and glass (Emissions Factor Documentation for Glass Manufacturing (EPA (1998))).

In general it is possible to use slags as primary material instead of natural rock for the production of rock wool. These are for example iron blast furnace slag, copper, lead, and phosphate slags. Generally rock wool products have one of the four following forms:

- "blowing" wool or "pouring" wool, which is injected into structures of buildings;
- batts, which may be covered by paper or foil as vapour sealant shaped to fit between structural members of buildings;
- industrial and commercial products such as high-density fibre felts and blankets, which are used for insulating boilers, ovens, pipes, refrigerators, and other process equipment;
- bulk fibre, which is used as a raw material in manufacturing other products, such as ceiling tile, wall board, spray-on insulation, cement, and mortar.

The module “rock wool, at plant” includes the melting process, defibration and polymerization. The raw material “basalt, at plant” is included in this study. It is a widespread rock of volcanic origin with a broad range of applications, mainly in the building industry. Information on basalt is mainly provided by Lagaly et al. (1997), inventory data is taken from Anonymous (1997) and Köhl & Frei (1994).

1.3 Foamglas

FOAMGLAS® is a thermal insulation product manufactured and distributed throughout the world by Pittsburgh Corning Europe (PCE) und Pittsburgh Corning Corporation. FOAMGLAS® is produced in the form of slabs and pre-cut shapes. It is chiefly used as an insulation material throughout the building industry in addition to being employed for a range of specialized industrial applications. The main application areas in buildings are: flat roofs, underground walls, floors and terraces/car park decks. For the European market, FOAMGLAS® is produced at three production sites in Tessenlo

(BE), Schmiedefeld (D) and Klasterec (CZ). In 2005, the production site in Tessenderlo produced approximately 320'000 m³ of FOAMGLAS[®] products of varying densities. This corresponds to 39'200'000 kg FOAMGLAS[®] products per year when calculating with an average density of 122.5 kg/ m³. Of this amount, approximately 70'000 m³ were delivered to Switzerland, including Lichtenstein, and 7'700 m³ were delivered to Austria.

In terms of quantity, the main product is the FOAMGLAS[®] board, T4 WDS, which is used for roof insulation (e.g. compact roof systems). Because of its industry prevalence, the T4 WDS production process can be regarded as being representative for all FOAMGLAS[®] products. 95% of the European consumption of this FOAMGLAS[®] board is produced in Tessenderlo. A negligible 5% is produced at the site in Germany. The market share of FOAMGLAS[®] products on the foam glass market for flat roof insulation in Europe is nearly 100%. Electricity use for the production volume delivered to Switzerland, France, the Netherlands and Austria, is covered by renewable energy certificates.

1.4 Polystyrene Foam

This report describes the production process of expanded and extruded polystyrene (or in short EPS and XPS). EPS and XPS find their applications mainly in the construction industry, either for flooring, roofing or walling. Extruded polystyrene shows resistance to moisture and has reliable strength and durability. The raw materials (production of ethylene and benzene) come up for the major part of the energy demand (90%). A mixture of CO₂ and Acetone (ca. 1:1), HFC-134a or HFC-152a is used as blowing agent in the extruder. Emissions resulting from the production of XPS are Acetone and Styrene for XPS from all blowing agents and R134a and HFC-152a from XPS with the respective blowing agent.

The data used in this report originates mainly from the information provided by the ECOBIS database, Germany (Starzner S. and Wurmer-Weiss P. 2000) and the company Jackson Insulation GmbH, Germany. Information on material input and the blowing agents are retrieved from Energie- und Umweltzentrum Allgäu (2002) and IPCC (2007).

2 Reserves, Resources and Raw Materials

2.1 Glass Wool

“Glass wool (wadding, fibres) are produced from mixtures of minerals and industrial chemicals that produce the required final composition. In addition to quartz sand, the minerals used include limestone, dolomite, phonolite, feldspar, nepheline-syenite, kernite, colemanite, ulexite, and heavy spar (barite). The industrial chemicals include sodium carbonate, sodium sulphate, and barium carbonate. The use of recycling glass has gained considerable importance.” (Fitzer et al. (1997))

2.2 Rock Wool / Basalt

“Basalt is the most common volcanic rock, occupying large areas of the Earth's crust. The basalts are the most important effusive rocks, and their formation can be observed even now (e.g., numerous volcanoes in Asia, the United States, Europe, and at the mid-ocean ridges). The most important types are the tholeiite basalts and the alkali basalts (feldspathoids, alkali pyroxene, alkali amphibole, and olivine). Medium to coarse grained basalts are known as dolerites, while olivine basalts (which contain feldspathoids as well as plagioclases) are known as basanites, and anchimetamorphic basalts of the Permian/Carboniferous era are known as melaphyres, diabases, and greenstones. The largest single basalt mass in Europe is in the Vogelsberg region of Germany. Other large deposits are in Hegau, the Eifel, Westerwald, and the Rhön mountains. Another deposit extends from the Oberpfalz (Bavaria), through the Erzgebirge, and as far as the Lausitz region” (Lagaly et al. (1997)).

The most important natural resources used for producing rock wool are dolomite, basalt, and limestone. The fuel used for the melting process is coke. An example of the transport distances of the base materials and fuel are given in Tab. 2.1. The production of the greatest Swiss rock wool factory Flumroc, is used as reference in this study.

Tab. 2.1 Base materials, fuel and transport distances (production of rock wool by Flumroc AG (Flumroc (2007)))

Raw Material	Origin	Distance [km] (train / road)
dolomite	surrounding	0 / 15
basalt	Germany	19 / 100
limestone	local rock	0 / 20
coke	France and Italy	500 / 0

A general comment on resources of primary materials in Switzerland has been made by the Swiss Geotechnical Commission: The resources used by the non-metallic minerals industry can be regarded as quarryable if its mining is possible with actual technology, the usage realizable with supportable financial and temporal expenditure and if the utilization does not collide with other interests. This last point causes a lot of problems in Switzerland as in the intensely populated landscape (Middle Lands) all different kinds of utilization collide. These are for example water protection, settlement area etc.. This circumstance reduces the real quarryable area fundamentally. This scenario is valid for all groups of resources in Switzerland.

For larger countries as Germany, France or Italy the mentioned scenario can also be applicable but in a longer time period.

2.3 Foamglas

FOAMGLAS® insulation materials are made from recycled, mechanically cleaned floatglass (68%) and feldspar (25%) in raw material quality. Small amounts of technical grade quality salt cake, soda ash, iron oxide, manganese oxide, sodium nitrate and carbon black are added to the raw material mix. Electricity is used for the melting process and raw mix production: natural gas is used for the production process.

Information on the reserves and resources can be found in part I on gravel products and part VII on lime products.

2.4 Polystyrene Foam

Polystyrene is a carbon based synthetic polymer, produced from Ethylbenzene. Thus the discussion is mainly revolving around oil (for benzene) and natural gas (for ethylene) resources. Depending on the process, coal is also considered as a raw material. Information on these raw materials can be found in ecoinvent report No. 5 on energy (see Dones et al. (2007)). The production of polystyrene is discussed in Hischier (2007).

2.5 Cellulose Fibre

Cellulose fibre for insulation material is made from waste paper. Thus resource availability is not critical.

2.6 Urea Formaldehyde Foam

Urea formaldehyde (UF) foam is made from UF resin. The production of the resin and the resources used therefore are discussed in Althaus et al. (2007).

3 Characterisation and Use

3.1 Glass Wool

3.1.1 Physical Properties

“Fibrous insulating materials of man-made mineral fibres are a highly disperse system of finely divided fibrous glass with a specific surface of 0.1 – 0.2 m²/g. The fibres are oriented predominantly in the horizontal direction. For this reason, the material is isotropic in two dimensions but not in the third. It is an open-pore system with little resistance to flow. This is the basis for its thermal and acoustic properties.

The fibre diameters vary in a broad spectrum. The logarithm of the fibre diameter approximately follows a normal distribution. The minimum fibre diameter is process dependent and is between 0.18 and 0.30 mm. However, the properties of the individual fibres are less important than those of the final insulating material, i.e., the fibres and binder as a whole.

The most important property of insulation is the thermal conductivity. The thermal conductivity of a particular fibrous material is determined by its bulk density ρ_B and its average fibre diameter. The mechanical properties are primarily determined by bulk density and binder content; in any case, however, fibrous insulating materials cannot be used as load-bearing structural unit.” (Fitzer et al. (1997))

The glass wool mat inventoried in this study is 80 to 200 mm thick and delivered in rolls. The thermal conductivity is 0.040 W/mK. The density is 40 kg/m³ (Richter et al. (1995)).

3.1.2 Chemical Properties

“Mineral wool insulation is inert to organic solvents, even at elevated temperatures. Extractable are mineral oils, emulsifiers, silicones, and low molecular mass components of binders.

Water and aqueous solutions attack the glass surface, the attack magnified by the large fibre surface. Attack by water also loosens the fibre–resin bond.” (Fitzer et al. (1997))

3.1.3 Use

“Mineral-fibre insulating materials are finished products for use in protection from heat, cold, fire, and noise. The major end use area is high-rise construction. Considerable quantities are used in industrial and machine construction for thermal insulation. Fibrous insulating materials are also important as noise absorbers in buildings and elsewhere.” (Fitzer et al. (1997))

3.2 Rock wool / Basalt

3.2.1 Characterisation

Basalt contains <52 % SiO₂ and is the volcanic equivalent of gabbroid rocks. Basalt has a dense, compact, sometimes granular or porphyric structure, and is dark to black in color. The main constituents of the matrix are plagioclase (anorthite 50 – 75 %) and pyroxene (mainly augite). Auxiliary minerals include olivine, magnetite, ilmenite, alkali feldspar, alkali pyroxene, alkali amphibole, feldspathoids, and glass. The physical properties required depend on the use intended for

the basalt. Typical properties are high compression strength (2500 – 4000 kg/cm², with a maximum of 5800 kg/cm²) and high transverse strength (150 – 250 kg/cm², DIN 52 100). To use basalt it requires a high homogeneity, and should be resistant to frost, acids, and waste gases (e.g., flue gas). The density of basalt varies between 3.00 and 3.15 g/cm³, the bulk density between 2.95 and 3.00 g/cm³. The porosity varies between 20 and 25 vol % (water absorption 0.1 - 0.3 wt %). The specific heat capacity for temperatures in the range zero to 770 °C is ca. 1.09 J g⁻¹ K⁻¹; the thermal conductivity (0 – 100 °C) ca. 0.028 W cm⁻¹ K⁻¹; the linear expansion coefficient is 9×10⁻⁶ K⁻¹. Anhydrous basalt starts melting at ca. 1280 °C, while basalt containing 1 % water starts melting at only ca. 1100 °C. Basalt wool is stable up to 900 °C. Basalt is affected by weathering very soon after it is mined, especially by the heat of strong sunlight in the presence of moisture. Characteristic white spots are formed, and the rock crumbles. Sunburnt basalt is unsuitable for many applications (Lagaly et al. (1997)).

Basaltic glass in form of fibres is known from a lot of basaltic volcanoes (Kündig et al. (1997)). Very fine basaltic fibres are formed when hot (1'100°C) and mobile lava is pressed through an already solidified but perforated crust. This defibration process is triggered by small gas-air explosions in the stream of lava. The chemical composition of the glass fibres used in the rock wool production is similar to the composition of naturally cooled volcano basalt and contains the minerals listed in Tab. 3.1:

Tab. 3.1 Chemical composition of basalt (Kündig et al. (1997))

Mineral	Chemical formula
olivine	(MgFe) ₂ SiO ₄
plagioclase feldspar	(CaAl ₂ SiO ₈)
augite	(Ca(MgFe)SiO ₆)
magnetite	(FeFe ₂ O ₄)

The composition of basalt in weight percent of the oxides is shown in Tab. 3.2:

Tab. 3.2 Chemical composition of basalt in weight percentage (Kündig et al. (1997))

Oxide	Weight percent
SiO ₂	50
TiO ₂	2
Al ₂ O ₃	15.9
FeO _{total}	11
MnO	0.2
MgO	8
CaO	10
Na ₂ O	3
K ₂ O	0.5
P ₂ O ₅	0.3
Total	100

Basaltic glass threads have the ideal characteristic in terms of elasticity, corrosion and heat resistance, based on their chemical composition (Si, Al, Mg, Fe, Ca, Na, O) and diameter around 0.01 mm.

Main advantages of rock wool are that it does not burn, rot, absorb humidity and odour.

Based on (Flumroc (2002)) rock wool contains:

A minimum of 96% artificially produced mineral fibres with non-crystalline structure which is extracted from natural basalt and dolomite as well as recycled rock wool products.

Modified phenolic resin varying in quantity between 0.5 and 3.0 weight percent depending on the specific product characteristics.

About 0.2 percent mineral oil to increase the hydrophobic characteristics of the final product and to decrease the amount of dust.

3.2.2 Use

Some basalts have high strength, and are therefore used in the building industry, e.g., as hard core, paving stones and as raw material for building materials. In another important application, molten basalt is converted into refractory fibres as mineral wool for insulation and fire protection (Lagaly et al. (1997)).

Flumroc AG is the leading company for rock wool production in Switzerland. It produces about 40% of the rock-wool products (mostly rock-wool mats) in the Swiss market.

The different products of Flumroc are generally applied for thermal insulation, sound insulation and/or fire protection of buildings (Flumroc (2002)).

The following products with different properties, production volumes and different technical characteristics are marketed by Flumroc and are included in the module „rock wool, at plant“:

Insulation board 1:	Thermal and sound insulation in cavity masonry, or as infilling in external timber walls.
Insulation board 2:	Thermal and sound insulation in ventilated façades with timber subframes, in sheet metal liner trays on industrial façades, and as sound insulation in partition walls.
DUO	Thermal and sound insulation in ventilated façades and in fair-faced cavity masonry.
ECCO	Thermal and sound insulation of ceiling end faces and high pitched roofs. As fire-protective cladding of combustible structural components.
NOVA	Thermal and sound insulation in steep-pitched roofs, insulation over rafters.
PRIMA	Thermal and sound insulation of flat roofs with a protective layer.

The form of the product can vary from boards to mats, stuffing wool and fine granulates. Tab. 3.3 presents main characteristics of Flumroc's different products and their production shares in year 2001.

Tab. 3.3 Most important products of Flumroc (CH) with their production volume in the year 2001 and main technical data

Product	Insulation board 1	Insulation board 3	DUO	ECCO	NOVA	PRIMA	Diverse
Production volume [%]	43.79	10.55	6.19	3.44	10.33	14.46	11.24
Mean apparent density ρ_a [kg/m ³]	32	60	50	75	90	120	60.7
Thermal conductivity λ_D [W/(mK)]	0.036	0.034	0.035	0.036	0.035	0.038	n.A.
Specific thermal capacity c [J/kgK]	830						n.A.
Coefficient of diffusion resistance μ (approx)	2						n.A.
Fire coding ⁵	6q.3						n.A.
Maximum temperature in use [°C]	250						n.A.

3.3 Foamglas

Physical properties:

The product inventoried in this study is an insulation slab 40 mm to 200 mm in thickness, a thermal conductivity λ_R of 0.040 W/mK and a density of 110 kg/m³.

Chemical properties:

FOAMGLAS® consists of glass (>99% by weight) and small amounts of hydrogen sulfide and carbon dioxide. The product is resistant to rodents, vermin and insects, as well as being rot proof, resistant to ageing, resistant towards acids (except hydrofluoric acid) and chemical substances, especially organic solvents.

3.4 Polystyrene Foam Slab

Background information on polystyrene and its production is given in Hischier (2007)

The thermal conductivity of polystyrene foam is typically between 0.035 and 0.04 W/mK, the density is around 30 kg/m³.

Polystyrene XPS foam insulation has a combination of properties that makes it a strong and efficient insulating material. It shows high resistance to moisture, reliable strength and durability — even without a facer or cover board. Polystyrene XPS with a heat conductivity of 0.035 to 0.040 W/mK is used as an insulating material for commercial, industrial and residential structures. Tab. 3.4 shows information on the material properties of extruded polystyrene (Starzner S. and Wurmer-Weiss P. 2000).

According to a publication issued by Climate Action Network Europe, about half of Polystyrene XPS is currently foamed with CO₂ as a blowing agent (CAN Europe (2004)). Canadian manufacturers of these foam products on the other hand argue that CO₂ “is not a technically or market viable option. The thermal insulation properties of CO₂-blown foam is said to be significantly compromised when compared to halocarbon-blown foam” (Environment Canada (2005)). In their 2007 Special report on

⁵ <http://bsronline.vkf.ch/D/Info5.asp>, access date 01.07.2003

“Safeguarding the Ozone Layer and the Global Climate System”, IPCC (2007) writes on this issue: “As a sole blowing agent or in conjunction with co-blowing agents (ethanol, hydrocarbons, acetone, isopropanol, water), CO₂ is principally used in XPS (extruded polystyrene) in regions which have already chosen to phase out HCFCs, and where the market considers the loss in thermal efficiency to be offset by the other inherent properties of XPS boards (moisture resistance, mechanical strength, compressive strength, dimensional stability and resistance to freeze-thaw deterioration). Higher densities or lower conversion may partially or wholly offset the low cost of CO₂ itself in some XPS applications. Nevertheless, the use of CO₂-based technology for XPS is likely to continue to grow as long as the cost of alternative HFC solutions remains high.

For extruded polystyrene (XPS) manufacturing processes, IPCC mentions HFC-134a and HFC-152a as the most common blowing agents in use, apart from CO₂.

Extruded polystyrene products only absorb 0.3% (w/w) moisture. With its closed-cell structure, XPS foam products have a natural “skin” surface which is not conducive to moisture absorbency. The closed-cell structure of XPS imparts long-term strength and durability. Products are available in a range of compressive strengths to suit varied application needs⁶.

Tab. 3.4 Material properties of extruded polystyrene

Technical information

Density (kg/m ³)	25 - 45
Heat conductivity (W/mK)	0.035 - 0.040 (CO ₂ -foamed, depending on density of the extruded polystyrene slab)
Melting point (°C)	>100 (deformation at temperatures over 85°C)
Water absorption (%)	0.1 – 0.5 (very low)
Pressure resistance (N/mm ²)	0.15 – 0.7 (medium)
Resistance	Resistant to leachate and non-oxidising acids, resistant to vermin, rotting, fungal decay; not resistant to solvents and UV-radiation. Has to be protected against rodents.

Polystyrene is a thermoplastic material which means that it is capable of being melted and re-inserted into the manufacture of new XPS insulation. XPS production creates virtually no “scrap” or waste material because all internal waste is recovered, ground up, re-pelletized, and fed back into the production system.

Use

Polystyrene XPS solutions are designed to cover a wide range of end use applications. Primarily as part of panels or other laminated solutions, they are both thermal and structural solutions. The following ranges of applications are some of the key areas of use of extruded polystyrene boards (KnaufInsulation (2005)):

- General panels (plasterboard, chipboard, cement boards etc.)
- Close tolerance panels (doors, plastisol faced etc.)
- Refrigerated vehicles.

The annual polystyrene XPS consumption in Switzerland is about 1.1 mio m³ (Hungerbühler (2004)). In Germany, the extruded polystyrene panels had a share of 3.2 % of all insulating materials in 1998 (Starzner S. and Wurmer-Weiss P. (2000)). 70% of the market is controlled by two companies in

⁶ <http://www.xpsa.com/tech/physical.html>; accessed June 2005

Europe, namely Dow France and BASF (Germany). Further big players on this market are Isobox Henry (UK), SIRAP GEMA (GER-ITA) and Polyglass (ES).

Like many construction materials, XPS is combustible. XPS insulation should be covered with a thermal barrier or otherwise installed in accordance with applicable building codes (XPSA (2005)).

Polystyrene as such is not poisonous. Though test results show that after production, ethylbenzene and styrene can fumigate from polystyrene products. These low emissions decrease rapidly within a few days after production (Starzner S. and Wurmer-Weiss P. (2000)). According to the Portland Cement Association and its recently published Technology Brief News article, attempts to find any emissions from polystyrene foams used in walls failed. No emissions could be detected by the most sensitive instruments. Extensive research on the polystyrenes concluded that when they were forced to burn, their emissions were “no more toxic” than the emissions of burning wood (PCA (2005)).

Other sources however note that there are low styrene gas emissions during the use-phase (eza (2004), Starzner S. and Wurmer-Weiss P. (2000)). If polystyrene XPS catches fire, carbon, carbon dioxide (CO₂), carbon monoxide, styrene and flame-retardant residue are mainly released. Hard sheets of extruded polystyrene foam may not be exposed to temperatures above 85 °C for long periods (Poliglas Dämmsysteme (1998)).

3.5 Cellulose Fibre

Cellulose fibre is used as thermal insulation in buildings. For this application, the fibres are blown into cavities e.g. between two planks of a wall to be insulated. Cellulose fibres in a typical application have a density of about 50 kg/m³ and a thermal conductivity of 0.04 W/mK.

Cellulose fibres are treated with borax and boric acid to enhance fire retarding properties.

3.6 Urea Formaldehyde Foam

UF foam is either available in prefabricated slabs or it is produced in situ to insulate cavities. The density of both foams is between 10 and 30 kg/m³ and the thermal conductivity is around 0.035 W/mK.

UF foam is mainly used as insulation material in buildings, as filler in mining and as soil conditioner (water and nutrition retention). Because UF foam can emit formaldehyde over a long time, it was banned in Canada in some States of the USA in the 1980ies⁷.

4 System Characterisation

4.1 Glass Wool

4.1.1 General

The data for glass wool are based on a study from EMPA which is based on data from Isover SA in Switzerland. It is based on the situation by year 1991 (Tab. 4.1). Unfortunately no current data were available.

Hutter et al. (1999) state, that the amount of waste glass used as raw material can be as high as 70%. For this a high purity of the waste glass is needed which is usually not the case in practice. The amount used in this study is about 65% and refers to the year 1991 (Richter et al. (1995)).

⁷ http://www.cmhc-schl.gc.ca/en/burema/gesein/abhose/abhose_ce06.cfm, access date 16.03.2004

4.1.2 Melting

“Two types of melting units are in use for the production of silicate melt: glass melting tank furnaces and cupola furnaces.

The glass melting tank furnaces are similar in construction to those used in other areas of glass manufacture. Preferred are cross-fired recuperative furnaces and U-shaped flame furnaces, but all-electric melters are also used if energy costs are high. Maximum performance is 300 t/d.

Cupola furnaces are used to produce molten rock. They require coarse feed and yield a less uniform melt. The traditional fuel is coke but oil- or gas-fired units are also used. Furnace productivity rarely exceeds 150 t/d.” (Fitzer et al. (1997)).

For this study the “cupola furnace” type has been used. The melting temperature is about 1’400°C.

4.1.3 Fibre Production

From the numerous processes and process variants for fiberizing silicate melt, the Owens process is inventoried in this study. Fitzer et al. (1997) describe this process as follows:

”In the Owens blowing process, the melt, in the form of a thick primary strand, emerges from holes at the bottom of a platinum crucible (the bushing) and is subsequently entrained by a downward-directed stream of gas from the blow-jet. The blowing medium may be steam or compressed air. The primary strand is drawn to form fibres with an average diameter of 3 – 6 mm. The length of the fibres is about 3 – 10 cm. The product contains slugs. Glass and rock melts are processed in this way.” (Fitzer et al. (1997))

Tab. 4.1 Input of raw materials for 1 kg of typical glass wool mat in this study (Richter et al. (1995))

Raw materials	Richter et al. (1995)
Silica sand	1.10E-01 kg
Sodium carbonate	8.36E-02 kg
Limestone	9.64E-03 kg
Dolomite	2.93E-03 kg
Kernite	5.76E-02 kg
Felspar (nepheline)	1.36E-01 kg
Fluorspar	2.20E-02 kg
Waste glass	6.42E-01 kg
Glass powder (not taken into account as assumed to be internally reused production waste)	1.55E-01 kg
Total raw materials	1.06E+00 kg
Binders	
Phenole, Formaldehyde and urea	1.60E-01 kg
Energy	
Electricity	2.32E+00 kWh
Natural gas	2.03E+00 kWh

4.1.4 Product Forming

“The fibres are formed into a continuous sheet by pneumatic processes, while an aqueous solution of binder is sprayed onto the freely falling individual fibres. The binders are slightly pre-condensed phenol – formaldehyde resins. They are often modified by aminoplastic resins (urea, melamine,

melamine ether, dicyanodiamide) to suppress the tendency of phenolics to exothermally autoxidize. In the industry these aminoplastic resins are called anti-punking agents.

Fig. 4.1 shows schematically the overall manufacturing process. The binder resins are cured at temperatures around 180 – 250 °C in a continuous flow dryer heated by circulated air. For shaped articles or pipe insulation, for example, special ovens that operate on a similar principle are used. On the other hand, presses with purely convective heat transfer are rarely used since they require long cycle times.

Mineral-fibre insulating materials can be cut to any desired dimension with knives, saws, shears, and milling cutters.” (Fitzer et al. (1997))

The binder used in this study is a mixture of phenol, formaldehyde and urea resins. The total amount of binder based on (Hutter et al. (1999)) is between 0.5 and 9 w-% and based on Richter et al. (1995) even 16 w-% from the weight of the glass wool product. Although the difference can not be explained due to too little information, a share of 16 w-% for the binders is taken into account because most other inventoried data are based on that study (Richter et al. (1995)).

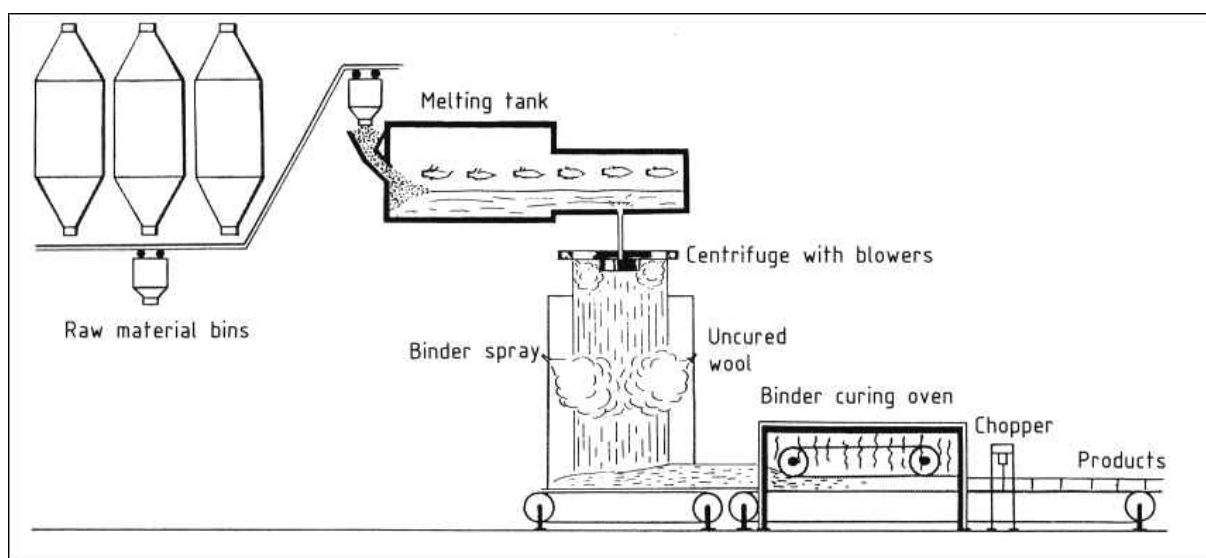


Fig. 4.1 Production process of mineral-fibre insulation (Rotary process is used to form fibres) (Fitzer et al. (1997))

4.1.5 Life Cycle Inventories

Glass wool mat, at plant

The ecoinvent database meta information of “glass wool mat, at plant” (Tab. 4.3) includes all important and necessary information to use the data correctly. Fig. 4.2 shows all in- and output flows of the product “glass wool mat, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Richter et al. (1995)) and the standard deviation for each datum.

Since no specific infrastructure information is available, the “rock wool plant” is used. It includes land-use, buildings and machines.

4.1.6 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

A comparison in Tab. 4.2 of the most important data used in this study (Richter et al. (1995)) with the data of a German study (Hutter et al. (1999)) shows large differences.

The total amount of raw materials is about the same in both studies (1.06 kg per kg product). In contrast the amount of waste glass differs strongly. The Swiss study reckons with about 65% and the German study with about 30%. The data from the Swiss study is clearly related to the product glass wool mat. The German study does not explicitly explain to which product it relates. It is assumed that this might be the reason for the different amount of used binder.

The electricity consumption for the Swiss data is higher than for the German data because Isover SA uses an electric heated melting furnace while in the German study the melting process is modelled with a natural gas heated melting furnace. The nevertheless higher amount of natural gas in the Swiss study could be attributed to the fact that waste glass is melted by heat from natural gas. Isover SA could not supply actual data and could not explain the difference to the German study.

There is no further information on the organic wet waste and the wet waste from filtering. For each waste a proxy is used in the study (shown in Fig. 4.2). Organic wet waste is inventoried as "disposal, hazardous waste, 25% water, to hazardous waste incineration" and wet waste from filtering as "disposal, average incineration residue, 0% water, to residual material landfill". The corresponding uncertainties have been raised by estimation.

Tab. 4.2 Comparison of raw materials, binders and energy per kg glass wool product between a German (Hutter et al. (1999)) and a Swiss study (Richter et al. (1995))

Raw materials	Hutter et al. (1999)	Richter et al. (1995)
Silica sand	3.20E-01 kg	1.10E-01 kg
Sodium carbonate (Soda)	1.25E-01 kg	
Sodium carbonate		8.36E-02 kg
Limestone	1.90E-02 kg	9.64E-03 kg
Dolomite	7.50E-02 kg	2.93E-03 kg
Borax (hydrated sodium borate)	8.20E-02 kg	
Kernite		5.76E-02 kg
Felspar (nepheline)	1.32E-01 kg	1.36E-01 kg
Sodium sulphate	4.00E-03 kg	
Fluorspar		2.20E-02 kg
Waste glass	3.10E-01 kg	6.42E-01 kg
Glass powder (not taken into account as assumed to be internally reused production waste)		1.55E-01 kg
Total raw materials	1.07E+00 kg	1.06E+00 kg
Binders		
Phenole, Formaldehyde and urea	7.00E-02 kg	1.60E-01 kg
Energy		
Electricity	1.40E+00 kWh	2.32E+00 kWh
Natural gas	1.05E+00 kWh	2.03E+00 kWh

The overall quality of the dataset "glass wool mat, at plant" is poor because data is relatively old (collection around 1993 to 1995; no exact information available in Richter et al. (1995)) and there are quite big inexplicable differences compared to the data in the German study. The Swiss market is not only supplied with glass wool from Isover SA but also from glass wool produced in foreign countries. As the German study shows big differences in the data compared to the Swiss company, it is expected that the companies delivering the Swiss market have quite different technology standards.

Tab. 4.3 ecoinvent meta information for the product “glass wool mat, at plant”

Name	glass wool mat, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Included processes: melting, fibre forming & collecting, hardening & curing and internal processes (workshop, etc.). Additionally transportation of raw materials and energy carrier for furnace, packing and infrastructure are included. For the heat needed, energy modules are used and the needed electricity is from Swiss grid.
Amount	1
Local Name	Glaswollematte, ab Werk
Synonyms	
General Comment to reference function	This module can be used for all different kind of glass wool mats. The density of the glass wool mat used as basis for the study is 40 kg/m ³ .
Start Date	1993
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Isover SA). For some exchanges DE-, RER- and GLO-modules are used as proxy
Technology text	The company worked on a very high technical level but the data refer to the situation before 1995. The energy for the melting process is mainly electricity and from natural gas. The amount of waste glass used as raw material is about 65%.
Representativeness [%]	
Production Volume	
Sampling Procedure	literature and environmental report
Extrapolations	see geography and technology
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
silica sand	glass wool mat, at plant (part 1)			raw material	construction materials	additives	No	DE	silica sand, at plant	1.10E-01	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
sodium carbonate (Soda)				raw material	chemicals	inorganics	No	RER	soda, powder, at plant	8.36E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
dolomite				raw material	construction materials	others	No	RER	dolomite, at plant	2.93E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
kernite				raw material, borax (hydrated sodium borate) has been used as proxy	chemicals	inorganics	No	US	sodium borates, at plant	5.76E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
feldspar (nepheline)				raw material	construction materials	others	No	RER	feldspar, at plant	1.36E-01	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
limestone				raw material	construction materials	others	No	CH	limestone, milled, packed, at plant	9.64E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fluorspar				raw material	chemicals	inorganics	No	GLO	fluorspar, 97%, at plant	2.20E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
waste glass				raw material	glass	packaging	No	RER	glass, cullets, sorted, at sorting plant	7.97E-01	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
manganese oxide				raw material, amount of manganese in manganese oxide: 80%	metals	extraction	No	RER	manganese, at regional storage	2.62E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
sodium nitrate				raw material, amount of N in NaNO3	agricultural means of production	mineral fertiliser	No	RER	calcium nitrate, as N, at regional storehouse	5.76E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fixing agent: phenol				raw material	chemicals	organics	No	RER	phenol, at plant	2.94E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fixing agent: formaldehyd				raw material	chemicals	organics	No	RER	formaldehyde, production mix, at plant	9.21E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fixing agent: urea				raw material	chemicals	organics	No	RER	urea, as N, at regional storehouse	3.86E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
caustic soda hydrated				raw material	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	3.50E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
sulphuric acid				raw material	chemicals	inorganics	No	RER	sulphuric acid, liquid, at plant	2.06E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
hydrosilicon				raw material	chemicals	inorganics	No	RER	silicone product, at plant	2.94E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
ammonia				raw material	chemicals	inorganics	No	CH	ammonia, liquid, at regional storehouse	2.47E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
ammonium sulphate				raw material	agricultural means of production	mineral fertiliser	No	RER	ammonium sulphate, as N, at regional storehouse	8.03E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
Mulrex (an oil-based coolant additive)				raw material, no chemical composition found; module "lubricating oil, at plant" used as proxy	chemicals	organics	No	RER	lubricating oil, at plant	1.11E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
Basonyl Red				raw material, basic dye for the printing links and specialty colorants industry, chlorine free, chemical nature: Xanthene; for of lack of alternative the module "acrylic dispersion, 65%, in H2O, at plant" is used as proxy.	paintings	production	No	RER	acrylic dispersion, 65% in H2O, at plant	5.70E-06	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
water				for whole production process, assumed that the whole water consumption is tap water	water supply	production	No	CH	tap water, at user	5.74E+00	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);

Fig. 4.2 Flows for "glass wool mat, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
propane and natural gas	glass wool mat, at plant (part 2)				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	7.33E+00	MJ	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
diesel oil in building machines					construction processes	machinery	No	GLO	diesel, burned in building machine	5.40E-02	MJ	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
electricity consumption					electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.32E+00	kWh	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
fireproof stones					construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	2.29E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
molybdenum, material of electrode					metals	extraction	No	RER	molybdenum, at regional storage	1.25E-05	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
road transports of raw materials to plant					transport systems	road	No	RER	transport, lorry >16t, fleet average	5.34E-02	tkm	Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
train transports of raw materials to plant					transport systems	train	No	RER	transport, freight, rail	3.82E-01	tkm	Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
ship transports of raw materials to plant					transport systems	ship	No	RER	operation, barge	5.29E-02	tkm	Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
cardboard, at plant					convenience product	paper & cardboard	No	RER	corrugated board, mixed fibre, single wall, at plant	1.14E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
packaging film, LDPE					convenience product	plastics	No	RER	packaging film, LDPE, at plant	1.32E-02	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
bonding agent					convenience product	chemicals	No	RER	rosin size, in paper production, at plant	3.44E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
varnish/colour					convenience product, for lack of alternative the module "acrylic varnish, 87.5%, in H2O, at plant" is used as proxy.	paintings	No	RER	acrylic varnish, 87.5% in H2O, at plant	3.50E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
silicone					convenience product	chemicals	No	RER	silicone product, at plant	2.87E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
ink					convenience product, for lack of better alternative the module "chemicals, organic, at plant" is used as proxy.	chemicals	No	GLO	chemicals organic, at plant	5.74E-05	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
solvent					convenience product, for lack of alternative the module "solvents, organix, unspecified, at plant" has been used as proxy.	chemicals	No	GLO	solvents, organic, unspecified, at plant	1.15E-05	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
road transports of different convenience products to plant					it is assumed that all transports are by lorry	transport systems	No	RER	transport, lorry >16t, fleet average	1.97E-03	tkm	Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
transports of paper bags and plastic bags to municipal incineration						waste management	No	CH	transport, municipal waste collection, lorry 21t	7.83E-04	tkm	Richter et al. (1995)	1	2.07	(2,4,2,1,1,5);
glass wool plant					"rock wool plant" used as proxy; normalized amount from rock wool production process	insulation materials	Yes	CH	rock wool plant	4.43E-10	unit	estimation and Flumroc (2002)	1	3.06	(2,4,2,1,1,5);

Fig. 4.2 Flows for "glass wool mat, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent									Uncertainty information			
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
glass wool mat, at plant (part 3)	→	municipal solid waste					waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	8.49E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→	wet waste (organic)				For lack of further information the module "disposal, used mineral oil, 10% water, to hazardous waste incineration" is used as proxy.	waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	2.81E-03	kg	Richter et al. (1995)	1	5.00	estimation
	→	wet waste from filtering				For lack of further information the module "disposal, average incineration residue, 0% water, to residual material landfill" is used as proxy.	waste management	residual material landfill	No	CH	disposal, average incineration residue, 0% water, to residual material landfill	1.39E-02	kg	Richter et al. (1995)	1	5.00	estimation
	→	cardboard, in municipal waste incineration				internally used cardboard (a part of the totally used cardboard)	waste management	municipal incineration	No	CH	disposal, packaging cardboard, 19.6% water, to municipal incineration	1.06E-03	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→	plastic foil, in municipal waste incineration				internally used plastic foil (a part of the totally used foil)	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	6.77E-04	kg	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→	waste heat from electricity consumption				waste heat from electricity consumption	air	low population density			Heat, waste	8.35E+00	MJ	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→	phenole				data from 1999	air	unspecified			Phenol	2.60E-04	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→	formaldehyde				data from 1999	air	unspecified			Formaldehyde	4.60E-04	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→	ammonia				data from 1999	air	unspecified			Ammonia	1.46E-03	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→	fluorine				data from 1999	air	unspecified			Fluorine	9.00E-05	kg	environmental report 1999 of Isover SA, Switzerland	1	2.02	(1,4,1,1,1,3);
	→	waste water to wastewater treatment					waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	1.52E-03	m3	Richter et al. (1995)	1	1.24	(2,4,2,1,1,5);
	→	packed glass wool mat, at plant					insulation materials	production	No	CH	glass wool mat, at plant	1.00E+00	kg	Richter et al. (1995)			

Fig. 4.2 Flows for "glass wool mat, at plant" and their representation in the ecoinvent database (cont.)

4.2 Basalt

4.2.1 General

In this study the inventory of one German quarry is considered, using its environmental report (Anonymous (1997)). The quarry exploits mainly diabas, a variant of basalt besides regular basalt. Exploitation of diabas and basalt can be considered comparable because these two hard rocks have similar properties. Furthermore it is assumed that the different materials produced need similar exploitation efforts, hence justifying the allocation of the environmental burdens by mass. No exact data for the output in 1997 was available. Based on personal communication from the manager of the company in concern, a yearly output of 250'000 t products was assumed, of which a third is supplied to rock wool production.

4.2.2 Process Description

Basalt is mined by quarrying. The first operation is to remove the overburden (i.e., the soil, clay, and bulk rock overlying the deposit). Various techniques and types of earth moving equipment are used for this operation.

The quarry is assumed to be covered and rehabilitated after ten years of operation.

The next stage is generally to drill the bedrock. Rotary and percussive drills are used to drill holes. These are then filled with controlled amounts of explosive. After blasting, the rock is loaded into dump trucks, generally by front-end shovels or hydraulic excavators, and transported to the crushing and screening plant.

In the crushing and screening plant, the larger lumps of rock are broken in a primary jaw crusher. Once obtained the desired size reduction, the rock is washed and classified using shaking screens with different meshes then it is stored. After washing, the muddy water is clarified in a waste water treatment plant. The products are partly sold to regional construction industry and partly to national rock wool producers.

4.2.3 Energy Consumption

The process consumes energy for mining (drill, excavator, trucks), crushing (conveyor, crusher, screens) and rehabilitation (trucks).

4.2.4 Transports

Transport occurs onsite from quarry to crusher by lorries and by conveyor belts during the crushing/washing process.

4.2.5 Emissions

Direct emissions to air and water reported in Anonymous (1997) originate mainly from combustion processes. These are already covered by the modules used for the energy carriers. Therefore these releases are omitted in the module for basalt production. Dust emissions for processing processes (exclusive mining operation) are filtered with modern technology and lie under 1 mg dust per m³ outgoing air.

4.2.6 Infrastructure

The infrastructure of the facilities is modelled using the infrastructure module for the production of limestone. The infrastructure of the mobile equipment is included in the module “diesel, burned in building machine” chosen to describe the use of fuel.

4.2.7 Assignment to ecoinvent datasets

The assignment of the data from the environmental report of a German quarry to the ecoinvent datasets is illustrated in Tab. 4.4.

Tab. 4.4 Transformation of data taken from the environmental report of one German quarry (Anonymous (1997)) to the inventory in ecoinvent.

Environmental Report (Anonymous (1997))			Inventoried in ecoinvent as		
Annual inputs			Input rates per kg of product		
Motor, gear and hydraulic oil ¹⁾	t	13.46	5.38E-05	kg	lubricating oil, at plant
Lacquer	t	0.75	3.00E-06	kg	alkyd paint, white, 60% in solvent, at plant
Paper	t	0.25	1.00E-06	kg	kraft paper, unbleached, at plant
Chemicals	t	0.09	3.60E-07	kg	chemicals organic, at plant
Technical oil and lubricants	t	0.04	1.60E-07	kg	lubricating oil, at plant
Light fuel oil	kWh	21373	3.13E-04 ²⁾	MJ	light fuel oil, burned in industrial furnace 1MW, non-modulating ⁶⁾
Diesel	kWh	2879800	4.14E-02 ³⁾	MJ	diesel, burned in building machine ⁶⁾
Petrol	kWh	1500	6.78E-06 ³⁾	km	operation, passenger car ⁶⁾
Electricity	kWh	1552954	6.21E-03	kWh	electricity, medium voltage, production UCTE, at grid ⁶⁾
Water	m ³	2100	8.40E-06	m ³	Water, lake
Annual outputs					
Waste paper	t	0.13	5.20E-07	kg	disposal, paper, 11.2% water, to municipal incineration
Compost	t	0.59	2.36E-06		(not inventoried, burden free)
Waste oil (motor and gear)	t	12.38	4.95E-05	kg	(recycled, burden free)
Waste oil (hydraulic)	t	1.08	4.32E-06	kg	(recycled, burden free)
Waste, contaminated with oil	t	1.22	4.88E-06	kg	disposal, paint remains, 0% water, to hazardous waste incineration
Miscellaneous waste	t	1.45	5.80E-06	kg	disposal, municipal solid waste, 22.9% water, to municipal incineration
Total waste	t	30.74	1.23E-04 ⁵⁾	kg	disposal, municipal solid waste, 22.9% water, to municipal incineration
Difference total waste to sum of reported wastes	t	13.89	5.56E-05 ⁵⁾	kg	
CO ₂ (	t	4.02	1.61E-05	kg	
SO ₂	kg	0.05	2.00E-10	kg	- ⁶⁾
NO _x	kg	6.43	2.57E-08	kg	- ⁶⁾
CO	kg	0.73	2.92E-09	kg	- ⁶⁾
Dust (not transparent which processes are included)	t	0.18	7.20E-07	kg	Inventoried as differential fractions PM<2.5, PM>2.5-<10 and PM>10 ⁷⁾
Effluents	m ³	24	9.60E-08	m ³	treatment, sewage, from residence, to wastewater treatment, class 2
Annual products					
Total hardstone	t	250000 ⁸⁾			
Basalt only	t	83333 ⁸⁾	1	kg	Basalt, at mine

¹⁾ Calculated input from reported waste oil (motor & gear and hydraulic); losses are not taken into account

²⁾ Transformation to volume with reported 10 kWh/l and back to energy with 0.86 kg/l and 42.6 MJ/kg from (Frischknecht et al. (2007)).

³⁾ Transformation to volume with reported 10 kWh/l and back to energy with 0.84 kg/l and 42.8 MJ/kg from (Frischknecht et al. (2007)).

⁴⁾ Transformation to volume with reported 10 kWh/l and back to energy with 0.75 kg/l and 42.5 MJ/kg and 2.82 MJ/km from (Frischknecht et al. (2007)).

⁵⁾ Reported wastes do not sum up to the reported total waste of 30.74 t/a. Difference between total waste and the sum of the reported wastes is inventoried as municipal waste.

⁶⁾ The modules chosen for the fuel combustion include the relevant emissions. Therefore the emissions reported in the environmental report, which are assumed to originate from combustion, are omitted.

⁷⁾ Particulate fractions for lime production are chosen from CEPMEIP-Database (Berdowski et al. (1995), SNAP 240400, lime production, low emission level: PM<2.5 = 10%; PM>2.5-<10 = 40%; PM>10 = 50%)

⁸⁾ Estimation based on personal communication with the manager of the German quarry of concern.

4.2.8 Life Cycle Inventory

The ecoinvent database meta information of “basalt, at plant” (Tab. 4.5) includes all important and necessary information to use the data correctly. Fig. 4.3 shows all in- and output flows of the product “basalt, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Anonymous (1997) and personal communication) and the standard deviation for each datum.

4.2.9 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Different sub-processes were considered of minor importance and therefore neglected, namely the transport of the material requirements from plants to the basalt quarry. The data in the environmental report used were inconsistent in many places.

The main uncertainty remains in the production volume, which features a big fluctuation from year to year and could not be determined exactly for the year considered.

The source of the data of dust and its size categories are of poor quality because it couldn't be found out which processes are included.

For all these reasons the data for basalt are of fair overall quality and should not be used for comparisons among different construction materials. In a larger context and if the importance of these products for a LCI study is not too high, the data can be used as such.

Tab. 4.5 ecoinvent meta information for the product “basalt, at plant”

Name	basalt, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The module includes a mining step and a subsequent beneficiation step comprising crushing, washing, and classification.
Amount	1
Local Name	Basalt, ab Abbau
Synonyms	
General Comment to reference function	The module is mainly designed for the use as raw-material for rock wool.
Start Date	2003
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data for one German quarry used as proxy for whole European situation
Technology text	Basalt is exploited in quarries.
Representativeness [%]	
Production Volume	nA
Sampling Procedure	Literature
Extrapolations	see Geography and Technology
Uncertainty Adjustments	-

General Flow information					Representation in ecoinvent 2000							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
basalt (resource)	mining of basalt (part 1)		total amount of mined basalt 1997: 250'000 t; the losses by dust emissions have been neglected.	resource	in ground			Basalt, in ground	1.00E+00	kg	estimation	1	1.26	(3,4,2,1,3,nA);	
land occupation, mine [m2a]			total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area; assumption: After 10 years the used land is recultivated	resource	land			Occupation, mineral extraction site	7.98E-03	m2a	Anonymous (1997)	1	1.64	(4,4,2,1,3,3);	
land transformation mine [m2]			total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, to mineral extraction site	7.98E-04	m2	Anonymous (1997)	1	2.11	(4,4,2,1,3,3);	
land transformation (from forest to mine)			total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, from forest	7.98E-04	m2	Anonymous (1997)	1	2.11	(4,4,2,1,3,3);	
land transformation mine [m2]			total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, from mineral extraction site	7.98E-04	m2	Anonymous (1997)	1	2.11	(4,4,2,1,3,3);	
land transformation (from mine to unknown)			total area: 200'000 m2 (20 ha), from this 400 m2 are quarry facilities and 199'600 m2 mining area	resource	land			Transformation, to unknown	7.98E-04	m2	Anonymous (1997)	1	2.11		
Infrastructure (mine) [unite]				construction materials	additives	Yes	CH	mine, limestone	8.00E-11	unit	Anonymous (1997)	1	3.28		
blasting [kg explosive]			amount of blasting copied from module "limestone, at mine"	construction processes	civil engineering	No	RER	blasting	7.73E-05	kg	data of KFN for the year 2001	1	1.32	(4,nA,3,1,3,nA);	
total machines used in mine for drilling, mining and transports			includes infrastructure and emissions	construction processes	machinery	No	GLO	diesel, burned in building machine	4.14E-02	MJ	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	

Fig. 4.3 Flows for “basalt, at plant” and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent 2000							Uncertainty					
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
petrol in cars	→	mining of basalt (part 2)			includes emissions	transport systems	road	No	CH	operation, passenger car	6.78E-06	km	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
combustion light fuel oil	→				used for heating of different buildings	oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	3.13E-04	MJ	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
electricity	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.21E-03	kWh	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Water	→					resource	in water			Water, lake	8.40E-07	m3	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
oil for motor, gear and hydraulics	→					chemicals	organics	No	RER	lubricating oil, at plant	5.38E-05	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Lacquer	→					paintings	production	No	RER	alkyd paint, white, 60% in solvent, at plant	3.00E-06	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Chemicals	→					chemicals	organics	No	GLO	chemicals organic, at plant	3.60E-07	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
Oil and lubricants	→					chemicals	organics	No	RER	lubricating oil, at plant	1.60E-07	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);	
			→	Waste heat		Same amount as electricity consumption	air	high population density			Heat, waste	2.24E-02	MJ		1	1.24	(1,4,2,1,3,3);
			→	Dust to air [kg]		Assumption: 10%wt of total	air	low population density			Particulates, < 2.5 um	1.60E-03	kg	Anonymous (1997)	1	3.11	(4,4,2,1,3,3);
			→	Dust to air [kg]		Assumption: 40%wt of total	air	low population			Particulates, > 2.5 um, and <	6.40E-03	kg	Anonymous (1997)	1	2.33	(4,5,2,3,4,3);
			→	Dust to air [kg]		Assumption: 50%wt of total	air	low population			Particulates, > 10 um	8.00E-03	kg	Anonymous (1997)	1	1.88	(4,5,2,3,4,3);
			→	waste paper			waste	municipal	No	CH	disposal, paper, 11.2% water, to	5.20E-07	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	waste,			waste	hazardous	No	CH	disposal, paint remains, 0%	4.88E-06	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	remaining waste			waste	municipal	No	CH	disposal, municipal solid waste, 22.9% water, to municipal	6.14E-05	kg	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	effluents			waste management	incineration	No	CH	treatment, sewage, from residence. to wastewater	9.60E-08	m3	Anonymous (1997)	1	1.24	(1,4,2,1,3,3);
			→	basalt, at mine		amount 1997: 250000 t	management construction materials	additives	No	RER	basalt, at mine	1.00E+00	kg		1	1.00	

Fig. 4.3 Flows for “basalt, at plant” and their representation in the ecoinvent database

4.3 Rock Wool

4.3.1 Process Description

The following summary of rock wool processing has been taken mainly from the homepage of the company Flumroc in Switzerland.⁸

The stone is liquefied in a smelting furnace at approximately 1500°C, spun into fibres, impregnated to provide water repellence then hardened again in a special furnace. The stone wool is then cut to size in boards or mats on the sawing line. The finished insulation products are packaged automatically and conveyed directly for transport to customers or the warehouse. Since the raw materials are smelted in a cupola furnace, coke is the most suitable energy source. Tab. 4.4 shows schematically the production process.

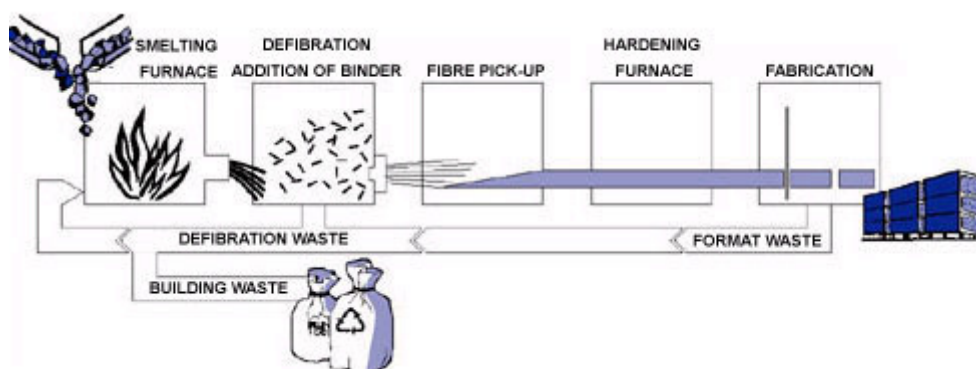


Fig. 4.4 Rock wool production process by Flumroc in Switzerland

A more detailed description of the production process of rock wool can be found in (EPA (1998)):

The first step in the process involves melting the mineral feed. The raw material (rock and sometimes slag) is loaded into a cupola melting furnace in alternating layers with coke at weight ratios of about 5 to 6 parts mineral to 1 part coke. As the coke is ignited and burned, the mineral charge is heated to the molten state at a temperature of 1300° to 1650°C. Combustion air is supplied through tuyeres located near the bottom of the furnace. Process modifications at some plants include oxygen enrichment and the use of natural gas auxiliary burners to reduce coke consumption.

The molten mineral charge exits the bottom of the cupola in a water-cooled trough and falls onto a fiberization device. Two variations of fiberization methods exist. The Powell process uses groups of rotors revolving at a high rate of speed to form the fibres. Molten material is distributed in a thin film on the surfaces of the rotors and then is thrown off by centrifugal force. As the material leaves the surface, small globules develop and form long, fibrous tails as they travel horizontally. Air or steam may be blown around the rotors to assist in fiberizing the material. A second fiberization method, the Downey process, uses a spinning concave rotor with air or steam attenuation. Molten material is distributed over the surface of the rotor, from which it flows up and over the edge and is captured by a high-velocity stream of air or steam.

During the spinning process, not all globules developed are converted into fibre (defibration waste). The non fiberized globules that remain are referred to as “shot”. In raw mineral wool, as much as half of the mass of the product may consist of shot. Shot is usually separated from the wool by gravity immediately following fiberization and then remelted. The produced amount of shot is always about the same and therefore treated as in an internal closed loop. Thus it is not part of the “final” product.

⁸ <http://www.flumroc.ch/txtengl/frame.htm>, access date: August 2002

Depending on the desired product, various chemical agents may be applied to the newly formed fibre immediately following the rotor. In almost all cases, oil is applied to suppress dust and, to some degree, anneal the fibre. This oil can be either a proprietary product or a medium weight fuel or lubricating oil. If the fibre is intended for use as loose wool or bulk products, no further chemical treatment is necessary. If the mineral wool product is required to have structural rigidity, as in batts and industrial felt, a binding agent is applied with or in place of the oil treatment. This binder is typically a phenol-formaldehyde resin that requires curing at elevated temperatures. Both the oil and the binder are applied by atomizing the liquids and spraying the agents to coat the airborne fibre.

After formation and chemical treatment, the fibre is collected in a blow chamber. Resin- and/or oil-coated fibres are drawn down on a wire mesh conveyor by fans located beneath the collector. The speed of the conveyor is set so that a wool blanket of desired thickness can be obtained.

Mineral wool containing the binding agent is carried by conveyor to a curing oven, where the wool blanket is compressed to the appropriate density and the binder is baked. Hot air at a temperature of 150° to 320°C is forced through the blanket until the binder has set. Curing time and temperature depend on the type of binder used and the mass rate through the oven. A cooling section follows the oven, where blowers force air at ambient temperatures through the wool blanket.

To make batts and industrial felt products, the cooled wool blanket is cut longitudinally and transversely to the desired size. Some insulation products are then covered with a vapour barrier of aluminium foil or asphalt-coated kraft paper on one side and untreated paper on the other side. The cutters, vapour barrier applicators, and conveyors are sometimes referred to collectively as a bat machine. Those products that do not require a vapour barrier, such as industrial felt and some residential insulation batts, can be packed for shipment immediately after cutting.

Loose wool products consist primarily of blowing wool and bulk fibre. For these products, no binding agent is applied, and the curing oven is eliminated. For granulated wool products, the fibre blanket leaving the blow chamber is fed to a shredder and pelletizer. The pelletizer forms small, 1-inch-diameter pellets and separates shot from the wool (shot is reused by feeding the furnace). A bagging operation completes the processes. For other loose wool products, fibre can be transported directly from the blow chamber to a baler or bagger for packaging.

Fig. 4.5 shows a detailed flow scheme for the manufacture of rock wool:

The data in ecoinvent are not separated by the mentioned sub-processes. In co-operation with Flumroc the processes packing and administration are included in a separate module called “rock wool, packed, at plant”.

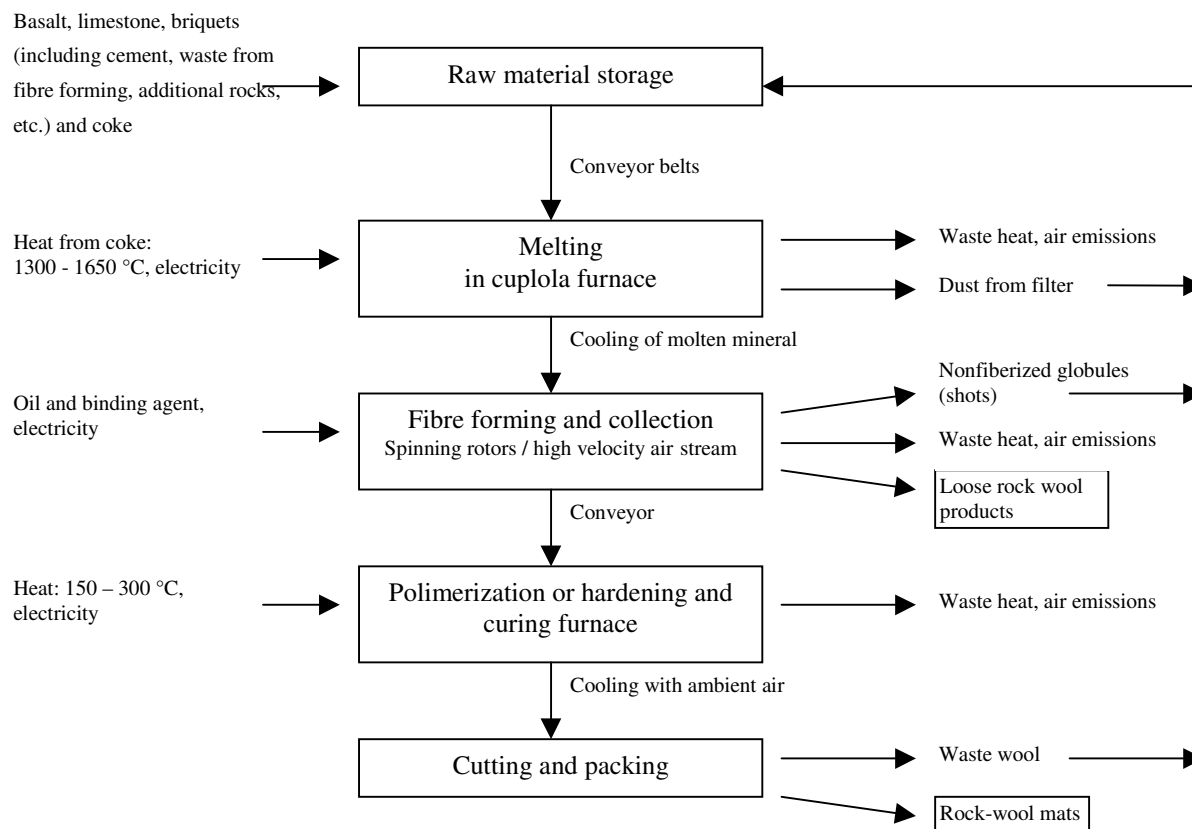


Fig. 4.5 Flow scheme of rock wool production process

4.3.2 Life Cycle Inventories

Rock wool, at plant

The dataset “rock wool, at plant” does not include any data for administration and solid waste. It is basis for the module “rock wool, packed, at plant” which includes in addition administration and solid waste. The infrastructure including its land-use, the buildings and the machines is described in the module “rock wool plant”.

The yearly output of Flumroc for the year 2007 in Switzerland is 62'756'300 kg. This includes the main products as mats and boards (61'798'500 kg) and fine granulated rock wool (957'800 kg).

The emissions of the melting furnace are reduced by carbon monoxide post combustion with recuperation of energy. Many emissions such as ammoniac, formaldehyde, volatile organic compounds, carbon monoxide and phenol are strongly reduced.

The effect of the monoxide post combustion is the conversion of carbon monoxide to carbon dioxide what raises the emission of carbon dioxide by about 25%. The nitrogen oxide emissions are also raised by the transformation of ammonia.

The allocation of the dust to the different categories of particulate sizes is made based on the “Emission Factor Documentation for Mineral Wool Manufacturing (EPA (1998)). The energy balance is taken from the environmental report of Flumroc (Flumroc (2008)) and shown in Tab. 4.6, a rough mass balance is shown in Tab. 4.7.

Tab. 4.6 Energy balance for Flumroc for the year 2007 (Flumroc (2008))

Energetic production input (Group)	Type	[KWh/a]	[kg/a]	gross calorific value [MJ/kg] (*)
Fossil fuels	Coke	67879310	8340120	29.3
	Heating oil	14827243	1250072	42.7
	Diesel	537014	45275	41.9
	Propane gas	2011965	156269	46.4
Regenerative fuel	Offal timber	273611	50000	19.7
Electric energy		14775900		
Total (without electric energy)		84789143		
Total (without electric energy and heat losses to river "Seez")		82525143		
Energetic production output (Group)				
Energy recuperation	Waste heat to internal room heating	4'000'000		
	Waste heat to cupola	1'315'000		
Water emissions	Waste heat cooling water in Seez	807'000		
Air emissions	Waste heat cupola	5'712'000		
	Waste heat cooling tank	21'139'000		
	Waste heat collecting tank	22'774'000		
	Waste heat hardening furnace	3'817'000		
	Waste heat cooling zone	5'177'000		
	Waste heat from hot iron / slag	3'273'000		
	Waste heat from basalt smelting	5'952'000		
	Waste heat product	1'154'000		
	Waste heat spinning wheel cooling water	6'732'000		
	Waste heat packing	1'157'000		
	Waste heat starting up and stopping	918'000		
	Convection / solar radiation	606'000		
	Waste heat from deacidification CO ₂ (Dolomite)	1'810'000		
	Waste heat from drying raw material	2'756'000		
Energy recuperation	District heating network	-1'116'424		
Difference	Not identified losses	17'274'320		
Total		105256896		

(*) energy content used in environmental report of Flumroc for 2007 (Flumroc (2008))

Tab. 4.7 Rough mass balance for rock wool production by Flumroc for the year 2007 (Flumroc (2008))

Input	[kg/a]
Mineral resources	
Dolomite	3'667'300
Basalt	42'718'070
Additives for briquette	8'684'100
Cement for briquette	9'419'680
Fixing agents	
Phenol resin	1'400'332
Formaldehyde resin	3'902'127
Urea resin	864'450
Ammonia	187'439
Ammonium hydrogen carbonate	91'550
Waterproofing oil	173'566
Calcium hydroxide	63'000
Output	
Main product	
Rock wool	62756300
By-product (not assessed)	
Raw iron	1711000

The ecoinvent database meta information of “rock wool, at plant” (Tab. 4.8) includes all important and necessary information to use the data correctly. Fig. 4.6 shows all in- and output flows of the product “rock wool, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Flumroc (2002), Flumroc (2007) and personal communication) and the standard deviation for each datum.

Tab. 4.8 ecoinvent meta information for the product "rock wool, at plant"

Name	rock wool, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.1
Included Processes	Included processes: melting, fibre forming & collecting, hardening & curing furnace, and internal processes (workshop, etc.). Transport of raw materials and energy carrier for furnace are also included. Not included are administration, packing and infrastructure.
Amount	1
Local Name	Steinwolle, ab Werk
Synonyms	
General Comment to reference function	This product is not available on the market as it is not packed. It has been defined for internal use in this database as input for the packaging module. The mass balance does not come out as some in- and outputs are included in the module "rock wool, packed, at plant". This module shall not be used to describe rock wool in other inventories.
Start Date	2000
End Date	2007
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Flumroc AG). For some exchanges RER-modules have been used as proxy
Technology text	The company works on a technically very high level. The fuel for the melting process is coke. The furnace is a cupola melting furnace. The emissions are abated by post-combustion with energy recuperation. The partition of the total dust emission to the categories of particulate sizes is based on an uncontrolled mineral wool cupola (EPA 1998) which only partly represents this case.
Representativeness [%]	40
Production Volume	112500t/a
Sampling Procedure	environmental report
Extrapolations	see geography and technology
Uncertainty Adjustments	Assumption: the production of Flumroc of 45'000 t per year is representative for the total consumed rock wool products in Switzerland.

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
dolomite	Rock wool production part 1				construction materials	others	No	RER	dolomite, at plant	5.84E-02	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
basalt					construction materials	additives	No	RER	basalt, at mine	6.81E-01	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
cement for briquet					construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	1.50E-01	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
additives for briquet: limestone powder (local rock)				60% of total additives	construction materials	others	No	CH	limestone, milled, packed, at plant	8.30E-02	kg	Flumroc (2007)	1	1.12	(3,3,1,1,1,1);
additives for briquet: bauxite powder				40% of total additives	metals	extraction	No	GLO	bauxite, at mine	5.54E-02	kg	Flumroc (2007)	1	1.12	(3,3,1,1,1,1);
recycling rock wool material from consumers				not balanced because it is internally recycled. Preparation (e.g. milling) is not considered						6.36E-03	kg	Flumroc (2007)			
coke for melting				input for heating (melting process), calculated with a gross calorific value of 29.3 MJ/kg	hard coal	fuels	No	RER	hard coal coke, at plant	5.02E+00	MJ	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
natural gas for heating and furnace					natural gas	fuels	No	CH	natural gas, high pressure, at consumer	1.08E+00	MJ	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);
Oxygen					chemicals	inorganics	No	RER	oxygen, liquid, at plant	5.66E-05	kg	Flumroc (2007)	1	1.22	(1,3,1,2,3,1);
diesel oil in building machines				total diesel consumption used for different internal transports; "diesel, burned in building machine" used as proxy	construction processes	machinery	No	GLO	diesel, burned in building machine	4.15E-02	MJ	Flumroc (2007)	1	1.22	(1,3,1,2,3,1);
waste wood for heating				wood used for inflaming the cupola melting furnace, not balanced						3.19E-04	kg	Flumroc (2007)			
electricity consumption (exkl. electricity consumption for packing)					electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.91E-01	kWh	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
fixing agent: phenol	Rock wool production part 2				chemicals	organics	No	RER	phenol, at plant	2.23E-02	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
fixing agent: formaldehyd					chemicals	organics	No	RER	formaldehyde, production mix, at plant	6.22E-02	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
fixing agent: urea					chemicals	organics	No	RER	urea, as N, at regional storehouse	1.38E-02	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
fixing agent: ammonia					chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	2.99E-03	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
fixing agent: ammonium bicarbonate					chemicals	inorganics	No	RER	ammonium bicarbonate, at plant	1.46E-03	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
fixing agent: waterproofing oil					chemicals	organics	No	RER	lubricating oil, at plant	2.77E-03	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
silane					construction materials	binder	No	CH	lime, hydrated, packed, at plant	2.06E-04	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
fixing agent: calcium hydroxide					construction materials	binder	No	CH	lime, hydrated, packed, at plant	1.00E-03	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
process water					resource	in water			Water, well, in ground	4.82E-03	m3	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);
Tap water					water supply	production	No	CH	tap water, at user	1.37E-01	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
material: aluminium foil for coating					metals	extraction	No	RER	aluminium, production mix, wrought alloy, at plant	5.67E-04	kg	Flumroc (2007)	1	1.22	(1,3,1,2,3,1);
process: coating with aluminium foil					metals	processing	No	RER	sheet rolling, aluminium	5.67E-04	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
lamination with paper					paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	2.88E-04	kg	Flumroc (2007)	1	1.22	(1,3,1,2,3,1);
glass fleece					insulation materials	production	No	CH	glass wool mat, at plant	2.88E-04	kg	Flumroc (2007)	1	1.22	(1,3,1,2,3,1);
fireproof stones				scone brick , total yearly amount: 2.232m3, density 2.3 t/m3 -> 5.1336 t/a	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	1.11E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
fireproof stones				chamotte grain, total yearly amount: 10'000kg	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	2.21E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
fireproof stones				burnt clay, total yearly amount: 12.24m3, density 1.7 t/m3 -> 20.808 t/a	construction materials	bricks	No	DE	refractory, fireclay, packed, at plant	4.61E-04	kg	Flumroc (2002)	1	1.22	(1,3,1,1,3,1);
Infrastructure rock wool plant				total produced amount per unit (50 years): 2257'988'550 kg	insulation materials	production	Yes	CH	rock wool plant	4.43E-10	unit	Flumroc (2007)	1	3.00	(1,3,1,1,1,1);
train transports of coke to furnace				coke: 50% from France (410 km) and 50% from Italy (535 km); average distance: 500 km	transport systems	train	No	RER	transport, freight, rail	8.57E-02	tkm	additional written information from Flumroc for the year 2000	1	2.00	(1,3,1,2,1,1);
road transports				transport of all input materials (including inputs for packaging) with lorry	transport systems	road	No	CH	transport, lorry >28t, fleet average	7.87E-02	tkm	Flumroc (2007)	1	2.05	(1,3,1,1,3,1);
train transports				transport of all input materials (including inputs for packaging) except coke with train	transport systems	train	No	RER	transport, freight, rail	6.73E-02	tkm	Flumroc (2007)	1	2.00	(1,3,1,2,1,1);

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont)

General Flow information					Representation in ecoinvent							Uncertainty information				
Input		Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Rock wool production part 3	→		Waste heat to air from all sources excluding electricity uses		total	air	low population density			Heat, waste	1.58E+00 MJ		Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
	→		Waste heat to water from all sources excluding electricity uses		total	air	low population density			Heat, waste	1.29E-02 MJ		Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
	→		dust to air [kg], > 10 um		share for uncontrolled mineral wool cupolas (6% of total dust) from EPA-report	air	low population density			Particulates, > 10 um	8.48E-06 kg		Flumroc (2007) and EPA (1998) (Mineral Wool Manufacturing)	1	1.78	(1,3,1,1,4,1);
	→		dust to air [kg], < 2.5 um		share for uncontrolled mineral wool cupolas (90% of total dust) from EPA-report	air	low population density			Particulates, < 2.5 um	1.27E-04 kg		Flumroc (2007) and EPA (1998) (Mineral Wool Manufacturing)	1	3.23	(1,3,1,1,4,1);
	→		dust to air [kg], > 2.5 um, and < 10um		share for uncontrolled mineral wool cupolas (4% of total dust) from EPA-report	air	low population density			Particulates, > 2.5 um, and < 10um	5.66E-06 kg		Flumroc (2007) and EPA (1998) (Mineral Wool Manufacturing)	1	2.24	(1,3,1,1,4,1);
	→		ammonia		Total measured emission from melting process	air	low population density			Ammonia	6.68E-04 kg		Flumroc (2007)	1	1.50	(1,3,1,1,1,1);
	→		lead		total measured emissions from Cupola furnace	air	low population density			Lead	5.89E-08 kg		Flumroc (2007)	1	5.00	(1,3,1,1,1,1);
	→		cadmium		total measured emissions from Cupola furnace	air	low population density			Cadmium	9.44E-09 kg		Flumroc (2007)	1	5.00	(1,3,1,1,1,1);
	→		hydrochloric acid		total measured emissions: 804.4kg/a	air	low population density			Hydrogen chloride	1.80E-05 kg		Flumroc (2007)	1	1.50	(1,3,1,1,1,1);
	→		chromium		total measured emissions from Cupola furnace	air	low population density			Chromium	4.73E-09 kg		Flumroc (2007)	1	5.00	(1,3,1,1,1,1);
	→		hydrogen fluoride		total measured emissions:113.4kg/a	air	low population density			Hydrogen fluoride	9.33E-08 kg		Flumroc (2007)	1	1.50	(1,3,1,1,1,1);
	→		formaldehyde		total measured emissions after	air	low population density			Formaldehyde	8.25E-05 kg		Flumroc (2007)	1	2.05	(1,3,1,1,3,1);

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
Rock wool production part 4	→	VOC		total measured emissions from different processes	air	low population density			NMVOC, non-methane volatile organic compounds, unspecified origin	4.87E-04	kg	Flumroc (2007)	1	2.24	(1,3,1,1,4,1);
	→	carbon dioxide		total calculated emissions	air	low population density			Carbon dioxide, fossil	5.35E-01	kg	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);
	→	carbon monoxide		total measured emissions	air	low population density			Carbon monoxide, fossil	4.14E-05	kg	Flumroc (2007)	1	5.00	(1,3,1,1,1,1);
	→	copper		total measured emissions from Cupola furnace	air	low population density			Copper	0.00E+00	kg	Flumroc (2007)	1	5.00	(1,3,1,1,1,1);
	→	nickel		total measured emissions from Cupola furnace	air	low population density			Nickel	0.00E+00	kg	Flumroc (2007)	1	5.00	(1,3,1,1,1,1);
	→	phenol		total measured emissions	air	low population density			Phenol	1.39E-04	kg	Flumroc (2007)	1	1.50	(1,3,1,1,1,1);
	→	sulfur dioxide		total measured emissions	air	low population density			Sulfur dioxide	4.13E-03	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
	→	hydrogen sulfide		total measured emissions	air	low population density			Hydrogen sulfide	0.00E+00	kg	Flumroc (2007)	1	1.50	(1,3,1,1,1,1);
	→	nitrogen oxide		total measured emissions	air	low population density			Nitrogen oxides	9.59E-04	kg	Flumroc (2007)	1	1.50	(1,3,1,1,1,1);
	→	zinc		total measured emissions from Cupola furnace	air	low population density			Zinc	2.51E-08	kg	Flumroc (2007)	1	5.00	(1,3,1,1,1,1);
	→	pig iron		not balanced						2.73E-02	kg	Flumroc (2007)			
	→	alu foil, to recycling		not balanced						7.33E-06	kg	Flumroc (2007)			

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent										Uncertainty information		
Input		Process Name		Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
waste to municipal incineration	→	Rock wool production part 5					waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.37E-03	kg	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);
waste solvents to treatment	→						waste management	hazardous waste incineration	No	CH	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	7.82E-06	kg	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);
waste oil to treatment	→						waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	3.43E-05	kg	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);
waste water to wastewater treatment	→						waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	7.08E-04	m3	Flumroc (2007)	1	1.22	(1,3,1,1,3,1);
								insulation materials	production	No	CH	rock wool, at plant	1.00E+00	kg	Flumroc (2007)		

Fig. 4.6 Flows for "rock wool, at plant" and their representation in the ecoinvent database (cont.)

Rock wool, packed, at plant

The ecoinvent database meta information of “rock wool, packed, at plant” (Tab. 4.9) includes all important and necessary information to use the data correctly. Fig. 4.7 shows all in- and output flows of the product “rock wool, packed, at plant”, the allocation to the modules in ecoinvent, the sources (mainly Flumroc (2002) and personal communication) and the standard deviation for each datum.

The inventories for packing and administration have been added to the module “rock wool, at plant”. For the amount of the different packaging materials (packaging film, pallet, etc.) a loading weight per pallet of 250 kg was assumed. The area of a pallet is 1m² (120×80cm)⁹, a load height of 2.5m is assumed. This gives a loading volume of 2.5m³. The average density of the packed rock wool product is assumed to be 100 kg/m³. Based on volume and the density the amount of the different packaging materials is calculated.

The machines and packing materials (pallets, cardboard, packaging film) and the waste treatment services necessary for packing are included in this module.

Tab. 4.9 ecoinvent meta information for the product “rock wool, packed, at plant”

Name	rock wool, packed, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.1
Included Processes	Includes mechanical packing and the administration of the rock wool factory.
Amount	1
Local Name	Steinwolle, verpackt, ab Werk
Synonyms	mineralisch//Isolation//Dämm//matte//platte//mineral//insulation//mat//board
General Comment to reference function	With functional unit kg, this module can be used for rock wool products with different densities. For the calculation of the packaging material demand, 100 kg/m ³ are assumed (volume: 2.4m ³ , load weight: 250 kg). Inputs and outputs are not balanced in the unit process since some are included in the unit process “rock wool, at plant”.
Start Date	2000
End Date	2007
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Flumroc AG). For some exchanges RER-modules have been used as proxy
Technology text	The company Flumroc AG works on a technically high level. The packing and loading process is automated.
Representativeness [%]	40
Production Volume	112500t/a
Sampling Procedure	environmental report
Extrapolations	see geography and technology
Uncertainty Adjustments	Assumption: the production of Flumroc of 45'000 t per year is representative for the total consumed rock wool products in Switzerland.

⁹ <http://www.rz.fh-ulm.de/labore/pwllabor/MARKTSP/PALSYSPalsys1.htm#EUR-Flachpalette>

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
rock wool, loose, at plant	packaging and administration of rock wool mat				insulation materials	production	No	CH	rock wool, at plant	1.00E+00	kg	Flumroc (2007)	1	1.07	(1,3,1,1,1,1);
infrastructure packing and loading of rock wool products				weight of packing- and loading-machines: assumption 2t (assumed lifetime: 20 years)	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	2.21E-06	kg	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.23	(5,3,1,2,1,1);
pallets				Weight of pallet: 24.6 kg	wooden materials	processing	No	RER	EUR-flat pallet	1.07E-03	unit	Flumroc (2007)	1	1.22	(1,3,1,2,3,1);
cardboard, at plant					paper & cardboard	cardboard & corrugated board	No	RER	corrugated board, mixed fibre, single wall, at plant	7.39E-04	kg	Flumroc (2007)	1	1.22	(1,3,1,2,3,1);
packaging film, LDPE					plastics	processing	No	RER	packaging film, LDPE, at plant	5.88E-03	kg	Flumroc (2007)	1	1.07	(1,3,1,2,1,1);
propan gas liquid				for heating of shrinking foil	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.60E-01	MJ	Flumroc (2002)	1	1.22	(1,3,1,2,3,1);
energy consumption for packing and loading				based on packing of lime products	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.09E-03	kWh	data of KFN for the year 2001	1	1.51	(1,2,1,1,4,1);
diverse material				mostly paint for coloring	paintings	production	No	RER	acrylic dispersion, 65% in H2O, at plant	5.55E-03	kg	Flumroc (2007)	1	2.01	(1,3,1,2,5,1);
				not balanced						2.64E-04	kg	Flumroc (2007)			
				not balanced						9.82E-05	kg	Flumroc (2007)			
					insulation materials	production	No	CH	rock wool, packed, at plant	1.00E+00	kg	Flumroc (2002)			
		PE foil, to recycling													
		Paper, to recycling													
		packed rock wool products, at plant													

Fig. 4.7 Flows for “rock wool, packed, at plant” and their representation in the ecoinvent database

Rock wool plant

The data are all based on personal communication and plans from Flumroc AG. The production volume is about 45'000 tons per year. It is assumed that the production site will have a life span of 50 years. This includes the land occupation and the building infrastructure. The life span of the machines is estimated to be 20 years. The life span of the conveyor belt is estimated to be 10 years as it is assumed that the belt is worn faster.

The infrastructure of all „packing and loading“ modules of building materials include only the specific machines and no buildings. The reason is to avoid double counting since in most cases the packing and loading process takes place in the same building as the production process.

The ecoinvent database meta information of “rock wool plant” (Tab. 4.10) includes all important and necessary information to use the data correctly. Fig. 4.8 shows all in- and output flows of the product “rock wool plant”, the allocation to the modules in ecoinvent, the sources (mainly Flumroc (2002) and personal communication) and the standard deviation for each datum.

Tab. 4.10 ecoinvent meta information for the product “rock wool plant”

Name	rock wool plant
Location	CH
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	Includes the area of the infrastructure (buildings, roads, etc.), the buildings and the machines used for the rock wool production
Amount	1
Local Name	Steinwollewerk
Synonyms	Fabrik//Betrieb//company//firm//mineralisch//Isolation//Dämm//matte//platte//mineral//insulation//mat//board
General Comment to reference function	The production volume of the plant is 45'000 t of rock wool products per year
Start Date	2000
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data are only from one company in Switzerland (Flumroc AG). For some exchanges RER-modules have been used as proxy
Technology text	The data for the infrastructure correspond to high level of technology. The furnace is a cupola melting furnace. The emissions are abated by post-combustion with energy recuperation. As no description of the used machines exists, a total weight for these machines has been estimated and the module "machine, heavy, unspecified" used.
Representativeness [%]	40
Production Volume	112500t/a
Sampling Procedure	personal communication
Extrapolations	see Geography
Uncertainty Adjustments	Assumption: the production of Flumroc of 45'000 t per year is representative for the total consumed rock wool products in Switzerland.

General Flow information					Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output		Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
land occupation for built area	infrastructure rock wool plant			86'250m2, assumption time occupation: 50 years	resource	land			Occupation, industrial area, built up	4.31E+06	m2a	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	1.50	(1,2,1,1,1,1);
land transformation to built area				total area with roof: 24'000m2	resource	land			Transformation, to industrial area, built up	2.40E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
land transformation to roads				total area with roads: 22'700m2	resource	land			Transformation, to traffic area, road network	2.27E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
land transformation to green area				total green area: 39'550m2	resource	land			Transformation, to industrial area, vegetation	3.96E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
land transformation from unknown					resource	land			Transformation, from unknown	8.63E+04	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	2.00	(1,2,1,1,1,1);
buildings multi storage				65% of total area built: 15'600 m2 , assumption average height: 12 m	construction processes	buildings	Yes	RER	building, multi-storey	1.87E+05	m3	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.06	(3,3,1,2,3,1);
buildings hall				35% of total area built: 8'400 m2	construction processes	buildings	Yes	CH	building, hall	8.40E+03	m2	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.06	(3,3,1,1,3,1);
Infrastructure cupola melting furnace				infrastructure (machines) for the melting process; total weight without fireproof stones: assumed 30 t; life-span assumption: 20 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	7.50E+04	kg	Written information M. Mebold, Flumroc, 02.09.02	1	3.05	(1,3,1,2,3,1);
Infrastructure rest				rest of infrastructure (machines) for the whole process: weight: 1'500t (mostly steel); life-span assumption: 20 years	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.75E+06	kg	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.05	(1,3,1,2,3,1);
infrastructure conveyor belt				length 4000 m, width 1.5 m, life-span 20 years; corresponding module in ecoinvent has a width of 3 m and life-span of 10 years	construction processes	machinery	Yes	RER	conveyor belt, at plant	1.00E+04	m	personal communication Flumroc, Mrs. Mebold, 16.09.02	1	3.05	(1,3,1,2,3,1);
		rock wool plant		total produced amount per unit (50 years): 2'257'988'550 kg	insulation materials	production	Yes	CH	rock wool plant	1.00E+00	unit				

Fig. 4.8 Flows for “rock wool plant” and their representation in the ecoinvent database

4.3.3 Comparison to other Data

Differences to other sources of data have been found. The following table shows the differences of the data according to four sources („Rock wool“ in Germany¹¹, the SIA declaration¹⁰, (Mötzl et al. (2000)) and the data of this study) and tries to explain the reasons for the differences:

Tab. 4.11 Comparison between different existing data and the data under study

Indicator	Data from company “Rock wool” Germany ¹¹	Data from Eyerer & Reinhardt (2000)	This study, calculations and information by Flumroc	Mötzl et al. 2000, Ökologie der Dämmstoffe
Primary energy (nuclear & fossil)	13 – 20 MJ/kg output ¹⁾	18.66 MJ / kg ²⁾	20.0 MJ /kg output ¹⁰ (unit process) ³⁾	22.12 MJ /kg output ⁴⁾
Dust and gaseous emissions in the production process (cupola melting furnace)	0.01 kg / kg output ⁵⁾		0.004 kg / kg output (unit process) ⁵⁾	
CO ₂ Emissions	0.02 kg / kg output ⁶⁾		0.8 kg / kg output ⁷⁾	
¹⁾ Includes the energy for mining, transport of the raw materials and for processing. The included energy carriers and their shares are not given. ²⁾ Based on an insulating board with a density of 160 kg/m ³ . Data of one major company in Germany. The system boundary is not given (not clear whether mining is included or not) ³⁾ Includes all energy carriers, the energy for the preliminary stages of the materials and its transports ⁴⁾ Only information provided: Includes renewable and non renewable primary energy of one producer in Austria ⁵⁾ No further information available ⁶⁾ Assumption: without post-combustion of carbon monoxide; Beside coke the energy carrier of the different companies producing rock wool have often gas instead of heating oil and there are still some companies which melt with an electric furnace ⁷⁾ Includes post-combustion of carbon monoxide				

In general the system boundaries of other studies are not transparent and therefore a comparison is difficult. Nevertheless all data concerning the primary energy vary between 13 and 22 MJ/kg. This shows that there is a quite substantial variation within the standards of the different production companies and/or within the system boundaries considered in different studies.

4.3.4 Data Quality

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

In general the quality of the dataset “Rock wool, at plant” is high although the share of the market in Switzerland is only 40%. It is assumed that the companies delivering to the Swiss market are located nearby and have a comparable high technology standard. The company Flumroc has been collecting data for several years and publishes them in their environmental reports at regular intervals.

¹⁰ <http://www.sia.ch/d/praxis/bauprodukte/index.cfm?Lang=d&ID=1095711>

¹¹ <http://www.rockwool.de/graphics/RW-CSE/CSE/pdf/Schwerpunkte/oeko-mappe.pdf>

4.4 Foamglas

4.4.1 System characterisation

Product

The inventory data for FOAMGLAS® is based on production values for 2005 and represents the product T4 WDS which is one of a number of different FOAMGLAS® products manufactured in Tessenderlo, Belgium with a density of 110 kg/m³. In terms of quantity, T4 WDS is the most important FOAMGLAS® product and its manufacturing process is representative for FOAMGLAS® products. 95% of T4 WDS sold on the European market is produced in Tessenderlo. A negligible 5% is produced at the site in Germany.

Due to the market monopoly of FOAMGLAS® products in Europe, the technology employed at the production site in Belgium is the de facto, current technology for producing FOAMGLAS® in Europe. An overview of the raw materials used to produce FOAMGLAS® is given in Tab. 4.12.

Tab. 4.12 Input of raw materials and energy input for 1kg of a FOAMGLAS® product

Raw materials	
Recycled glass	8.14E-01 kg
Feldspar	2.58E-01 kg
Iron oxide	2.77E-02 kg
Sodium nitrate	2.46E-03 kg
Salt cake	5.38E-03 kg
Carbon black	6.15E-03 kg
Energy	
Natural gas	1.16E+01 MJ
Electricity	1.83E+00 kwh

System boundary and process description

The system boundary considered for this inventory is shown in Fig. 4.9.

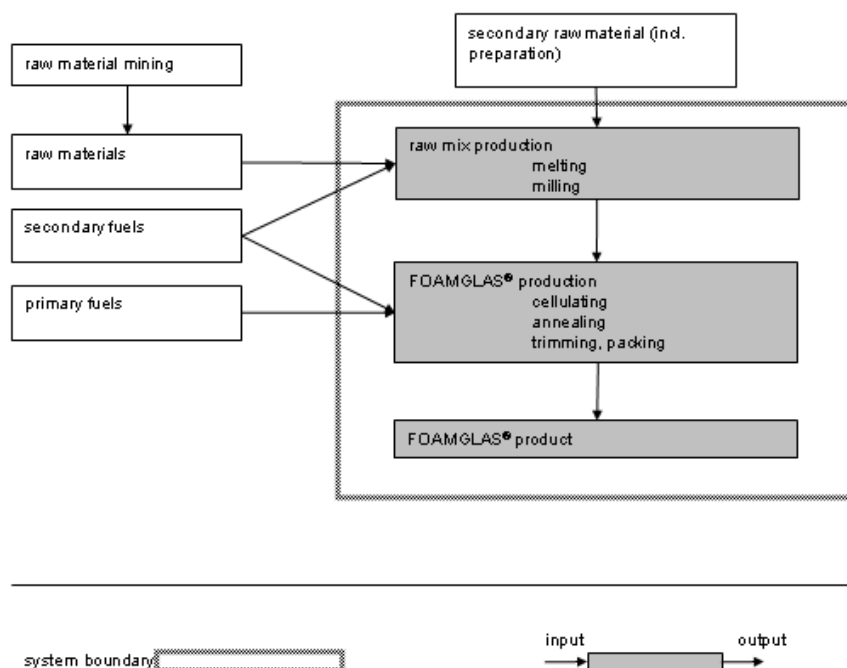
FOAMGLAS® Manufacturing

Fig. 4.9 System boundaries considered of the FOAMGLAS® production process

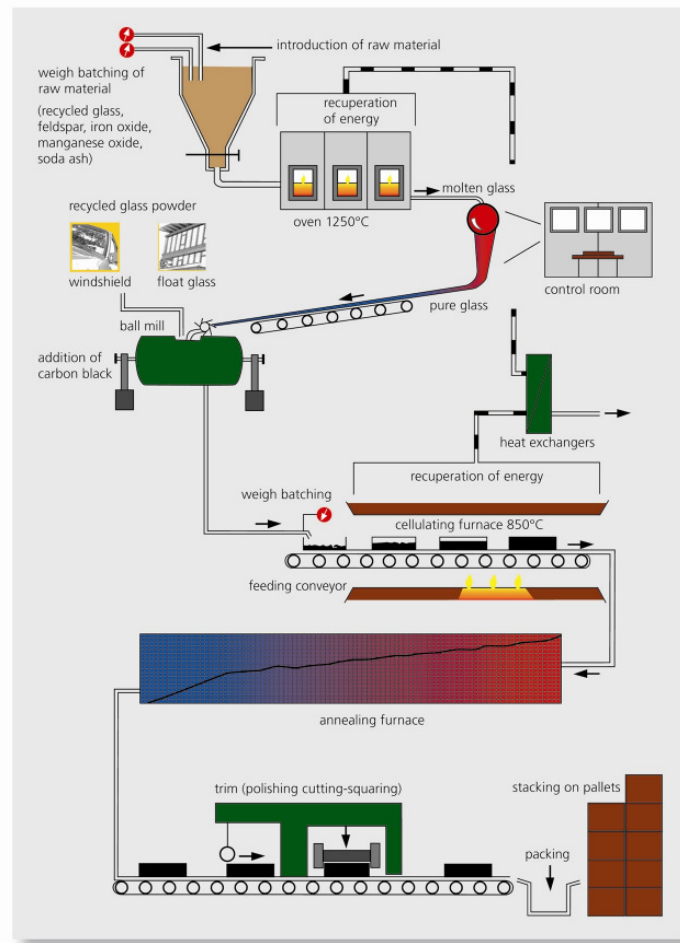
Over the past several years, the use of sand has been completely, if gradually, superseded by the use of recycled glass. Today, more than half of the recycled glass employed is directly injected in the ball mill (milling) without melting, allowing for a large reduction in energy input.

Details of the present manufacturing process are schematically shown in Fig. 4.10. The first step of the FOAMGLAS® production process consists of the raw mix production, a raw material conditioning process which includes the process-storage of raw material in a silo, weigh batching of raw material, melting of raw material in an oven and milling in a ball mill. Raw materials such as: recycled glass, feldspar, iron oxide, salt cake, soda ash, manganese oxide, sodium carbonate and sodium nitrate are added to the melting furnace and heated to approximately 1250 °C. The melting furnace is an electric furnace equipped with glass melting molybdenum electrodes. The molten glass flows from the melting oven into the continuously fed ball mill where the glass is rapidly ground to colloidal fineness (approximately 10 micron or finer) with the help of corundum cylinders as a grind medium. Another batch of crushed recycled glass is added to the ball mills without melting.

During the second step - the virtual production process - the milled raw material mix undergoes a foaming process in steel moulds coated with clay and aluminium hydroxide at 850°C in the cellulating gas furnace (> 1000 kW) and is subsequently cooled down in the annealing furnace. Finally the cold foam glass cuboid is cut into pieces, packed and stored.

Production of FOAMGLAS®

(Plant Tessenderlo, Belgium)

**Fig. 4.10: Production process of FOAMGLAS® insulation material****Energy consumption**

Electricity and natural gas are the fuels used in the production of FOAMGLAS®. 63% of the electricity is consumed by the melting oven (compare Tab. 4.13). About 18.5 % is used for the grinding or milling process in the ball mills and some 18.5% is used for supplementary systems and processes such as trimming, packing and for storage. Natural gas is used solely for the cellulating and annealing furnace.

Tab. 4.13: Electricity use for the production of 650 kg FOAMGLAS® T4WDS

Production step		Electricity consumption
Raw mix production	Oven (melting)	613 kwh
	Ball mill (milling)	180 kwh
FOAMGLAS® production	Supplementary systems and trimming, packing, storage	180 kwh
		Natural gas consumption
FOAMGLAS® production	Cellulating and annealing furnace, packing	208 Nm ³

Heat emissions from the melting oven and annealing furnace are reused in the production process. The electricity used for FOAMGLAS® production is provided via both the local electricity grid and via renewable energy certificates which consist of 2.2% wind energy and 97.8% hydro power. These certificates are provided and controlled by the independent Renewable Energy Certificate System¹² (RECS) and sold in Tessenderlo by Fa. Electrabel¹³. The production facility in Tessenderlo currently has certificates for production for Switzerland, Austria, France and the Netherlands. Certificates which are reserved for products delivered to Switzerland and Austria are valid until at least 2009 and cover the use of electric power up to 20 000 000 kWh, corresponding to a production quantity of 90'000 m³ FOAMGLAS® products with an average density of 122.5 kg/m³. In 2005, approximately 70'000 m³ FOAMGLAS® products were delivered to Switzerland and 7700 m³ to Austria¹⁴.

At the production facility in Tessenderlo, production lines are split up as shown in Fig. 4.11 in accordance with the grinding process in the ball mill. The FOAMGLAS® production-line 11 produces solely for Switzerland and Austria which makes it easy to attribute the quantity of used, certified energy to respective products and countries.

¹² <http://www.recs.org>

¹³ Renewable Energy Certificates are RECS (Renewable Energy Certificate System)-certified and delivered by Fa. Electrabel, Belgium, Contact: Mr. Minnen 0032 (0) 25192782

¹⁴ Personal communication, Email, Hans Strauven, Pittsburgh Corning Europe, 1. Feb. 2006

FOAMGLAS® Production-lines

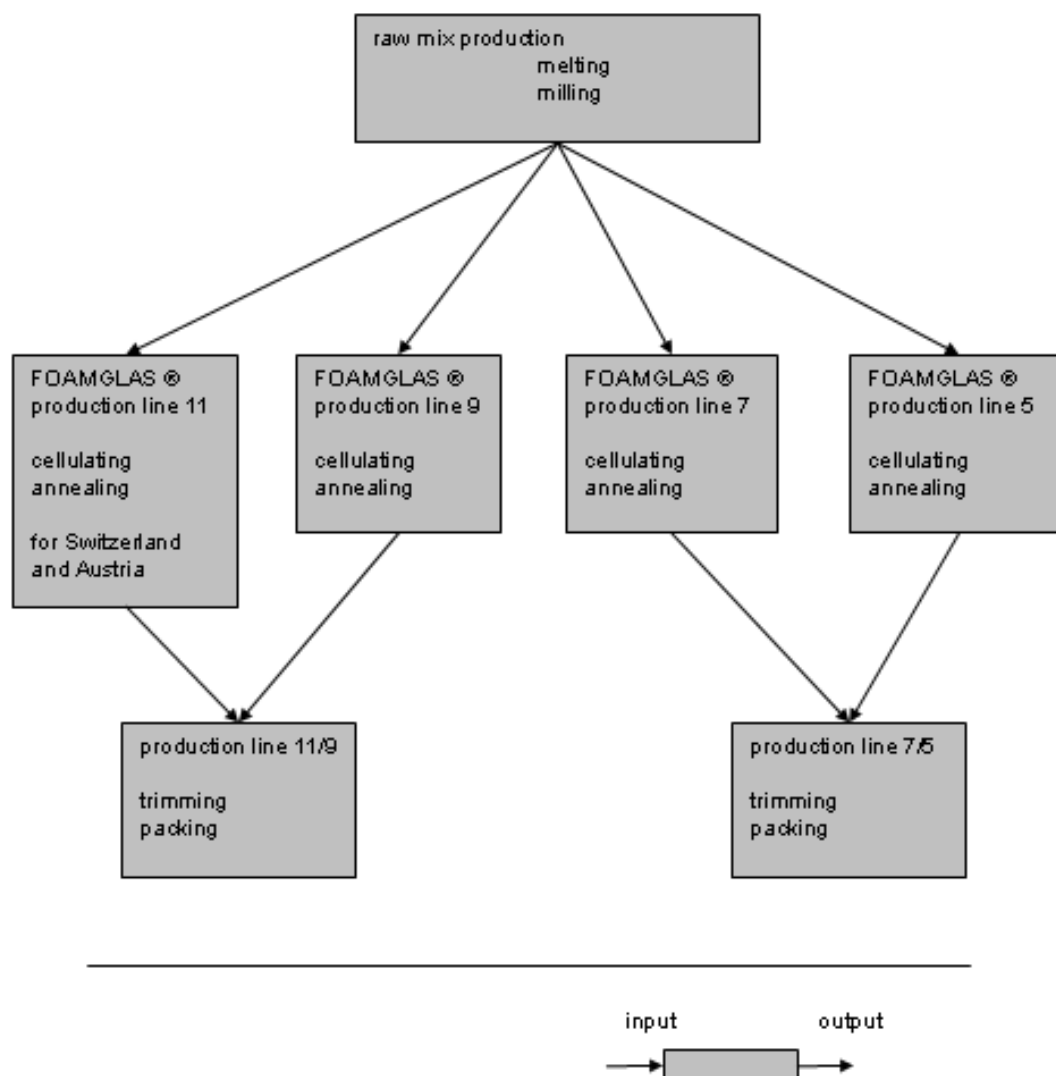


Fig. 4.11: Common and separated production steps for all countries at the production facility in Tessenderlo

Transports

Most raw materials are produced by and purchased from suppliers in Europe. As a result, the transport of raw materials is chiefly done by 40 t lorries. A negligible amount is transported by train and ship, e.g. carbon black which comes from overseas. Transport distances for road transports are estimated and calculated by the use of a route planning system¹⁵. An overview of the transport distances is given in Tab. 4.14.

¹⁵ www.reiseplanung.de

Tab. 4.14: Overview of transport distances

Raw materials	Approximate transport distance with lorries
Recycled glass	100 km
Feldspar	350 km
Iron oxide	25 km
Sodium nitrate	75 km
Salt cake	1000 km
Carbon black	550 km

Parts: wear-and-tear

In the FOAMGLAS[®] production process, a number of parts are subject to abrasion or corrosion and must, therefore, be periodically replaced. Parts which need to be commonly replaced include: molybdenum electrodes which are used for the melting process, corundum cylinders which serve as cylinder material for the milling process and the chromium steel moulds which are used for the foaming process. Abrasion of saw blades (97% Fe, 3% Cr) in the trimming process is included in the inventory but is of minor significance. The wear-and-tear of conveyors is too minimal to be of note¹⁶.

Packing

The packing facility consists of conveyor belts, a foil machine and a shrink tunnel run on electric power. FOAMGLAS[®] T4 WDS are delivered to construction sites as shrink wrapped, PE-foil (LDPE), 0.135 m³ parcels (measuring 0.5 m x 0.45 m x 0.6 m) with a corrugated cardboard cover element (0.45 m*0.6 m). The cardboard quality is a simple corrugated cardboard made from recycled fibres. The approximate net weight of a parcel is 14.85 kg. A delivery pallet commonly includes 12 parcels with a net weight of 178.2 kg which are covered with a PE-foil sheet on the top and are wrapped with PE-foil (LDPE). For deliveries, pure softwood, 10 kg, one-way pallets are used instead of 22 kg EUR-flat pallets. Every fourth pallet comes back to the factory in good shape and can be reused, which results in a reuse frequency of 0.25 or a multiplication of the pallet use by 0.75. Tab. 4.15 shows the packing materials used for one ready-for-delivery pallet.

Tab. 4.15: Packing materials used for one ready for delivery pallet of FOAMGLAS[®] T4 WDS measured by weight

Packing Material	kg per parcel	packing film per parcel	Thickness	Total per pallet
Shrink foil for parcel	0.133 kg	1.8 m ²	80E-6 m	1.596 kg
Cardboard	0.1015 kg	0.45m * 0.6m		1.218 kg
PE-foil sheet			100E-6 m	0.196 kg
Shrink foil for pallet			100E-6 m	0.786 kg
10 kg pure softwood one way pallet				7.5 kg

Electrical heating is used for shrink wrapping of FOAMGLAS[®] parcels. For the pallet packing elastic PE-foil is applied without heating.

¹⁶ Personal communication, Pittsburgh Corning Europe, Hans Strauven, 2006

Water and air use

Processed air is used at every production step for supplementary systems. A mix of tap water and ground-water-derived industrial water pumped at plant are used in the oven, the ball mill and the cellulating furnace. Tab. 4.16 shows the air and water use for the production of 650 kg FOAMGLAS®.

Tab. 4.16: Air and water use for the production of 650 kg FOAMGLAS® T4WDS

Water and air use	
Process air	2.75 Nm ³
Tap water	0.024 Nm ³
Industrial water	0.154 Nm ³

Land-use

The area used for the FOAMGLAS® production in Tessenderlo is approximately 0.17 km². Of that, 62'926 m² is vegetation, 44'197 m² is traffic area and 63'939 m² is industrial space with buildings. Land occupation is calculated by dividing the total area (0.17 km²) by the total amount of FOAMGLAS® products per year on the occupied area (39'200'000 kg/a). Land transformation is calculated by dividing the total area by the total life time production assuming a production of 39'200'000 kg/a over an estimated period of 50 years. Production halls occupy an area of 35'826 m² and the cubature of the multi-storey buildings is estimated to be about 5'400 m³.

Emissions to air, water and soil

Particulate matter (PM) is produced at different production steps: in the melting process, in the ball mill, in the cellulating furnace and during trimming. In the ball mill and the cellulating furnace no particulate matter emissions occur due to the re-circulation of unfiltered air in a closed loop system. The melting oven and trim station are equipped with sinter plate fabric filters from Herding¹⁷ for emission control which perform better than 1 mg/m³ air. Sinter plate fabric filters from Herding perform better than F9 Filters according to information from Herding. F9 Filters have a filtration efficiency for particulates > 2.5 µm higher than 99% in compliance with EN 779. Therefore pollutants of concern in the FOAMGLAS® production are particulate matter less than 2.5 micrometers and all emissions to air are assumed to be smaller than 2.5 µm. The particulates > 2.5 µm retained by the filter from the melting oven go into the brick production. The particulates retained by the filter of the trimming process which are mostly bigger than 20 µm are reused as raw material in the production process.

Tab. 4.17 shows data on maximum particulate matter emissions calculated from filter efficiency, retained amount of particulates which go to the brick factory or go back to the oven and the production volume in Tessenderlo. For the calculated particle size distribution in Tab. 4.17 granulomorphometric analysis data from the oven filter was used.

Gaseous emissions to air occur either by the processing of raw materials from the melting of raw materials, cellulation and trimming or by emissions from combustion processes (i.e. heating and burning of fuels). Combustion emissions are the primary gaseous pollutants. Calculated emissions from processing raw materials are small and carbon dioxide emissions are in the order of 1-2% by weight. Emissions from processing raw materials are indicated in the flow chart in the annex and do not include the emissions from burning processes as they are included there.

¹⁷ Herding Filtertechnik, <http://www.herding.ch>, 2006

Waste water is treated at the communal waste water treatment facility with the exception of the elimination of aluminiumoxide which is filtered out at the production facility.

Tab. 4.17: Calculated particulate matter

Particulate matter (PM)	Oven		Trim	
	kg		kg	
PM >10	9.34E-02	output to brick factory	2.29E-01	input to oven
PM10-PM2.5	6.24E-02	output to brick factory	1.53E-01	input to oven
<PM2.5 (fine dust)	1.47E-6	emissions to air	4.42E-05	emissions to air
Totale particulate matter (TPM)	1.56E-01		3.82E-01	

In the production process, solid waste accrues at different production steps. General waste such as steel, other metals, electronics, plastics, batteries and paper is gathered in different containers in the factory and recycled. Damaged moulds go back to the steel supplier, where they are repaired or recycled.

4.4.2 Foamglas in ecoinvent

The ecoinvent database meta information (Tab. 4.18, Tab. 4.19) includes all important and necessary information to use the data correctly. Fig. 4.12 to Fig. 4.15 show all in- and output flows of the foamglas products, the allocation to the modules in ecoinvent, the sources and the standard deviation for each datum.

Foam glass, packed, at plant (RER)

The dataset "foam glass, packed, at plant" includes data for administration, packing, solid waste, emissions from processing of raw material and the infrastructure, including its land-use, the buildings and the machines used.

Foam glass, packed, at plant ch and austria

The ecoinvent module Austria (A) and Switzerland (CH) takes into account the use of renewable energy certificates which are attributed to the production volume of these countries. The certificates are reserved for the aforementioned countries and are not credited to the general module foam glass, packed, at plant (BE). The additional transport distances to the centre of distribution in Switzerland and Austria are taken into account as following. From BE-3980 Tessenderlo to Austria AT-4050 Traun 877 km was brought to account and 587 km to Switzerland CH-4702 Oensingen.

Tab. 4.18 ecoinvent meta information for the foamglas production

Name	foam glass, at plant	foam glass, at regional storage
Location	RER	CH
Infrastructure Process	0	0
Unit	kg	kg
Data Set Version	2	2
Included Processes	Included processes: melting, forming to a pipe, cooling down, smashing, crushing, mixing with carbon powder. Foaming by heating in a mould, cooling, cutting to slabs and packing. Internal processes (workshop, administration etc.), transport of raw materials and energy carrier for furnace, packing and infrastructure are also included.	Included processes: melting, forming to a pipe, cooling down, smashing, crushing, mixing with carbon powder. Foaming by heating in a mould, cooling, cutting to slabs and packing. Internal processes (workshop, administration etc.), transport of raw materials and energy carrier for furnace, packing and infrastructure are also included. Renewable energy certificates are accounted and transport to Switzerland, respectively Lichtenstein is included.
Amount	1	1
Local Name	Schaumglas, ab Werk	Schaumglas, ab Regionallager
Synonyms	T4 WDS	T4 WDS
General Comment to reference function	Refers to foam glass of a density of 110 kg/m ³ and a thermal conductivity of 0.040 W/mK. Foam glass products sold to Switzerland and Lichtenstein are produced with renewable energy certificates from RECS (www.recs.org).	Refers to foam glass of a density of 110 kg/m ³ and a thermal conductivity of 0.040 W/mK. Foam glass products sold to Switzerland and Lichtenstein are produced with renewable energy certificates from RECS (www.recs.org).
Start Date	2005	2005
End Date	2005	2005
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted
Technology text	All data refer to the year 2005. Today foam glass is made of 68% recycled glass and only recycled glass is used instead of silica sand. Recycled glass is to some extent directly injected to the cellulating furnace without melting, resulting in less energy use. The energy for the melting process is electricity. Natural gas is used for the cellulating process. All emissions from processing raw materials and heating are taken into account.	All data refer to the year 2005. Today foam glass is made of 68% recycled glass and only recycled glass is used instead of silica sand. Recycled glass is to some extent directly injected to the cellulating furnace without melting, resulting in less energy use. The energy for the melting process is electricity. Natural gas is used for the cellulating process. All emissions from processing raw materials and heating are taken into account.
Representativeness [%]	100	100
Production Volume	A volume of 30625 tons per year is produced	A volume of 30625 tons per year is produced
Sampling Procedure	survey	survey
Extrapolations	none	none
Uncertainty Adjustments		

Tab. 4.19 ecoinvent meta information for the foamglas production

Name	foam glass, at regional storage	foam glass plant
Location	AT	BE
Infrastructure Process	0	1
Unit	kg	unit
Data Set Version	2	2
Included Processes	Included processes: melting, forming to a pipe, cooling down, smashing, crushing, mixing with carbon powder. Foaming by heating in a mould, cooling, cutting to slabs and packing. Internal processes (workshop, administration etc.), transport of raw materials and energy carrier for furnace, packing and infrastructure are also included. Renewable energy certificates are accounted and transport to Austria is included.	Includes the buildings, and machines in Belgium
Amount	1	1
Local Name	Schaumglas, ab Regionallager	Schaumglaswerk
Synonyms	T4 WDS	
General Comment to reference function	Refers to foam glass of a density of 110 kg/m ³ and a thermal conductivity of 0.040 W/mK. Foam glass products sold to Switzerland and Lichtenstein are produced with renewable energy certificates from RECS (www.recs.org).	
Start Date	2005	2005
End Date	2005	2005
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted	Data are from one company in Belgium which covers nearly 100% of the European foam glass market. For some exchanges RER-, CH- and GLO-modules are used as proxy. No energy certificates are accounted
Technology text	All data refer to the year 2005. Today foam glass is made of 68% recycled glass and only recycled glass is used instead of silica sand. Recycled glass is to some extent directly injected to the cellulating furnace without melting, resulting in less energy use. The energy for the melting process is electricity. Natural gas is used for the cellulating process. All emissions from processing raw materials and heating are taken into account.	All data refer to the year 2005.
Representativeness [%]	100	100
Production Volume	A volume of 30625 tons per year is produced	A volume of 30625 tons per year is produced
Sampling Procedure	survey	survey
Extrapolations	none	none
Uncertainty Adjustments		

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
recycled floatglass	foam glass, at plant: Inputs part 1		raw material to oven (37.6%) and ball mill (62.4%), recycled glass: mechanically cleaned window and windshield glass	glass	packaging	No	RER	glass, cullets, sorted, at sorting plant	8.14E-01	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
FOAM GLAS scrap			processed raw material to oven, not taken into account as assumed to be burden free						3.85E-01	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
feldspar			unprocessed raw material to oven	construction materials	others	No	RER	feldspar, at plant	2.58E-01	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
soda ash			raw material to oven, from total 50% unprocessed and 50% processed raw material, "soda powder" is used as proxy for both qualities	chemicals	inorganics	No	RER	soda, powder, at plant	4.62E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
iron oxide			processed raw material to oven, "iron ore at beneficiation" used as proxy	metals	extraction	No	GLO	iron ore, 65% Fe, at beneficiation	2.77E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
manganese oxide			processed raw material to oven	metals	extraction	No	GLO	manganese concentrate, at beneficiation	2.62E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
carbon black			processed raw material to ball mill	chemicals	inorganics	No	GLO	carbon black, at plant	6.15E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
salt cake			processed raw material to oven	chemicals	inorganics	No	RER	sodium sulphate, from natural sources, at plant	5.38E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
sodium nitrate			processed raw material to oven, potassium nitrate used as proxy. Calculation of N content according to stoichiometry (KNO ₃ =101.10 g/mol; N=14.007 g/mol)	agricultural means of production	mineral fertiliser	No	RER	potassium nitrate, as N, at regional storehouse	3.41E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
aluminiumhydroxide			processed raw material to cellulating furnace,	chemicals	inorganics	No	RER	aluminium hydroxide, at plant	6.40E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
coating material			processed raw material to cellulating furnace	construction materials	additives	No	CH	clay, at mine	3.49E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
process air			use of electricity for the compressor is considered in electricity input						4.23E-03	Nm3	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);

Fig. 4.12 Flows for "foamglas, at plant" (RER) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
tap water	foam glass, at plant: Inputs part 2		tap water for all processes	water supply	production	No	RER	tap water, at user	3.69E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
industrial water			groundwater for all processes, pumped at the production facility	resource	in water			Water, well, in ground	2.37E-04	m3	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
packaging film PE			packaging film to trim for shrink wrapping	plastics	processing	No	RER	packaging film, LDPE, at plant	1.46E-02	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
packaging cardboards			cardboard for packing	paper & cardboard	cardboard & corrugated board	No	RER	corrugated board, recycling fibre, double wall, at plant	7.31E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
packaging paletts (softwood)			pure softwood paletts for packing, "EUR-flat pallet" used as proxy	wooden materials	processing	No	RER	EUR-flat pallet	1.60E-03	unit	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
natural gas for cellulating and annealing			natural gas for cellulating and annealing furnace	natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.16E+01	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
electricity			electricity for oven, ball mill and other machines and processes	electricity	supply mix	No	BE	electricity, medium voltage, at grid	1.50E+00	kWh	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
diesel for fork lifts			diesel used for operating fork lifts	construction processes	machinery	No	GLO	diesel, burned in building machine	9.62E-02	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
heating oil			oil for heating administration buildings	oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	4.23E-02	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
molybdenum electrodes			molybdenum electrodes as wear parts for the oven, production process of the electrodes is not included	metals	extraction	No	RER	molybdenum, at regional storage	7.69E-05	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
corundum cylinders			corundum (sintered) cylinders as wear parts for the ball mill, "sinter iron" used as proxy	metals	extraction	No	GLO	sinter, iron, at plant	4.00E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
moulds			chromium moulds as wear parts for the cellulation and annealing furnace	metals	extraction	No	RER	chromium steel 18/8, at plant	1.15E-03	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
saw blades			chromium blades as wear parts for the trimming machine	metals	extraction	No	RER	chromium steel 18/8, at plant	1.17E-04	kg	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
raw material transport by barge			only carbon black is transported by barge over a distance of 1150km	transport systems	ship	No	OCE	transport, transoceanic freight ship	7.08E-03	tkm	Thalmann P, 2006	1	2.00	(1,3,1,1,1,1);
raw material transport by lorry			transportation of all raw materials, packing materials and wear parts to the factory, assumed that all transport by a lorry 40t	transport systems	road	No	CH	transport, lorry >28t, fleet average	1.94E-01	tkm	Thalmann P, 2006	1	2.00	(1,2,1,1,1,1);
infrastructure			includes machines, buildings, land occupation and land transformation, total foamglass production (all products) over 50 years: 1'960'000 tons	insulation materials	production	Yes	BE	foam glass plant	5.10E-10	unit	Thalmann P, 2006	1	3.05	(1,2,1,1,1,5);

Fig. 4.12 Flows for "foamglas, at plant" (RER) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
foam glass, at plant: Outputs	→ FOAM GLAS scrap	production waste to brick factory not taken into account as it is reused							1.57E-01	kg	Thalmann P, 2006	1	1.21	(1,2,1,1,1,5);
	→ packaging film PE			waste management	municipal incineration	No	CH	disposal, PE sealing sheet, 4% water, to municipal incineration	1.46E-02	kg		1	1.07	(1,3,1,1,1,1);
	→ packaging cardboards			waste management	municipal incineration	No	CH	disposal, packaging cardboard, 19.6% water, to municipal incineration	7.31E-03	kg		1	1.07	(1,3,1,1,1,1);
	→ packaging paletts (softwood)	1 palett = 22 kg (atro)		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	4.21E-02	kg		1	1.07	(1,3,1,1,1,1);
	→ carbon dioxide process emissions	process emission (no combustion): carbon dioxide from oxidation of raw material, 89.5% is emitted by the ball mill, 5.25% by the cellulating furnace and 5.25% by the trim process		air	low population density			Carbon dioxide, fossil	2.94E-02	kg	Thalmann P, 2006	1	1.12	(3,3,1,1,1,1);
	→ sulfur dioxide process emissions	process emissions: sulfur dioxide emissions from oven		air	low population density			Sulfur dioxide	2.31E-04	kg	Thalmann P, 2006	1	1.12	(3,3,1,1,1,1);
	→ nitrogen oxide emissions	nitrogen emissions from oven		air	low population density			Nitrogen oxides	1.00E-03	kg	Thalmann P, 2006	1	1.52	(3,3,1,1,1,1);
	→ Filter dust to recycling	in average about 99.25% of the total weight of the emitted particulates is bigger than 2.5 µm and is retained by dust filters and used in brick production							5.38E-01	kg	Thalmann P, 2006	1		
	→ dust emissions	in average about 0.75% of the total weight of the emitted particulates is smaller than 2.5 µm		air	low population density			Particulates, < 2.5 µm	4.57E-05	kg	Thalmann P, 2006	1	3.05	(1,2,1,1,1,5);
	→ waste water treated	waste water from all processes		waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 3	2.08E-04	m3	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
	→ waste heat	waste heat from electricity consumption		air	low population density			Heat, waste	5.39E+00	MJ	Thalmann P, 2006	1	1.07	(1,3,1,1,1,1);
	→ FOAMGLASS			insulation materials	production	No	RER	foam glass, at plant	1.00E+00	kg	Thalmann P, 2006			

Fig. 4.12 Flows for "foamglas, at plant" (RER) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
foam glass	foam glass, at regional storage (CH): Inputs			insulation materials	production	No	RER	foam glass, at plant	1.00E+00	kg	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
subtraction of total electricity consumption to replace it by green energy from certificates				electricity	supply mix	No	BE	electricity, medium voltage, at grid	-1.50E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate wind power		2.2% wind power		wind power	power plants	No	RER	electricity, at wind power plant	3.36E-02	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate hydro power		97.8% hydro power		hydro power	power plants	No	CH	electricity, hydropower, at power plant	1.49E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
transmission network medium voltage			transformation, transport and distribution of electricity	electricity	distribution	Yes	CH	transmission network, electricity, medium voltage	4.85E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network high voltage				electricity	distribution	Yes	CH	transmission network, electricity, high voltage	1.28E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network long distance				electricity	distribution	Yes	UCTE	transmission network, long-distance	4.79E-10	km	calculation based on ecoinvent DB v1.2	1	3.38	(5,5,4,5,1,5); estimates
SF6				chemicals	inorganics	No	RER	sulphur hexafluoride, liquid, at plant	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.07	(2,1,1,3,1,na); based on emissions

Fig. 4.13 Flows for "foamglas, at regional storage" (CH) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information				
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
transport to CH → foam glass, at regional storage (CH): Outputs			→	Heat, waste		air	unspecified			Heat, waste	6.06E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses
			→	Heat, waste		soil	unspecified			Heat, waste	1.75E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses
			→	Dinitrogen monoxide		air	unspecified			Dinitrogen monoxide	7.56E-06	kg	calculation based on ecoinvent DB v1.2	1	4.60	standard deviation based on variation reported in literature
			→	Ozone		air	unspecified			Ozone	6.80E-06	kg	calculation based on ecoinvent DB v1.2	1	5.00	standard deviation based on variation reported in literature
			→	Sulfur hexafluoride		air	unspecified			Sulfur hexafluoride	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.51	(2,1,1,3,1,na); national statistics
			→		transport from BE-3980 Tessenderlo to CH-4702 Oensingen, distance 587 km	transport systems	road	No	CH	transport, lorry >28t, fleet average	5.87E-01	tkm	Thalmann P, 2006	1	2.00	(1,2,1,1,1,1);
			→	FOAMGLASS in Switzerland		insulation materials	production	No	CH	foam glass, at regional storage	1.00E+00	kg	Thalmann P, 2006			

Fig. 4.13 Flows for "foamglas, at regional storage" (CH) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
foam glass	foam glass, at regional storage (AT): Inputs			insulation materials	production	No	RER	foam glass, at plant	1.00E+00	kg	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
subtraction of total electricity consumption to replace it by green energy from certificates				electricity	supply mix	No	BE	electricity, medium voltage, at grid	-1.50E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate wind power		2.2% wind power		wind power	power plants	No	RER	electricity, at wind power plant	3.36E-02	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
green energy certificate hydro power		97.8% hydro power		hydro power	power plants	No	CH	electricity, hydropower, at power plant	1.49E+00	kWh	Thalmann P, 2006	1	1.05	(1,2,1,1,1,1);
transmission network medium voltage			transformation, transport and distribution of electricity	electricity	distribution	Yes	CH	transmission network, electricity, medium voltage	4.85E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network high voltage				electricity	distribution	Yes	CH	transmission network, electricity, high voltage	1.28E-08	km	calculation based on ecoinvent DB v1.2	1	3.10	(3,1,1,3,3,5); based on consumption statistics
transmission network long distance				electricity	distribution	Yes	UCTE	transmission network, long-distance	4.79E-10	km	calculation based on ecoinvent DB v1.2	1	3.38	(5,5,4,5,1,5); estimates
SF6				chemicals	inorganics	No	RER	sulphur hexafluoride, liquid, at plant	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.07	(2,1,1,3,1,na); based on emissions

Fig. 4.14 Flows for "foamglas, at regional storage" (AT) and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
transport to A foam glass, at regional storage (AT): Outputs		Heat, waste		air	unspecified			Heat, waste	6.06E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses	
		Heat, waste		soil	unspecified			Heat, waste	1.75E-02	MJ	calculation based on ecoinvent DB v1.2	1	1.32	(4,1,3,3,1,5); estimation based on losses	
		Dinitrogen monoxide		air	unspecified			Dinitrogen monoxide	7.56E-06	kg	calculation based on ecoinvent DB v1.2	1	4.60	standard deviation based on variation reported in literature	
		Ozone		air	unspecified			Ozone	6.80E-06	kg	calculation based on ecoinvent DB v1.2	1	5.00	standard deviation based on variation reported in literature	
		Sulfur hexafluoride		air	unspecified			Sulfur hexafluoride	1.38E-07	kg	calculation based on ecoinvent DB v1.2	1	1.51	(2,1,1,3,1,na); national statistics	
		FOAMGLASS in Austria	transport from BE-3980 Tessenderlo to the AT-4050 Traun, distance 877 km	transport systems	road	No	CH	transport, lorry >28t, fleet average	8.77E-01	tkm	Thalmann P, 2006	1	2.00	(1,2,1,1,1,1);	
				insulation materials	production	No	AT	foam glass, at regional storage	1.00E+00	kg	Thalmann P, 2006				

Fig. 4.14 Flows for "foamglas, at regional storage" (AT) and their representation in the ecoinvent database (cont.)

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
machines	foam glass plant			construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	3.00E+06	kg	Thalmann P, 2006	1	3.28	(3,2,1,1,4,5);
conveyor belts installed				construction processes	machinery	Yes	RER	conveyor belt, at plant	1.00E+03	m	Thalmann P, 2006	1	3.28	(2,2,1,1,4,5);
production halls				construction processes	buildings	Yes	CH	building, hall	3.58E+04	m2	Thalmann P, 2006	1	3.28	(3,2,1,1,4,5);
administration buildings				construction processes	buildings	Yes	RER	building, multi-storey	5.40E+03	m3	Thalmann P, 2006	1	3.28	(3,2,1,1,4,5);
fabrication area				resource	land			Transformation, from unknown	1.71E+05	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
industrial area with buildings				resource	land			Transformation, to industrial area, built up	6.39E+04	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
vegetation area				resource	land			Transformation, to industrial area, vegetation	6.29E+04	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
traffic area				resource	land			Transformation, to traffic area, road network	4.42E+04	m2	Thalmann P, 2006	1	2.29	(3,2,1,1,4,5);
occupation area				resource	land			Occupation, industrial area, built up	1.71E+05	m2a	Thalmann P, 2006	1	1.84	(3,2,1,1,4,5);
production plant Tessenderlo														
		→ Foamglass production plant	Total foamglass production (all products) over 50 years: 1'960'000 tons	insulation materials	production	Yes	BE	foam glass plant	1.00E+00	unit	Thalmann P, 2006			

Fig. 4.15 Flows for "foamglas plant" and their representation in the ecoinvent database

4.4.3 Data quality

Data ascertainties were conducted by collaborators of Pittsburgh Corning Europe in Tessenderlo in cooperation with BAU- UND UMWELTCHEMIE Beratungen + Messungen AG. An accompanying external critical review according to ISO 14040:1997 §7.3.1 was carried out by Empa (H.J. Althaus). Empa had full access to the data considered in this study which are required for data validation and for a future integration in the ecoinvent database.

The presented data collection on raw material is based chiefly on measurements (e.g. raw materials and packing materials) at the production facility in Tessenderlo. Energy consumption is measured by gas and electricity meters and is cross-checked with theoretical calculations. Data on the wear-and-tear of parts and material quality were obtained from delivery orders and from their yearly consumption across the entire production volume. Transport distances include transport from the supplier to the production facility. All data are average values for the overall production of this product per year.

Modelling principles according to ecoinvent-report 1 are used. All data, assumptions, units, energy and emission considerations and naming rules are implemented as proposed in this report. Transport distances are calculated according the chapter Transport services in ecoinvent-report No. 1 Frischknecht et al. (2007).

The uncertainties are estimated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta-information.

In general, the data quality of the dataset "foam glass, packed, at plant" is high due to the monopoly situation of the product FOAMGLAS® in Europe. The company Pittsburgh Corning Europe has been collecting data for the last three years and has been given full access to all data for this inventory update.

4.5 Polystyrene Foam Slab (EPS)

4.5.1 Process Description

Polystyrene foam slabs are produced by thermoforming of expandable polystyrene (EPS). This process is described in Hischier (2007).

Basically the expandable polystyrene is heated and formed. The pentane evaporates in this process and builds the foam.

4.5.2 Polystyrene Foam Slab (EPS) in ecoinvent

The production of polystyrene foam slabs is modelled by combining the modules "polystyrene, expandable, at plant" and "foaming, expanding" from Hischier (2007).

Since in Hischier (2007) 1 kg "foaming, expanding" relates to the input of 1 kg EPS and 4.15% of the input is disposed as waste, 1.0415 kg of EPS has to be inventoried to produce 1 kg polystyrene slab. The recycling or disposal of the 0.0415 kg polystyrene waste is inventoried in the "forming, expanding" module.

The ecoinvent meta-information for the polystyrene foam slab production is given in Tab. 4.20 and the inputs and output flows for "polystyrene foam slab, at plant" are shown in Fig. 4.16.

Data Quality

Since only one material and one processing modules of ecoinvent are combined, the values are certain.

The data quality for the polystyrene foam is thus the same as the data quality for EPS and for the thermoforming process (see Hischier (2007)).

Tab. 4.20 ecoinvent meta information for the polystyrene foam slab production

Name	polystyrene foam slab, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes production and thermoforming of EPS
Amount	1
Local Name	Polystyrolplatte expandiert, ab Werk
Synonyms	
General Comment to reference function	Combination of material and processing module. EPS foam slab has a density of 30 kg/m ³ and a thermal conductivity of 0.035-0.04 W/mK.
Start Date	2003
End Date	2003
Data Valid For Entire Period	1
Other Period Text	
Geography text	
Technology text	European average EPS production; thermoforming from 2 Factories in Switzerland
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	none
Extrapolations	none
Uncertainty Adjustments	none

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
EPS	polystyrene foam slab, at plant	EPS foam	foaming relates to 1 kg output. Thus input is higher to account for waste (which is disposed in the thermoforming module)	plastics	polymers	No	RER	polystyrene, expandable, at plant	1.04E+00	kg	Hischier 2003	1	1.00	uncertainties in the combined modules - > no uncertainty
thermoforming EPS				plastics	processing	No	RER	foaming, expanding	1.00E+00	kg	Hischier 2003	1	1.00	uncertainties in the combined modules - > no uncertainty
				insulation materials	production	No	RER	polystyrene foam slab, at plant	1.00E+00	kg				

Fig. 4.16 Flows for "polystyrene foam slab, at plant" and their representation in the ecoinvent database

4.6 Polystyrene extruded (XPS)

4.6.1 Process Description

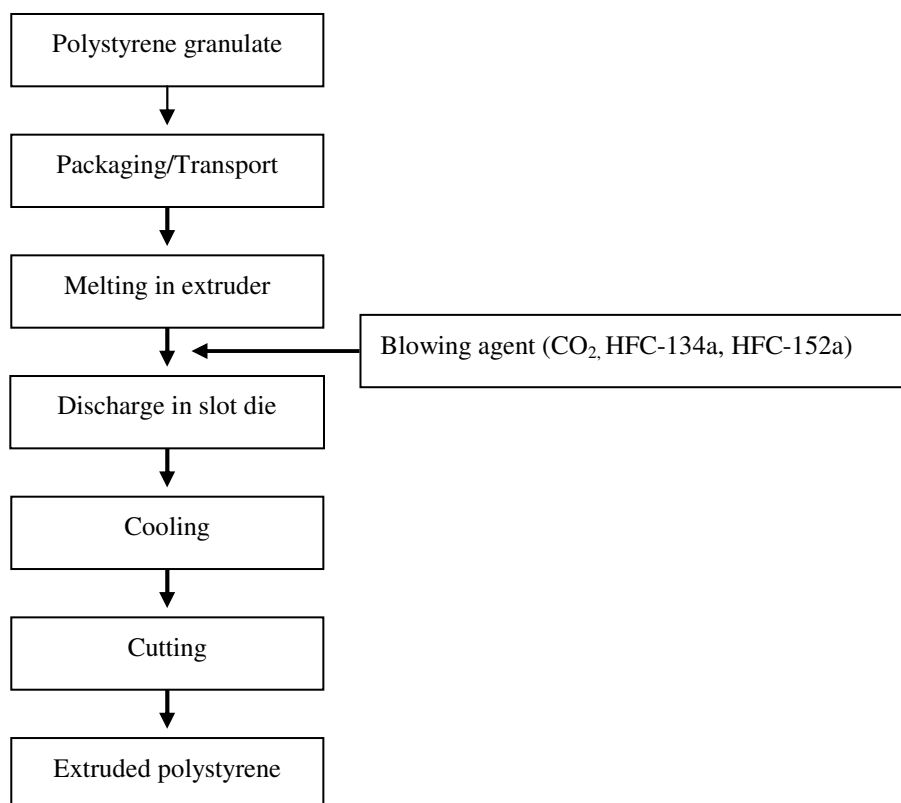
Raw Materials

XPS consists to 92% of polystyrene. The main raw material used for the production of polystyrene is benzene and ethylene. These two substances are mixed with a catalyst to produce styrene, which again is polymerised to Polystyrene (eza (2004)). Benzene and ethylene are described in the ecoinvent report No. 11 Part II on Plastics (Hischier et al. (2007)). About 6% of the XPS is blowing agent and the remaining 2% are flame retardants (bromine compounds) and other additives.

Production Process

Extruded polystyrene foam is made from solid polystyrene crystals (granulates). The polystyrene granulates, along with special additives and a blowing agent (nowadays often CO₂ with acetone as a co-blowing agent, but also HFC-134a and HFC-152a), are fed into an extruder. Within the extruder the mixture is combined and melted, under controlled conditions of high temperature and pressure, into a viscous plastic fluid. The hot, thick liquid is then forced in a continuous process through a die. As it emerges from the die it expands, is shaped, cooled, and trimmed to dimension. This continuous extrusion process results in a foam product with a uniform closed-cell structure, a smooth continuous skin, and consistent product qualities¹⁸. Fig. 4.17 shows the process chain of polystyrene XPS (Starzner S. and Wurmer-Weiss P. (2000)).

Fig. 4.17 Production process of extruded polystyrene



¹⁸ <http://www.diversifoam.com/xeps.htm>; access: 07.08.2007

Emissions

The production of XPS creates a certain amount of heat waste. This value is calculated from the electricity demand.

For this study, the styrene emission values of XPS extrusion are estimated at 0.1‰ of the polystyrene input independent of the blowing agent used.

For the CO₂ blown XPS, no direct CO₂-emission during the production phase is accounted for. For the two HFC blown XPS, 20% of the HFC input is accounted as an emission into air. Current production losses are 15-25% for Europe and 10-25% for North America (IPCC 2007, p.28).

Energy Consumption

The production step “melting of polystyrene pearls in extruder” consumes 1 kWh of electrical energy per kg (value taken from Jackson Insulation GmbH).

Transports

One explicit transport step is required for this process, when the polystyrene granulates, the flame retardants, the blowing agents plus the rock salts are moved from the manufacturing company to the place of extrusion. For this dataset, the standard distances ofecoinvent (100/50 km by 32 t lorry and 200/100 km by train in Europe) are applied.

4.6.2 Polystyrene, extruded (XPS) in ecoinvent

Three Datasets with different blowing agents are established. The dataset “Polystyrene, extruded” represents a consumption mix from 50% (w/w) CO₂ blown XPS¹⁹ and equally 25% (w/w) of two HFC blown XPS. HFCs will most likely completely vanish in European XPS production within the next 10 to 20 years but they will survive longer in North America (Environment Canada (2005)).

The meta information of “polystyrene, extruded (XPS), at plant” and of the three datasets with different blowing agents (Tab. 4.21- Tab. 4.24) include all important and necessary information to use the data correctly.

Fig. 4.18 – Fig. 4.21 shows the system of the dataset “polystyrene, extruded (XPS), at plant”. It shows all in- and output flows, the allocation to the datasets in the ecoinvent database database V2.0, the sources and the standard deviation of all in- and output flows.

The production steps include the extrusion of expandable polystyrene into polystyrene XPS. The discharge process through a slot die is part of the extrusion. The dataset relates to the production of XPS polystyrene using CO₂, HFC-134a and HFC-152a as blowing agents.

¹⁹ Supported by information of http://www.climnet.org/pubs/FoamsFACTSHEET_March2004.pdf

Tab. 4.21 ecoinvent database meta information of the product “polystyrene, extruded (XPS), at plant”

Name	polystyrene, extruded (XPS), at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The dataset provides a mix of XPS produced with different blowing agents.
Amount	1
Local Name	Polystyrol, extrudiert (XPS), ab Werk
Synonyms	XPS
General Comment to reference function	This dataset can be used in the construction building sector. The cutting of the extruded polystyrene blocks is not included in the process. Blowing agent shares are 50% (w/w) CO ₂ , 25% (w/w) HFC-134a and 25% (w/w) HFC-152a.
Start Date	1994
End Date	2006
Data Valid For Entire Period	1
Other Period Text	Technology in 1994 was still using CFC or HCFC as blowing agent. In 2005, already more than 50% of companies in Europe are using CO ₂ as blowing agent. Though mainly in the USA, HFC-134a and HFC-152a as a blowing agent are still common.
Geography text	Data for leading polystyrene XPS producers in Germany and the USA. Applicable for Europe and North America.
Technology text	Extrusion process of expandable polystyrene including discharge via slot die. CO ₂ and HCF-134a/HFC-152a as blowing agents.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, databank
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that Germany is representative for Central Europe is considered in the uncertainties.

Tab. 4.22 ecoinvent database meta information of the product “polystyrene, extruded (XPS) HFC-134a blown, at plant”

Name	polystyrene, extruded (XPS)_HFC-134a blown, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The dataset describes the production of extruded polystyrene. Included processes are the melting of polystyrene pearls in the extruder, the discharge through a slot die, as well as the cooling with water.
Amount	1
Local Name	Polystyrol, extrudiert (XPS)_HFC-134a geschäumt, ab Werk
Synonyms	XPS
General Comment to reference function	This dataset can be used in the construction building sector. The cutting of the extruded polystyrene blocks is not included in the process. Blowing agent is HFC-134a. Process emissions are 20% (w/w)
Start Date	1994
End Date	2006
Data Valid For Entire Period	1
Other Period Text	Technology in 1994 was still using CFC or HCFC as blowing agent. In 2005, already more than 50% of companies in Europe are using CO ₂ as blowing agent. Though mainly in the USA, HFC-134a and HFC-152a as a blowing agent are still common.
Geography text	Data for leading polystyrene XPS producers in Germany and the USA. Applicable for Europe and North America.
Technology text	Extrusion process of expandable polystyrene including discharge via slot die. HCF-134a as blowing agent.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, databank
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that Germany is representative for Central Europe is considered in the uncertainties.

Tab. 4.23 ecoinvent database meta information of the product “polystyrene, extruded (XPS) HFC-152a blown, at plant”

Name	polystyrene, extruded (XPS)_HFC-152a blown, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The dataset describes the production of extruded polystyrene. Included processes are the melting of polystyrene pearls in the extruder, the discharge through a slot die, as well as the cooling with water.
Amount	1
Local Name	Polystyrol, extrudiert (XPS)_HFC-152a geschäumt, ab Werk
Synonyms	XPS
General Comment to reference function	This dataset can be used in the construction building sector. The cutting of the extruded polystyrene blocks is not included in the process. Blowing agent is HFC-152a. Process emissions are 20% (w/w)
Start Date	1994
End Date	2006
Data Valid For Entire Period	1
Other Period Text	Technology in 1994 was still using CFC or HCFC as blowing agent. In 2005, already more than 50% of companies in Europe are using CO ₂ as blowing agent. Though mainly in the USA, HFC-134a and HFC-152a as a blowing agent are still common.
Geography text	Data for leading polystyrene XPS producers in Germany and the USA. Applicable for Europe and North America.
Technology text	Extrusion process of expandable polystyrene including discharge via slot die. HCF-152a as blowing agent.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, databank
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that Germany is representative for Central Europe is considered in the uncertainties.

Tab. 4.24 ecoinvent database meta information of the product “polystyrene, extruded (XPS) CO₂ blown, at plant”

Name	polystyrene, extruded (XPS)_CO ₂ blown, at plant
Location	RER
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	The dataset describes the production of extruded polystyrene. Included processes are the melting of polystyrene pearls in the extruder, the discharge through a slot die, as well as the cooling with water.
Amount	1
Local Name	Polystyrol, extrudiert (XPS)_CO ₂ geschäumt , ab Werk
Synonyms	XPS
General Comment to reference function	This dataset can be used in the construction building sector. The cutting of the extruded polystyrene blocks is not included in the process. Blowing agent is CO ₂ (50.7% w/w). Co-blowing agent ist acetone (49.3% w/w). Process emissions are 25% (w/w) for acetone.
Start Date	1994
End Date	2006
Data Valid For Entire Period	1
Other Period Text	Technology in 1994 was still using CFC or HCFC as blowing agent. In 2005, already more than 50% of companies in Europe are using CO ₂ as blowing agent. Though mainly in the USA, HFC-134a and HFC-152a as a blowing agent are still common.
Geography text	Data for leading polystyrene XPS producers in Germany and the USA. Applicable for Europe and North America.
Technology text	Extrusion process of expandable polystyrene including discharge via slot die. CO ₂ as blowing agent.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature, databank
Extrapolations	see Geography
Uncertainty Adjustments	The assumption that Germany is representative for Central Europe is considered in the uncertainties.

General Flow information				Representation in ecoinvent 2000								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	Stdv 95%	General Comment
XPS CO2 blown	polystyrene, extruded (XPS), at plant	Polystyrene, extruded (XPS), at plant	0.5	insulation materials	production	No	RER	polystyrene, extruded (XPS) CO2 blown, at plant	5.00E-01	kg	Starzner & Wurmer-Weiss (2000)	1	140	(4,4,4,3,1,5,3)
XPS HFC 134a blown			0.25	insulation materials	production	No	RER	polystyrene, extruded (XPS), HFC-134a blown, at plant	2.50E-01	kg		1	140	(4,4,4,3,1,5,3)
XPS HFC 152a			0.25	insulation	production	No	RER	polystyrene, extruded (XPS),	2.50E-01	kg	Energie- &	1	140	(4,4,4,3,1,5,3)
				insulation materials	production	No	RER	polystyrene, extruded (XPS), at plant	1.00E+00	kg				

Fig. 4.18 Flows for “polystyrene, extruded (XPS), at plant” and their representation in the ecoinvent database

4. System Characterisation

General Flow information				Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Polystyrene	polystyrene, extruded (XPS), HFC-134a blown, at plant		92.4% (w/w)	plastics	polymers	No	RER	polystyrene, expandable, at plant	9.24E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)
Flame retardant			1% (w/w) bromated inorganic chemical	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	1.00E-02	kg		1	1.24	(2,4,1,3,1,5,3)
Fluorspar				chemicals	inorganics	No	GLO	fluorspar, 97%, at plant	5.00E-03	kg	Energie- &	1	1.26	(3,4,1,3,1,5,3)
Rock salt				chemicals	inorganics	No	RER	sodium chloride, powder, at plant	5.00E-03	kg		1	1.26	(3,4,1,3,1,5,3)
Transport lorry 32 t				transport systems	road	No	RER	transport, lorry >16t, fleet average	1.01E-01	tkm	ecoinvent standard distances	1	2.05	(2,3,1,1,1,5,5)
Transport train				transport systems	train	No	RER	transport, freight, rail	2.31E-01	tkm		1	2.05	(2,3,1,1,1,5,5)
Electricity			For melting of polystyrene perls in extruder and discharge through slot die	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.00E+00	kWh	Jackson Insulation GmbH (2008)	1	1.24	(2,4,1,3,1,5,2)
Blowing agent			Assumption: Blowing agent is HFC-134a, so 7% (w/w)	chemicals	organics	No	RER	refrigerant R134a, at plant	7.00E-02	kg	IPCC (2007)	1	1.25	(2,3,1,2,3,4,4)
		R134a emission	Emission to air: 20% losses during production	air	low population density			Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	1.40E-02	kg	IPCC (2007)	1	1.14	(2,3,1,2,1,4,4)
		Polystyrene waste arising from extrusion lines	10g/kg XPS (assumed 99% closed loop recycling)	waste management	municipal incineration	No	CH	disposal, polystyrene, 0.2% water, to municipal incineration	1.00E-04	kg	Jackson Insulation GmbH (2008)	1	1.24	(2,4,1,3,1,5,4)
		Styrene (gas)	Assumption that styrene emissions for XPS are 0.01% of input	air	unspecified			Styrene	9.24E-05	kg	Own calculations and EPSASA (2002)	1	1.59	(3,4,1,3,1,5,31)
		Heat waste		air	unspecified			Heat, waste	3.60E+00	MJ	Calculation	1	1.26	(3,4,1,3,1,5,13)
Cooling			value for extrusion of plastic pipe	resource	in water			Water, cooling, unspecified natural origin	1.97E-02	m3	estimation	1	1.26	(3,4,1,3,1,5,4)
Infrastructure			explosive production plant as proxy	construction materials	others	Yes	CH	explosive production plant	6.67E-09	unit	estimation	1	2.33	(5,5,1,1,5,5,11)
		Polystyrene, extruded (XPS), at plant		insulation materials	production	No	RER	polystyrene, extruded (XPS), HFC-134a blown, at plant	1.00E+00	kg				

Fig. 4.19 Flows for “polystyrene, extruded (XPS) HFC-134a blown, at plant” and their representation in the ecoinvent database

4. System Characterisation

General Flow information				Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Polystyrene	polystyrene, extruded (XPS), HFC-152a blown, at plant		92.4% (w/w)	plastics	polymers	No	RER	polystyrene, expandable, at plant	9.24E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)
Flame retardant			1% (w/w) bromated inorganic chemical	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	1.00E-02	kg		1	1.24	(2,4,1,3,1,5,3)
Fluorspar				chemicals	inorganics	No	GLO	fluorspar, 97%, at plant	5.00E-03	kg	Energie- &	1	1.26	(3,4,1,3,1,5,3)
Rock salt				chemicals	inorganics	No	RER	sodium chloride, powder, at plant	5.00E-03	kg		1	1.26	(3,4,1,3,1,5,3)
Transport lorry 32 t				transport systems	road	No	RER	transport, lorry >16t, fleet average	1.01E-01	tkm	ecoinvent standard distances	1	2.05	(2,3,1,1,1,5,5)
Transport train				transport systems	train	No	RER	transport, freight, rail	2.31E-01	tkm		1	2.05	(2,3,1,1,1,5,5)
Electricity			For melting of polystyrene perls in extruder and discharge through slot die	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.00E+00	kWh	Jackson Insulation GmbH (2008)	1	1.24	(2,4,1,3,1,5,2)
Blowing agent			Assumption: Blowing agent is HFC-152a, so 7% (w/w)	chemicals	organics	No	US	1,1-difluoroethane, HFC-152a, at plant	7.00E-02	kg	IPCC (2007)	1	1.25	(2,3,1,2,3,4,4)
		HFC-152a emission	Emission to air: 20% losses during production	air	low population density			Ethane, 1,1-difluoro-, HFC-152a	1.40E-02	kg	IPCC (2007)	1	1.14	(2,3,1,2,1,4,4)
		Polystyrene waste arising from extrusion lines	10g/kg XPS (assumed 99% closed loop recycling)	waste management	municipal incineration	No	CH	disposal, polystyrene, 0.2% water, to municipal incineration	1.00E-04	kg	Jackson Insulation GmbH (2008)	1	1.24	(2,4,1,3,1,5,4)
		Styrene (gas)	Assumption that styrene emissions for XPS are 0.01% of input	air	unspecified			Styrene	9.24E-05	kg	Own calculations and EPSASA (2002)	1	1.59	(3,4,1,3,1,5,31)
		Heat waste		air	unspecified			Heat, waste	3.60E+00	MJ	Calculation	1	1.26	(3,4,1,3,1,5,13)
Cooling			value for extrusion of plastic pipe	resource	in water			Water, cooling, unspecified natural origin	1.97E-02	m3	estimation	1	1.26	(3,4,1,3,1,5,4)
Infrastructure			explosive production plant as proxy	construction materials	others	Yes	CH	explosive production plant	6.67E-09	unit	estimation	1	2.33	(5,5,1,1,5,5,11)
		Polystyrene, extruded (XPS), at plant		insulation materials	production	No	RER	polystyrene, extruded (XPS), HFC-152a blown, at plant	1.00E+00	kg				

Fig. 4.20 Flows for “polystyrene, extruded (XPS) HFC-152a blown, at plant” and their representation in the ecoinvent database

4. System Characterisation

General Flow information				Representation in ecoinvent 2000							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent 2000	Mean value	Unit	Source	Type	StDv 95%	General Comment
Polystyrene	polystyrene, extruded (XPS) CO ₂ blown, at plant		92.1% (w/w)	plastics	polymers	No	RER	polystyrene, expandable, at plant	9.21E-01	kg	Starzner & Wurmer-Weiss (2000)	1	1.24	(2,4,1,3,1,5,3)
Flame retardant			1% (w/w) bromated inorganic chemical	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	1.00E-02	kg		1	1.24	(2,4,1,3,1,5,3)
Fluorspar				chemicals	inorganics	No	GLO	fluorspar, 97%, at plant	5.00E-03	kg	Energie- &	1	1.26	(3,4,1,3,1,5,3)
Rock salt				chemicals	inorganics	No	RER	sodium chloride, powder, at plant	5.00E-03	kg		1	1.26	(3,4,1,3,1,5,3)
Transport lorry 32 t				transport systems	road	No	RER	transport, lorry >16t, fleet average	9.87E-02	tkm	ecoinvent standard distances	1	2.05	(2,3,1,1,1,5,5)
Transport train				transport systems	train	No	RER	transport, freight, rail	1.98E-01	tkm		1	2.05	(2,3,1,1,1,5,5)
Electricity			For melting of polystyrene perls in extruder and discharge through slot die	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.00E+00	kWh	Jackson Insulation GmbH (2008)	1	1.24	(2,4,1,3,1,5,2)
Blowing agent			Assumption: Blowing agent is CO ₂ , 50.7 % (w/w)	chemicals	inorganics	No	RER	carbon dioxide liquid, at plant	3.40E-02	kg	Jackson Insulation GmbH (2008)	1	1.22	(2,3,1,2,1,5,4)
Co-blowing agent			Assumption: Co-blowing agent is Aceton, 49.3% Aceton (w/w)	chemicals	organics	No	RER	acetone, liquid, at plant	3.30E-02	kg	Jackson Insulation GmbH (2008)	1	1.24	(2,4,1,3,1,5,4)
		Acetone emission	Emission to air: 25% (w/w) losses during production	air	unspecified			Acetone	8.25E-03	kg	Jackson Insulation GmbH (2008)	1	1.58	(2,4,1,3,1,5,31)
		Polystyrene waste arising from extrusion lines	10g/kg XPS (assumed 99% closed loop recycling)	waste management	municipal incineration	No	CH	disposal, polystyrene, 0.2% water, to municipal incineration	1.00E-04	kg	Jackson Insulation GmbH (2008)	1	1.24	(2,4,1,3,1,5,4)
		Styrene (gas)	Assumption that styrene emissions for XPS are 0.01% of input	air	unspecified			Styrene	9.21E-05	kg	Own calculations and EPSASA (2002)	1	1.59	(3,4,1,3,1,5,31)
		Heat waste	value for extrusion of plastic pipe	air	unspecified			Heat, waste	3.60E+00	MJ	Calculation	1	1.26	(3,4,1,3,1,5,13)
Cooling			explosive production plant as proxy	resource	in water			Water, cooling, unspecified natural origin	1.97E-02	m3	estimation	1	1.26	(3,4,1,3,1,5,4)
Infrastructure				construction materials	others	Yes	CH	explosive production plant	6.67E-09	unit	estimation	1	2.33	(5,5,1,1,5,5,11)
		Polystyrene, extruded (XPS), at plant		insulation materials	production	No	RER	polystyrene, extruded (XPS) CO ₂ blown, at plant	1.00E+00	kg				

Fig. 4.21 Flows for “polystyrene, extruded (XPS) CO₂ blown, at plant” and their representation in the ecoinvent database

Data quality

The uncertainties are evaluated with the help of the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent meta information.

Most data of this process are based on the Starzner S. and Wurmer-Weiss P. (2000), though important input was also retrieved from other sources. Information on the emissions related to the extrusion process is based on IPCC (2007), however, some assumptions had to be made (share of different blowing agents during production process).

4.7 Cellulose Fibre (inclusive blowing in)

4.7.1 Process Description

Waste paper is shredded and mixed with boric acid and borax. The mixture is milled in a hammer mill where the fibres in the paper are separated and impregnated with the boron. The fibres are separated from dust in a cyclone and packed in paper bags (Richter et al. (1995)).

Because the application of cellulose fibres as insulation material for buildings requires some electricity for blowing, this process has to be inventoried to allow for a comparison of cellulose fibres with other insulation materials such as EPS foam, rock wool or glass wool.

Infrastructure

Beside some storage capacity in the factory the main equipment is the hammer mill. It is assumed that infrastructure needed for producing 1 kg cellulose fibres insulation is about the same as for 1 kg wood pellets. Thus the wood pellet manufacturing plant is chosen as proxy for the cellulose fibre production plant.

Emissions

Dust is the only relevant direct emission from handling and milling of the paper.²⁰

Waste

The solid wastes from the process include the dust from separating of the dust from the fibres in the cyclone and the packaging material.

Richter et al. (1995)) indicate a waste water stream in the process flow chart but does not mention any waste water in the inventory data. It is plausible that some of the water input remains in the product and in the solid waste but some waste water is nevertheless expected. The composition of the wastewater is expected to include boron and organic matter. However, no measurements are available.

4.7.2 Cellulose Fibre Production and Application in ecoinvent

The production process is based on the data given in Richter et al. (1995). For the lack of data on air emissions and wastewater they must be neglected.

²⁰ Weibel & Stritz (1995) interpreted the amount of paper dust from the cyclone given in Richter et al. (1995) as air emission instead of waste.

The energy demand for the application of cellulose fibre as building insulation is taken from Eyerer & Reinhardt (2000)). This source indicates only the primary energy demand. As exclusively electricity is used for this process, the electricity consumption is calculated using the conversion factor primary energy to electricity of 1/3.

The ecoinvent meta-information for the cellulose fibre production is given in Tab. 4.25 and the input and output flows for "cellulose fibre, inclusive blowing in, at plant" are shown in Fig. 4.22.

Data Quality

The inventory is based on one source only and no direct emissions are known. On the other hand, the process is rather uncomplicated and the shortcomings of the inventory are not believed to be of major relevance. Thus the overall quality of the data is considered sufficient.

Tab. 4.25 ecoinvent meta information for the cellulose fibre production and application

Name	cellulose fibre, inclusive blowing in, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	includes the input energy and material to the production processes, transports of the materials and the available process emissions. Also included is the packaging and the energy for the application of the fibres as insulation materials in buildings.
Amount	1
Local Name	Zellulosefasern, inklusive Einblasen, ab Werk
Synonyms	
General Comment to reference function	Cellulose fibre in building applications has a density of 50 kg/m ³ and a thermal conductivity of 0.04 W/mK when applied. No emissions are inventoried besides the heat waste from electricity use for lack of data.
Start Date	1995
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	production data from one major producer in Switzerland
Technology text	mechanical separation of fibres from waste paper
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	literature
Extrapolations	none
Uncertainty Adjustments	none

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Waste paper	cellulose fibre, inclusive blowing in, at plant			paper & cardboard	pulps	No	CH	waste paper, mixed, from public collection, for further treatment	8.01E-01	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,3)
boric acid				chemicals	inorganics	No	RER	boric acid, anhydrous, powder, at plant	8.87E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
borax				chemicals	inorganics	No	RER	Borax, anhydrous, powder, at plant	1.09E-01	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
paper bags			for packaging	paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	3.70E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
pallets			for packaging	wooden materials	processing	No	RER	EUR-flat pallet	1.21E-03	unit	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
plastic			for packaging	plastics	processing	No	RER	packaging film, LDPE, at plant	3.25E-04	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
wire			for packaging	metals	extraction	No	RER	reinforcing steel, at plant	1.81E-03	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
water				water supply	production	No	RER	tap water, at user	2.38E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,4)
Electricity			for production	electricity	supply mix	No	CH	electricity, low voltage, at grid	1.02E-01	kWh	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,2)
Electricity			0.1 MJ primary energy for application	electricity	supply mix	No	CH	electricity, low voltage, at grid	1.85E-04	kWh	calculated based on Eyrer et al. 2000	1	1.22	(1,3,2,2,1,5,2)
natural gas				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace low-NOx >100kW	6.41E-02	MJ	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,1)
Dielsel for internal transports				construction processes	machinery	No	GLO	diesel, burned in building machine	1.12E-02	MJ	Richter et al. 1995	1	2.05	(1,3,2,1,1,5,5)
Transport rail			600 km for chemicals, 100 km for paper and pallets	transport systems	train	No	RER	transport, freight, rail	2.05E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Transport lorry			everything 50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.32E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
plant			wood pellet plant as proxy (1E8 kg output in 50 years)	wood energy	fuels	Yes	RER	wood pellet manufacturing, infrastructure	1.00E-08	unit	estimated	1	3.23	(5,n.A.,n.A.,n.A.,n.A.,n.A.,9)
		cellulose fibre (incl. blowing)		insulation materials	production	No	CH	cellulose fibre, inclusive blowing in, at plant	1.00E+00	kg	Richter et al. 1995			
		Waste heat into air		air	unspecified			Heat, waste	3.66E-01	MJ	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,13)
		Paper dust to incineration	dust from cyclone	waste management	municipal incineration	No	CH	disposal, paper, 11.2% water, to municipal incineration	1.12E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)
		packaging material to incineration		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.04E-02	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)
		pallets to incineration		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	2.80E-03	kg	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)
		pallets to recycling	not inventoried						1.08E-03	unit	Richter et al. 1995	1	1.22	(1,3,2,1,1,5,6)

Fig. 4.22 Flows for "cellulose fibre, inclusive blowing in, at plant" and their representation in the ecoinvent database

4.8 Urea Formaldehyde Foam

4.8.1 Process description

“A surfactant solution containing a hardener (e.g., phosphoric acid) is foamed with compressed air and blended with an aqueous 30 – 40 % solution of urea–formaldehyde resin. The resin is distributed throughout the surfactant foam, and cross-linking is initiated by the acidic hardener. A light, whipped-cream-like foam is formed which can be shaped easily and used to fill any cavity. The foam hardens rapidly to a water-insoluble, rot-resistant, foamed plastic without creating expansion pressure. A hardening time of 15 – 40 s is generally selected, which can be controlled within certain limits by changing external variables (temperature, hardener concentration).” (Weber (1997))

Problems caused by formaldehyde release can be avoided largely by using more urea or improved resin formulations.

4.8.2 Infrastructure

A special compressor is needed for the in situ production. According to information given on a producers web page²¹, its weight is estimated to 50 kg at a production rate of 1 m³/15 min.

The industrial production infrastructure is approximated by the module for a generic chemical production plant. It is assumed that the output of the plant is twice the output of a comparable plant for organic chemicals in order to account for the relatively simple processes involved in the UF-foam production.

4.8.3 Emissions

A direct formaldehyde emission is estimated on the basis of emissions in the production of wooden boards with urea-formaldehyde binders. It is assumed that the industrial prefabrication process is better controlled than the in situ process and thus leads to less emissions.

4.8.4 Waste

2% of the produced slabs is inventoried as waste. This accounts for cut off in the slab production. 5% of the in situ foam is inventoried as waste to account for material loss by cleaning of the tools.

4.8.5 Urea Formaldehyde Foam Production in ecoinvent

For lack of data the material inputs, the relevant emissions and the process energy are estimated. For details see Tab. 4.26, Fig. 4.23 and Fig. 4.24.

Data Quality

The inventory is based on theoretical process description and calculations. The direct formaldehyde emission is estimated. Since this emission might be of high relevance, the overall quality of the data is considered poor, even though the process is rather uncomplicated.

²¹ <http://www.schaum-chemie.de/produkte/isoschaum.php>; access date: 15.03.2004

Tab. 4.26 ecoinvent meta information for the UF foam production

Name	urea formaldehyde foam, in situ foaming, at plant	urea formaldehyde foam slab, hard, at plant
Location	CH	CH
Infrastructure Process	0	0
Unit	kg	kg
Data Set Version	2.0	2.0
Included Processes	Includes raw materials and in situ production of foam. Transports to application site are not included.	Includes raw materials and production of foam.
Amount	1	1
Local Name	Ortschaum, Harnstoffformaldehyd, ab Werk	Hartschaumplatte, Harnstoffformaldehyd, ab Werk
Synonyms		
General Comment to reference function	Foam has a density of 10 - 30 kg /m3 and a thermal conductivity of 0.035 W/mK. Data is based on crude assumptions.	Foam has a density of 10 - 30 kg /m3 and a thermal conductivity of 0.035 W/mK. Data is based on crude assumptions.
Start Date	1997	1997
End Date	2003	2003
Data Valid For Entire Period	1	1
Other Period Text		
Geography text		
Technology text	On site production of foam from two components using a compressor and a portable injection device.	Production of slabs in large manufacturing plant using stationary machinery.
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	none	literature
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow information				Representation in ecoinvent							Uncertainty			
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
UF resin	urea formaldehyde foam, in situ foaming, at plant			paintings	production	No	RER	urea formaldehyde resin, at plant	9.56E-01	kg	estimated	1	2.56	(5,5,5,5,5,3)
water				water supply	production	No	CH	tap water, at user	4.10E-01	kg	estimated	1	2.56	(5,5,5,5,5,4)
surfactant				washing agents	tensides	No	RER	alkylbenzene sulfonate, linear, petrochemical, at plant	5.25E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
phosphoric acid			hardener	chemicals	inorganics	No	RER	phosphoric acid, industrial grade, 85% in H ₂ O, at plant	5.25E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
compressed air														
electricity			for compressor: Power consumption 1 kW; 1 m ³ (=20 kg) foam / 15 min	electricity	supply mix	No	CH	electricity, low voltage, at grid	1.25E-02	kWh	estimated, based on http://www.schaum-chemie.de (access 15.3.2004)	1	2.56	(5,5,5,5,5,2)
Transport rail			600 km for chemicals	transport systems	train	No	RER	transport, freight, rail	6.36E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Transport lorry			50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.30E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Compressor			Assumption: compressor weight 50 kg, life time foam production 100'000 kg	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	5.00E-04	kg	estimated, based on http://www.schaum-chemie.de (access 15.3.2004)	1	4.25	(5,5,5,5,5,9)
		waste heat	only from electricity consumption. Heat of reaction is not inventoried.	air	unspecified			Heat, waste	4.50E-02	MJ	estimated	1	2.56	(5,5,5,5,5,13)
		formaldehyd	1‰ of UF resin input	air	high population density			Formaldehyde	9.56E-04	kg	estimated	1	3.22	(5,5,5,5,5,23)
		water	drying; not inventoried						4.29E-01	kg	calculated			
		Waste	5% of produced foam	waste management	municipal incineration	No	CH	disposal, polyurethane, 0.2% water, to municipal incineration	5.00E-02	kg	estimated	1	2.56	(5,5,5,5,5,6)
		UF foam, in situ		insulation materials	production	No	CH	urea formaldehyde foam, in situ foaming, at plant	1.00E+00	kg				

Fig. 4.23 Flows for "urea formaldehyde foam, in situ foaming, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty				
Input		Process Name		Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
UF resin	→	urea formaldehyde foam slab, hard, at plant				paintings	production	No	RER	urea formaldehyde resin, at plant	9.28E-01	kg	estimated	1	2.56	(5,5,5,5,5,3)
water	→					water supply	production	No	CH	tap water, at user	3.98E-01	kg	estimated	1	2.56	(5,5,5,5,5,4)
surfactant	→					washing agents	tensides	No	RER	alkylbenzene sulfonate, linear, petrochemical, at plant	5.10E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
phosphoric acid	→				hardener	chemicals	inorganics	No	RER	phosphoric acid, industrial grade, 85% in H2O, at plant	5.10E-02	kg	estimated	1	2.56	(5,5,5,5,5,3)
compressed air	→															
electricity	→				Power consumption 1 kW --> 1 m3 (=20 kg) foam / 5 min	electricity	supply mix	No	CH	electricity, low voltage, at grid	4.17E-03	kWh	estimated, based on http://www.schaum-chemie.de (access 15.3.2004)	1	2.56	(5,5,5,5,5,2)
Transport rail	→				600 km for chemicals	transport systems	train	No	RER	transport, freight, rail	6.18E-01	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
Transport lorry	→				50 km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.15E-02	tkm	estimated	1	2.09	(4,5,n.A.,n.A.,n.A.,n.A.,5)
plant	→				Proxy: chemical production plant	chemicals	organics	Yes	RER	chemical plant, organics	2.00E-10	unit	estimated	1	4.25	(5,5,5,5,5,9)
	→			waste heat	only from electricity consumption. Heat of reaction is not inventoried.	air	unspecified				Heat, waste	1.50E-02	MJ	estimated	1	2.56
	→		formaldehyd	0.1‰ of UF resin input	air	high population density				Formaldehyde	9.28E-05	kg	estimated	1	3.22	(5,5,5,5,5,23)
	→		water	drying; not inventoried							4.17E-01	kg	calculated			
	→		Waste	2% of produced foam	waste management	municipal incineration	No	CH	disposal, polyurethane, 0.2% water, to municipal incineration	2.00E-02	kg	estimated	1	2.56	(5,5,5,5,5,6)	
	→		UF foam slab		insulation materials	production	No	CH	urea formaldehyde foam slab, hard, at plant	1.00E+00	kg					

Fig. 4.24 Flows for “urea formaldehyde foam slab, hard, at plant” and their representation in the ecoinvent database

5 Cumulative Results and Interpretation

The cumulative results can be downloaded from the ecoinvent Database (www.ecoinvent.org).

6 References

- Althaus et al. (2007) Althaus H.-J., Chudacoff M., Hischier R., Jungbluth N., Osses M. and Primas A. (2007) Life Cycle Inventories of Chemicals. Final report ecoinvent Data v2.0 No. 8. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch.
- Anonymous (1997) Anonymous (1997) Herhof-Umwelterklärung. Herhof-Basalt und Diabas-Werke GmbH, Online-Version under: <http://www.herhof.de/>.
- Berdowski et al. (1995) Berdowski J., Visschedijk A., Creemers E., Pulles T., Pacyna J., Fudala J. and Querreveld D. (1995) Co-ordinated European Programme on Particulate Matter Emission Inventories, Projections and Guidance (CEPMEIP) from <http://www.air.sk/tno/cepmeip/>.
- Dones et al. (2007) Dones R., Bauer C., Bolliger R., Burger B., Faist Emmenegger M., Frischknecht R., Heck T., Jungbluth N. and Röder A. (2007) Life Cycle Inventories of Energy Systems: Results for Current Systems in Switzerland and other UCTE Countries. Final report ecoinvent data v2.0 No. 5. Paul Scherrer Institut Villigen, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch.
- CAN Europe (2004) Climate Action Network Europe (March 2004) Fact Sheet: HFCs in Foams. Online-Version under: http://www.climnet.org/pubs/FoamsFACTSHEET_March2004.pdf
- Environment Canada (2005) Environment Canada (2005) Stratospheric Ozone. Halocarbon Emissions and Control Technologies in Foam Manufacturing. Online-Version under: <http://www.ec.gc.ca/ozone/DOCs/SandS/FOAMS/EN/body.cfm>
- EPA (1998) EPA (1998) Mineral Wool Manufacturing, Emissions Factor Documentation for AP-42, Background Document, Section 11.18. U.S. Environmental Protection Agency (EPA), Office of Air Quality Planning and Standards, Emission Inventory Branch,, Research Triangle Park/NC, Online-Version under: <http://www.epa.gov/ttn/chief/ap42/>.
- EPSASA (2002) Expanded Polystyrene Association South Africa (2002) Selection Guide introducing expanded polystyrene, administered by the Association of Architectural Aluminium Manufacturers of South Africa. Online-Version under: <http://www.aaamsa.com/Documents/publications/publication14.htm>
- eza (2004) Energie- und Umweltzentrum Allgäu (2004) Bau- und Dämmstoffe. Polystyrol XPS. Kempten, Germany. Online-Version under: <http://www.eza-allgaeu.de/bau-%20und%20d%C3%A4mmstoffe,47.html>
- Eyerer & Reinhardt (2000) Eyerer P. and Reinhardt H. W. (2000) Ökologische Bilanzierung von Baustoffen und Gebäuden: Wege zu einer ganzheitlichen Bilanzierung. In: Baupraxis. Birkhäuser Verl., Basel, Boston, Berlin.
- Fitzer et al. (1997) Fitzer E., Kleinholz R., Tiesler H., Hugh Stacey M., De Bruyne R., Lefever I. and Heine M. (1997) Fibers, 5. Synthetic Inorganic. In: Ullmann's encyclopedia of industrial chemistry (ed. Anonymous). 5th edition on CD-ROM Edition. Wiley & Sons, London.

- Flumroc (2002) Flumroc (2002) Vierter Flumroc-Ökologiebericht. Flumroc AG, Flums (CH).
- Flumroc (2007) Flumroc (2007) Flumroc-Ökologiebericht. Flumroc AG, Flums (CH).
- Friskhnecht et al. (2007) Friskhnecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Hirschier R., Hellweg S., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent Data v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch.
- Hirschier (2007) Hirschier R. (2007) Life Cycle Inventories of Packaging and Graphical Paper. Final report ecoinvent Data v2.0 No. 11. EMPA St. Gallen, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch.
- Hungerbühler (2004) Hungerbühler K. (2004) Vorlesungsreihe Risikoanalyse chemischer Prozesse und Produkte, SS 2004. Ökobilanz LCA Fallbeispiel. Online-Version under: http://ltcmail.ethz.ch/hungerb/teaching/courses/Vorlesung_Dok/LCA_part1&2_S04.pdf
- Hutter et al. (1999) Hutter C., Köhler D., Lilleike J. and Schwärzer M. (1999) Ganzheitliche Bilanzierung von Grundstoffen und Halbzeugen, Teil II Baustoffe. Forschungsstelle für Energiewirtschaft, der Gesellschaft für praktische Energiekunde e.V., München, Online-Version under: <http://www.ffe.de/download/gabie/baust.pdf>.
- IPCC (2007) IPCC/TEAP Special Report: Safeguarding the Ozone Layer and the Global Climate System. Chapter 7 Foams. Online-Version under: <http://www.ipcc.ch/activity/sroc/SROC07.pdf>
- Käufer (1994) Prof. Dr. H. Käufer et al. (1994), Vergleichende normierende Bewertung der Umweltanalyse von Trinkwasserinstallationssystemen, Berlin, unveröffentlicht (unpublished).
- KnaufInsulation (2005) Polyfoam Fabrication boards. Product data sheet. Online-Version under: <http://www.knaufinsulation.co.uk/cgi-bin/files/pdf/Datasheet%20-%20Polyfoam%20Fabrication%20Board%20Feb%2007%20PD88607.pdf>
- Köhl & Frei (1994) Köhl K. and Frei A. (1994) Umweltjahresbericht der Firma Flumroc AG. Flumroc AG.
- Kündig et al. (1997) Kündig R., Mumenthaler T., Eckardt P., Keusen H. R., Schindler C., Hofmann F., Vogler R. and Guntli P. (1997) Die mineralischen Rohstoffe der Schweiz. Schweizerische Geotechnische Kommission, Zürich.
- Lagaly et al. (1997) Lagaly G., Klose D., Tufar W., Minihan A. and Lovell A. (1997) Silicates. In: Ullmann's encyclopedia of industrial chemistry (ed. Anonymous). 5th edition on CD-ROM Edition. Wiley & Sons, London.
- Mötzl et al. (2000) Mötzl H., Zelger T., Gann M., Liebinger A. and Lipp B. (2000) Ökologie der Dämmstoffe; Grundlagen der Wärmedämmung, Lebenszyklusanalyse von Wärmedämmstoffen, Optimale Dämmstandards. Springer Verl., Wien, New York.
- PCA (2005) Portland Cement Association (2005) Technology Briefs No. 4. Plastic Foams for Concrete Homes. Skokie, Illinois. Online-Version under: <http://www.cement.org/homes/brief04.asp>
- Richter et al. (1995) Richter K., Fischer M., Gahlmann H. and Menard M. (1995) Energie- und Stoffbilanzen bei der Herstellung von Wärmedämmstoffen. Eidg. Materialprüfungs- und Forschungsanstalt (EMPA), Dübendorf.
- Starzner & Wurmer-Weiss (2000) Starzner S. and Wurmer-Weiss P. (2000) ECOBIS 2000, ökologisches Baustoffinformationssystem.

Weber (1997)

Weber H. (1997) Foams from Urea–Formaldehyde (UF) Resins. In: Ullmann's encyclopedia of industrial chemistry (ed. Anonymous). 5th edition on CD-ROM Edition. Wiley & Sons, London.

Annex 6

[updated Part IX of Kellenberger et al. 2007]

Fibre Cement Products and Processes

Data v2.1 (2009)

Authors: Daniel Kellenberger, EMPA Dübendorf
Hans-Jörg Althaus, EMPA Dübendorf

Reviewer: Roberto Dones, Paul Scherrer Institut

Summary

This part deals with three different types of fibre cement products and its packing process. The products are roof slates, corrugated slabs and facing tiles.

The production of facing tiles was updated with version 2.1 on the basis of an in depth study of the production at Eternit in Niederurnen. The study was performed by T. Kull, Ecosense and reviewed by Hans-Jörg Althaus, Empa. The implementation into ecoinvnet was done by Hans-Jörg Althaus.

The production of fibre cement products mainly requires the raw materials fibre, cement, limestone meal and water.

The main part of the energy consumption in the production process for all three products is electrical energy. The cumulated energy demand is dominated by the energy for the cement production. Also the emissions mainly stem from the cement production process.

Table of Contents

SUMMARY	III
TABLE OF CONTENTS	V
1 INTRODUCTION	1
2 RESERVES AND RESOURCES	1
3 USE.....	1
4 CHARACTERISATION	1
5 SYSTEM CHARACTERISATION	2
5.1 Raw Materials	3
5.2 Energy Consumption.....	4
5.3 Transports.....	5
5.4 Air Emissions	5
5.5 Wastewater	6
5.6 Land-use	6
6 LIFE CYCLE INVENTORIES	6
7 DATA QUALITY	29
8 REFERENCES	29

1 Introduction

Fibre cement is a composite material made out of cement and fibres for reinforcement (Starzner & Wurmer-Weiss (2000)). In former times asbestos was used. Since 1991 asbestos containing materials are not allowed to be produced, sold and used anymore in Switzerland. Therefore today only cement products reinforced with synthetic fibres are on the market.

The products described in this report are “fibre cement roof slate”, “fibre cement corrugated slab” and “fibre cement facing tile”. These are the most used fibre cement products in the building industry. Other products are tubes, plant vessels, and pre-forms.

Fibre cement products are waterproof, frost resistant and humidity regulative. Generally facing tiles and roof slates are coated on the exposed side. Uncoated fibre cement slabs have only a very small market share and are not recommended for application on facades.

2 Reserves and Resources

Swiss Portland cement is proportionally the most important raw material. It is mainly made of clinker which uses mainly limestone and marly clay as raw materials. Its manufacture is described in the corresponding report (Part 2, Cement Products and Processes).

3 Use

The most important uses of fibre cement products in the building sector are (Starzner & Wurmer-Weiss (2000)):

Facing tile:	- cladding for external walls ventilated at rear and infills - siding for pillars - framework constructions, bottom of roof, parapet cladding, etc.
Roof slates:	Roof covering tiles instead of clay roof tiles
Corrugated slab:	large-size fibre cement slabs with wave profile for roof covering

4 Characterisation

Some technical specifications for the studied products are taken from the SIA building materials declaration²² and the ecological building material information system called ECOBIS (Starzner & Wurmer-Weiss (2000)), „ökologisches Baustoffinformationssystem“, which includes environmental- and health-relevant information concerning the different building material categories for all life cycle stages (raw material, production and processing, usage and after treatment) from “cradle to grave”. It is an information system for people from the building sector and should support the choice of building products based on an ecological view. Tab. 4.1 shows some of the offered information.

²² <http://www.sia.ch/cf/oekoinfofoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=>, access date: Mai 2003

Tab. 4.1 Technical specification of the studied products (SIA)²², Starzner & Wurmer-Weiss (2000) and Weibel & Stritz (1995))

	Fibre cement roof slates	Fibre cement corrugated slab	Fibre cement facing tile
Bulk density ρ [kg/m ³]	1800	1700	1900
Thermal conductivity λ [W/mK]	about 0.5	about 0.5	about 0.5
Weight per area [kg/m ²]	23.7 – 24.6 (used: 24.2)	17.8 – 20.1 (used: 19)	15.1 – 15.2 (used: 15.2)
Thickness [cm]	1.34	1.12	0.80
Maximum size [mm]	600 * 300	No Information	2'500/2'800/3'100 * 1'250/1'500

5 System Characterisation

Fibre cement products are composed of the materials shown in Tab. 5.1. The mixture of the materials before hardening is called mush.

Tab. 5.1 Composition of the studied products²³

[%] (w/w)	Fibre cement roof slates	Fibre cement corrugated slab	Fibre cement facing tile
Process fibres: new pulp fibres	1.1	1.8	1
Process fibres: recycled pulp fibres	1.7	1.8	1.5
Additive: limestone meal	13.9	11.1	3.7
Binding agent: portland cement	69.8	72.4	78.6
Reinforcing fibres: polyvinyl alcohol	2.6	1.8	3.7
Water: constitutional water	10.9	11.1	11.5
Total	100	100	100

A more detailed description from the ecological building material information system is reported in the following.

“The additives (in fibre cement) are used as filling material and to improve the material binding. All types of fibres are added to guarantee for high enough tensile strength and need chemical resistance to the alkaline environment of the cement. It is also important to use fibres which are not absorbable by the lung and therefore not carcinogenic. Such fibres are e.g. polyvinyl alcohol fibres which have a diameter between 3 μm and 12 to 14 μm , polyacrylnitrile fibres with a diameter between 3 μm and 14 to 18 μm , and polyethylene fibres with a length of 200 to 400 μm and a diameter between 6 μm and 3 μm . For corrugated slabs also milled fibre cement products (recycling material) can be used.

The cement particles must adhere to the fibres within the mush. For fibres with very smooth surfaces, adhesion is poor compared to the asbestos fibres with their fine and frayed structure. For this reason pulp fibres are added. They promote adhesion between the fibres and the cement and help to avoid the flush of cement particles with the surplus water when exuded.

Water as raw material is needed to hydrate the cement and assure the workability of the mush.

Air within the hardened fibre cement slab gives the freezing water expansion space thus helping to avoid destruction by frost.

²³ <http://www.sia.ch/cf/oekoinfofoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=,> access date: Mai 2003

For weathering and UV-radiation protection a colouring surface coating on the visible side is usually added. To adjust the diffusion resistance to the resistance of the front side, the backside is sealed.

One producer states that in most cases the the visible side is hot-coated using all-acrylate coating. It is based on inorganic colouring pigments (solvents: 2 to 4%). Some solvent and heavy metal emissions from this coating process are taken into account.

The colourless **backside sealing** is composed of 36% polyvinylidene chloride butyl acrylate, 60% water and 4% paraffin wax. For this study such sealing has not be taken into account as no data are available and the amount is very small.” (Starzner & Wurmer-Weiss (2000), translation by the author)

Fig. 5.1 shows the production process of fibre cement slabs.

The following more detailed explanation is taken from Starzner & Wurmer-Weiss (2000). The production occurs with a generally automated winding mode. The raw materials are mixed with water to obtain a homogeneous paste. Rotating sieving cylinders dip in this fibre cement mush, which drain to the inside. Thus the sieve surface gets covered with a layer of fleece of about 1 mm which is transported with a conveyor belt. Suction nozzles on the conveyor belt (fleece belt) dewater the layer.

The produced endless mat is winded up on a format-cylinder until the demanded thickness is reached. With this process the fibres of the neighbour layers are interconnecting.

After cutting up the mat cylinder, the mat is detached from the format-cylinder and by punching and pressing automatically formed to small format slates or plain and corrugated slabs. From the still soft raw slabs also ventilating tubes and special shapes are formed.

After some hours the strength of the products is high enough for them to be taken out of the form. Followed by a natural post-curing process in a store room or a vapour curing process, usually a surface coating process finishes the production process.

5.1 Raw Materials

The amounts of raw materials used for the different fibre cement products are given in the SIA building materials declaration²⁴ and refer only to the company ”Eternit AG” with branches in Niederurnen and Payerne in Switzerland. The data in the declaration describe the composition of the different products, without taking into account the cutting waste. As the waste is internally reused it is not taken into account.

For the fibre cement facing tiles, data from a new LCA study commissioned by Eternit were used in ecoinvent (Kull et al. 2008a, b).

²⁴ <http://www.sia.ch/cf/oekoinfoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=,> access date: Mai 2003

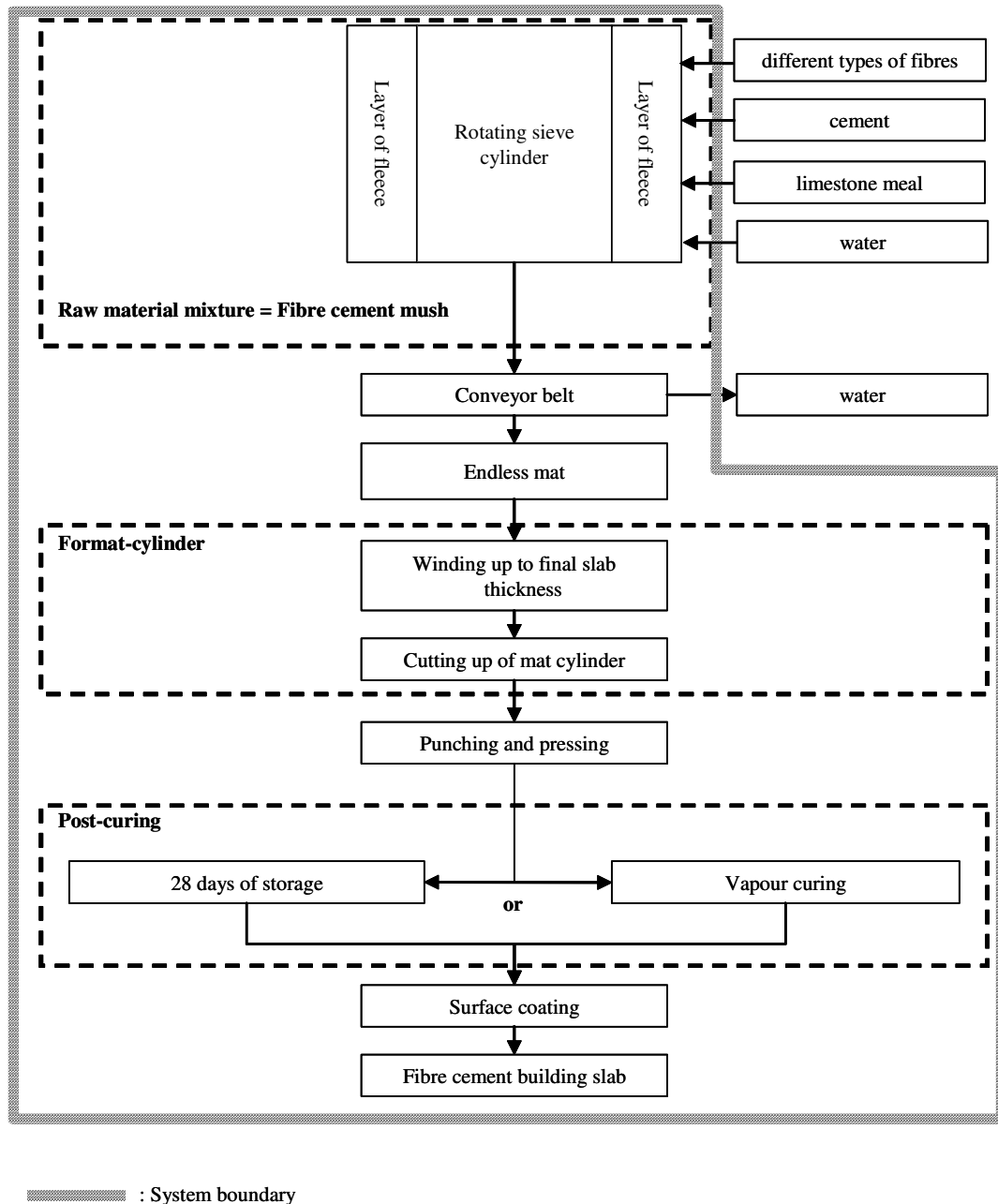


Fig. 5.1 Production process for fibre cement building slabs (Starzner & Wurmer-Weiss (2000))

5.2 Energy Consumption

The energy requirements listed in the SIA building materials declaration²⁴ refer to the total amount energy consumed (including the upstream processes) for the three products fibre cement roof slates, fibre cement corrugated slabs and fibre cement facing tiles. Kasser & Pöll (1998) state that about 50 to 70% of the “grey energy” (more or less the CED) derives from the upstream processes. As no exact amount is available the worst case for the production process (without upstream processes) of fibre cement products is assumed here. A rough check with the CED for the clinker production shows that direct energy consumption even could be higher. As no better data are available for this study this means that 50% of the energy consumption indicated in the SIA building materials declaration²⁴ is allocated to the upstream processes and 50% to the production process.

The total amount of energy used for the production process is based on the description in the previous section. The share of the different energy carriers concerning these two products (fibre cement roof slate and fibre cement corrugated slab) are based on (Weibel & Stritz (1995)). For the fibre cement facing tiles, data from a new LCA study commissioned by Eternit were used in ecoinvent (Kull et al. 2008a, b).

The thermal energy used for forming and drying fibre cement products at Eternit AG in Switzerland in the year 1991 is produced in three different furnaces with different sizes and using different fuels (Weibel & Stritz (1995)). For the fibre cement facing tiles, data from a new LCA study commissioned by Eternit were used in ecoinvent (Kull et al. 2008a, b).

Tab. 5.2 shows the share of electric and thermal energy consumption.

Tab. 5.2 Share of electrical and thermal energy consumption (based on (Weibel & Stritz (1995))

	fibre cement roof slates [%]	fibre cement corrugated slab [%]	fibre cement facing tile [%]
CED of electricity consumption	82.6	91.4	87
Light fuel oil in furnace 1500 kW	7.1	8.1	7.6
Light fuel oil in furnace 400 kW	0.4	0.4	0.4
Propane gas in furnace 230 kW	9.8	0	4.9
Total	100	100	100

An important part of the energy (10%) is used for the neutralisation of the alkaline wastewater with the combustion gases (Weibel & Stritz (1995)).

The heating of the building and the energy consumption of administrative related actions are not included.

5.3 Transports

The transport distances are taken from (Weibel & Stritz (1995)) and presented in Tab. 5.3.

Tab. 5.3 Transport distances (Weibel & Stritz (1995))

Transport distances [km]					
	Additive	Reinforcing fibre	Polyvinyl alcohol	Pulp fibre	Cement
Transport lorry 28t	4				
Transport by freight train	132	860	36	400	90
Transport freight ship		14000			

These distances are based on one Swiss company and the year 1994. It is assumed that they have not changed significantly and that they are approximately valid for all three fibre cement products, included in this study. For the fibre cement facing tiles, data from a new LCA study commissioned by Eternit were used in ecoinvent (Kull et al. 2008a, b).

5.4 Air Emissions

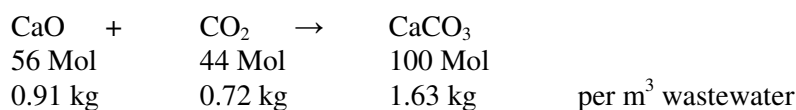
Usually the particulates arising from the production process are collected in filters and reused in the process (Starzner & Wurmer-Weiss (2000)). Due to a lack of data no dust emissions are taken into account in this study.

Most other emissions to air stem from the combustion of fuel. These emissions are taken into account in the corresponding modules.

5.5 Wastewater

The wastewater from the production process is usually very alkaline because of a high amount of calcium oxide (quicklime) from the cement, and has to be neutralised before being released to the public sewage system (Weibel & Stritz (1995)). The alkaline wastewater with pH 13.5 is corrected to pH 8.5. To neutralise the wastewater, an unspecified amount of acid is added, then CO₂ emissions from combustion are scrubbed through the wastewater. Since no data concerning the type and amount of acid is available it is not included in the inventory.

The CO₂ binds with the calcium oxide to give lime (CaCO₃) as shown in the following stoichiometric equation:



The CO₂ emissions from combustion included in the upstream processes must be corrected to take into account CO₂ binding. The amount of CO₂ bound in the wastewater is 0.72 kg/m³ (Weibel & Stritz (1995)) and is included as a negative emission to compensate the emission inventoried in the energy production module. The amount is very small because the amount of wastewater is very small (0.068 kg/kg fibre cement roof slate).

The produced amount of CaCO₃ is included in the output module “disposal of decarbonising waste”.

The total amount of solvent emissions indicated in the SIA building materials declaration²⁵ is assumed to be 90% NMVOC and 10% halogenated hydrocarbons.

For the fibre cement facing tiles, data from a new LCA study commissioned by Eternit were used in ecoinvent (Kull et al. 2008a, b).

5.6 Land-use

No data for the fibre cement plant of Eternit AG in Switzerland are available. Therefore the module “rockwool plant” is used as proxy.

6 Life Cycle Inventories

Ecoinvent database meta information (tables), flows and their representation in the ecoinvent database are shown in (figures):

Fibre cement roof slate, at plant	Tab. 6.1 and Fig. 6.1
Fibre cement corrugated slab, at plant	Tab. 6.2 and Fig. 6.2
Fibre cement facing tile, large format, at plant	Tab. 6.3 and Fig. 6.3
Fibre cement facing tile, small format, at plant	

²⁵ <http://www.sia.ch/cf/oekoinfofoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=,> access date: Mai 2003

Fibre cement facing tile, at plant
Packing fibre cement products

Tab. 6.1 Ecoinvent meta information for the product “fibre cement roof slate, at plant”

Name	fibre cement roof slate, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce fibre cement products (raw material provision, raw material mixing, forming, post curing process coating process, and storage), transports to plant processes, and infrastructure. No acid for neutralisation of the wastewater and no requirements for administration are included.
Amount	1
Local Name	Faserzementdachschiefer, ab Werk
Synonyms	tile
General Comment to reference function	The amounts of raw materials are calculated based on the composition of fibre cement roof slate. Air emission data are lacking but emissions from the boilers are imported via the relevant modules. A certain amount of CO ₂ produced by the boiler is bound in the waste water. The amount of this entrained CO ₂ is inventoried with a negative value because it is an avoided emission. The total yearly is assumed to be 20'000 tonnes. The lifespan of the plant and the conveyor belt are assumed to be 50 years and 20 years, respectively.
Start Date	1991
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from one company in Switzerland (Eternit AG), for some exchanges RER- and OCE-modules have been used as proxy
Technology text	The raw materials are mixed with water to a homogeneous paste. Rotating sieving cylinders dip in this fibre cement mush which drain to the inside. Therefore the sieve surface gets covered with a layer of fleece of about 1mm which is transported with a conveyor belt. The produced endless mat is winded up on a format-cylinder until the demanded thickness is reached. With that process the fibres of the neighbour layers are interconnecting. After cutting up the mat cylinder the mat is detached from the format-cylinder and by punching and pressing automatically formed to small format slates, even and corrugated slabs.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature and "ecological building material declaration" (a sort of environmental report)
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
process fibres: pulp fibres	Production of "fibre cement roof slate" (Part 1)		raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	1.10E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
process fibres: waste paper			raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	CH	waste paper, mixed, from public collection, for further treatment	1.70E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
additive: limestone meal			raw material, losses are neglected as internally reused.	construction materials	others	No	CH	limestone, milled, packed, at plant	1.39E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
fixing agent: cement			raw material, losses are neglected as internally reused.	construction materials	binder	No	CH	portland calcareous cement, at plant	6.98E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
reinforcing fibre: polyvinyl alcohol			raw material, losses are neglected as internally reused.	plastics	monomers	No	RER	vinyl acetate, at plant	2.60E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
water: constitutional water			raw material, losses are neglected as internally reused.	water supply	production	No	RER	tap water, at user	1.09E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
total electricity consumption for complete process				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.69E-01	kWh	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 1MW				oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	2.42E-01	MJ	SIA building product declaration	1	1.12	
energy consumption: light fuel oil in industrial furnace 100kW				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.47E-02	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: gas in industrial furnace				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	3.33E-01	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
tap water			raw material	water supply	production	No	RER	tap water, at user	3.20E+00	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
acid for neutralisation of waste water			no data available;								Weibel & Stritz (1995)			
dispersive acrylate			used for coating	paintings	production	No	RER	acrylic varnish, 87.5% in H2O, at plant	1.77E-02	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
packing				construction materials	others	No	CH	packing, fibre cement products	1.00E+00	kg	estimation	1	1.40	(3,5,1,1,3,5);
infrastructure plant			assumption: same infrastructure for 1kg fibre cement product is inquired as for 1kg rock wool product	insulation materials	production	Yes	CH	rock wool plant	4.44E-10	unit	estimation	1	3.50	(5,1,1,1,4,5);
conveyor belt			estimation of length: 100m, estimation of lifespan 20years. Yearly production amount: 20'000t	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E-07	m	estimation	1	3.45	(5,1,1,1,4,1);
transportation of limestone meal by lorry				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	5.56E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of limestone meal by train				transport systems	train	No	CH	transport, freight, rail	1.83E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);

Fig. 6.1 Flows for "fibre cement roof slate, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
transportation of polyvinyl alcohol by train	Production of "fibre cement roof slate" (Part 2)			transport systems	train	No	CH	transport, freight, rail	9.36E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of reinforcing fibre by train				transport systems	train	No	RER	transport, freight, rail	2.24E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of pulp fibres by train				transport systems	train	No	RER	transport, freight, rail	4.40E-03	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of cement by train				transport systems	train	No	CH	transport, freight, rail	6.28E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
transportation of reinforcing fibres by freight ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	3.64E-01	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);	
lubricant oil				chemicals	organics	No	RER	lubricating oil, at plant	1.10E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		carbon dioxide reacted in waste water	Calcium oxide in waste water reacts with carbon dioxide from exhaust gas to calcium carbonate (limestone). It must be subtracted (that's why negative) from CO2 emissions from combustion which are included in the modules for boilers. 0.72 kg of CO2 per m3 waste water. Amount of waste water: 0.00154 m3/kg	air	unspecified			Carbon dioxide, fossil	-1.10E-03	kg	calculation	1	1.38	(4,5,1,1,1,5);	
		calcium carbonate to disposal	Calcium carbonate from reaction in waste water (compare above)	waste management	residual material landfill	No	CH	disposal, decarbonising waste, 30% water, to residual material landfill	2.50E-03	kg	calculation	1	1.22	(2,1,1,1,1,5);	
		waste water internally clarified by precipitating calcium carbonate	unpolluted as clarified	waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater treatment, class 3	1.54E-03	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		waste water in sewage treatment plant	concrete production effluent used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	6.80E-06	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		paint in municipal waste incineration		waste management	hazardous waste incineration	No	CH	disposal, emulsion paint remains, 0% water, to hazardous waste incineration	3.41E-03	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		lubricating oil to disposal		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	1.10E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);	
		90% of total solvent: NMVOC to air	share estimated	air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	8.10E-05	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);	
		10% of solvent: chlorinated halogenated hydrocarbons to air	share estimated	air	unspecified			Hydrocarbons, chlorinated	9.00E-06	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);	
		fibre cement roof slate		construction materials	coverings	No	CH	fibre cement roof slate, at plant	1.00E+00	kg					

Fig. 6.1 Flows for "fibre cement roof slate, at plant" and their representation in the ecoinvent database (cont)

Tab. 6.2 Ecoinvent meta information for the product “fibre cement corrugated slab, at plant”

Name	fibre cement corrugated slab, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the whole manufacturing process to produce fibre cement products (raw material provision, raw material mixing, forming, post curing process coating process, and storage), transports to plant processes, and infrastructure. No acid for neutralisation of the wastewater and no requirements for administration are included.
Amount	1
Local Name	Faserzementwellplatte, ab Werk
Synonyms	roofing//sheet//well//dach//tile
General Comment to reference function	The amounts of raw materials are calculated based on the composition of fibre cement roof slate. Air emission data are lacking but emissions from the boilers are imported via the relevant modules. A certain amount of CO ₂ produced by the boiler is bound in the waste water. The amount of this entrained CO ₂ is inventoried with a negative value because it is an avoided emission. The total yearly is assumed to be 20'000 tonnes. The lifespan of the plant and the conveyor belt are assumed to be 50 years and 20 years, respectively.
Start Date	1991
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from one company in Switzerland (Eternit AG), for some exchanges RER- and OCE-modules have been used as proxy
Technology text	The raw materials are mixed with water to a homogeneous paste. Rotating sieving cylinders dip in this fibre cement mush which drain to the inside. Therefore the sieve surface gets covered with a layer of fleece of about 1mm which is transported with a conveyor belt. The produced endless mat is wound up on a format-cylinder until the demanded thickness is reached. With that process the fibres of the neighbour layers are interconnecting. After cutting up the mat cylinder the mat is detached from the format-cylinder and by punching and pressing automatically formed to small format slates, even and corrugated slabs.
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Literature and "ecological building material declaration" (a sort of environmental report)
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
process fibres: pulp fibres	Production of "fibre cement corrugated slab" (Part 1)		raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	1.80E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
process fibres: waste paper			raw material, losses are neglected as internally reused.	paper & cardboard	pulps	No	CH	waste paper, mixed, from public collection, for further treatment	1.80E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
additive: limestone meal			raw material, losses are neglected as internally reused.	construction materials	others	No	CH	limestone, milled, packed, at plant	1.11E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
fixing agent: cement			raw material, losses are neglected as internally reused.	construction materials	binder	No	CH	portland calcareous cement, at plant	7.24E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
reinforcing fibre: polyvinyl alcohol			raw material, losses are neglected as internally reused.	plastics	monomers	No	RER	vinyl acetate, at plant	1.80E-02	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
tap water			raw material, losses are neglected as internally reused.	water supply	production	No	RER	tap water, at user	1.11E-01	kg	SIA building product declaration	1	1.05	(1,1,1,1,1,1);
total electricity consumption for complete process				electricity	supply mix	No	CH	electricity, medium voltage, at grid	2.67E-01	kWh	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 1MW				oil	heating systems	No	CH	light fuel oil, burned in industrial furnace 1MW, non-modulating	2.48E-01	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
energy consumption: light fuel oil in industrial furnace 100kW				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	1.35E-02	MJ	SIA building product declaration	1	1.12	(2,1,3,1,1,1);
tap water			raw material	water supply	production	No	RER	tap water, at user	2.30E+00	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
dispersive acrylate			used for coating	paintings	production	No	RER	acrylic varnish, 87.5% in H2O, at plant	5.20E-03	kg	Weibel & Stritz (1995)	1	1.12	(2,1,3,1,1,1);
packing				construction materials	others	No	CH	packing, fibre cement products	1.00E+00	kg	estimation	1	1.40	(3,5,1,1,3,5);
infrastructure plant			assumption: same infrastructure for 1kg fibre cement product is inquired as for 1kg rock wool product	insulation materials	production	Yes	CH	rock wool plant	4.44E-10	unit	estimation	1	3.50	(5,1,1,1,4,5);
conveyor belt			estimation of length: 100m, estimation of lifespan 20years.	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E-07	m	estimation	1	3.45	(5,1,1,1,4,1);
transportation of limestone meal by lorry				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	4.44E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of limestone meal by train				transport systems	train	No	CH	transport, freight, rail	1.47E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);

Fig. 6.2 Flows for "fibre cement corrugated slab, at plant" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
transportation of polyvinyl alcohol by train	Production of "fibre cement corrugated slab" (Part 2)			transport systems	train	No	CH	transport, freight, rail	6.48E-04	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of reinforcing fibre by train				transport systems	train	No	RER	transport, freight, rail	1.62E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of pulp fibres by train				transport systems	train	No	RER	transport, freight, rail	7.20E-03	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of cement by train				transport systems	train	No	CH	transport, freight, rail	6.52E-02	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
transportation of reinforcing fibres by freight ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	2.52E-01	tkm	Weibel & Stritz (1995)	1	2.06	(2,1,3,1,1,5);
lubricant oil				chemicals	organics	No	RER	lubricating oil, at plant	7.79E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		carbon dioxide reacted in waste water	Calcium oxide in waste water reacts with carbon dioxide from exhaust gas to calcium carbonate (limestone). It must be subtracted (that's why negative) from CO2 emissions from combustion which are included in the modules for boilers. 0.72 kg of CO2 per m3 waste water. Amount of waste water: 0.00115 m3/kg	air	unspecified			Carbon dioxide, fossil	-8.28E-04	kg	calculation	1	1.38	(4,5,1,1,1,5);
		calcium carbonate to disposal	Calcium carbonate from decarbonising in waste water (compare above). The modul is a proxy as the waste is carbonate-rich waste obtained during decarbonisation of water and not waste water.	waste management	residual material landfill	No	CH	disposal, decarbonising waste, 30% water, to residual material landfill	1.96E-03	kg	calculation	1	1.22	(2,1,1,1,1,5);
		waste water Internally clarified by precipitating calcium carbonate	unpolluted as clarified	waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater treatment, class 3	1.15E-03	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		waste water in sewage treatment plant	concrete production effluent used as proxy	waste management	wastewater treatment	No	CH	treatment, concrete production effluent, to wastewater treatment, class 3	1.10E-05	m3	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		paint in sewage treatment plant		waste management	hazardous waste incineration	No	CH	disposal, emulsion paint remains, 0% water, to hazardous waste incineration	5.08E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		lubricating oil to disposal		waste management	hazardous waste incineration	No	CH	disposal, used mineral oil, 10% water, to hazardous waste incineration	7.79E-04	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
		solvent: NMVOC to air		air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	2.52E-04	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);
		solvent: chlorinated halogenated hydrocarbons to air		air	unspecified			Hydrocarbons, chlorinated	2.80E-05	kg	SIA building product declaration	1	1.52	(2,1,3,1,1,1);
		fibre cement corrugated slab		construction materials	coverings	No	CH	fibre cement corrugated slab, at plant	1.00E+00	kg				

Fig. 6.2 Flows for "fibre cement corrugated slab, at plant" and their representation in the ecoinvent database (cont)

Tab. 6.3 Ecoinvent meta information for the product “fibre cement facing tile, large format, at plant”

Name	fibre cement facing tile, large format, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.1
Included Processes	All processes relating to the product at the Eternit production plant in Niederurnen and in Payern are included. Also included are the material and energy input and the waste treatment services corresponding to these processes.
Amount	1
Local Name	Faserzementfassadenplatte, grossformatig, ab Werk
Synonyms	
General Comment to reference function	Data are based on an in depth analysis of the production of Eternit Switzerland. Different products (base colours & coatings) are averaged.
Start Date	
End Date	2007
Data Valid For Entire Period	2008
Other Period Text	1
Geography text	
Technology text	Production in Switzerland for World wide market
Representativeness [%]	Raw tile production, drying, coating
Production Volume	
Sampling Procedure	
Extrapolations	literature
Uncertainty Adjustments	none

General Flow Information			Representation in ecoinvent							Uncertainty Informations					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment	
PE Fleece	fibre cement facing tile, large format, at plant, rough product	waste heat		plastics	polymers	No	RER	fleece, polyethylene, at plant	0.001239	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
CO2 (liquid)				chemicals	inorganics	No	RER	carbon dioxide liquid, at plant	0.002134	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
				air	unspecified			Heat, waste	0.4040034	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Tap water				water supply	production	No	CH	tap water, at user	0.1888	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Inorganic chemicls				chemicals	inorganics	No	GLO	chemicals inorganic, at plant	0.052121305	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	0.007519	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Iron sulphate				metals	extraction	No	RER	iron sulphate, at plant	0.001134	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Polyethylene				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	0.01712175	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Bleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, ECF bleached, at plant	0.0218	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Unbleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	0.0246	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
PVAC				plastics	monomers	No	RER	vinyl acetate, at plant	0.0171225	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
			PE to MWI		waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Inert waste to landfill		waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	0.02212	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Unspecified waste to MWI		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	0.00533	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Hazardous waste to incineration		waste management	hazardous waste incineration	No	CH	disposal, hazardous waste, 25% water, to hazardous waste incineration	0.00014	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Wood to MWI		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	0.00002	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)

Fig. 6.3 Flows for “fibre cement facing tile, large format, at plant” and their representation in the ecoinvent database

General Flow Information			Representation in ecoinvent							Uncertainty Informations				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment
Ground water	fibre cement facing tile, large format, at plant, rough product	➡ Chromium VI ➡ COD		resource	in water			Water, well, in ground	0.002742	m3	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Infrastructure WWT plant				waste management	wastewater treatment	Yes	CH	wastewater treatment plant, class 5	1.825E-08	unit	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Thermal energy from oil				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	0.02341	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	0.48145	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport train Europe				transport systems	train	No	RER	transport, freight, rail	0.21446	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport train Switzerland				transport systems	train	No	CH	transport, freight, rail	0.00367925	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport road				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	0.04843425	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
				water	river			Chromium VI	0.000000073	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Electricity				electricity	supply mix	No	CH	electricity, medium voltage, at grid	0.1122245	kWh	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
				water	river			COD, Chemical Oxygen Demand	0.00037676	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	0.000835	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Portland cement Z42.5				construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	0.821833333	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Portland cement Z52.5				construction materials	binder	No	CH	portland cement, strength class Z 52.5, at plant	0.8157	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Limestone				construction materials	additives	No	CH	limestone, milled, loose, at plant	0.03930459	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Production plant				insulation materials	production	Yes	CH	rock wool plant	4.44E-10	unit	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Limestone				construction materials	others	No	CH	limestone, milled, packed, at plant	0.06797	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Thermal energy from gas				natural gas	heating systems	No	RER	natural gas, burned in boiler modulating >100kW	0.043093895	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Internal transports				transport systems	road	No	RER	operation, van < 3,5t	0.00008	vk	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)

Fig. 6.3 Flows for “fibre cement facing tile, large format, at plant” and their representation in the ecoinvent database (cont)

General Flow Information			Representation in ecoinvent							Uncertainty Informations					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment	
Groundwater	fibre cement facing tile, large format, at plant, rough product			resource	in water			Water, well, in ground	0.0002111	m3	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
PE Fleece				plastics	polymers	No	RER	fleece, polyethylene, at plant	0.000463	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Tap water				water supply	production	No	CH	tap water, at user	0.2128	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Inorganic chemicals				chemicals	inorganics	No	GLO	chemicals inorganic, at plant	0.00283	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	0.00040808	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Polyethylene				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	0.00092888	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Sodium hydroxide				chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	0.0003786	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Bleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, ECF bleached, at plant	0.00118055	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Unbleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	0.0013355	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
PVAC				plastics	monomers	No	RER	vinyl acetate, at plant	0.00092888	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Water to external treatment				waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater treatment, class 3	0.00006464	m3	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
			➡ PE to MWI		waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	0.00154	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			➡ Inert waste to landfill		waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	0.0587	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			➡ Unspecified waste to MWI		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	0.01132	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			➡ Hazardous waste to incineration		waste management	hazardous waste incineration	No	CH	disposal, hazardous waste, 25% water, to hazardous waste incineration	0.00113	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			➡ Wood to MWI		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	0.00003	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)

Fig. 6.3 Flows for “fibre cement facing tile, large format, at plant” and their representation in the ecoinvent database (cont)

General Flow Information			Representation in ecoinvent							Uncertainty Informations				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment
Infrastructure WWT plant	→			waste management	wastewater treatment	Yes	CH	wastewater treatment plant, class 5	0	unit	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Thermal energy from oil				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	0.00484	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Iron chloride				chemicals	inorganics	No	CH	iron (III) chloride, 40% in H ₂ O, at plant	0.0005056	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Transport ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	0.028286	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Transport train Europe				transport systems	train	No	RER	transport, freight, rail	0.0106504	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Transport train Switzerland				transport systems	train	No	CH	transport, freight, rail	0.000204824	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Transport road				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	0.003521898	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Electricity				electricity	supply mix	No	CH	electricity, medium voltage, at grid	0.110375	kWh	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Portland cement Z42.5				construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	0.044876835	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Portland cement Z52.5				construction materials	binder	No	CH	portland cement, strength class Z 52.5, at plant	0.04427	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Limestone				construction materials	additives	No	CH	limestone, milled, loose, at plant	0.00213313	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Limestone				construction materials	others	No	CH	limestone, milled, packed, at plant	0.0036892	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Thermal energy from gas				natural gas	heating systems	No	RER	natural gas, burned in boiler modulating >100kW	1.28483	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Internal transports				transport systems	road	No	RER	operation, van < 3,5t	0.0004729	vk	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Acrylic dispersion				component of colour	paintings	No	RER	acrylic dispersion, 65% in H ₂ O, at plant	0.015978102	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Parafines				component of colour	chemicals	No	RER	paraffin, at plant	0.001091224	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Inorganic chemics				component of colour	chemicals	No	GLO	chemicals inorganic, at plant	0.00160464	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Tap water				component of colour	water supply	No	CH	tap water, at user	0.016352556	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		→ Ammonia to air		air	unspecified			Ammonia	0.0000024	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		→ Heat waste to air		air	unspecified			Heat, waste	0.397	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		→ NMVOC to air		air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	0.000543592	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		→ Large format fibre cement facing tile	unpacked	construction materials	coverings	No	CH	fibre cement facing tile, large format, at plant	1	kg				

Fig. 6.3 Flows for “fibre cement facing tile, large format, at plant” and their representation in the ecoinvent database (cont)

Tab. 6.4 Ecoinvent meta information for the product “fibre cement facing tile, small format, at plant”

Name	fibre cement facing tile, small format, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.1
Included Processes	All processes relating to the product at the Eternit production plant in Niederurnen and in Payern are included. Also included are the material and energy input and the waste treatment services corresponding to these processes.
Amount	1
Local Name	Faserzementfassadenplatte, kleinformig, ab Werk
Synonyms	
General Comment to reference function	Data are based on an in depth analysis of the production of Eternit Switzerland. Different products (base colours & coatings) are averaged.
Start Date	
End Date	2007
Data Valid For Entire Period	2008
Other Period Text	1
Geography text	
Technology text	Production in Switzerland for World wide market
Representativeness [%]	Raw tile production, drying, coating
Production Volume	
Sampling Procedure	
Extrapolations	literature
Uncertainty Adjustments	none

General Flow Information			Representation in ecoinvent							Uncertainty Informations					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment	
PE Fleece	fibre cement facing tile, small format, at plant, rough product	waste heat		plastics	polymers	No	RER	fleece, polyethylene, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
CO2 (liquid)				chemicals	inorganics	No	RER	carbon dioxide liquid, at plant	0.002134	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
				air	unspecified			Heat, waste	0.424	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Tap water				water supply	production	No	CH	tap water, at user	0.1977	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Inorganic chemicls				chemicals	inorganics	No	GLO	chemicals inorganic, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Iron sulphate				metals	extraction	No	RER	iron sulphate, at plant	0.001134	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Polyethylene				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	0.01103	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Bleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, ECF bleached, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Unbleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	0.03056	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
PVAC				plastics	monomers	No	RER	vinyl acetate, at plant	0.016127	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
			PE to MWI		waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Inert waste to landfill		waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	0.0151	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Unspecified waste to MWI		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	0.00533	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Hazardous waste to incineration		waste management	hazardous waste incineration	No	CH	disposal, hazardous waste, 25% water, to hazardous waste incineration	0.00014	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
			Wood to MWI		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	0.00002	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)

Fig. 6.4 Flows for “fibre cement facing tile, small format, at plant” and their representation in the ecoinvent database

General Flow Information			Representation in ecoinvent							Uncertainty Informations				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment
Ground water	fibre cement facing tile, small format, at plant, rough product	Chromium VI		resource	in water			Water, well, in ground	0.002872	m3	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Infrastructure WWT plant				waste management	wastewater treatment	Yes	CH	wastewater treatment plant, class 5	0	unit	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Thermal energy from oil				oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	0.02372	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport ship				transport systems	ship	No	OCE	transport, transoceanic freight ship	0.52601	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport train Europe				transport systems	train	No	RER	transport, freight, rail	0.04557	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport train Switzerland				transport systems	train	No	CH	transport, freight, rail	0.00369	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Transport road				transport systems	road	No	CH	transport, lorry 20-28t, fleet average	0.05038	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
				water	river			Chromium VI	0.000000073	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Electricity				electricity	supply mix	No	CH	electricity, medium voltage, at grid	0.11766	kWh	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
				water	river			COD, Chemical Oxygen Demand	0.00037668	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	0.000835	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Portland cement Z42.5				construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	0.8547	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Portland cement Z52.5				construction materials	binder	No	CH	portland cement, strength class Z 52.5, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Limestone				construction materials	additives	No	CH	limestone, milled, loose, at plant	0.02122	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Production plant				insulation materials	production	Yes	CH	rock wool plant	2.22E-10	unit	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Limestone				construction materials	others	No	CH	limestone, milled, packed, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Thermal energy from gas				natural gas	heating systems	No	RER	natural gas, burned in boiler modulating >100kW	0.10569	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)
Internal transports				transport systems	road	No	RER	operation, van < 3,5t	0.000708	vkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)

Fig. 6.3 Flows for “fibre cement facing tile, large format, at plant” and their representation in the ecoinvent database (cont)

General Flow Information			Representation in ecoinvent							Uncertainty Informations					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment	
Groundwater	fibre cement facing tile, small format, at plant, rough product			resource	in water			Water, well, in ground	0.002339	m3	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
PE Fleece				plastics	polymers	No	RER	fleece, polyethylene, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,4)	
Tap water				water supply	production	No	CH	tap water, at user	0.256143	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Inorganic chemicals				chemicals	inorganics	No	GLO	chemicals inorganic, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Organic chemicals				chemicals	organics	No	GLO	chemicals organic, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Polyethylene				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	0.00126175	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Sodium hydroxide				chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	0.000546	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Bleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, ECF bleached, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Unbleached pulp				paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	0.003497	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
PVAC				plastics	monomers	No	RER	vinyl acetate, at plant	0.00184556	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
			Water to external treatment		waste management	wastewater treatment	No	CH	treatment, sewage, unpolluted, to wastewater treatment, class 3	0.00010664	m3	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			PE to MWI		waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			Inert waste to landfill		waste management	inert material landfill	No	CH	disposal, inert waste, 5% water, to inert material landfill	0.1314	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			Unspecified waste to MWI		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	0.0115	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			Hazardous waste to incineration		waste management	hazardous waste incineration	No	CH	disposal, hazardous waste, 25% water, to hazardous waste incineration	0.00113	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
			Wood to MWI		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	0.00003	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)

Fig. 6.3 Flows for “fibre cement facing tile, large format, at plant” and their representation in the ecoinvent database (cont)

General Flow Information			Representation in ecoinvent							Uncertainty Informations				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment
Infrastructure WWT plant	fibre cement facing tile, small format, at plant, rough product			waste management	wastewater treatment	Yes	CH	wastewater treatment plant, class 5	0	unit	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Thermal energy from oil			oil	heating systems	No	CH	light fuel oil, burned in boiler 100kW, non-modulating	0.00484	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Iron chloride			chemicals	inorganics	No	CH	iron (III) chloride, 40% in H2O, at plant	0.0007288	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Transport ship			transport systems	ship	No	OCE	transport, transoceanic freight ship	0.06006	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Transport train Europe			transport systems	train	No	RER	transport, freight, rail	0.00568	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Transport train Switzerland			transport systems	train	No	CH	transport, freight, rail	0.00042156	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Transport road			transport systems	road	No	CH	transport, lorry 20-28t, fleet average	0.00732526	tkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Electricity			electricity	supply mix	No	CH	electricity, medium voltage, at grid	0.168586	kWh	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Portland cement Z42.5			construction materials	binder	No	CH	portland cement, strength class Z 42.5, at plant	0.09781	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Portland cement Z52.5			construction materials	binder	No	CH	portland cement, strength class Z 52.5, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Limestone			construction materials	additives	No	CH	limestone, milled, loose, at plant	0.00242837	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Limestone			construction materials	others	No	CH	limestone, milled, packed, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Thermal energy from gas			natural gas	heating systems	No	RER	natural gas, burned in boiler modulating >100kW	0.69906	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Internal transports			transport systems	road	No	RER	operation, van < 3,5t	0.003088	vkm	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)	
Acrylic dispersion			component of colour	paintings	production	No	RER	acrylic dispersion, 65% in H2O, at plant	0.012393386	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Parafines			component of colour	chemicals	organics	No	RER	paraffin, at plant	0.000879243	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Inorganic chemicils			component of colour	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
Tap water			component of colour	water supply	production	No	CH	tap water, at user	0.014060284	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		Ammonia to air		air	unspecified			Ammonia	0	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		Heat waste to air		air	unspecified			Heat, waste	0.607	MJ	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		NM VOC to air		air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	0.000667595	kg	Eternit 2008	1	1.05	(1,1,1,1,1,1,2)
		Small format fibre cement facing tile		unpacked	construction materials	coverings	No	CH	fibre cement facing tile, small format, at plant	1	kg			

Fig. 6.3 Flows for “fibre cement facing tile, large format, at plant” and their representation in the ecoinvent database (cont)

Tab. 6.5 Ecoinvent meta information for the product “fibre cement facing tile, at plant”

Name	fibre cement facing tile, at plant
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.1
Included Processes	Establishes a mix of small and large fibre cement facing tiles produced by Eternit Switzerland according to the Swiss consumption mix.
Amount	1
Local Name	Faserzementfassadenplatte, ab Werk
Synonyms	
General Comment to reference function	Swiss consumption mix
Start Date	
End Date	2007
Data Valid For Entire Period	2008
Other Period Text	1
Geography text	
Technology text	Production in Switzerland for World wide market
Representativeness [%]	Raw tile production, drying, coating
Production Volume	
Sampling Procedure	
Extrapolations	literature
Uncertainty Adjustments	none

General Flow Information			Representation in ecoinvent								Uncertainty Informations				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment	
Large scale facing tiles	fibre cement facing tile, at plant	market mix, fibre cement facing tile	0.7	construction materials	coverings	No	CH	fibre cement facing tile, large format, at plant	0.7	kg	Personal communication Mr. Gätzi, Eternit AG	1	1.05	(1,1,1,1,1,1,4)	
small scale facing tiles			0.3	construction materials	coverings	No	CH	fibre cement facing tile, small format, at plant	0.3	kg	Personal communication Mr. Gätzi, Eternit AG	1	1.05	(1,1,1,1,1,1,4)	
				construction materials	coverings	No	CH	fibre cement facing tile, at plant	1	kg					

Fig. 6.5 Flows for “fibre cement facing tile, at plant” and their representation in the ecoinvent database

Tab. 6.6 Ecoinvent meta information for the product “packing fibre cement products”

Name	packing fibre cement products
Location	CH
Infrastructure Process	0
Unit	kg
Data Set Version	2.0
Included Processes	Includes the process of packing, transportation, and loading of the bags on paletts. The machines, the waste treatment after use by the end-user of the fibre cement products and the transports of the packing materials are included.
Amount	1
Local Name	Verpacken, Faserzementprodukte
Synonyms	
General Comment to reference function	The production amount used for normalization is 20'000t/a. This module is assumed to be valid also for plants up to 100'000t/a. The lifespan of the infrastructure and the land occupation are assumed to be 50 years.
Start Date	2000
End Date	2001
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data from a company producing lime products in Switzerland are adjusted
Technology text	Assumed that packaging is the same as for lime products
Representativeness [%]	
Production Volume	Unknown
Sampling Procedure	Assumption
Extrapolations	See geography
Uncertainty Adjustments	None

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
infrastructure packing and loading of fibre cement products	packing and loading of fibre cement products (including energy consumption) (Part 1)		assumption: same machines as for "packing of lime products": 30t, estimated lifespan: 25years; total amount of packed material (2001): 20000 t/a.	construction processes	machinery	Yes	RER	industrial machine, heavy, unspecified, at plant	6.00E-05	kg	estimation based on observation	1	3.23	(5,na,na,na,na,na);
conveyor belt			assumption: same length of conveyor belt as for "packing of lime products": 10m	construction processes	machinery	Yes	RER	conveyor belt, at plant	2.50E-08	m	estimation based on observation	1	3.23	(5,na,na,na,na,na);
electricity consumption for packing and loading			assumption: same amount as used for module "packing lime products"	electricity	supply mix	No	CH	electricity, medium voltage, at grid	3.09E-03	kWh	data of KFN for the year 2001	1	1.05	(1,2,1,1,1,1);
pallets, at plant			assumption: same amount as used for module "packing lime products", estimated total weight of load on one pallet: 1 t, weight of pallet: 22 kg, number of applications: 10	wooden materials	processing	No	RER	EUR-flat pallet	1.00E-04	unit	estimation	1	1.44	(4,5,1,1,3,5);
steel band for packing (steel)				metals	extraction	No	RER	steel, low-alloyed, at plant	1.10E-03	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
steel band for packing (rolling)				metals	processing	No	RER	section bar rolling, steel	1.10E-03	kg	Weibel & Stritz (1995)	1	1.24	(2,1,3,1,1,5);
paperbag, at plant				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	7.87E-03	kg	estimation	1	1.24	(1,4,1,2,1,5);
plastic foil, at plant			assumption: same amount as used for the module "packing lime products" has been adapted to the loading weight of fibre cement products (1'000kg), circumference of pallet: 4m; height: 1m, 2 layer of plastic with a weight of 0.0123 kg/m2	plastics	processing	No	RER	packaging film, LDPE, at plant	9.84E-05	kg	estimation	1	1.21	(4,na,na,na,na,na);
transports of pallets, paper bags and plastic bags to plant by train			assumption: same amount as used for the module "packing lime products", standard distance: transport by train with distance of: 200km	transport systems	train	No	CH	transport, freight, rail	6.00E-03	tkm	standard distance	1	2.23	(5,na,na,na,na,na);
transports of pallets, paper bags and plastic bags to plant by lorry			assumption: same amount as used for the module "packing lime products", standard distance: transport by lorry 28t with distance of: 50km	transport systems	road	No	CH	transport, lorry 20-28t, fleet average	1.50E-03	tkm	standard distance	1	2.23	(5,na,na,na,na,na);
transports of paper bags and plastic bags to municipal incineration			assumption: same amount as used for the module "packing lime products", assumption: transports all by lorry 16t and total of transportdistance: 10km	transport systems	road	No	RER	transport, lorry 3.5-16t, fleet average	7.99E-05	tkm	data of KFN for the year 2001	1	1.50	(5,na,na,na,na,na);

Fig. 6.6 Flows for "packing fibre cement products" and their representation in the ecoinvent database

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
packing and loading of fibre cement products (including energy consumption) (Part 2)		→ waste heat	assumption: same amount as used for the module "packing lime products", same amount as electricity used	air	unspecified			Heat, waste	1.11E-02	MJ		1	1.05	(1,2,1,1,1,1);
		→ disposal pallets		waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	2.93E-03	kg		1	1.24	(1,4,1,2,1,5);
		→ paperbag, in municipal waste incineration	assumption: same amount as used for the module "packing lime products"	waste management	municipal incineration	No	CH	disposal, paper, 11.2% water, to municipal incineration	7.87E-03	kg	data of KFN for the year 2001	1	1.24	(1,4,1,2,1,5);
		→ plastic foil, in municipal waste incineration	assumption: same amount as used for the module "packing lime products"	waste management	municipal incineration	No	CH	disposal, polyethylene, 0.4% water, to municipal incineration	1.23E-04	kg	data of KFN for the year 2001	1	1.21	(4,na,na,na,na,na);
		→ packing and loading of 1 kg fibre cement product	process of packing 1 kg of fibre cement product	construction materials	others	No	CH	packing, fibre cement products	1.00E+00	kg				

Fig. 6.6 Flows for “packing fibre cement products” and their representation in the ecoinvent database (cont)

In general this dataset has been copied from the lime packing dataset. Some adjustments were made to account for different densities.

7 Data Quality

The uncertainties are estimated using the pedigree matrix (see Frischknecht et al. (2007)). For detailed results compare the different flows and the ecoinvent database meta information.

Most data for fibre cement roof slate and fibre cement corrugated slab production are directly taken from Weibel & Stritz (1995). These data are based on an inventory of one Swiss company for year 1991 (with some additional data for 1994). They give very little background information to their inventories. Thus a check whether the adaptation to the ecoinvent database modules is made correctly is not possible. The total amount of energy consumption and the raw materials for the different fibre cement products are adjusted based on the SIA building materials declaration²⁶. The year of publication of this information is 2002. No data are available on the composition of the wastewater.

In general both used reports do not show details on the sub-processes included in the data. Therefore a combination of the two above-mentioned reports may not be fully consistent. It was not possible to get up-to-date data from the only Swiss producer Eternit AG.

For all these reasons the data these types of fibre cement products are of poor overall quality and should not be used to cross compare these different materials or for direct comparison to other materials in the same function. In a larger context (e.g., use for LCA of buildings) and if the importance of these products is not too high for the application, the data may be used.

Data for fibre cement facing tile based on an in depth study of the production process of Eternit (Niederurnen) in 2008. These data are of excellent quality.

8 References

- | | |
|--------------------------------|--|
| Frischknecht et al. (2007) | Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Hirschier R., Hellweg S., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent 2000 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch . |
| Kasser & Pöll (1998) | Kasser U. and Pöll M. (1998) Graue Energie von Baustoffen; Daten zu Baustoffen, Bauchemikalien, Verarbeitungs- und Transportprozessen mit Erläuterungen und Empfehlungen für die Baupraxis. 2. vollständig neu überarbeitete Aufl. Edition. Büro für Umweltchemie, Zürich. |
| Kull et al. 2008a | Kull, T., Althaus, H.-J., Reinhard, J., von Däniken, A. (2008) Ökobilanz von Faserzement Fassadenplatten. Bericht. Ecosense AG, Wallisellen |
| Kull et al. 2008b | Kull, T., Althaus, H.-J., Reinhard, J., von Däniken, A. (2008) Ökobilanz von Faserzement Fassadenplatten. Anhänge. Ecosense AG, Wallisellen |
| Starzner & Wurmer-Weiss (2000) | Starzner S. and Wurmer-Weiss P. (2000) ECOBIS 2000, ökologisches Baustoffinformationssystem. |
| Weibel & Stritz (1995) | Weibel T. and Stritz A. (1995) Ökoinventare und Wirkungsbilanzen von Baumaterialien; Grundlagen für den ökologischen Vergleich von Hochbaukonstruktionen. In: ESU-Reihe Nr. 1/95. Laboratorium für Energiesysteme, Eidg. Technische Hochschule (ETH), Zürich. |

²⁶ <http://www.sia.ch/cf/oekoinfoliste.cfm?ID=2943702&Lang=d&Mfg=72&Grp=0&Produkt=>, access date: Mai 2003

7 Report "Life Cycle Inventories for Rail Transport"

Autor: Matthias Tuchschnid, SBB

Review: Christian Bauer, PSI

Last Changes: 2010 [updated chapter 6 of Spielmann et al. 2007]

In this chapter we present and discuss the data employed for the rail transportation modules and outline the foundations and arguments for methodological choices made. In some cases we also give further information on alternative data available or choices possible.

7.1 Goal and Scope

The overall objective of the transport modeling is to supply sets of highly aggregated environmental interventions due to rail transport services to complete the assessment of energy and materials life cycles.

7.1.1 Functional Unit

In order to relate transport models to life cycles of other products and services, the environmental interventions are related to the reference unit of one tonne kilometer [tkm]. A tonne kilometer by rail is defined as follows: "Unit of measure of goods transport which represents the transport of one tonne by a train over one kilometer" EUROSTAT (2000). Exchanges and environmental interventions of passenger transport are expressed in passenger kilometer [pkm].

Goods transportation by rail: Europe, Switzerland, Germany, France, Italy, Austria, Belgium

In this study goods transportation is modeled to represent the average Swiss conditions and also the average conditions in the countries of Germany, France, Italy, Austria and Belgium. Additionally average goods transportation in Europe is shown.

Passenger rail transport in Switzerland

Three types of passenger rail transport are distinguished in this project: a) Swiss long distance rail traffic b) Swiss regional rail traffic and c) Swiss metropolitan train (S-Bahn). Long distance rail traffic is performed by trains as follows: Intercity, Eurocity, CIS/ICE/TGV, Interregio, and Express trains. For regional traffic the following types of trains are in operation: S-Trains; RegioExpress and various regional trains. Due to the significant higher load factor of the metropolitan train (S-Bahn), this subtype of regional rail traffic is also modeled.

Passenger rail transport in Germany, France, Italy, Austria and Belgium

The passenger transport in the five European countries is modeled as average train passenger transport, there is no differentiation between regional rail traffic and long distance rail traffic. Additionally the high speed rail traffic in Germany, Italy and France is included in the ecoinvent database. If not stated explicitly, data and assumptions for the high speed rail traffic are directly taken from von Rozycki (2003).

Thus, the following components of passenger transport are available from this project:

- a) Regional train transport with SBB electricity mix in Switzerland
- b) Metropolitan train transport with SBB electricity mix in Switzerland
- c) Long distance transport with SBB electricity mix in Switzerland
- d) Average train transport in Germany (electricity mix from Germany, not DB)
- e) High Speed train transport in Germany (electricity mix from Germany, not DB)
- f) Average train transport in France (electricity mix from France, not SNCF)
- g) High Speed train transport in France (electricity mix from France, not SNCF)
- h) Average train transport in Italy (electricity mix from Italy, not FS)
- i) High Speed train transport in Italy (electricity mix from Italy, not FS)
- j) Average train transport in Austria (electricity mix from Austria, not ÖBB)
- k) Average train transport in Belgium (electricity mix from Belgium, not SNCB)

7.1.2 System Boundaries

Environmental interventions of rail transportation can be classified in interventions due to direct processes and indirect processes. In this project we used this classification as a point of departure for the structuring the system under investigation and the data collection. Direct processes merely address environmental interventions due to rail operation. Indirect processes are further divided into two components a) rail equipment and b) rail infrastructure expenditures. Consequently, the system „rail transport“ is divided in three components:

- Rail Operation
- Rail Equipment
- Rail Infrastructure

Rail Operation: This component contains all processes that are directly connected with the operation of trains including shunting processes.

Goods Transport Equipment: Rail equipment is further divided in locomotives and wagons (often referred to as cars) and contains all processes that are connected with the vehicle life cycle (excluding the operation) such as manufacturing, maintenance as well as disposal.

Rail Infrastructure: This transport component comprises rail track construction, maintenance and disposal.

The second and third component summarise processes with numerous interfaces to other ecoinvent modules (materials and energy). Thus, the actual environmental interventions of these processes are calculated when these data are linked to the referring processes in the ecoinvent database, and are often referred to as indirect environmental interventions of transportation.

7.1.3 Reference Units and Key Figures

Values for energy consumption and environmental emissions are available for various reference units. In general a distinction can be made between load independent and load dependent reference units. In Fig. 7.1 different types of reference units are illustrated.

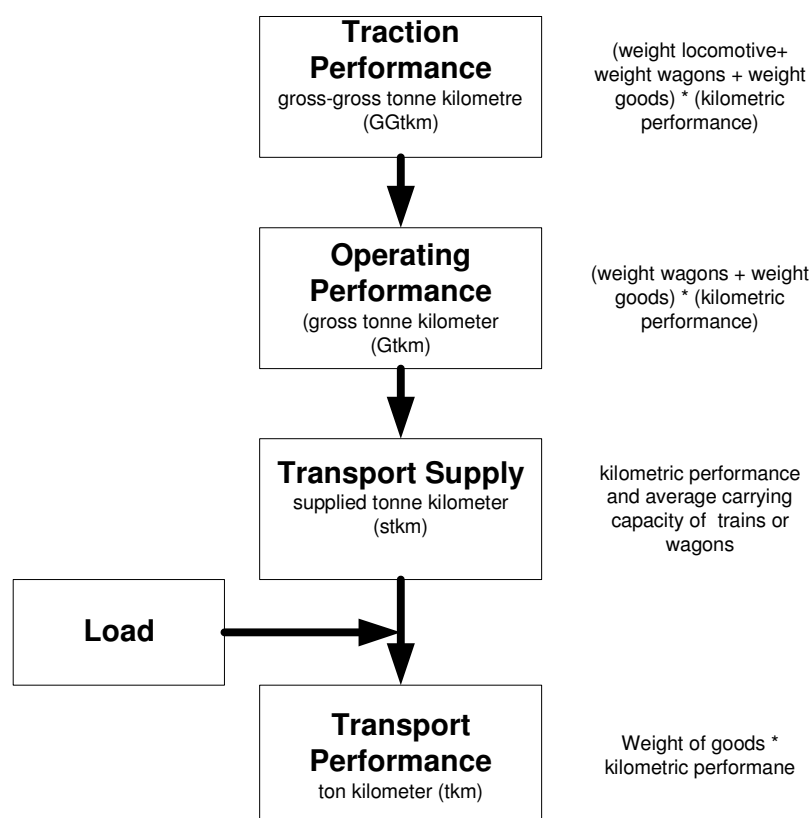


Fig. 7.1 Steps in the transport performance formation and referring reference units

Furthermore, a distinction between fare-tkm and net-tkm must be made. The latter refers to the actual kilometric performance, whilst fare-tkm refer to the actually paid kilometric performance. Between these two reference values a difference up to 100% is possible²⁷. In this project, we use the term tkm and refer to the net tkm.

Switzerland

In Tab. 7.1 and Tab. 7.2 key figures and relevant to calculate the specific environmental performance of the Swiss rail transport system are summarized.

Tab. 7.1 Key figures (2000) of Swiss goods transportation excluding private rail companies ((SBB, 2002) and (BfS, 2000))

Indicator	Unit	Value
Train kilometre (passenger and goods transportation)	Mio. vkm	130.995
Traction performance all trains (passenger and goods transportation)	Mio. Gtkm	59449.535
Traction performance goods transportation	Mio. Gtkm	24331.407
Goods train kilometric performance (train kilometer)	Mio. vkm	28
Transport performance good trains	Mio. tkm	10786
Gross tonne per freight train	Gt	824
Ratio gross tonne/carried goods	Gt/t	2.40

²⁷ Oral communication with Helmut Kuppelwieser, SBB Environmental Management

Tab. 7.2 Key figures (2008) of Swiss passenger transportation excluding private rail companies, the calculation of the infrastructure is based on the performance values of the year 2000 (oral communication with H. Kuppelwieser (SBB), the values in brackets are not used for calculation)

Indicator	Unit	Value for the year 2000	Value for the year 2008
Regional passenger train kilometric performance	Mio. vkm	(53)	59.89
Long distance passenger train kilometric performance	Mio. vkm	(49)	71.53
Traction performance regional trains	Mio. Gtkm/a	11498.682	
Traction performance long distance trains	Mio. Gtkm/a	23833.449	
Operating performance regional trains	Mio. Gtkm/a	7172.375	
Operating performance long distance trains	Mio. Gtkm/a	17873.245	
Supplied Seats regional train	Seats/train	(315)	309
Supplied Seats long distance train	Seats/train	(627)	616
Average load factor regional train	%	(17)	18.4
Average load factor long distance train	%	(28)	32.7
Transport performance regional train ¹⁾	Mio. pkm/a	2839.275	(4069)
Transport performance long distance train ¹⁾	Mio. pkm/a	8522.560	(12047)
Ratio grosstonne / passenger regional train ²⁾			3.39
Ratio grosstonne / passenger long-distance train ²⁾			2.42

1) own calculations

2) Oral communication with M. Halder (Environmental Center SBB)

Tab. 7.3 Key figures (2008) of Swiss passenger metropolitan train (S-Bahn Zürich)

Indicator	Unit	Value	Source
Transport performance	millions pkm	1916	ZVV, 2008
train performance	millions vkm	20.2	ZVV, 2008
average load factor ¹⁾	people in train	94.9	
number of trains	Fahrzeuge	251	ZVV, 2008
average train performance per year	vkm/a	165000	Menth, 2007
weight of metropolitan train (Re450 / DPZ)	Tonnen	220	Menth, 2007
operation performance ¹⁾	Btkm	4444000000	
load per passenger ¹⁾		2.32	
energy consumption	Wh / Btkm	32.4	Menth, 2007
energy consumption per vkm ¹⁾	kWh/Fzkm	7.128	
energy consumption per pkm ¹⁾	kWh/pkm	0.075	
losses due to electricity transport ¹⁾		10%	
operation of metropolitan train ("S-Bahn") ¹⁾	kWh/pkm	0.0837	

1) own calculations

Europe

The availability of data required for the calculation of specific environmental interventions of the European Rail system is limited and the data available fairly uncertain. In Tab. 7.4 some key figures are summarised.

Tab. 7.4 Key figures for European goods transportation

Country	National	International	Total	National share on EU transport	Total
	Mio tkm/a	Mio tkm/a	Mio tkm/a	%	Mio Gtkm/a ¹⁾
Belgium	2166	4903	7069	3.2	14138
Denmark	476	n.a.	476	0.2	952
Germany	35763	30084	65847	30.0	131694
Greece	121	191	312	0.1	624
Spain	9174	2148	11322	5.2	22644
France	32267	16241	48508	22.1	97016
Ireland	469	n.a.	469	0.2	938
Italy	11415	10992	22407	10.2	44814
Luxembourg	104	n.a.	104	0.0	208
Netherlands	763	3015	3778	1.7	7556
Austria	3093	7990	11083	5.1	22166
Portugal	1638	410	2048	0.9	4096
Finland	6313	3572	9885	4.5	19770
Sweden	18635	4803	23438	10.7	46876
UK	12440	n.a.	12440	5.7	24880
EU-15	134837	84349	219186	100.0	438372

1: Calculated with a gross-tons/net-tons ratio of 2.

n.a.: for the reference year 1998 no data was available

7.2 Operation of trains

7.2.1 Operation of Goods Trains in Switzerland

The module operation of goods trains comprises both, the actual performance of goods trains as well as required shunting processes. For a technical and ecological consideration of rail transport services, a distinction between the type of traction, diesel and electricity is important. In Switzerland, however, diesel locomotives are almost exclusively used for shunting purposes.

Traction Energy

The data presented in the database represents average environmental interventions, calculated top down from the yearly energy consumption and yearly transport performance. For electricity consumption of freight trains, yearly average data is available from the SBB (SBB, 2002). These data is linked to reference values as presented in Tab. 7.1 in order to calculate specific electricity consumption. In Tab. 7.5 these figures are summarised. More specific data, addressing different types of train compositions is available from other studies. (Borken et al. (1999); Maibach et al. (1999)).

Tab. 7.5 Specific electricity consumption at current collector for traction of goods trains

Indicator	Unit	Value
End-electricity consumption per train	kWh/(Mio vkm)	1.25E+07
Total electricity consumption	kWh	1.64E+09
Good transport total electricity consumption	kWh	6.70E+08
Specific electricity consumption goods transportation in 2000	kWh/tkm	(0.0620)
Specific electricity consumption goods transportation in 2006 ²⁸	kWh/tkm	(0.0669)
Specific electricity consumption goods transportation in 2008 ²⁹	kWh/tkm	0.0709

The actual consumption depends on numerous impact factors such as:

- Gradient of the tracks
- Type of good, i.e. relation between operation performance and transport performance
- Train length and total weight
- Speed and acceleration

In an alpine country as Switzerland, this can result in a significantly higher specific electricity consumption compared to fairly flat countries such as the Netherlands or Germany. Borken (2003), for instance, calculate a final energy consumption for an average train (1000 Gt) and average load factor (0.5 net tons-per gross tons) of 0.034 kWh/tkm in a flat area and 0.051 kWh/tkm in a mountain region. The remaining difference between the latter value and the SBB value may be explained with the lower average load factor of the SBB in 2000 (0.42 t/Gt).

Transmission losses (transmission ex traction substation 4% and transmission and conversion ex high voltage network 6%) are accounted for with 10%, resulting in a final energy consumption of 0.07804 kWh/tkm for the SBB (reference year 2006).

Airborne Emissions

Due to the fact that on the Swiss rail network only very few diesel locomotives are in operation emissions to air are limited to particulate emissions due to abrasion and emission of sulphur hexafluoride (SF₆) occurring during conversion at traction substations.

Source of abrasion are rail tracks, wheels, brakes, overhead contact line and exhaust emissions of shunting locomotives. In Tab. 7.6 PM₁₀ emission factors as available from literature are summarised. Particulate emission due to abrasion are currently issue of intensive studies in Switzerland³⁰. The figures presented here, therefore should only be considered as rough estimates. For the determination of PM₁₀ emission factors for brakes, wheels, rail and overhead contacting line, we employ the figures as presented in Tab. 7.6. For the determination of PM_{>10} we assume a share of 25 % for wheel and rail abrasion. The abrasion of brakes and the resulting share of PM₁₀ depend on the wheel load and thus differ between fully loaded goods trains and empty goods train (see Tab. 7.7). The figure employed in this project represents an average based on the load factor of 0.42 for the Swiss average freight train.

²⁸ Email from Markus Halder, BahnUmwelt-Center SBB, 17th of april 2007

²⁹ Oral communication with Markus Halder, BahnUmwelt-Center SBB, 17th of november 2009

³⁰ Oral communication with Helmut Kuppelwieser, SBB Environmental Management

Tab. 7.6 PM10 emission of rail transportation

Source of Abrasion	Buwal (2002)		(SBB, 2001)		This project	
	Total abrasion	Assumed Share PM10	Total abrasion	Assumed Share PM10	Total abrasion	Assumed Share PM10 (goods transport)
	t/a	%	t/a	%	t/a	
Brakes	2066	100	250-475	12 ¹⁾ – 21 ²⁾	363	17
Wheels	546	50	-		546	50
Rail	124	50	-		124	50
overhead contact line	36	100	-		36	100
Shunting locomotive exhaust emissions	45	100	-			
Total	2816				1069	

1: figure for a fully loaded freight train (SBB, 2002)

2: figure for an empty freight train (SBB, 2002)

Tab. 7.7 Size distribution of braking line abrasion

Abrasion	Size distribution empty goods train and passenger train (wheel load: 2.5t) ¹⁾	Size distribution fully loaded goods train (wheel load: 11.25t) ¹⁾	This project (0.42 load factor)
	%	%	%
Non airborne emissions	56	28	44
PM10	21	12	17
PM>10	23	60	39
Total Abrasion	100	100	100

1: (SBB, 2002)

The essential allocation between goods and passenger transport is performed by employing the traction performance as allocation factor. The specific emission factor is derived by division of the overall transport performance of goods trains (10786 millionen tkm). In Tab. 7.8 the resulting figures are summarised.

Tab. 7.8 Specific emission factors of particulates due to abrasion

	unit	Total abrasion	PM 10 ¹⁾	PM >10
Yearly load	t/a	1069		
Allocated yearly road to passenger transport	t/a	633		
Allocated yearly load goods transport		436	74.1	170.0
Specific emission factor	Kg/tkm	4.04E-05	6.87E-06	1.58E-05

1) the entire load of PM10 is classified as coarse particulates (a diameter > 2.5 µm and < 10µm)

For sulphur hexafluoride (SF6) emitted during conversion at traction substations we assume an emission index of 4.4E-08 kg/kWh Frischknecht (2003) resulting in a specific emission of 3.79E-09 kg/tkm.

Emissions to Soil

Abrasion (from braking lining, rail and wheel) is predominately composed of iron and mineral components BUWAL (2002). In this study we assume that all non-airborne emissions are emitted as iron to soil. In consequence, we assume a yearly load of 191.8 t/a iron emitted to soil. Emission of lubricates due to traction are not accounted for, since measures are in operation to avoid such emissions.

Diesel Consumption and Emissions of Shunting Processes

Diesel Consumption

A total diesel consumption of 8.695 Mio litres (in the year 2000) is reported for the SBB (SBB, 2002). In this study we assume that all diesel is consumed for shunting, neglecting the consumption of diesel for the lorry fleet and diesel aggregates. Furthermore we assume that 100% of the shunting processes for goods trains is performed with diesel locomotives. Consequently, we obtain a specific diesel consumption of 0.68 g/tkm. In Tab. 7.9 a summary of specific diesel consumption values for shunting as available in literature are presented.

Tab. 7.9 Summary of specific diesel consumption values for shunting processes

	Frischknecht et al. (1996) ¹	Knörr et al. (2000) ²	This project
Specific Diesel Consumption for shunting [g/tkm]	0.93	0.68	0.68

1: in this study Gtkm is used as a reference value. The data presented in this table has been calculations based on the assumptions made in the referring study and SBB data from 1985. 100% of the diesel shunting is allocated to goods transportation.

2: in this study, tkm (supplied tonne kilometre) is used as a reference value. The given value (0.25 g/tkm) has been transformed based on the assumption of an average load factor of 37%. 90% of the diesel shunting is allocated to goods transportation.

We assume the same diesel consumption of 0.68 g/tkm for shunting processes also for the goods transport in the other European countries and the average European transport.

Emissions to Air

In Knörr (2000), shunting specific emission indices are presented. These emission factors are representative for German conditions in 1995 and demonstrated in Tab. 7.10. Obviously these indices merely differ marginal from those of diesel traction in Germany J. Borken et al. (2003), apart from SO₂ emissions, which strongly depend on the sulphur of the fuel. In addition, other emission indices as available from literature are presented in Tab. 7.10.

In this project CO₂ and SO₂ indices are estimated assuming that all carbon and sulphur in the fuel is transformed completely into CO₂ and SO₂, respectively. In line with assumptions made for road transport we assume a sulphur content of 300 ppm resulting in a emission index of 0.6 g/kg. For CO₂ we employ an emission index of 73.5 g/MJ or 3146 g/kg. The list of emission indices is enhanced with pollutants derived from heavy good vehicle emission factors for urban traffic conditions. These values reflect the Swiss average heavy good vehicle in urban traffic conditions and are derived from INFRAS (1999). The figures in the last column of Tab. 7.10 present the figures employed in this study.

Tab. 7.10 Emission indices for diesel locomotives

Emission	Unit	Knörr ¹⁾	Borken ²⁾	Joergensen ⁶⁾	Ecoinvent 96 ³⁾	DB ⁴⁾	SNCF ⁴⁾	HGV ⁵⁾	This project
CO ₂	g/kg	3175	3170	3180	3145	3.175	3150	3117.00	3146.00
CO	g/kg	15.8	n.a.	22	10	n.a.	n.a.	6.61	15.80
HC ⁷⁾	g/kg	2.5	5.0	11	4	1.9	4.9	5.37	-
Nox	g/kg	55	55	53	44	55.4	39.6	35.10	55.00
PM	g/kg	1.4	1.5	3	3.8	2	1.5	1.82	1.39
SO ₂	g/kg	3	0.7	4	2.6	0.08	0.7	0.60	0.6
N ₂ O	g/kg	n.a.	n.a.	n.a.	0.1	n.a.	n.a.	0.10	0.10
NMVO ⁸⁾	g/kg	n.a.	4.9	n.a.	3.8	1.8	4.7	5.25	5.25
CH ₄	g/kg	n.a.	0.1	n.a.	0.2	0.1	0.2	0.13	0.13
Benzene	g/kg	n.a.	n.a.	n.a.	0.001	n.a.	n.a.	0.10	0.10
NH ₃	g/kg	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.02	0.02
Toluol	g/kg	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.04	0.04
Xylol	g/kg	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.04	0.04

1) figures represent shunting specific emission indices Knörr et al. (2000)

2) average emission indices for diesel traction J. Borken et al. (2003)

3) Frischknecht et al. (1996)

4) country specific emission factors taken from J. Borken et al. (2003). DB: Germany; DSB: Denmark; SNCF: France.

5) average emission factors for Swiss heavy good vehicles (HGV) for urban traffic conditions in 2000 derived from INFRAS (1999).

6) average emission factors for diesel railway locomotives including passenger and freight locomotives Joergensen & Sorenson (1997)

7) the presented figures of HC emissions comprise CH₄ and NMVOC emissions

8) the presented figures of NMVOC emissions still contain emissions of Benzene, Toluol and Xylol

In Tab. 7.11 the assumptions and resulting figures for particulate emissions employed in this project are summarized.

Tab. 7.11 Particle size distributions of exhaust emissions and exhaust emission factors [g/kg] for diesel locomotives without particle filters (EU-Norm IIIa)

Vehicle Category	Assumptions			Emission factors for different size classes [µm]		
	PM ₁₀ emission factor [g/kg] ¹	Fraction of PM ₁₀ with a diameter < 2.5 µm	Fraction of TSP with diameter < 10 µm	Fine Particles (< 2.5)	Coarse Particles (2.5-10)	Large Particles (>10)
Diesel locomotive	1.39	92.3	96.2	1.28E+00	5.42E-02	5.28E-02

Particle filters in locomotives

Within the diesel fleet of the Swiss Federal railway (SBB) one third of the diesel lokomotives are equipped with particle filters according to the EU-legislation Nr. 2004/26/EG (EU, 2004). These

locomotives³¹ are responsible for 65% of the needed traction power. The legislation determines a value of 0.025g PM / kWh or 0.3g PM / kg Diesel.

Tab. 7.12 Particle size distributions of exhaust emissions and exhaust emission factors [g/kg] for diesel locomotives with particle filters (EU-Norm IIIb)

Vehicle Category	Assumptions			Emission factors for different size classes [µm]		
	PM10 emission factor [g/kg] ¹	Fraction of PM10 with a diameter < 2.5 µm	Fraction of TSP with diameter < 10 µm	Fine Particles (< 2.5)	Coarse Particles (2.5-10)	Large Particles (>10)
Diesel locomotive	0.297	92.3	96.2	2.74E-01	1.16E-02	1.13E-02

Tab. 7.13 Particle size distributions of exhaust emissions and exhaust emission factors [g/kg] for average SBB diesel locomotives with a share of 65% equipped locomotives with particle filters

Vehicle Category	Assumptions			Emission factors for different size classes [µm]		
	PM10 emission factor [g/kg] ¹	Fraction of PM10 with a diameter < 2.5 µm	Fraction of TSP with diameter < 10 µm	Fine Particles (< 2.5)	Coarse Particles (2.5-10)	Large Particles (>10)
Diesel locomotive	0.680	92.3	96.2	6.27E-01	2.65E-02	2.58E-02

Tab. 7.14 presents emission factors expressed in mg/kg fuel for cadmium, copper, chromium, nickel, selenium and zinc, as provided by the Expert Panel on Heavy Metals and POPs of the UNECE Task Force on Emission Inventories EEA (2000). However, these emission factors have to be considered as preliminary estimates only. More measurements are needed in order to confirm these values. Values for mercury and lead are taken from Jungbluth (2003).

Tab. 7.14 Heavy metal emission indices for Diesel locomotives

Cadmium [g/kg fuel]	Copper [g/kg fuel]	Chromium [g/kg fuel]	Nickel [g/kg fuel]	Selenium [g/kg fuel]	Zinc [g/kg fuel]	Lead [g/kg fuel]	Mercury [g/kg fuel]
1,0E-05	1,7E-03	5,0E-05	7,0E-05	1,0E-05	1,0E-03	1,1E-07	2,0E-08

³¹ Mainly the locomotives of Am843, Am843d and Tm 232

Life Cycle Inventory Input Data

In Tab. 7.15 the input data for the entire unit process operation is presented.

Tab. 7.15 Life cycle inventory table for operation of Swiss goods trains

Explanations	Name	Location	Infrastructure-Process	Unit	operation, freight train		uncertainty type	StandardDeviation95%	GeneralComment
					CH 0 tkm	CH 0 tkm			
Technosphere	diesel, at regional storage	CH	0	kg	6.80E-4	1.07E-2	1	1.09	(2,3,1,3,1,1); International Rail study
	electricity, high voltage, SBB, at grid	CH	0	kWh	8.61E-2	0	1	1.09	(2,3,1,3,1,1); International Rail study
air, unspecified	Benzene	-	-	kg	6.80E-8	1.07E-6	1	2.06	(2,3,1,3,3,1); handbook emission factors for HGV
	Methane, fossil	-	-	kg	8.84E-8	1.39E-6	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
	Carbon monoxide, fossil	-	-	kg	1.07E-5	1.69E-4	1	5.01	(2,3,1,3,1,1); International Rail study
	Carbon dioxide, fossil	-	-	kg	2.14E-3	3.37E-2	1	1.09	(2,3,1,3,1,1); International Rail study
	Dinitrogen monoxide	-	-	kg	6.80E-8	1.07E-6	1	1.57	(2,3,1,3,3,1); International Rail study
	Ammonia	-	-	kg	1.36E-8	2.13E-7	1	1.57	(2,3,1,3,3,1); International Rail study
	NM VOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	3.45E-6	5.42E-5	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
	Nitrogen oxides	-	-	kg	3.74E-5	5.88E-4	1	1.51	(2,3,1,3,1,1); International Rail study
	Particulates, > 10 um	-	-	kg	1.58E-5	1.59E-5	1	1.52	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, < 2.5 um, and < 10um	-	-	kg	6.89E-6	7.00E-6	1	2.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, < 2.5 um	-	-	kg	4.27E-7	2.94E-6	1	3.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Sulfur dioxide	-	-	kg	4.08E-7	6.42E-6	1	1.09	(2,3,1,3,1,1); European railway study
	Toluene	-	-	kg	2.72E-8	4.28E-7	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
	Xylene	-	-	kg	2.72E-8	4.28E-7	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
	Mercury	-	-	kg	1.36E-14	2.13E-13	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Lead	-	-	kg	7.48E-14	1.18E-12	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Cadmium	-	-	kg	6.80E-12	1.07E-10	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Copper	-	-	kg	1.16E-9	1.82E-8	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Chromium	-	-	kg	3.41E-11	5.36E-10	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Nickel	-	-	kg	4.76E-11	7.49E-10	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Selenium	-	-	kg	6.80E-12	1.07E-10	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Zinc	-	-	kg	6.80E-10	1.07E-8	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Sulfur hexafluoride	-	-	kg	3.79E-9	0	1	1.63	(3,5,1,2,3,1); literature
	Heat, waste	-	-	MJ	3.10E-1	4.58E-1	1	1.07	(2,1,1,1,1,1); standard value
soil, unspecified	Iron	-	-	kg	1.78E-5	1.78E-5	1	3.00	(-, -, -, -, -, 1); own calculation based on literature
products	operation, freight train	CH	0	tkm	1.00E+0				
	operation, freight train, diesel, with particle filter	CH	0	tkm		1.00E+0			

Data Quality and uncertainties

The quality of the electricity value is considered as good. Due to the fact that it has been obtained top down, uncertainties due to assumption about load factors are excluded. The diesel consumption of shunting processes is considered as less good, since further assumptions have been made that pose additional uncertainties due to choices on the diesel consumption figure. The quality of emission factors for shunting varies, and is addressed with basic uncertainties and data quality indicators such as reliability, technology correlation and representativeness.

7.2.2 Operation of Goods Trains in selected European countries (Germany, France, Italy, Austria, Belgium)

The module operation of goods trains comprises both; the actual performance of goods trains as well as required shunting processes. In contrast to operation in Switzerland, a distinction between the type of traction, diesel and electricity is important since in most European countries diesel and electro locomotives are in operation for both goods transportation and passenger transportation.

Energy consumptions of goods trains

The international railway statistics from UIC (UIC, 2007) shows the energy consumption for good traffic as well as the typical performance figures.

Tab. 7.16 Energy consumption for traction and typical performance figures

	Diesel	electricity	Traffic performance	Transport performance	Train performance
	[1000t]	[GWh]	[Mio. Btkm]	[Mio. tkm]	[1000 Zugkm]
Austria	14.0	908.0	44'232	19'840	46'491
Italy	8.0	832.0	39'674	24'432	59'980
France	0.0 ^{a)}	3'122.6	97'615	42'434	95'385
Germany	130.0	3'173.0	205'113	92'077	205'234
Belgium	28.0	268.0	18'450	9'357	14'308

- a) In the UIC-statistics, no differentiation for diesel consumption has been made for goods and passenger transport. In consideration of the shown traffic performance, one assumed that regular goods trains operate only with electricity. Nevertheless, the diesel consumption for shunting is considered.

With the transport performance factors, one may calculate the specific energy consumption per tkm. The typical diesel consumption of 0.68g/tkm for shunting has been added to the specific energy demand.

Tab. 7.17 Specific energy consumption

	electricity energy consumption	Diesel energy consumption for traction	Diesel energy consumption for shunting	total energy consumption, incl. shunting ^{a)}	total energy consumption, incl. shunting per Gtkm
Unit	[kWh/tkm]	[kg/tkm]	[kg/tkm]	[kWh-equ./tkm]	[kWh-equ./Gtkm]
Austria	0.046	7.1E-04	6.8E-04	0.062	0.028
Italy	0.034	3.3E-04	6.8E-04	0.046	0.028
France	0.074	0.0E+00	6.8E-04	0.082	0.036
Germany	0.034	1.4E-03	6.8E-04	0.059	0.027
Belgium	0.029	3.0E-03	6.8E-04	0.072	0.037
Switzerland	0.078	0.0E+00	6.8E-04	0.086	0.038

- a) The aggregation in the last two columns has been made for illustrative purposes, only the values in the first three columns are used in the life cycle inventory data in Table 6-17. The aggregation assumed energy content of 42.8 MJ/kg Diesel, one has to consider the lower efficiency of diesel locomotives.

Airborne Emissions

Airborne emission indices due to diesel traction as available from literature are summarised in Tab. 7.10. The indices employed in this project are presented in the last column. For heavy metal emissions we employ the figures as presented in section 7.2.1.

Emissions to Soil

Iron emissions to soil are accounted for employing the figure for Swiss operation services. Emission of lubricates due to traction are not accounted for, since measures are in operation to avoid such emissions.

Diesel Consumption and Emissions of Shunting Processes

For energy consumption of shunting processes assumptions and data as presented in chapter 7.2.1 is applied.

Life Cycle Inventory Input Data

In Tab. 7.18 the inventory tables for the five modules in from each country are summarised.

Tab. 7.18 Life cycle inventory table for operation of goods trains in Europe

Explanations	Name	Location	Unit	operation, freight train					uncertainty Type Standard Deviation %	General Comment
				operation, freight train	operation, freight train	operation, freight train	operation, freight train	operation, freight train		
				AT 0 tkm	IT 0 tkm	FR 0 tkm	DE 0 tkm	BE 0 tkm		
Technosphere	diesel, at regional storage	RER	0	1.39E-3	1.01E-3	6.80E-4	2.09E-3	3.67E-3	1	1.09 (2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	AT	0	6.22E-2					1	1.09 (2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	IT	0		4.60E-2				1	1.09 (2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	FR	0			8.17E-2			1	1.09 (2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	DE	0				5.93E-2		1	1.09 (2,3,1,3,1,1); International Rail study
air, unspecified	electricity, high voltage, at grid	BE	0					7.23E-2	1	1.09 (2,3,1,3,1,1); International Rail study
	Benzene	-	kg	1.39E-7	1.01E-7	6.80E-8	2.09E-7	3.67E-7	1	2.06 (2,3,1,3,1,1); handbook emission factors for HGV
	Methane, fossil	-	kg	1.80E-7	1.31E-7	8.84E-8	2.72E-7	4.77E-7	1	1.57 (2,3,1,3,1,1); handbook emission factors for HGV
	Carbon monoxide, fossil	-	kg	2.19E-5	1.59E-5	1.07E-5	3.31E-5	5.80E-5	1	5.01 (2,3,1,3,1,1); International Rail study
	Carbon dioxide, fossil	-	kg	4.36E-3	3.17E-3	2.14E-3	6.58E-3	1.16E-2	1	1.09 (2,3,1,3,1,1); International Rail study
	Dinitrogen monoxide	-	kg	1.39E-7	1.01E-7	6.80E-8	2.09E-7	3.67E-7	1	1.57 (2,3,1,3,1,1); International Rail study
	Ammonia	-	kg	2.76E-8	2.01E-8	1.36E-8	4.17E-8	7.32E-8	1	1.57 (2,3,1,3,1,1); International Rail study
	NM VOC, non-methane volatile organic compounds, unspecified origin	-	kg	7.02E-6	5.10E-6	3.45E-6	1.06E-5	1.86E-5	1	1.57 (2,3,1,3,1,1); handbook emission factors for HGV
	Nitrogen oxides	-	kg	7.61E-5	5.54E-5	3.74E-5	1.15E-4	2.02E-4	1	1.51 (2,3,1,3,1,1); International Rail study
	Particulates, > 10 um	-	kg	1.58E-5	1.58E-5	1.58E-5	1.59E-5	1.60E-5	1	1.52 (3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, > 2.5 um, and < 10um	-	kg	6.95E-6	6.93E-6	6.91E-6	6.98E-6	7.07E-6	1	2.02 (3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, < 2.5 um	-	kg	1.78E-6	1.29E-6	8.72E-7	2.68E-6	4.71E-6	1	3.02 (3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Sulfur dioxide	-	kg	8.31E-7	6.04E-7	4.08E-7	1.25E-6	2.20E-6	1	1.09 (2,3,1,3,1,1); European railway study
	Toluene	-	kg	5.55E-8	4.03E-8	2.72E-8	8.37E-8	1.47E-7	1	2.06 (3,3,1,3,1,1); handbook emission factors for HGV
	Xylene	-	kg	5.55E-8	4.03E-8	2.72E-8	8.37E-8	1.47E-7	1	2.06 (3,3,1,3,1,1); handbook emission factors for HGV
	Mercury	-	kg	2.76E-14	2.01E-14	1.36E-14	4.17E-14	7.32E-14	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
	Lead	-	kg	1.52E-13	1.11E-13	7.48E-14	2.30E-13	4.04E-13	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
	Cadmium	-	kg	1.39E-11	1.01E-11	6.80E-12	2.09E-11	3.67E-11	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
	Copper	-	kg	2.35E-9	1.71E-9	1.16E-9	3.55E-9	6.24E-9	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
	Chromium	-	kg	6.94E-11	5.04E-11	3.41E-11	1.05E-10	1.84E-10	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
	Nickel	-	kg	9.70E-11	7.05E-11	4.76E-11	1.46E-10	2.57E-10	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
soil, unspecified	Selenium	-	kg	1.39E-11	1.01E-11	6.80E-12	2.09E-11	3.67E-11	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
	Zinc	-	kg	1.39E-9	1.01E-9	6.80E-10	2.09E-9	3.67E-9	1	5.10 (4,5,1,2,1,1); expert estimate quoted from literature
	Sulfur hexafluoride	-	kg	2.74E-9	2.03E-9	3.59E-9	2.61E-9	3.18E-9	1	1.63 (3,5,1,2,3,1); literature
	Heat, waste	-	MJ	2.24E-1	1.66E-1	2.94E-1	2.14E-1	2.60E-1	1	1.07 (2,1,1,1,1,1); standard value
	Iron	-	kg	1.78E-5	1.78E-5	1.78E-5	1.78E-5	1.78E-5	1	3.00 (-,-,-,-,-,1); own calculation based on literature
products	operation, freight train	AT	0	1.00E+0						
	operation, freight train	IT	0		1.00E+0					
	operation, freight train	FR	0			1.00E+0				
	operation, freight train	DE	0				1.00E+0			
	operation, freight train	BE	0					1.00E+0		

Data Quality and Uncertainty

The data quality of the energy consumption for the modeled train types is considered as good. However, some uncertainties arise due to the selected load factor of 0.5. As demonstrated in the

previous chapter, for Switzerland this figure is not realistic. Further uncertainties arise due to the application of the average country specific electricity mixes for electro traction. Some rail companies have their own electricity generation plants (e.g. about 20% of the German traction energy is produced by the DB itself). The emission factors are based on emission indices obtained from literature studies and complemented with average lorry emissions. The different quality for various emissions to air is addressed with the basic uncertainty for each pollutant and reliability and representativeness quality indicator scores.

7.2.3 Operation of average Goods Trains in Europe

The module operation of goods trains comprises both; the actual performance of goods trains as well as required shunting processes. In contrast to operation in Switzerland, a distinction between the type of traction, diesel and electricity is important since in most European countries diesel and electro locomotives are in operation for both goods transportation and passenger transportation.

First, we determine the energy consumption for electric traction and diesel tractions separately. Second, we estimate specific energy consumption for a mixed electricity and diesel traction representing European goods transportation conditions in 2000.

Energy Consumptions of Electric trains

In a recent study, specific energy consumption figures for electric trains are calculated for different train weights and geographical conditions J. Borken et al. (2003). A typical average gross weight for international trains is assumed to be 1000 Gt. In addition, specific electricity consumption for a short train and a long train with an average weight of 500 Gt and 1500 Gt has been calculated. In Tab. 7.19 the resulting figures for each country are summarized.

Tab. 7.19 Specific final energy consumption for electric goods trains of different length

	Short (500 t)	Average (1000t)	Long (1500 t)
	Wh/tkm	Wh/tkm	Wh/tkm
Belgium	60.4	42.7	34.9
Denmark	48.3	34.2	27.9
Germany	60.4	42.7	34.9
Greece	60.4	42.7	34.9
Spain	60.4	42.7	34.9
France	60.4	42.7	34.9
Ireland	60.4	42.7	34.9
Italy	60.4	42.7	34.9
Luxembourg	60.4	42.7	34.9
Netherlands	48.3	34.2	27.9
Austria	72.4	51.2	41.8
Portugal	60.4	42.7	34.9
Finland	60.4	42.7	34.9
Sweden	48.3	34.2	27.9
UK	60.4	42.7	34.9
EU-15	58.9	41.6	34.0

In this study we use the EU-15 figure for an average train and further assume an energy loss due to transformation of 15% (see section 0). Thus, we achieve a final energy consumption including losses of 0.0478 KWh/tkm.

Energy Consumption of Diesel Trains

Borken (2003), stated that the available energy data for diesel traction ranges between 2.6 and 9.7 g/Gtkm. For a train of 1000 Gt in a hilly country they proposed a specific energy consumption of approximately 5 g/Gtkm or 10g/tkm. The later value is used in this study.

European Mix of Diesel and Electro Traction

Data for a European average goods train, comprising a share of both types of energy, diesel and electricity, was not readily available. Thus, some assumptions have been essential, which are described in this section.

Data for the energy consumption for rail traction is available from EUROSTAT (2001) and displayed in Tab. 7.20.

Tab. 7.20 Energy consumption of main fuels for European rail transport

	diesel	electricity	electricity	Changes diesel (85-98)	Changes electricity (85-98)
	tons/a	tons/a ¹⁾	GWh/a ¹⁾	%	%
Belgium	6.30E+04	1.17E+05	1.36E+03	-45	15
Denmark	7.90E+04	2.60E+04	3.02E+02	-32	119
Germany	6.24E+05	1.38E+06	1.61E+04	30	44
Greece	4.20E+04	1.50E+04	1.74E+02	-19	433
Spain	4.55E+05	3.22E+05	3.74E+03	150	33
France	4.54E+05	9.55E+05	1.11E+04	-8	46
Ireland	1.02E+05	2.00E+03	2.33E+01	135	100
Italy	1.92E+05	6.76E+05	7.86E+03	0	62
Luxembourg	5.00E+03	7.00E+03	8.14E+01	-44	85
Netherlands	0.00E+00	1.40E+05	1.63E+03	0	47
Austria	0.00E+00	1.88E+05	2.19E+03	0	-1
Portugal	4.70E+04	3.10E+04	3.60E+02	-18	37
Finland	5.40E+04	4.40E+04	5.12E+02	-25	43
Sweden	1.60E+04	2.41E+05	2.80E+03	-81	7
UK	5.89E+05	6.45E+05	7.50E+03	-21	54
EU-15	2.72E+06	4.79E+06	5.57E+04	-1	49

1: conversion factor used: 1GWh = 86 tons diesel

Whilst the data distinguishes between electricity consumption and diesel consumption, no distinction between goods and passenger transport for each energy type is available. Furthermore, the electricity data also includes urban transport systems. In consequence, these data cannot be easily used to determine the specific energy consumption for European freight rail transport. Data for the share of electricity and diesel traction in goods transportation, however, is available for the German rail company (DB), which is responsible for 30% of the total European goods transportation, as demonstrated in Tab. 7.4. In the recent environmental report DB (2000) figures for the primary energy consumption for electric and diesel traction are published and summarized in Tab. 7.21.

Tab. 7.21 Total primary energy consumption for electric and diesel traction in Germany

Type of traction	Unit	Type of Transport	1998	1999	2000	Ratio passenger/goods transport
Electric traction	GWh/a	passenger transport	23135	23165	23151	0.69
		goods transport	10189	10269	9911	0.31
		total	33324	33434	33062	1.00
Diesel traction	GWh/a	passenger transport	5576	4982	5170	0.75
		goods transport	1876	1924	1670	0.25
		total	7452	6906	6840	1.00

Primary energy consumption is treated as a key indicator of the DB and accounts for the generation of electricity and fuels besides the utilizable final energy, which is required for this project. Thus, the share of final energy on the primary energy for both diesel and electricity has to be determined. As stated in DB (2000), the primary energy is derived using the TREMOD computer model Knörr et al. (2000). In a recent publication, the final energy consumption share of the primary energy is published J. Borken et al. (2003) and demonstrated in Tab. 7.22. Based on these figures the specific final energy consumption can be determined. The resulting figures are also summarized in Tab. 7.22.

Tab. 7.22 Final energy consumption for diesel and electrify consumption and specific consumption factors

	Unit	Electric traction	Unit	Diesel traction
Ratio final energy/primary energy ¹⁾	%	0.27	%	0.89
Total final energy consumption (2000)	GWh/a	2775	kg/a	127821
Specific final energy consumption ²⁾	KWh/tkm	0.0344	kg/tkm	0.0016
Spec. final energy (incl. Transform. Losses) ³⁾	KWh/tkm	0.0396	kg/tkm	0.0016

1: derived from J. Borken et al. (2003)

2: based on a transport performance of 80634 Mio. tkm/a in the reference year 2000 DB (2000)

3: loss of 15 %

In the absence of further information for the remaining European countries we employ the German values for the representation of goods transportation conditions in Europe. However, it should be bear in mind that the share of diesel and electricity may vary significantly in the various European countries.

Airborne Emissions

Airborne emission indices due to diesel traction as available from literature are summarised in Tab. 7.10. The indices employed in this project are presented in the last column. For heavy metal emissions we employ the figures as presented in section 7.2.1.

Emissions to Soil

Iron emissions to soil are accounted for employing the figure for Swiss operation services. Emission of lubricates due to traction are not accounted for, since measures are in operation to avoid such emissions.

Diesel Consumption and Emissions of Shunting Processes

For energy consumption of shunting processes assumptions and data as presented in chapter 7.2.1 is applied.

Life Cycle Inventory Input Data

In Tab. 7.23 the inventory tables for the three modules, "operation freight train diesel", "operation freight train electricity" and "operation freight train" are summarised. The latter module comprises both diesel and electro traction, and represents European average conditions.

Tab. 7.23 Life cycle inventory table for operation of goods trains in Europe

Explanations	Name	Location	Infrastructure-Process	Unit	operation, freight train, diesel	operation, freight train, electricity	operation, freight train	uncertainty Type	StandardDeviation 95%	GeneralComment
					RER 0 tkm	RER 0 tkm	RER 0 tkm			
Technosphere	diesel, at regional storage	RER	0	kg	1.07E-2	6.77E-4	2.26E-3	1	1.30	(2,3,1,3,3,1); International Rail study
	electricity, high voltage, production UCTE, at grid	UCTE	0	kWh	0	4.78E-2	3.96E-2	1	1.30	(2,3,1,3,3,1); International Rail study
air, unspecified	Benzene			kg	1.07E-6	6.77E-8	2.26E-7	1	2.06	(2,3,1,3,3,1); handbook emission factors for HGV
	Methane, fossil			kg	1.39E-6	8.80E-8	2.94E-7	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
	Carbon monoxide, fossil			kg	1.69E-4	1.07E-5	3.57E-5	1	5.01	(2,3,1,3,1,1); International Rail study
	Carbon dioxide, fossil			kg	3.36E-2	2.13E-3	7.12E-3	1	1.09	(2,3,1,3,1,1); International Rail study
	Dinitrogen monoxide			kg	1.07E-6	6.77E-8	2.26E-7	1	1.57	(2,3,1,3,3,1); International Rail study
	Ammonia			kg	2.14E-7	1.35E-8	4.52E-8	1	1.57	(2,3,1,3,3,1); International Rail study
	NM VOC, non-methane volatile organic compounds, unspecified origin			kg	5.41E-5	3.43E-6	1.15E-5	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
	Nitrogen oxides			kg	5.87E-4	3.72E-5	1.24E-4	1	1.51	(2,3,1,3,1,1); International Rail study
	Particulates, > 10 um			kg	1.63E-5	1.58E-5	1.59E-5	1	1.52	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, > 2.5 um, and < 10um			kg	7.44E-6	6.91E-6	6.99E-6	1	2.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, < 2.5 um			kg	1.37E-5	8.69E-7	2.90E-6	1	3.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Sulfur dioxide			kg	6.41E-6	4.06E-7	1.36E-6	1	1.09	(2,3,1,3,1,1); European railway study
	Toluene			kg	4.27E-7	2.71E-8	9.05E-8	1.00	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
	Xylene			kg	4.27E-7	2.71E-8	9.05E-8	1.00	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
	Mercury			kg	2.14E-13	1.35E-14	4.52E-14	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Lead			kg	1.17E-12	7.45E-14	2.49E-13	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Cadmium			kg	1.07E-10	6.77E-12	2.26E-11	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Copper			kg	1.82E-8	1.15E-9	3.85E-9	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Chromium			kg	5.34E-10	3.39E-11	1.13E-10	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Nickel			kg	7.47E-10	4.74E-11	1.58E-10	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Selenium			kg	1.07E-10	6.77E-12	2.26E-11	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Zinc			kg	1.07E-8	6.77E-10	2.26E-9	1.00	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Sulfur hexafluoride			kg	0	2.10E-9	1.74E-9	1.00	1.63	(3,5,1,2,3,1); literature
	Heat, waste			MJ	4.85E-1	2.03E-1	2.45E-1	1.00	1.07	(2,1,1,1,1,1); standard value
soil, unspecified	Iron			kg	6.02E-5	6.02E-5	6.02E-5	1.00	3.00	; own calculation based on literature
products	operation, freight train, diesel	RER	0	tkm	1.00E+0					
	operation, freight train, electricity	RER	0	tkm		1.00E+0				
	operation, freight train	RER	0	tkm			1.00E+0			

Data Quality and Uncertainty

The data quality of the energy consumption for the modeled train types is considered as good. However, some uncertainties arise due to the selected load factor of 0.5. As demonstrated in the previous chapter, for Switzerland this figure is not realistic. Further uncertainties arise due to the application of an European electricity mix for electro traction. Some rail companies have their own electricity generation plants (e.g. about 20% of the German traction energy is produced by the DB itself). The emission factors are based on emission indices obtained from literature studies and complemented with average lorry emissions. The different quality for various emissions to air is

addressed with the basic uncertainty for each pollutant and reliability and representativeness quality indicator scores.

7.2.4 Operation of Goods Trains in China

Three different types of freight transport are defined to reflect the use of electric, diesel, and steam locomotives in the coal industry and in China in general:

- "operation, coal freight train, diesel (CN)", 70% of total coal transport by train;
- "operation, coal freight train, electricity (CN)", 25% of total coal transport by train; and,
- "operation, coal freight train, steam (CN)", 5% of total coal transport by train; mostly own mine trains, for transports within large coal mining areas and/or from mines to shunting/loading centres for long-range transports by national railways.

The 2004 China official statistics on the number of locomotives in operation in all railway Companies (national and local) give the shares 69.7%, 28.7% and 1.5%, respectively. These shares have been modified to reflect the likely specific use of steam locomotives for coal transport for some tens kilometres on the average.

Tab. 7.24 through Tab. 7.26 show the three datasets corresponding to the different transport systems. The diesel technology is identical as described in the corresponding dataset for RER but assuming the consumption of diesel per tkm derived from Chinese statistics. However, the original diesel consumption per tkm has been corrected because it is most likely incompatible being only one quarter of the value taken from European statistics assumed inecoinvent. The correction factor of 2.67 for fuel intensity is the product of two separate sub-factors: 2 to take into account of the fact that coal is transported only one way, and 1/3 to consider the weight of the freight car. In fact the Chinese statistics give on the same page the energy use per tkm, the "Utilization Rate of Loading Capacity of Freight Cars" at an astonishing 98% (definitively unrealistic because coal makes 40% of the total transported goods), and the total mass of loaded freight car "Static Load of Freight Cars (Standard Gauge)" of nearly 60 tonnes of which roughly 20 tonnes must be the car itself. With this correction, also applied to the electric and steam locomotive, the fuel utilization factor comes closer to the RER data. The uncertainty in the value due to the mentioned discrepancy has been signalled with an increased uncertainty factor compared to the corresponding item in the corresponding European dataset.

The electric train operation is also based upon the corresponding RER dataset in. However, also in this case the Chinese official statistics give a much lower energy use than the European ones, 3.6 times lower electricity consumption per tkm. Again, the original value has been increased by a factor of 2.67 and the uncertainty factor has been increased compared to the corresponding item in (Spielmann et al. 2004). In this dataset, the diesel consumption for diesel locomotives used in shunting has accordingly been decreased from the European value.

The steam locomotive dataset is modelled in first approximation by assuming a factor of 4 in fuel (equivalent) utilization compared with a diesel locomotive, based on consideration of likely actual efficiencies (mix of first generation steam locomotives with typical drawbar thermal efficiency of approximately 7% and some second generation with superheating with efficiency up to 15%). Further correction has been applied using the factor 2.67 explained above.

Emissions from the steam locomotive have been roughly estimated adjusting the emission rates (per MJ) from "hard coal, burned in coal mine power plant (CN)" by the average LHV (24 MJ/kg bituminous coal assumed to be used for locomotives vs. 12 MJ/kg poor quality coal for small power plants). Additionally, to somewhat take into account the less efficient combustion characteristics, emissions rates have been modified upwards using a factor 2 greater than the corresponding ones (per MJ) from a small power plants, except CO₂, CH₄, NO_x, and PM. For the latter, although coal for steam locomotives has a presumably lower ash content (say approximately 25% of coal weight)

compared with coal for small coal mine plants (up to 50% of coal weight (Dones et al. 2003)), the retention efficiency is likely lower. Considering also that these locomotives are used mostly within Coal Bureaus, therefore in already highly polluted areas, and that pictures of some steam locomotive show dark and abundant plumes, the PM emission is likely high. Due to the assumptions on ash content, the disposal of ashes has been halved for the locomotive compared to small plants.

Due to the several approximations described above, the CN datasets for transport systems cannot be used for comparison of transport means in different countries, and should be used only when transport is not a major contributor to total environmental burdens of a Chinese process or product being investigated with LCA.

Tab. 7.24 Dataset "operation, coal freight train, diesel (CN)".

Name	Location	Unit	operation, coal freight train, diesel	Uncertainty Type	Standard Deviation 95%	General Comment
Location			CN			
InfrastructureProcess			1			
Unit			tkm			
diesel, at regional storage	RER	kg	2.50E-03	1	3	own assumption; original dataset for Europe 1.09 (2,3,1,3,1,1); International Rail study
Heat, waste		MJ	1.13E-01	1	1.07	(2,1,1,1,1,1); standard value
Ammonia		kg	5.00E-08	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Benzene		kg	2.50E-07	1	2.06	(2,3,1,3,3,1); handbook emission factors for HGV
Cadmium		kg	2.50E-11	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Carbon dioxide, fossil		kg	7.85E-03	1	1.09	(2,3,1,3,1,1); International Rail study
Carbon monoxide, fossil		kg	3.95E-05	1	5.01	(2,3,1,3,1,1); International Rail study
Chromium		kg	1.25E-10	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Copper		kg	4.25E-09	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Dinitrogen monoxide		kg	2.50E-07	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Lead		kg	2.73E-13	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Mercury		kg	5.00E-14	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Methane, fossil		kg	3.25E-07	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Nickel		kg	1.75E-10	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Nitrogen oxides		kg	1.37E-04	1	1.51	(2,3,1,3,1,1); International Rail study
NMVOC, non-methane volatile organic compounds, unspecified origin		kg	1.26E-05	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Particulates, < 2.5 um		kg	3.20E-06	1	3.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
Particulates, > 10 um		kg	7.22E-06	1	1.52	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
Particulates, > 2.5 um, and < 10um		kg	9.91E-06	1	2.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
Selenium		kg	2.50E-11	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Sulfur dioxide		kg	1.50E-06	1	1.09	(2,3,1,3,1,1); International Rail study
Sulfur hexafluoride		kg	0.00E+00	1	1.63	(3,5,1,2,3,1); literature
Toluene		kg	9.98E-08	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
Xylene		kg	9.98E-08	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
Zinc		kg	2.50E-09	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Iron		kg	1.41E-05	1	3	(-, -, -, -, -, 1); own calculation based on literature

Tab. 7.25 Dataset "operation, coal freight train, electricity (CN)".

Name	Location	Unit	operation, coal freight train, electricity	Uncertainty Type	Standard Deviation 95%	General Comment
Location			CN			
InfrastructureProcess			1			
Unit			tkm			
electricity, medium voltage, at grid	CN	kWh	1.11E-02	1	2	own assumption (original dataset for Europe 1.09)
diesel, at regional storage	RER	kg	6.77E-04	1	2	own assumption (original dataset for Europe 1.09)
Heat, waste		MJ	7.07E-02	1	1.07	(2,1,1,1,1,1); standard value
Ammonia		kg	1.35E-08	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Benzene		kg	6.77E-08	1	2.06	(2,3,1,3,3,1); handbook emission factors for HGV
Cadmium		kg	6.77E-12	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Carbon dioxide, fossil		kg	2.13E-03	1	1.09	(2,3,1,3,1,1); International Rail study
Carbon monoxide, fossil		kg	1.07E-05	1	5.01	(2,3,1,3,1,1); International Rail study
Chromium		kg	3.38E-11	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Copper		kg	1.15E-09	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Dinitrogen monoxide		kg	6.77E-08	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Lead		kg	7.40E-14	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Mercury		kg	1.35E-14	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Methane, fossil		kg	8.79E-08	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Nickel		kg	4.73E-11	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Nitrogen oxides		kg	3.71E-05	1	1.51	(2,3,1,3,1,1); International Rail study
NM VOC, non-methane volatile organic compounds, unspecified origin		kg	3.42E-06	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
Particulates, < 2.5 um		kg	8.67E-07	1	3.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
Particulates, > 10 um		kg	1.96E-06	1	1.52	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
Particulates, > 2.5 um, and < 10um		kg	2.68E-06	1	2.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
Selenium		kg	6.77E-12	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Sulfur dioxide		kg	4.06E-07	1	1.09	(2,3,1,3,1,1); International Rail study
Sulfur hexafluoride		kg	2.10E-09	1	1.63	(3,5,1,2,3,1); literature
Toluene		kg	2.70E-08	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
Xylene		kg	2.70E-08	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
Zinc		kg	6.77E-10	1	5.1	(4,5,1,2,1,1); expert estimate quoted from literature
Iron		kg	3.81E-06	1	3	(3,3,1,3,1,1); own calculation based on literature

Tab. 7.26 Dataset "operation, coal freight train, steam (CN)".

Name	Location	Unit	operation, coal freight train, steam	Uncertainty Type	Standard Deviation 95%	General Comment
<i>Location</i>			<i>CN</i>			
<i>InfrastructureProcess</i>			<i>1</i>			
<i>Unit</i>			<i>tkm</i>			
hard coal, at mine	CN	kg	2.63E-02	1	2	own assumption (original dataset for Europe 1.09)
Heat, waste		MJ	4.65E-01	1	1.39	increased by a factor 1.3 from the corresponding uncertainty in "hard coal, burned in coal mine power plant"
Antimony		kg	2.49E-09	1	7.80	"
Arsenic		kg	1.52E-08	1	7.80	"
Barium		kg	1.53E-07	1	7.80	"
Benzene		kg	2.73E-07	1	3.25	"
Benzo(a)pyrene		kg	2.52E-13	1	3.25	"
Boron		kg	8.83E-07	1	7.80	"
Bromine		kg	7.19E-07	1	3.51	"
Butane		kg	2.39E-08	1	3.25	"
Cadmium		kg	1.46E-09	1	7.80	"
Carbon dioxide, fossil		kg	5.86E-02	1	1.45	"
Carbon monoxide, fossil		kg	2.52E-05	1	7.80	"
Chromium		kg	1.44E-08	1	7.80	"
Chromium VI		kg	1.78E-09	1	5.20	"
Cobalt		kg	5.95E-09	1	7.80	"
Copper		kg	1.92E-08	1	7.80	"
Dinitrogen monoxide		kg	3.15E-07	1	2.60	"
Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin		kg	8.82E-15	1	5.20	"
Ethane		kg	5.17E-08	1	3.25	"
Formaldehyde		kg	7.31E-08	1	3.25	"
Hydrocarbons, aliphatic, alkanes, unspecified		kg	2.76E-07	1	3.25	"
Hydrocarbons, aliphatic, unsaturated		kg	2.72E-07	1	3.25	"
Hydrogen chloride		kg	3.85E-05	1	3.90	"
Hydrogen fluoride		kg	3.44E-06	1	3.90	"
Iodine		kg	3.68E-07	1	3.90	"
Lead		kg	7.97E-08	1	7.80	"
Lead-210		kBq	1.53E-04	1	7.80	"
Manganese		kg	7.60E-08	1	7.80	"
Mercury		kg	4.03E-09	1	7.80	"
Methane, fossil		kg	6.30E-07	1	3.25	"
Molybdenum		kg	2.71E-09	1	7.80	"
Nickel		kg	4.20E-08	1	7.80	"
Nitrogen oxides		kg	2.57E-04	1	2.60	"
PAH, polycyclic aromatic hydrocarbons		kg	1.26E-09	1	5.20	"
Particulates, < 2.5 um		kg	6.33E-03	1	5.20	"
Particulates, > 10 um		kg	1.58E-03	1	3.25	"
Particulates, > 2.5 um, and < 10um		kg	7.45E-04	1	4.55	"
Pentane		kg	1.85E-07	1	3.25	"
Polonium-210		kBq	2.80E-04	1	6.50	"
Potassium-40		kBq	3.76E-05	1	6.50	"
Propane		kg	4.41E-08	1	3.25	"
Propene		kg	2.02E-08	1	3.25	"
Radium-226		kBq	3.95E-05	1	6.50	"
Radium-228		kBq	1.17E-05	1	6.50	"
Radon-220		kBq	8.19E-04	1	6.50	"
Radon-222		kBq	4.61E-04	1	6.50	"
Selenium		kg	1.25E-08	1	7.80	"
Strontium		kg	1.39E-07	1	7.80	"
Sulfur dioxide		kg	1.58E-03	1	2.60	"
Thorium-228		kBq	6.30E-06	1	6.50	"
Thorium-232		kBq	9.90E-06	1	6.50	"
Toluene		kg	1.37E-07	1	3.25	"
Uranium-238		kBq	3.29E-05	1	6.50	"
Vanadium		kg	3.51E-08	1	7.80	"
Xylene		kg	1.16E-06	1	3.25	"
Zinc		kg	9.70E-08	1	7.80	"
disposal, hard coal ash, 0% water, to residual material landfill	PL	kg	8.39E-03	1.00	3.00	own assumption

7.2.5 Operation of Goods Trains in the USA

According to (Facanha & Horvath 2006) it can be assumed that US freight trains for coal transport are all operated using diesel engines. As a first approximation the corresponding European diesel freight transport dataset is used.

7.2.6 Operation of Swiss Passenger Trains

In Switzerland, passenger transport is performed almost exclusively by electric powered trains. In contrast to goods transportation no diesel locomotives are used for shunting processes in passenger transport.

Traction Energy

The data presented in the database represents average environmental interventions, calculated top down from the yearly energy consumption and yearly transport performance. For electricity consumption of regional and long distance trains, yearly average data is available from the SBB (SBB, 2002, see also Footnote 32). These data is linked to reference values as presented in Tab. 7.2 in order to calculate specific electricity consumption. Tab. 7.27 and Tab. 7.28 these figures are summarised.

Tab. 7.27 Electricity consumption of regional traffic

Indicator	Unit	Value
Total consumption regional transport	Mio. kWh	486.3
Specific consumption operation performance	kWh/Gtkm	0.042
Specific consumption kilometric performance	kWh/vkm	9.174
Specific consumption transport performance in 2000	kWh/pkm	0.171
Specific consumption transport performance in 2006 ³²	kWh/pkm	0.161
Specific consumption transport performance in 2008 ³³	kWh/pkm	0.149

Values refer to electricity consumption at traction substation

Tab. 7.28 Electricity consumption of long-distance traffic

Indicator	Unit	Value
Total consumption long-distance transport	Mio. kWh	743.4
Specific consumption operation performance	kWh/Gtkm	0.031
Specific consumption kilometric performance	kWh/vkm	15.318
Specific consumption transport performance in 2000	kWh/pkm	0.087
Specific consumption transport performance in 2006	kWh/pkm	0.077
Specific consumption transport performance in 2008	kWh/pkm	0.068

Values refer to electricity consumption at traction substation

³² Email from Markus Halder, BahnUmwelt-Center SBB, 17th of april 2007

³³ Oral communication with Markus Halder, BahnUmwelt-Center SBB, octubre 2009

Tab. 7.29 Key figures and electricity consumption (2008) of Swiss passenger metropolitan train ("S-Bahn Zürich")

Indicator	Unit	Value	Source
Transport performance	millions pkm	1916	ZVV, 2008
train performance	millions vkm	20.2	ZVV, 2008
average load factor ¹⁾	people in train	94.9	
number of trains	Fahrzeuge	251	ZVV, 2008
average train performance per year	vkm/a	165000	Menth, 2007
weight of metropolitan train (Re450 / DPZ)	Tonnen	220	Menth, 2007
operation performance ¹⁾	Btkm	4444000000	
load per passenger ¹⁾		2.32	
energy consumption	Wh / Btkm	32.4	Menth, 2007
energy consumption per vkm ¹⁾	kWh/Fzkm	7.128	
energy consumption per pkm ¹⁾	kWh/pkm	0.075	
losses due to electricity transport ¹⁾		10%	
operation of metropolitan train ("S-Bahn") ¹⁾	kWh/pkm	0.0837	

1) Own calculations, the average load factor is calculated as follows: transport performance divided to train performance.

Transmission losses (transmission and conversion ex high voltage network) are accounted for with 10%, resulting in a final energy consumption for the year 2008 of 0.0748 kWh/pkm and 0.1639kWh/pkm for the operation of long distance trains and regional trains, respectively. The metropolitan train has a specific energy consumption of 0.0837 kWh/pkm.

Airborne Emissions

Due to the fact that on the Swiss rail network only very few diesel locomotives are in operation emissions to air are limited to particulate emissions due to abrasion and emission of sulphur hexafluoride (SF₆) occurring during conversion at traction substations.

Source of abrasion are rail tracks, wheels, brakes and overhead contact line. In Tab. 7.6 PM₁₀ emission factors as available from literature are summarised. Particulate emission due to abrasion are currently issue of intensive studies in Switzerland³⁴. The figures presented here, therefore should only be considered as rough estimates. For the determination of PM₁₀ emission factors for wheels, rail and overhead contacting line, we employ the same figures as used for goods transportation (see Tab. 7.6). For the determination of PM > 10 we assume a share of 25 % for wheel and rail abrasion. The abrasion of brakes and the resulting share of PM₁₀ depend on the wheel load and thus differ between fully loaded goods trains and passenger trains (see Tab. 7.30).

³⁴ Oral communication with Helmut Kuppelwieser, SBB Environmental Management

Tab. 7.30 Size distribution of braking line abrasion for passenger trains

Abrasion	Size distribution empty goods train and passenger train (wheel load: 2.5t) ¹⁾	Size distribution fully loaded goods train (wheel load: 11.25t) ¹⁾	This project for passenger trains
	%	%	%
Non airborne emissions	56	28	56
PM10	21	12	21
PM>10	23	60	23
Total Abrasion	100	100	100

1: (SBB, 2002)

The essential allocation is performed by employing the traction performance as allocation factor. In Tab. 7.31 the resulting figures are summarised.

Tab. 7.31 Specific emission factors of particulates due to abrasion

	unit	Total abrasion	PM 10 ¹⁾	PM >10
Yearly load	t/a	1069		
Allocated yearly road to freight transport	t/a	436		
Allocated yearly load regional transport, included S-Bahn Zürich	t/a	206	43.3	47.4
Allocated yearly load, only S-Bahn Zürich	t/a	70	14.8	16.2
Allocated yearly load long distance transport	t/a	427	89.7	98.2
Specific emission factor regional train	kg/pkm	7.25E-05	1.52E-05	1.67E-05
Specific emission factor S-Bahn	kg/pkm	3.67E-05	7.71E-06	8.44E-06
Specific emission factor long distance train	kg/pkm	5.02E-05	1.05E-05	1.16E-05

1) the entire load of PM10 is classified as coarse particulates (a diameter > 2.5 µm and < 10µm)

For sulphur hexafluoride (SF₆) emitted during conversion at traction substations we assume an emission index of 4.4E-08 kg/kWh Frischknecht (2003) resulting in a specific emission of 3.29E-09 kg/pkm and 7.21E-09 kg/pkm for long-distance transport and regional transport, respectively. The metropolitan train has a specific emission factor of 3.64E-09 kg SF₆/pkm

Emissions to Soil

Abrasion (from braking lining, rail and wheel) is predominately composed of iron and mineral components BUWAL (2002). In this study we assume that all non-airborne abrasion emissions are emitted as iron to soil. In consequence, we assume a yearly iron load of 115.4 t/a and 239.1 t/a for regional and long distance transport, respectively.

Emission of lubricates due to traction are not accounted for, since measures are in operation to avoid such emissions.

Life Cycle Inventory Input Data

Tab. 7.32 Life cycle inventory table for operation of Swiss average long distance train

Explanations	Name	Location	Infrastructure-Process	Unit	operation, long-distance train, SBB mix	uncertaintyType	StandardDeviation 95%	GeneralComment
	Location InfrastructureProcess Unit				CH 0 pkm			
Technosphere	electricity, high voltage, SBB, at grid	CH	0	kWh	7.48E-2	1	1.05	(1,1,1,1,1,1); SBB calculations, top down approach
air, unspecified	Particulates, > 2.5 um, and < 10um			kg	1.05E-5	1	2.05	(4,2,2,1,1,2); own estimates based on PM10 studies
	Particulates, > 10 um			kg	1.16E-5	1	1.56	(4,2,2,1,1,2); own estimates based on PM10 studies
	Sulfur hexafluoride			kg	3.29E-9	1	1.63	(3,5,2,2,3,1); literature
	Heat, waste			MJ	2.69E-1	1	1.22	(2,1,1,1,1,5); standard value
soil, unspecified	Iron			kg	2.81E-5	1	1.56	(4,2,2,1,1,2); own estimates based on PM10 studies
product	operation, long-distance train, SBB mix	CH	0	pkm	1.00E+0			

Tab. 7.33 Life cycle inventory table for operation of Swiss average regional train

Explanations	Name	Location	Infrastructure-Process	Unit	operation, regional train, SBB mix	uncertaintyType	StandardDeviation 95%	GeneralComment
	Location InfrastructureProcess Unit				CH 0 pkm			
Technosphere	electricity, high voltage, SBB, at grid	CH	0	kWh	1.64E-1	1	1.05	(1,1,1,1,1,1); SBB calculations, top down approach
air, unspecified	Particulates, > 2.5 um, and < 10um			kg	1.52E-5	1	2.05	(4,2,2,1,1,2); own estimates based on PM10 studies
	Particulates, > 10 um			kg	1.67E-5	1	1.56	(4,2,2,1,1,2); own estimates based on PM10 studies
	Sulfur hexafluoride			kg	7.21E-9	1	1.63	(3,5,2,2,3,1); literature
	Heat, waste			MJ	5.90E-1	1	1.22	(2,1,1,1,1,5); standard value
soil, unspecified	Iron			kg	4.06E-5	1	1.56	(4,2,2,1,1,2); own estimates based on PM10 studies
product	operation, regional train, SBB mix	CH	0	pkm	1.00E+0			

Tab. 7.34 Life cycle inventory table for operation of Swiss metropolitan train

Explanations	Name	Location	Infrastructure-Process	Unit	operation, metropolitan train, SBB mix	uncertaintyType	StandardDeviation 95%	GeneralComment
	Location InfrastructureProcess Unit				CH 0 pkm			
Technosphere	electricity, high voltage, SBB, at grid	CH	0	kWh	8.27E-2	1	1.05	(1,1,1,1,1,1); SBB calculations, top down approach
air, unspecified	Particulates, > 2.5 um, and < 10um			kg	7.71E-6	1	2.05	(4,2,2,1,1,2); own estimates based on PM10 studies
	Particulates, > 10 um			kg	8.44E-6	1	1.56	(4,2,2,1,1,2); own estimates based on PM10 studies
	Sulfur hexafluoride			kg	3.64E-9	1	1.63	(3,5,2,2,3,1); literature
	Heat, waste			MJ	2.98E-1	1	1.22	(2,1,1,1,1,5); standard value
soil, unspecified	Iron			kg	2.05E-5	1	1.56	(4,2,2,1,1,2); own estimates based on PM10 studies
product	operation, metropolitan train, SBB mix	CH	0	pkm	1.00E+0			

7.2.7 Operation of passenger rail transport in Germany, France, Italy, Austria and Belgium

Traction Energy

The international railway statistics from UIC (UIC, 2007) shows the energy consumption for average rail transport as well as the typical performance figures. No differentiation has been made to regional or long-distance-traffic.

Tab. 7.35 Energy consumption for traction and typical performance figures

	Diesel	electricity	Traffic performance	Transport performance	Train performance
	[1000t]	[GWh]	[Mio. Btkm]	[Mio. tkm]	[1000 Zugkm]
Austria	26	945	26'707	8'758	93'661
Italy	77	3'607	75'327	47'114	284'245
France	183	4'676	146'191	81'664	394'919
Germany	211	6'588	162'567	74'677	694'092
Belgium	11	1'121	26'558	9'932	80'696
Switzerland	n.a.	n.a.	37'790	15'771	144'609

- b) In the UIC-statistics, no differentiation for diesel consumption has been made for goods and passenger transport in France. In consideration of the shown traffic performance, one assumed that goods trains operate only with electricity, so the full value of 183'000 t of diesel is considered for passenger transport.

With the transport performance factors, one may calculate the specific energy consumption per tkm.

Tab. 7.36 Specific energy consumption

	electricity energy consumption	Diesel energy consumption for traction	total energy consumption	total energy consumption per Gtkm
Unit	[kWh/pkm]	[kg/pkm]	[kWh-equ./pkm]	[kWh-equ./Gtkm]
Austria	0.108	3.0E-03	0.115	0.038
Italy	0.077	1.6E-03	0.080	0.050
France	0.057	2.2E-03	0.063	0.035
Germany	0.088	2.8E-03	0.095	0.044
Belgium	0.113	1.1E-03	0.116	0.043
Switzerland	0.097	o	0.097	0.041

- b) The aggregation in the last two columns has been made for illustrative purposes, only the values in the first two columns are used in the life cycle inventory data in Table 6-17. The aggregation assumed energy content of 42.8 MJ/kg Diesel, one has to consider the lower efficiency of diesel locomotives.

Airborne Emissions

For airborne emissions we employ the figures we calculated for Swiss long distance transport (particulate emissions due to abrasion and emission of sulphur hexafluoride (SF₆) occurring during conversion at traction substations). For sulphur hexafluoride (SF₆) emitted during conversion at traction substations we assume an emission index of 4.4E-08 kg/kWh Frischknecht (2003).

Emissions to Soil

Abrasion (from braking lining, rail and wheel) we employ the same figure as for the Swiss long distance train.

Life Cycle Inventory Input Data

Tab. 7.37 Life cycle inventory table for operation average trains in selected European countries

Explanations	Name	Location	Infrastructure-Process	Unit	operation, average train, SBB mix	operation, average train	operation, average train	operation, average train	operation, average train	operation, average train	uncertainty type	Standard Deviation 95%	GeneralComment
					CH 0 pkm	AT 0 pkm	IT 0 pkm	FR 0 pkm	DE 0 pkm	BE 0 pkm			
Technosphere	diesel, at regional storage	RER	0	kg	0	2.97E-3	1.63E-3	2.24E-3	2.83E-3	1.11E-3	1	1.09	(2,3,1,3,1,1); International Rail study
	electricity, high voltage, SBB, at grid	CH	0	kWh	9.73E-2						1	1.09	(2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	AT	0	kWh		1.08E-1					1	1.09	(2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	IT	0	kWh			7.66E-2				1	1.09	(2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	FR	0	kWh				5.73E-2			1	1.09	(2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	DE	0	kWh					8.82E-2		1	1.09	(2,3,1,3,1,1); International Rail study
	electricity, high voltage, at grid	BE	0	kWh	0					1.13E-1	1	1.09	(2,3,1,3,1,1); International Rail study
air, unspecified	Benzene			kg	0	2.97E-7	1.63E-7	2.24E-7	2.83E-7	1.11E-7	1	2.06	(2,3,1,3,3,1); handbook emission factors for HGV
	Methane, fossil			kg	0	3.86E-7	2.12E-7	2.91E-7	3.67E-7	1.44E-7	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
	Carbon monoxide, fossil			kg	0	4.69E-5	2.58E-5	3.54E-5	4.47E-5	1.75E-5	1	5.01	(2,3,1,3,1,1); International Rail study
	Carbon dioxide, fossil			kg	0	9.34E-3	5.14E-3	7.05E-3	8.89E-3	3.48E-3	1	1.09	(2,3,1,3,1,1); International Rail study
	Dinitrogen monoxide			kg	0	2.97E-7	1.63E-7	2.24E-7	2.83E-7	1.11E-7	1	1.57	(2,3,1,3,3,1); International Rail study
	Ammonia			kg	0	5.92E-8	3.26E-8	4.47E-8	5.63E-8	2.21E-8	1	1.57	(2,3,1,3,3,1); International Rail study
	NMVO, non-methane volatile organic compounds, unspecified origin			kg	0	1.50E-5	8.28E-6	1.14E-5	1.43E-5	5.61E-6	1	1.57	(2,3,1,3,3,1); handbook emission factors for HGV
	Nitrogen oxides			kg	0	1.63E-4	8.98E-5	1.23E-4	1.55E-4	6.09E-5	1	1.51	(2,3,1,3,1,1); International Rail study
	Particulates, > 10 um			kg	1.16E-5	1.17E-5	1.16E-5	1.17E-5	1.17E-5	1.16E-5	1	1.52	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, > 2.5 um, and < 10um			kg	1.05E-5	1.07E-5	1.06E-5	1.07E-5	1.07E-5	1.06E-5	1	2.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Particulates, < 2.5 um			kg	0	3.81E-6	2.10E-6	2.87E-6	3.63E-6	1.42E-6	1	3.02	(3,3,1,3,1,1); own calculation base on PM figures of int. Rail study
	Sulfur dioxide			kg	0	1.78E-6	9.80E-7	1.34E-6	1.69E-6	6.64E-7	1	1.09	(2,3,1,3,1,1); European railway study
	Toluene			kg	0	1.19E-7	6.54E-8	8.97E-8	1.13E-7	4.43E-8	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
	Xylene			kg	0	1.19E-7	6.54E-8	8.97E-8	1.13E-7	4.43E-8	1	2.06	(3,3,1,3,3,1); handbook emission factors for HGV
	Mercury			kg	0	5.92E-14	3.26E-14	4.47E-14	5.63E-14	2.21E-14	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Lead			kg	0	3.27E-13	1.80E-13	2.47E-13	3.11E-13	1.22E-13	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Cadmium			kg	0	2.97E-11	1.63E-11	2.24E-11	2.83E-11	1.11E-11	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Copper			kg	0	5.04E-9	2.78E-9	3.81E-9	4.80E-9	1.88E-9	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Chromium			kg	0	1.49E-10	8.18E-11	1.12E-10	1.41E-10	5.55E-11	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Nickel			kg	0	2.08E-10	1.14E-10	1.57E-10	1.98E-10	7.75E-11	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Selenium			kg	0	2.97E-11	1.63E-11	2.24E-11	2.83E-11	1.11E-11	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Zinc			kg	0	2.97E-9	1.63E-9	2.24E-9	2.83E-9	1.11E-9	1	5.10	(4,5,1,2,1,1); expert estimate quoted from literature
	Sulfur hexafluoride			kg	4.28E-9	4.75E-9	3.37E-9	2.52E-9	3.88E-9	4.97E-9	1	1.63	(3,5,1,2,3,1); literature
	Heat, waste			MJ	3.50E-1	3.88E-1	2.76E-1	2.06E-1	3.18E-1	4.06E-1	1	1.07	(2,1,1,1,1,1); standard value
soil, unspecified	Iron			kg	2.81E-5	2.81E-5	2.81E-5	2.81E-5	2.81E-5	2.81E-5	1	3.00	(-,-,-,-,-,-,-,-,-,-); own calculation based on literature
product	operation, average train, SBB mix	CH	0	pkm	1.00E+0								
	operation, average train	AT	0	pkm		1.00E+0							
	operation, average train	IT	0	pkm			1.00E+0						
	operation, average train	FR	0	pkm				1.00E+0					
	operation, average train	DE	0	pkm					1.00E+0				
	operation, average train	BE	0	pkm						1.00E+0			

7.2.8 Operation of High Speed Trains in Europe (Germany, France, Italy)

Data for the German ICE is used in this project as representative for high speed transport in Germany, Italy and France. All high speed trains operate with electricity, the electricity mix is chosen to the respective country. If not stated explicitly, the data is taken from von Rozycki (2003).

Traction Energy

In Tab. 7.38 electricity consumption figures are summarised. In order to calculate the specific consumption per pkm we assume 309 passengers per train (Rozycki, 2003).

Tab. 7.38 Electricity consumption of an ICE (Rozycki, 2003), we assume a typical load of 309 passengers.

	unit	Traction energy (ICE I/II)	Train board energy (ICE I/II)	Total	Overhead energy and losses in distribution (10%) ^{a)}	Overall energy
Energy consumption per train-kilometer	kWh/trainkm	22.5	1.35	23.85	2.385	26.24
Energy consumption per passenger-km	kWh/pkm	0.073	0.004	0.077	0.008	0.085

a) Identical values as in Switzerland are assumed for losses in distribution of electricity

Airborne Emissions

For airborne emissions we employ the figures we calculated for Swiss long distance transport (particulate emissions due to abrasion and emission of sulphur hexafluoride (SF₆) occurring during conversion at traction substations). For sulphur hexafluoride (SF₆) emitted during conversion at traction substations we assume an emission index of 4.4E-08 kg/kWh Frischknecht (2003) resulting in a specific emission of 3.74 E-9.

Emissions to Soil

Abrasion (from braking lining, rail and wheel) we employ the same figure as for the Swiss long distance train. Emission of lubricates due to traction are not accounted for, since measures are in operation to avoid such emissions.

Life Cycle Inventory Input Data

Tab. 7.39 Life cycle inventory table for operation of high speed train in France, Italy and Germany

Explanations	Name	Location	Infrastructure-Process	Unit	operation, high speed train	operation, high speed train	operation, high speed train	uncertaintyType	Standard Deviation 95%	GeneralComment
					DE 0 pkm	FR 0 pkm	IT 0 pkm			
Technosphere	electricity, high voltage, at grid	DE	0	kWh	8.49E-2			1	1.20	literature studies; uncertainties are an estimate of the study team.
	electricity, high voltage, at grid	FR	0	kWh		8.49E-2		1	1.20	literature studies; uncertainties are an estimate of the study team.
	electricity, high voltage, at grid	IT	0	kWh			8.49E-2	1	1.20	literature studies; uncertainties are an estimate of the study team.
air, unspecified	Particulates, > 10 um			kg	1.16E-5	1.16E-5	1.16E-5	1	2.50	own estimates based on recent PM10 studies addressing Swiss conditions. Data represent average Swiss conditions resulting in an additional uncertainty compared to Swiss modules.
	Particulates, > 2.5 um, and < 10um			kg	1.05E-5	1.05E-5	1.05E-5	1	2.50	own estimates based on recent PM10 studies addressing Swiss conditions. Data represent average Swiss conditions resulting in an additional uncertainty compared to Swiss modules.
	Sulfur hexafluoride			kg	3.74E-9	3.74E-9	3.74E-9	1	1.63	(3,5,1,2,3,1); literature
	Heat, waste			MJ	3.06E-1	3.06E-1	3.06E-1	1	1.20	(2,1,1,1,1,1); standard value
soil, unspecified	Iron			kg	2.81E-5	2.81E-5	2.81E-5	1	2.50	own estimates based on recent PM10 studies addressing Swiss conditions. Data represent average Swiss conditions resulting in an additional uncertainty compared to Swiss modules.
	operation, high speed train	DE	0	pkm	1.00E+0					
	operation, high speed train	FR	0	pkm		1.00E+0				
products	operation, high speed train	IT	0	pkm			1.00E+0			
	operation, high speed train									

7.3 Rail Transport Equipment

In this section, the environmental interventions of rail transport equipment due to manufacturing, maintenance and disposal are documented. In Tab. 7.40 the key figures for the Swiss freight transport equipment are presented.

Tab. 7.40 Key figures for the swiss rail transport equipment

	Unit	Total CH 1997 ¹⁾	SBB 1999 ²⁾	SBB 2000 ²⁾
Goods transport performance	tkm	8.63E+09	9.80E+09	1.08E+10
Wagons	unit	22229	20119	19791
Wagons per transport unit	unit/tkm	2.58E-06	2.05E-06	1.83E-06
Locomotives goods transportation ³⁾	unit	431	307	307
Locomotives per transport unit	unit/tkm	1.22E-07	7.66E-08	6.95E-08

1: figures are taken from the Swiss national transport statistics Bundesamt für Statistik (BfS) (2000) and represent the total goods transport rail equipment in Switzerland, including all private rail companies.

2: figures are taken from an SBB report (SBB, 2002).

3: In the SBB report as well as the Swiss national statistic no further distinction between locomotives used for goods transportation and passenger transportation has been made. The figures presented here are based on the share of gross tonne kilometre of goods transportation of the SBB.

A comparison between the SBB figures of two years reveals a higher efficiency in the use of wagons in 2000. In this project we employ the SBB figures for the year 2000.

7.3.1 Manufacturing of Locomotives and Goods Transportation Rail Wagons

For manufacturing of locomotives and rail wagons, no new data has been collected; i.e. the data is based on Elsener & Strub (1993).

Manufacturing of locomotives

The environmental interventions of the locomotive manufacturing are derived from an assessment of the "RE 460". The "Re 460" is a locomotive which currently is used for both rail and passenger transportation in Switzerland. In the future this type will be replaced with a specific "goods transportation" locomotive the "Re 482". In the year 2000, however, none of the latter was in operation. The life span of the locomotive "Re 460" is assumed to be 40 years resulting in a lifetime performance of 9.6 Mio. Kilometres Frischknecht et al. (1996). The materials with a weight above 200 kg are summarised in the in Tab. 7.48. Expenditures due to vehicle manufacturing are limited to energy consumption (electricity and light fuel oil, including administration).

Manufacturing of goods transportation rail wagons

In 1997, 22229 rail wagons are recorded in the Swiss statistic Bundesamt für Statistik (BfS) (2000). The vehicle park of rail wagons is extremely inhomogeneous (up to 70 different types). The SBB reports 19791 wagons in its wagon park for the year 2000, including 6770 private wagons. The share of 4 axial wagons is about 48%, with a dominating share of 4 axial wagons in the private wagon park (SBB, 2002). The available statistics, however, do not give information about the performance of different wagon types. Furthermore in the SBB statistics no tank wagons are listed since they are under control of a private company. In contrast, to a locomotive, the material composition of wagons is less comprehensive. The dominating materials are steel and aluminium. In Tab. 7.41 the material composition three types of goods wagons are summarised.

Tab. 7.41 Material composition of goods transport rail wagons

Material	Closed wagon		Tank wagon (not isolated)		Tank wagon isolated		Model average wagon
	2 axes ¹⁾	4 axes ²⁾	2 axes ¹⁾	4 axes ¹⁾	2 axes ¹⁾	4 axes ¹⁾	
	Kg/unit	Kg/unit	Kg/unit	Kg/unit	Kg/unit	Kg/unit	Kg/unit
Steel	14830	30300	11895	24350	11577	23676	20952
Aluminum	1060	2160	-	-	135	305	1072
Wood	1050	2140	-	-	-	-	1037
Rubber	10	10	10	10	10	10	10
Paint	80	120	95	140	95	140	106
Mineral Fibre ³⁾	-	-	-	-	183	369	32

1: data are based on oral communication with a Swiss wagon manufacturing company (Mr. Soder, Meier AG, mechanical engineering and wagon manufacturing, Rheinfelden 1993).

2: data has been extrapolated from the closed wagon 2 axes using the ratio (tank wagon 4 axes/tank wagon 2 axes) as factor.

3: mineral fibre is modelled as mineral wool.

The specific energy consumption per wagon is available from Frischknecht et al. (1996). The electricity consumption is 9224 kWh and the consumption of oil is 46000 MJ. We further assume that the oil is used for heating purposes.

As stated above the SBB operates various other types of wagons, for which no information has been available. Consequently, for further aggregation, we subsumed the not represented wagon types in the represented categories. In the absence of statistic information of the actual kilometric performance of the different types of rail wagons we model an average rail wagon as follows:

- Closed and open wagons are treated as closed wagons.
- 65 % of the tank wagons feature a thermal isolated cistern, which are mainly used for the transport of chemicals
- 65% of the transport is performed by closed wagons
- 35% of the transport performance is allocated to tank wagons
- 50% of the wagons in each category are 2 axis vehicles.

The resulting material consumption for an average wagon is given in Tab. 7.52.

Manufacturing of Regional Trains

In this study, a common SBB-regional transport train the "Kolibri" is modelled. The total weight of a Kolibri is 171 t. The material composition is summarised in Tab. 7.50. For a composition of such weight, 271 seats are offered. The average load is 17% (46 passengers). The yearly vehicle kilometre performance is 150'000 vkm. In this project we assume a life span of 40 years, resulting in a total lifetime kilometric performance of 6'000'000 vkm and a life time transport performance of 276'000'000 pkm. Estimates for energy consumption for manufacturing is taken from Frischknecht (1996) assuming an electricity consumption and light fuel oil consumption of 2.7 MJ/kg material and 3.8 MJ/kg material, respectively. The resulting values for the entire train are given in Tab. 7.50.

Manufacturing of Long-Distance Passenger Trains

In this study, a new SBB-intercity/interregio train composition, the IC 2000 (Dosto) is modelled. The IC 2000 usually is equipped with the Re 460 and a train composition comprises 7 cars offering about

1400 seats. The maximum speed is 200 km/h. In the year 2000 the SBB had 237 wagons in operation. In the near future the stock will be expanded.

The data presented here is taken from (Maibach, 1999). The total weight of a train composition is 317 t. The material composition is summarised in Tab. 7.51. In contrast to older train compositions using conventional passenger cars (e.g. EW 4) dominated by steel, the share of aluminium of the IC 2000 is significantly higher. The aim of the substitution of steel is to decrease the weight and hence the electricity consumption. The average load is assumed with 28 % (392 passengers). The yearly vehicle kilometre performance is 500'000 vkm (Maibach, 1999). In this project we assume a life span of 40 years, resulting in a total life time kilometric performance of 20'000'000 vkm and a life time transport performance of 7'840'000'000 pkm. Estimates for energy consumption for manufacturing is taken from Frischknecht (1996) assuming an electricity consumption and light fuel oil consumption of 2.7 MJ/kg material and 3.8 MJ/kg material, respectively. The resulting values for the entire train are given in Tab. 7.51.

Manufacturing of ICE trains

Train manufacturing data is available from Rozycki (2003) mainly based on (Maibach, 1999). The resulting values for the entire train are given in Tab. 7.52.

7.3.2 Maintenance of Rail Equipment

Maintenance of Locomotives and Goods Transportation Rail Wagons

In Frischknecht et al. (1996) material expenditures for maintenance have been estimated separately for locomotives and goods transportation rail wagons pursuing a bottom up approach. The estimates are based on the assumption that break blocks are substituted after a performance of 40000 km and a comprehensive check every 4 - 6 years where amongst other things the breaking system is greased. The energy consumption for the wagon maintenance is assumed to be the same as for wagon manufacturing. For the maintenance of locomotives no energy values are available in Frischknecht et al. (1996). The material consumption merely reflects the scheduled maintenance of mechanical parts.

In this project new SBB data has been available for the stationary energy and material consumption of goods transportation equipment maintenance. We assume that these data comprises expenditures for both, locomotives and wagons. In Tab. 7.42 and Tab. 7.43 energy consumption and material expenditures, respectively, for maintenance are summarised.

Tab. 7.42 Yearly energy consumption for rail equipment maintenance

Energy carriers	Unit	2001	2000 ¹⁾	1999 ¹⁾	1998 ¹⁾	1997 ¹⁾	1996 ¹⁾
Electricity	kWh	5.68E+06	4.20E+06	3.78E+06	4.22E+06	5.51E+06	3.84E+06
Light fuel oil	kWh	1.69E+07	9.15E+06	8.90E+06	9.96E+06	9.09E+06	1.02E+07
Natural gas	kWh	1.04E+07	6.48E+06	7.03E+06	5.18E+06	6.12E+06	7.36E+06
Coal	kWh	1.04E+06	-	-	-	-	-
Total	kWh	3.40E+07	1.98E+07	1.97E+07	1.94E+07	2.07E+07	2.14E+07

1: The figures for the years 1996-2000 do not include the consumption of the service region Tessin, Basel, Zurich)

Due to the fact that the data for 2000 do not comprise all service garages, we employ the figures for 2001, representing expenditures of all SBB goods transportation service garages.

In order to allocate the maintenance expenditures between locomotives and wagons, we allocate 95% of the maintenance expenditures to wagon maintenance and 5% to locomotive maintenance. This allocation is based on the average vehicle weight (the weight of a locomotive is 3 times higher than an average wagon) and the number of vehicles in the rail equipment. Furthermore we assume that the

frequency of maintenance is the same for either type. In Tab. 7.43 specific energy consumption and material expenditures for wagons and locomotives are presented.

Tab. 7.43 Specific energy consumption for maintenance of goods transport wagons and locomotives

Energy Carrier	Unit ¹⁾	Locomotive	Wagon	Unit	Locomotive	Wagon
Electricity	kWh/unit	3.41E+04	1.10E+04	kWh/tkm	5.92E-05	5.02E-04
Light fuel oil	MJ/unit	3.65E+05	1.17E+05	MJ/tkm	6.34E-04	5.38E-03
Natural gas	MJ/unit	2.25E+05	7.22E+04	MJ/tkm	3.90E-04	3.31E-03
Coal	MJ/unit	2.25E+04	7.22E+03	MJ/tkm	3.90E-05	3.31E-04

1: figures represent the total life span

In 2000 the SBB goods wagons are predominately equipped with gray-cast-iron brake shoes. The replacement of the latter with synthetic break shoes starts in 2004³⁵. In 2001 186'578 units of break shoes have been replaced in goods wagon. The weight of one break shoe is about 10 kg. The specific material consumption for maintenance of goods transport wagons and locomotives is summarised in Tab. 7.44. In Tab. 7.45 the energy and figures used in this project are summarised and compared with data of Frischknecht et al. (1996).

Tab. 7.44 Specific material consumption for maintenance of goods transport wagons and locomotives

Material	Unit	Locomotive	Wagon	Unit	Locomotive	Wagon
Wheels	Unit/unit	n.a.	4.35E+00	Unit/unit	n.a.	2.00E-07
Cast iron ¹⁾	Unit/unit	n.a.	3.77E+02	Unit/unit	n.a.	1.73E-05
Paint	kg/unit	1.21E+02	3.89E+01	kg/tkm	5.56E-06	1.79E-06
Cleaning agents	kg/unit	1.02E+03	3.27E+02	kg/tkm	4.66E-05	1.50E-05
Lubricates non biodegradable)	kg/unit	5.63E+02	1.81E+02	kg/tkm	2.58E-05	8.30E-06
Biodegradable lubricates	kg/unit	1.28E+02	4.11E+01	kg/tkm	5.86E-06	1.88E-06

1) Break shoes: Original material composition of gray-cast-iron brake shoes: 93% iron; 3.1% carbon; 1.7% silicon; 0.6% Mn; 0.9 % P.

³⁵ Oral communication with Helmut Kuppelwieser, SBB Environmental Management

Tab. 7.45 Material and energy consumption for the maintenance of goods transportation rail equipment

	Unit ¹⁾	This project		Frischknecht et al. (1996) (bottom up)		Tab. 7.43 and Tab. 7.44 (Top Down)	
		Loco-motive	Wagon	Loco-motive	Wagon	Loco-motive	Wagon
Steel low alloyed/iron	Kg/vehicle	29700		29700	-	-	
Sintered metal	Kg/vehicle	1330 ³⁾	-	1330	-	-	-
Cast iron		-	3510	-	-	-	3510
Rubber	Kg/vehicle	840	-	840	-	-	-
Lubricate	Kg/vehicle	690	222	730	31.5 ²⁾	691	222
PVC	Kg/vehicle	40	-	40	-	-	-
Paint	Kg/vehicle	121	39	-	-	121	39
Cleaning Agents	Kg/vehicle	1020	327	-	-	1020	327
Electricity	KWh/vehicle	34100	11000	-	9224	34100	11000
Heating Oil	MJ/vehicle	365000	117000	-	46000	365000	117000
Natural Gas	MJ/vehicle	225000	72200	-	-	225000	72200
Coal	MJ/vehicle	22500	7220	-	-	22500	7220

1: values are given for the entire lifespan of a vehicle. For both locomotives and wagons a lifespan of 40 years is assumed.

2: In the quoted report, 21 kg/unit for two axis wagon and 42 kg/unit four axis vehicle are assumed. The value presented in the table assumes that 50% of the entire fleet are two axis and 50% are four axis vehicles.

3: for the sintered metal a composition as followed has been assumed in this project: 75% cast iron, 20% copper, 5% coke

Waste of maintenance processes

For SBB service garages waste disposal indicators are available. The figures used in this study represent the conditions for 2001, since the available data for 2000 does not include all SBB service garages. The specific waste due to maintenance activities has been calculated by pursuing the same approach as for the calculation of energy and material expenditures. In Tab. 7.46 the resulting figures are presented. The sludge is assumed to contain lubricates which result from cleaning activities. For dismantled metals, for instance used break shoes, we apply a cut off allocation, since they are recycled.

Tab. 7.46 Waste due to maintenance activities

Waste Type	unit	Total 2001	unit	wagon	locomotive
Wood	kg/a	959260	kg/unit	1849	5754
Sludge	kg/a	153440	kg/unit	296	920

Maintenance of Swiss Passenger Trains

For passenger transport equipment no SBB data has been available for the stationary energy and material consumption. Thus literature data is employed in this project, representing expenditures for one major revision (Maibach, 1999) as well as a regular substitution of break shoes (40 kg break shoes/car (Maibach, 1999)). The resulting input data for maintenance of regional and long-distance passenger trains is presented Tab. 7.54 and Tab. 7.55, respectively.

Maintenance of ICE Trains

A comprehensive inventory of ICE maintenance is available from Rozycki (2003). He further distinguishes two types of maintenance; a) regular maintenance and b) repair (Tab. 7.47).

In Tab. 7.56 the actual input data used in this project is summarized.

Tab. 7.47 Maintenance activities for a German ICE train

Regular maintenance	electricity (German mix)	KWh/train	1.48E+06
	heating (district heating)	MJ/train	2.38E+07
	drinking water	kg/train	4.00E+07
	natriumhydroxid, in water	kg/train	4.00E+04
	brake lining	kg/train	3.12E+04
	paper	kg/train	1.87E+04
	wastewater	m3/train	7.00E+00
Repair	electricity (German mix)	KWh/train	9.45E+04
	heating energy (district heating)	MJ/train	1.29E+06
	drinking water	kg/train	4.50E+05
	steel	kg/train	4.50E+04
	non-iron metals ¹⁾	kg/train	2.50E+02
	paint	kg/train	5.00E+03
	grease/solvents ²⁾	kg/train	1.00E+03
	paper/wood etc. ³⁾	kg/train	2.00E+03
	municipal solid waste	kg/train	3.00E+03
	wastewater	m3/train	4.50E+02

1: in the absence of further information we assume a split of 50% aluminum and 50% cooper

2: in the absence of further information we assume a split of 50% lubricating oil and 50% solvent

3: in the absence of further information wood is modeled as paper.

7.3.3 Disposal of Locomotives and Goods Transportation Rail Wagons

Disposal Locomotive

More than 80% of the materials of modelled locomotive are metals. For these metals we assume a complete recycling and thus make a cut of allocation. For the remaining materials, merely the disposal of plastics, glass and paint as well as used oil are taken into account. For the transportation, standard distances are applied for the disposal of glass and plastics. For the remaining materials, a distance of 50 km has been assumed.

Disposal Goods Wagon

The discarded wagons are scrapped and recycled. The bogies are reused and merely the body is renewed. Fractions of rubber and paint are very small and are neglected. Thus, the disposal of wagons is not modelled in the inventory.

Disposal regional and long distance trains

The majority of materials of the train compositions are metals. For these metals we assume a complete recycling and thus make a cut of allocation. For the remaining materials, merely the disposal of plastics, glass and paint are taken into account. For the transportation, standard distances are applied

for the disposal of glass and plastics. For the remaining materials, a distance of 50 km has been assumed.

7.3.4 Life Cycle Inventory Input Data

Tab. 7.48 Life cycle inventory input data for manufacturing of a locomotive of the type "Re 462"

	Name	Location	Unit	locomotive	Uncertainty Type	Standard Deviation 95 %	General Comment
product	locomotive	RER	unit	1.00E+0			
technosphere	reinforcing steel, at plant	RER	kg	3.42E+4	1	1.27	(4,3,2,1,1,3); oral communication
	steel, low-alloyed, at plant	RER	kg	1.64E+4	1	1.27	(4,3,2,1,1,3); oral communication
	chromium steel 18/ 8, at plant	RER	kg	2.81E+3	1	1.27	(4,3,2,1,1,3); oral communication
	aluminium, production mix, at plant	RER	kg	5.55E+3	1	1.27	(4,3,2,1,1,3); oral communication
	copper, at regional storage	RER	kg	6.81E+3	1	1.27	(4,3,2,1,1,3); oral communication
	polyethylene, HDPE, granulate, at plant	RER	kg	6.34E+3	1	1.27	(4,3,2,1,1,3); oral communication
	flat glass, uncoated, at plant	RER	kg	2.35E+2	1	1.27	(4,3,2,1,1,3); oral communication
	alkyd paint, white, 60% in solvent, at plant	RER	kg	3.85E+2	1	1.27	(4,3,2,1,1,3); oral communication
	polyvinylchloride, at regional storage	RER	kg	3.51E+2	1	1.27	(4,3,2,1,1,3); oral communication
	lubricating oil, at plant	RER	kg	3.56E+3	1	1.27	(4,3,2,1,1,3); oral communication
	lead, at regional storage	RER	kg	2.63E+2	1	1.27	(4,3,2,1,1,3); oral communication
	glued laminated timber, outdoor use, at plant	RER	m ³	5.94E-1	1	1.27	(4,3,2,1,1,3); oral communication
	rock wool, packed, at plant	CH	kg	3.20E+1	1	1.27	(4,3,2,1,1,3); oral communication
	transport, lorry 32t	RER	tkm	3.53E+3	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	transport, freight, rail	RER	tkm	1.44E+4	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.52E+5	1	1.27	(4,3,2,1,1,3); oral communication
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	MJ	1.44E+6	1	1.27	(4,3,2,1,1,3); oral communication
	Heat, waste	-	MJ	5.47E+5	1	1.27	(4,3,2,1,1,3); standard value

Tab. 7.49 Life cycle inventory input data for the module manufacturing of goods rail wagons

	Name	Location	Unit	goods wagon	Uncertainty Type	Standard Deviation 95 %	General Comment
product	goods wagon	RER	unit	1.00E+0			
technosphere	steel, low-alloyed, at plant	RER	kg	2.10E+4	1	1.36	(4,3,3,3,1,4); own calculations based on oral communication
	aluminium, production mix, at plant	RER	kg	1.07E+3	1	1.36	(4,3,3,3,1,4); own calculations based on oral communication
	synthetic rubber, at plant	RER	kg	1.00E+1	1	1.36	(4,3,3,3,1,4); own calculations based on oral communication
	alkyd paint, white, 60% in solvent, at plant	RER	kg	1.06E+2	1	1.36	(4,3,3,3,1,4); own calculations based on oral communication
	glued laminated timber, outdoor use, at plant	RER	m ³	1.92E+0	1	1.36	(4,3,3,3,1,4); own calculations based on oral communication
	transport, lorry 32t	RER	tkm	2.21E+3	1	2.28	(4,5,na,na,na,na); standard distance
	transport, freight, rail	RER	tkm	4.43E+3	1	2.28	(4,5,na,na,na,na); standard distance
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	9.22E+3	1	1.36	(4,3,3,3,1,4); own calculations based on oral communication
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	MJ	4.60E+4	1	1.36	(4,3,3,3,1,4); own calculations based on oral communication
	Heat, waste	-	MJ	3.32E+4	1	1.36	(4,3,3,3,1,4); standard value

Tab. 7.50 Life cycle inventory input data for the module manufacturing of a regional train (Kolibri)

	Name	Location	Infra	Unit	regional train	Uncertainty Type	Standard Deviation 95%	General Comment
product	Location				CH			
	Infrastructure Process				1			
	Unit				unit			
	regional train	CH	1	unit	1.00E+0			
technosphere	reinforcing steel, at plant	RER	0	kg	1.14E+5	1	1.27	(4,3,2,1,1,3); literature studies
	aluminium, production mix, at plant	RER	0	kg	1.00E+4	1	1.27	(4,3,2,1,1,3); literature studies
	copper, at regional storage	RER	0	kg	7.00E+3	1	1.27	(4,3,2,1,1,3); literature studies
	polyethylene, HDPE, granulate, at plant	RER	0	kg	1.30E+4	1	1.27	figure comprises various plastics used.
	flat glass, uncoated, at plant	RER	0	kg	7.23E+3	1	1.27	(4,3,2,1,1,3); literature studies
	alkyd paint, white, 60% in solvent, at plant	RER	0	kg	1.28E+3	1	1.27	(4,3,2,1,1,3); literature studies
	glued laminated timber, outdoor use, at plant	RER	0	m ³	8.67E+0	1	1.27	(4,3,2,1,1,3); literature studies
	rock wool, packed, at plant	CH	0	kg	1.28E+3	1	1.27	(4,3,2,1,1,3); literature studies
	lead, at regional storage	RER	0	kg	1.45E+3	1	1.27	(4,3,2,1,1,3); literature studies
	transport, lorry 28t	CH	0	tkm	7.70E+3	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	transport, freight, rail	CH	0	tkm	3.08E+4	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	electricity, medium voltage, at grid	CH	0	kWh	1.28E+5	1	2.08	(4,3,2,1,1,3); literature studies; value represents a rough estimate (basic uncertainty =2)
	heat, light fuel oil, at industrial furnace 1MW	CH	0	MJ	1.81E+5	1	2.08	(4,3,2,1,1,3); literature studies; value represents a rough estimate (basic uncertainty =2)
	Heat, waste	-	-	MJ	4.62E+5	1	2.08	(4,3,2,1,1,3); standard value

Tab. 7.51 Life cycle inventory input data for the module manufacturing of a long distance train (IC 2000)

	Name	Location	Infra	Unit	long-distance train	Uncertainty Type	Standard Deviation 95%	General Comment
product	Location				CH			
	Infrastructure Process				1			
	Unit				unit			
	long-distance train	CH	1	unit	1.00E+0			
technosphere	reinforcing steel, at plant	RER	0	kg	5.42E+4	1	1.27	(4,3,2,1,1,3); literature studies
	aluminium, production mix, at plant	RER	0	kg	1.60E+5	1	1.27	(4,3,2,1,1,3); literature studies
	copper, at regional storage	RER	0	kg	9.98E+3	1	1.27	(4,3,2,1,1,3); literature studies
	polyethylene, HDPE, granulate, at plant	RER	0	kg	7.70E+4	1	1.27	(4,3,2,1,1,3); literature studies
	flat glass, coated, at plant	RER	0	kg	9.51E+3	1	1.27	(4,3,2,1,1,3); literature studies
	tempering, flat glass	RER	0	kg	9.51E+3	1	1.27	(4,3,2,1,1,3); literature studies
	alkyd paint, white, 60% in solvent, at plant	RER	0	kg	6.34E+3	1	1.27	(4,3,2,1,1,3); literature studies
	transport, lorry 28t	CH	0	tkm	1.59E+4	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	transport, freight, rail	CH	0	tkm	6.34E+4	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	electricity, medium voltage, at grid	CH	0	kWh	2.40E+5	1	1.27	(4,3,2,1,1,3); literature studies
	heat, light fuel oil, at industrial furnace 1MW	CH	0	MJ	1.20E+6	1	1.27	(4,3,2,1,1,3); literature studies
	Heat, waste	-	-	MJ	8.64E+5	1	1.27	(4,3,2,1,1,3); standard value

Tab. 7.52 Life cycle inventory input data for the module manufacturing of a German ICE

	Name	Location	Infra	Unit	ICE	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				DE			
	Infrastructure Process				1			
	Unit				unit			
product	ICE	DE	1	unit	1.00E+0			
technosphere	reinforcing steel, at plant	RER	0	kg	2.34E+05	1	1.27	(4,3,2,1,1,3); literature studies
	steel, low-alloyed, at plant	RER	0	kg	8.88E+04	1	1.27	(4,3,2,1,1,3); literature studies
	aluminium, production mix, at plant	RER	0	kg	9.00E+04	1	1.27	(4,3,2,1,1,3); literature studies
	copper, at regional storage	RER	0	kg	3.44E+04	1	1.27	(4,3,2,1,1,3); literature studies
	polyethylene, HDPE, granulate, at plant	RER	0	kg	1.55E+05	1	1.27	(4,3,2,1,1,3); literature studies
	tempering, flat glass	RER	0	kg	2.50E+04	1	1.27	(4,3,2,1,1,3); literature studies
	flat glass, coated, at plant	RER	0	kg	2.50E+04	1	1.27	(4,3,2,1,1,3); literature studies
	alkyd paint, white, 60% in solvent, at plant	RER	0	kg	1.26E+04	1	1.27	(4,3,2,1,1,3); literature studies
	transport, lorry 32t	RER	0	tkm	2.76E+04	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	transport, freight, rail	RER	0	tkm	1.10E+05	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	electricity, medium voltage, at grid	DE	0	kWh	6.11E+05	1	2.08	(4,3,2,1,1,3); literature studies; value represents a rough estimate (basic uncertainty =2)
	heat, light fuel oil, at industrial furnace 1MW	RER	0	MJ	1.94E+06	1	2.08	(4,3,2,1,1,3); literature studies; value represents a rough estimate (basic uncertainty =2)
	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	1.94E+06	2	2.08	(4,3,2,1,1,3); literature studies; value represents a rough estimate (basic uncertainty =2)
	Heat, waste	-	-	MJ	2.20E+06	1	2.08	(4,3,2,1,1,3); standard value

Tab. 7.53 Life cycle inventory input data for goods transport rail equipment maintenance

	Name	Location	Unit	maintenance, locomotive	maintenance, goods wagon	Uncertainty Type	Standard Deviation 95%	General Comment
product	maintenance, locomotive	RER	unit	1.00E+0				
	maintenance, goods wagon	RER	unit		1.00E+0			
technosphere	reinforcing steel, at plant	RER	kg	2.91E+4	-	1.00	1.36	(4,3,3,3,1,4); oral communication
	cast iron, at plant	RER	kg	9.98E+2	3.51E+3	1.00	1.36	(4,3,3,3,1,4); oral communication
	alkyd paint, white, 60% in solvent, at plant	RER	kg	1.21E+2	3.90E+1	1.00	1.26	(3,1,1,3,1,4); oral communication
	polyvinylchloride, at regional storage	RER	kg	4.00E+1	4.00E+1	1.00	1.26	(,,3,3,1,4); oral communication
	lubricating oil, at plant	RER	kg	6.90E+2	2.22E+2	1.00	1.26	(3,1,1,3,1,4); oral communication
	hard coal coke, at plant	RER	MJ	2.33E+3		1.00	1.26	(3,1,1,3,1,4); oral communication
	transport, lorry 32t	RER	tkm	3.20E+3	3.81E+2	1.00	2.05	(4,na,na,na,na,na); standard distance
	transport, freight, rail	RER	tkm	6.41E+3	7.62E+2	1.00	2.05	(4,na,na,na,na,na); standard distance
	natural gas, burned in industrial furnace >100kW	RER	MJ	2.25E+4	7.22E+4	1.00	1.26	(3,1,1,3,1,4); oral communication
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	3.41E+4	1.10E+4	1.00	1.26	(3,1,1,3,1,4); oral communication
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	MJ	3.65E+5	3.65E+5	1.00	1.26	(3,1,1,3,1,4); oral communication
	hard coal, burned in industrial furnace 1-10MW	RER	MJ	2.25E+4	7.22E+3	1.00	1.26	(3,1,1,3,1,4); oral communication
	disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	kg	9.20E+2	1.85E+3	1.00	1.26	(3,1,1,3,1,4); oral communication
	disposal, wood untreated, 20% water, to municipal incineration	CH	kg	5.75E+3	2.96E+2	1.00	1.26	(3,1,1,3,1,4); oral communication
	Heat, waste	-	MJ	1.23E+5	3.96E+4	1.00	1.26	(3,1,1,3,1,4); standard value

Tab. 7.54 Input data for maintenance of regional passenger train equipment

	Name	Location	Infra	Unit	maintenance, regional train	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				CH			
	Infrastructure Process				1			
	Unit				unit			
product	maintenance, regional train	CH	1	unit	1.00E+0			
technosphere	reinforcing steel, at plant	RER	0	kg	5.00E+3	1	1.21	(2,2,2,3,3,3); literature studies
	aluminium, production mix, at plant	RER	0	kg	3.32E+2	1	1.21	(2,2,2,3,3,3); literature studies
	polyethylene, HDPE, granulate, at plant	RER	0	kg	1.97E+3	1	1.21	(2,2,2,3,3,3); literature studies
	synthetic rubber, at plant	RER	0	kg	1.02E+3	1	1.21	(2,2,2,3,3,3); literature studies
	flat glass, uncoated, at plant	RER	0	kg	6.60E+2	1	1.21	(2,2,2,3,3,3); literature studies
	alkyd paint, white, 60% in solvent, at plant	RER	0	kg	1.55E+2	1	1.21	(2,2,2,3,3,3); literature studies
	glued laminated timber, outdoor use, at plant	RER	0	m ³	8.84E+1	1	1.21	(2,2,2,3,3,3); literature studies
	rock wool, packed, at plant	CH	0	kg	8.12E+1	1	1.21	(2,2,2,3,3,3); literature studies
	lead, at regional storage	RER	0	kg	2.03E+2	1	1.21	(2,2,2,3,3,3); literature studies
	solvents, organic, unspecified, at plant	GLO	0	kg	1.25E+2	1	1.21	(2,2,2,3,3,3); literature studies; estimate based on ICE maintenance figures (assuming 25% of the ICE expenditures).
	lubricating oil, at plant	RER	0	kg	1.25E+2	1	1.27	(4,3,2,1,1,3); literature studies; estimate based on ICE maintenance figures (assuming 25% of the ICE expenditures).
	paper, woodfree, uncoated, at regional storage	RER	0	kg	2.07E+4	1	1.27	(4,3,2,1,1,3); literature studies; estimate based on ICE maintenance figures (assuming 25% of the ICE expenditures).
	tap water, at user	CH	0	kg	1.14E+5	1	1.27	(4,3,2,1,1,3); literature studies; estimate based on ICE maintenance figures (assuming 25% of the ICE expenditures).
	bitumen, at refinery	CH	0	kg	3.75E+2	1	1.21	(2,2,2,3,3,3); literature studies
	transport, lorry 28t	CH	0	tkm	4.34E+2	1	2.28	(4,5,na,na,na,na); standard transport distance
	transport, freight, rail	CH	0	tkm	1.74E+3	1	2.28	(4,5,na,na,na,na); standard transport distance
	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	9.74E+4	1	2.05	(2,2,2,3,3,3); literature studies; rough estimate (basic uncertainty =2)
	electricity, medium voltage, at grid	CH	0	kWh	1.39E+4	1	2.05	(2,2,2,3,3,3); literature studies; rough estimate (basic uncertainty =2)
	heat, light fuel oil, at industrial furnace 1MW	CH	0	MJ	9.74E+4	1	2.05	(2,2,2,3,3,3); literature studies; rough estimate (basic uncertainty =2)
	disposal, wood untreated, 20% water, to municipal incineration	CH	0	kg	6.32E+2	1	1.21	(2,2,2,3,3,3); own assumptions based on the new input
	disposal, emulsion paint, 0% water, to municipal incineration	CH	0	kg	1.39E+1	1	1.21	(2,2,2,3,3,3); literature studies
	disposal, glass, 0% water, to municipal incineration	CH	0	kg	6.60E+2	1	1.21	(2,2,2,3,3,3); own assumptions based on the new input
	disposal, plastics, mixture, 15.3% water, to sanitary landfill	CH	0	kg	1.97E+3	1	1.21	(2,2,2,3,3,3); own assumptions based on the new input
	treatment, sewage, to wastewater treatment, class 1	CH	0	m ³	114.25	1	1.27	(4,3,2,1,1,3); literature studies; estimate based on ICE maintenance figures (assuming 25% of the ICE expenditures).
emissions to air	Heat, waste	-	-	MJ	4.99E+4	1	1.27	(4,3,2,1,1,3); standard value
emissions to air	NM VOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	7.83E+1	1	2.08	(4,3,2,1,1,3); Derived from the used paint and solvent, assuming all solvent is emitted to air.

Tab. 7.55 Input data for maintenance of long distance passenger train equipment

	Name	Location	Infra	Unit	maintenance, long-distance train	UncertaintyType	StandardDeviation95%	GeneralComment
product	Location				CH			
	InfrastructureProcess				1			
	Unit				unit			
	maintenance, long-distance train	CH	1	unit	1.00E+0			
technosphere	reinforcing steel, at plant	RER	0	kg	9.90E+4	1	1.27	(4,3,2,1,1,3); literature studies
	aluminium, production mix, at plant	RER	0	kg	1.31E+4	1	1.27	(4,3,2,1,1,3); literature studies
	copper, at regional storage	RER	0	kg	1.25E+2	1	1.27	(4,3,2,1,1,3); literature studies; estimate based on ICE maintenance figures
	polyethylene, HDPE, granulate, at plant	RER	0	kg	7.82E+4	1	1.27	(4,3,2,1,1,3); literature studies
	synthetic rubber, at plant	RER	0	kg	1.32E+4	1	1.27	(4,3,2,1,1,3); literature studies
	flat glass, coated, at plant	RER	0	kg	2.61E+4	1	1.21	(2,2,2,3,3,3); literature studies; estimate based on ICE maintenance figures
	tempering, flat glass	RER	0	kg	2.61E+4	1	1.21	(2,2,2,3,3,3); literature studies; estimate based on ICE maintenance figures
	alkyd paint, white, 60%in solvent, at plant	RER	0	kg	6.18E+3	1	1.21	(2,2,2,3,3,3); literature studies; estimate based on ICE maintenance figures
	sodium hydroxide, 50%in H2O, production mix, at plant	RER	0	kg	4.00E+4	1	1.21	(2,2,2,3,3,3); literature studies; estimate based on ICE maintenance figures
	bitumen, at refinery	CH	0	kg	4.85E+3	1	1.21	(2,2,2,3,3,3); literature studies
	transport, lorry 28t	CH	0	tkm	1.08E+4	1	2.28	(4,5,na,na,na,na); standard transport distances
	transport, freight, rail	CH	0	tkm	4.33E+4	1	2.28	(4,5,na,na,na,na); standard transport distances
	solvents, organic, unspecified, at plant	GLO	0	kg	5.00E+2	1	1.21	(2,2,2,3,3,3); literature studies; estimate based on ICE maintenance figures
	lubricating oil, at plant	RER	0	kg	5.00E+2	1	1.27	(4,3,2,1,1,3); literature studies; estimate based on ICE maintenance figures
	paper, woodfree, uncoated, at regional storage	RER	0	kg	2.07E+4	1	1.27	(4,3,2,1,1,3); literature studies; estimate based on ICE maintenance figures
	tap water, at user	CH	0	kg	4.57E+5	1	1.27	(4,3,2,1,1,3); literature studies, rough estimate
	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	6.47E+4	1	2.08	(4,3,2,1,1,3); literature studies, rough estimate
	electricity, medium voltage, at grid	CH	0	kWh	5.42E+5	1	2.08	(4,3,2,1,1,3); literature studies, rough estimate
	heat, light fuel oil, at industrial furnace 1MW	CH	0	MJ	4.38E+6	1	2.08	(4,3,2,1,1,3); literature studies, rough estimate
	disposal, emulsion paint, 0%water, to municipal incineration	CH	0	kg	6.45E+2	1	1.27	(4,3,2,1,1,3); literature studies
	disposal, plastics, mixture, 15.3%water, to sanitary landfill	CH	0	kg	1.31E+4	1	1.27	(4,3,2,1,1,3); literature studies
	disposal, glass, 0%water, to municipal incineration	CH	0	kg	2.61E+4	1	1.27	(4,3,2,1,1,3); literature studies
	treatment, sewage, to wastewater treatment, class 1	CH	0	m3	4.57E+2	1	1.27	(4,3,2,1,1,3); literature studies
emissions to air	Heat, waste	-	-	MJ	1.95E+6	1	1.27	(4,3,2,1,1,3); standard value
emissions to air	NM VOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	3.11E+3	1	1.21	(2,2,2,3,3,3); Derived from the used paint and solvent, assuming all solvent is emitted to air.

Tab. 7.56 Life cycle inventory input data for maintenance of a German ICE

	Name	Location	Infra	Unit	maintenance, ICE	UncertaintyType	StandardDeviation95%	GeneralComment
product	Location				DE			
	InfrastructureProcess				1			
	Unit				unit			
	maintenance, ICE	DE	1	unit	1.00E+0			
technosphere	reinforcing steel, at plant	RER	0	kg	4.50E+4	1	1.17	(2,2,1,3,1,3); literature study (repair)
	aluminium, production mix, at plant	RER	0	kg	1.25E+2	1	1.17	(2,2,1,3,1,3); literature studies (repair)
	copper, at regional storage	RER	0	kg	1.25E+2	1	1.17	(2,2,1,3,1,3); literature studies (repair)
	alkyd paint, white, 60% in solvent, at plant	RER	0	kg	5.00E+3	1	1.17	(2,2,1,3,1,3); literature studies (repair)
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	0	kg	4.00E+4	1	1.17	(2,2,1,3,1,3); literature studies (regular maintenance)
	transport, lorry 32t	RER	0	tkm	5.60E+3	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	transport, freight, rail	RER	0	tkm	2.24E+4	1	2.28	(4,5,na,na,na,na); calculation based on standard distances
	heat, at cogen 1MWs lean burn, allocation exergy	RER	0	MJ	2.51E+7	1	1.17	(2,2,1,3,1,3); literature studies (repair & regular maintenance)
	electricity, medium voltage, at grid	DE	0	kWh	1.58E+6	1	1.17	(2,2,1,3,1,3); literature studies (repair & regular maintenance)
	tap water, at user	RER	0	kg	4.57E+5	1	1.17	(2,2,1,3,1,3); literature studies (repair & regular maintenance)
	paper, woodfree, uncoated, at regional storage	RER	0	kg	2.07E+4	1	1.17	(2,2,1,3,1,3); literature studies (repair & regular maintenance)
	solvents, organic, unspecified, at plant	GLO	0	kg	5.00E+2	1	1.17	(2,2,1,3,1,3); literature studies (repair)
	lubricating oil, at plant	RER	0	kg	5.00E+2	1	1.17	(2,2,1,3,1,3); literature studies (repair)
	treatment, sewage, to wastewater treatment, class 1	OH	0	m ³	4.57E+2	1	1.17	(2,2,1,3,1,3); literature studies (repair & regular maintenance)
	disposal, emulsion paint, 0% water, to municipal incineration	OH	0	kg	2.00E+3	1	2.03	(2,2,1,3,1,3); based on own calculations, assuming that 0.6 of the paint are emitted as NMVOCs
emissions to air	Heat, waste	-	-	MJ	5.69E+6	1	1.17	(2,2,1,3,1,3); standard value
emissions to air	NMVOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	3.50E+3	1	2.03	(2,2,1,3,1,3); Derived from the used paint and solvent, assuming all solvent is emitted to air

Tab. 7.57 Input data for disposal of goods train equipment

	Name	Location	Unit	disposal, locomotive	UncertaintyType	StandardDeviation95%	GeneralComment
product	disposal, locomotive	RER	unit	1.00E+0			
technosphere	transport, lorry 32t	RER	tkm	5.44E+2	1.00E+0	2.28E+0	(4,5,na,na,na,na); standard distance
	disposal, used mineral oil, 10% water, to hazardous waste incineration	OH	kg	3.56E+3	1.00E+0	1.26E+0	(3,1,1,3,1,4); oral communication
	disposal, emulsion paint, 0% water, to municipal incineration	OH	kg	3.85E+2	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition locomotive
	disposal, glass, 0% water, to municipal incineration	OH	kg	2.35E+2	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition locomotive
	disposal, plastics, mixture, 15.3% water, to sanitary landfill	OH	kg	6.69E+3	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition locomotive
	disposal, polyvinylchloride, 0.2% water, to municipal incineration	OH	kg	3.51E+2	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition locomotive

Tab. 7.58 Input data for disposal of regional passenger train equipment

	Name	Location	Infra	Unit	disposal, regional train	UncertaintyType	StandardDeviation95%	GeneralComment
	Location				CH			
	InfrastructureProcess				1.00E+0			
	Unit				unit			
product	disposal, regional train	CH	1	unit	1.00E+0			
technosphere	transport, lorry 28t	CH	0	tkm	1.32E+3	1.00E+0	2.28E+0	(4,5,na,na,na,na); standard distance
	disposal, emulsion paint, 0%water, to municipal incineration	CH	0	kg	5.57E+0	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition train composition
	disposal, glass, 0%water, to municipal incineration	CH	0	kg	7.23E+3	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition train composition
	disposal, plastics, mixture, 15.3%water, to sanitary landfill	CH	0	kg	1.30E+4	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition train composition
	disposal, wood untreated, 20%water, to municipal incineration	CH	0	kg	6.20E+3	1.00E+0	1.26E+0	(3,1,1,3,1,4); material composition train composition

Tab. 7.59 Input data for disposal of long-distance passenger train equipment

	Name	Location	Infra	Unit	disposal, long-distance train	UncertaintyType	StandardDeviation95%	GeneralComment
	Location				CH			
	InfrastructureProcess				1.00E+0			
	Unit				unit			
product	disposal, long-distance train	CH	1	unit	1.00E+0			
technosphere	transport, lorry 28t	CH	0	tkm	6.58E+2	1.00E+0	2.28E+0	(4,5,na,na,na,na); standard distance
	disposal, emulsion paint, 0%water, to municipal incineration	CH	0	kg	6.45E+2	1.00E+0	1.26E+0	(3,1,1,3,1,4); derived from the material composition of the modelled IC (unit process IC manufacturing)
	disposal, glass, 0%water, to municipal incineration	CH	0	kg	9.51E+3	1.00E+0	1.26E+0	(3,1,1,3,1,4); derived from the material composition of the modelled IC (unit process IC manufacturing)
	disposal, plastics, mixture, 15.3%water, to sanitary landfill	CH	0	kg	3.00E+3	1.00E+0	1.26E+0	(3,1,1,3,1,4); derived from the material composition of the modelled IC (unit process IC manufacturing)

Tab. 7.60 Life cycle inventory input data for disposal of a German ICE

	Name	Location	Infra	Unit	disposal, ICE	UncertaintyType	StandardDeviation95%	GeneralComment
	Location				DE			
	InfrastructureProcess				1.00E+0			
	Unit				unit			
product	disposal, ICE	DE	1	unit	1.00E+0			
technosphere	transport, lorry 32t	RIER	0	tkm	1.50E-1	1.00E+0	2.28E+0	(4,5,na,na,na,na); standard distance
	disposal, emulsion paint, 0%water, to municipal incineration	CH	0	kg	1.00E+0	1.00E+0	1.26E+0	(3,1,1,3,1,4); derived from the material composition of the modelled ICE (unit process ICE manufacturing)
	disposal, glass, 0%water, to municipal incineration	CH	0	kg	1.00E+0	1.00E+0	1.26E+0	(3,1,1,3,1,4); derived from the material composition of the modelled ICE (unit process ICE manufacturing)
	disposal, plastics, mixture, 15.3%water, to sanitary landfill	CH	0	kg	1.00E+0	1.00E+0	1.26E+0	(3,1,1,3,1,4); derived from the material composition of the modelled ICE (unit process ICE manufacturing)

Data Quality and Uncertainties

The quality of the data documented for the "Re 462" is considered as good, however the locomotive is a fairly new locomotive and thus not representative for the current locomotive park of the SBB. The share of aluminium and plastic materials is assumed to be lower for older locomotives. Thus, the data is considered of medium quality. For wagons the quality of the data for a specific wagon is fairly good. Uncertainties result mainly from assumptions made to calculate an average wagon.

For the disposal the data quality is considered as sufficient. Since the actual amount of disposed material is directly linked to the material composition the reliability of the data is based on the quality of the manufacturing model.

7.4 Rail Infrastructure

In this project, generic data for the Swiss rail network has been determined. Environmental interventions and inputs from the technosphere are referred to 1-meter and year [ma] rail track. The Swiss rail network includes normal gauge and narrow gauge tracks. It comprises approximately 10200 km rail tracks (SBB, 2002). In addition, 1400 km siding are reported. As demonstrated in Tab. 7.40 more than 98% of the goods transport in Switzerland is performed by the SBB. Thus, in line with the inventories for operation and manufacturing, for the calculation of the interventions of rail infrastructure we merely take the SBB network into account. The SBB rail network predominately comprises one-way tracks and two-way tracks with a length of 1391 km and 1609 km (in the year 2000). One-way tracks are converted in two-way tracks, resulting in approximately 2300 km two way tracks. The material expenditures, if not stated differently, refer to a two-way track. In order to relate the long-term material investments for infrastructure to the reference unit, assumption about the life span are required and are documented in the following sections. All data presented in this section refers to the total network.

7.4.1 Infrastructure Demand and Allocation

Similar as road infrastructure, rail infrastructure is not solely used by goods trains. Thus, allocation between passenger and goods transportation is unavoidable. As outline in the chapter road transport, various allocation principles are available to allocate the environmental interventions of infrastructure processes.

Allocation of Construction and Disposal Expenditures

In line with the approach of road transport infrastructure we use the gross tonne kilometre as allocation factor for construction and disposal activities. For the Swiss rail network in the year 2000 we apply a total Gt_{km}-performance of 5.94×10^{10} Gt_{km} resulting in a specific rail demand per Gt_{km} of 3.87×10^{-5} (m*a)/Gt_{km}. Assuming a ratio gross tonne/carried goods of 2.26 Gt_{km}/tkm we obtain a specific rail demand per functional unit of 8.7×10^{-5} (m*a)/tkm. For the European rail network we apply a typical demand factor of 2.4 Gt_{km}/tkm.

For the determination of passenger train infrastructure demand, for regional and long-distance transport we employ a gross tonne/transported passenger demand of 3.39^{36} Gt/p and 2.42 Gt/p, respectively. Thus we obtain a specific rail demand per functional unit of 1.31×10^{-4} and 9.37×10^{-5} (m*a)/pkm for regional and long-distance transport, respectively.

For the German ICE, allocation is performed employing the kilometric train performance of freight and passenger trains. According to von Rozycki (2003), we assume a kilometric performance of the ICE of 10950 vkm/year (11% of total traffic). If we take into account an average load of 309 passenger per train we achieve a demand factor of 3.25×10^{-8} m*a/pkm.

³⁶ Oral communication with Markus Halder, Environmental Center Swiss Federal Railways, octobre 2009

Allocation of Operation and Maintenance Expenditures (Including Land Use)

For operation services, including land use, we use the yearly vehicle kilometre performance as allocation factor. For the Swiss road network in the year 2000 we account for a total vkm-performance of $1.31\text{E}+08$ vkm resulting in a specific road demand per $1.76\text{E}-02$ (m^*a)/vkm. Assuming a ratio kilometric performance/transport performance of $2.60\text{E}-03$ vkm/tkm we obtain a specific rail demand per functional unit of $4.57\text{E}-05$ (m^*a)/tkm. For the European rail network we apply the same demand factor as for Switzerland.

For the determination of passenger train infrastructure demand, for regional and long-distance transport we employ a ratio kilometric performance/transport performance of 4.0 vkm/pkm and 2.8 vkm/pkm, respectively. Thus we obtain a specific rail demand per functional unit of $3.29\text{E}-04$ and $1.00\text{E}-04$ (m^*a)/pkm for regional and long-distance transport, respectively.

For the German ICE, allocation is performed employing the kilometric train performance of freight and passenger trains. According to von Rozycki (2003) we assume a kilometric performance of the ICE of 10950 vkm/year (11% of total traffic). If we take into account an average load of 309 passenger per train we achieve an demand factor of $3.25\text{E}-8\text{m}^*\text{a}/\text{pkm}$.

7.4.2 Rail Track Construction and Renewal

The construction of rail track infrastructure includes the actual construction of the rail track (substructure and track bedding) as well as expenditures for the construction of tunnels and bridges. In Fig. 7.2 the different levels of a rail track are illustrated. Due to the fact that the life span of different components of the infrastructure varies significantly, material and energy expenditures are referred to one year. We assume a life span for the infrastructure of 100 years. Thus, all expenditures are aggregated for a life span of 100 years and referred to one year. It should be noted that the modelled energy consumption and material expenditures represent the conditions of the reference year 2000, even though some of these processes have been taken place in the past.

Further components of rail track infrastructure, such as signalling infrastructure, train overtaking stations, sound insulation walls and buildings (stations, service garages) are not included in this study. In a recent research von Rozycki et al. (2003), pointed out that these components do not contribute significantly to a life cycle resource consumption of a rail transport system. SF_6 -emissions of electric transformer stations are accounted for in the train operation unit process.

Construction of Rail Track

For the construction of the rail driveway part material consumption due to track bedding (rails and railway sleepers (concrete reinforced with steel) and gravel) and substructure (bulldozed and compressed soil) are taken into account. In Fig. 7.2 a simplified model of the layers of rail conventional rail track infrastructure is illustrated.

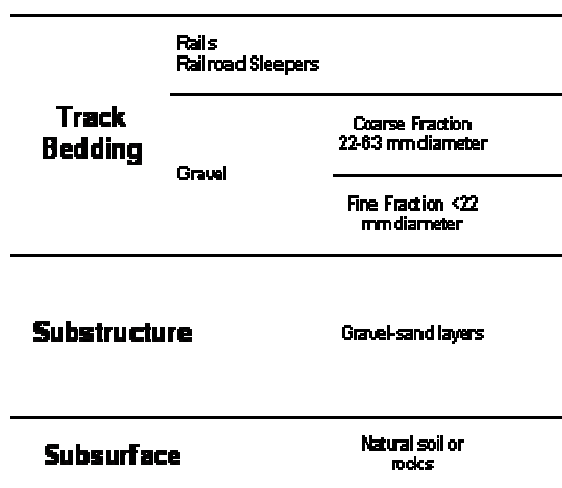


Fig. 7.2 Layers of rail track infrastructure

In addition, material expenditures for construction of the catenary system (overhead contact line and steel masts) are accounted for. In Tab. 7.61 data available is summarised.

Tab. 7.61 Materials and energy used in the construction of for a rail track

Materials/Energy	Use	Unit	Elsener & Strub (1993)	von Rozycki et al. (2003)
Gravel, crushed	track bedding	kg/(m*a)	113.000	530.000
Concrete	railroad sleeper	kg/(m*a)	20.900	34.300
Steel	railroad sleeper		n.a	1.300
Wood	railroad sleeper	kg/(m*a)	4100.000	0
Steel	rail	kg/(m*a)	4.100	9.400
Steel low-alloyed	mast	kg/(m*a)	0.560	0.740
Zinc	protective coating mast	kg/(m*a)	0.006	0.005
Copper	overhead contact line	kg/(m*a)	0.055	0.420
Transport lorry 28t	track bedding	Tkm/(m*a)	423.862	

Timber railroad sleepers are not modelled in this study, since they are continuously replaced with concrete railroad sleepers. In Tab. 7.69 the figures as employed in this project are presented.

Construction of rail tunnels

Tunnels require a significant higher material input, primarily in form of concrete and steel for reinforcement. In addition, large quantities of soil and rocks, have to be excavated and removed. In the year 2000 the length of tunnels in Switzerland accounted for 218 km (SBB 2002). This results in a share of tunnels on the total rail track network of approximately 7% if we simply sum up one way and multi way tracks. Thus, per km rail, 70 m tunnel are accounted for in this project

For the construction of rail tunnels data is available from Maibach et al. (1999). These data is based on studies of the "Lötschberg-Basistunnel" and supplemented with information of the tunnel construction specialist of a Rothpelz-Lienhardt. The data represents a one-track tunnel with a diameter of 10m and a concrete layer of 0.5m. Von Rozycki (2003), further distinguishes two types of tunnel

constructions approaches and supplies the referring basic material consumption, referring to a two-way track per pipe tunnel. In **Fehler! Verweisquelle konnte nicht gefunden werden.** this data is summarised.

Construction of Bridges

For material expenditures data is available from von Rozycki et al. (2003) and displayed in Tab. 7.63. In the year the length of bridges is accounted for 92 km. This results in a share of bridges on the total rail track network of approximately 3% if we simply sum up one way and multi way tracks. Thus, per km rail, 30 m bridge is accounted for in this project. In Tab. 7.69 the figures as employed in this project are presented.

Tab. 7.62 Material and Energy Consumption for construction of rail tunnels in Switzerland

Materials/Energy	Unit	Maibach (1999) including track	Maibach (1999) excl. Track ¹⁾	Von Rozycki (2003)
gravel, crushed	kg/(m*a)	696	583	
concrete	kg/(m*a)	465	444.1	575
steel low-alloyed/iron	kg/(m*a)	1.58	1.02	
steel	kg/(m*a)	14.7	10.6	24.5
ceramics	kg/(m*a)	1.6	0	-
zinc	kg/(m*a)	0.006	0	-
aluminum	kg/(m*a)	0.05	0	-
copper	kg/(m*a)	0.055	0	-
PVC	kg/(m*a)	0.0653	0	-
electricity	kWh/(m*a)	-	569	-
diesel, building machine	MJ/(m*a)	-	135	-
transport lorry building materials	Tkm/(m*a)	-	103.872	-
excavation soil	kg/(m*a)	-	-	4850
transport lorry landfill	Tkm/(m*a)	-	-	48500

1: In Maibach (1999) the expenditures for tunnel construction also account for expenditures for track construction. In order to avoid double counting, these expenditures are subtracted.

Tab. 7.63 Material Expenditures for the construction of rail bridges

	absolute material consumption		life span	specific material consumption	
Material	concrete	steel		concrete	steel
Unit	t/km bridge	t/km bridge	years	kg/(m*a) bridge	kg/(m*a) bridge
Glen bridges	55000	3000	100	550	30
Road/railway bridges	89000	4900	50	1780	98
Average bridge this project ¹⁾				673	36.8

1: The average bridge in this study comprises 90% glen bridge and 10% road rail bridges

7.4.3 Operation and Maintenance of Rail Infrastructure

Operation and Maintenance of SBB Rail Tracks

The renewal of materials is already addressed in the construction module. Important components not addressed in the construction module are points or switches. Points are required to change tracks. For high-speed railway routes 2.1 points per average two-way rail track kilometre are reported von Rozycki et al. (2003). For the SBB-network about 18000 points are in operation. For main lines, on average a point is installed every 830 metres, on the remaining lines the average point density is 270 metres Oggier (2002). In von Rozycki et al. (2003) a heating energy consumption to avoid icing in winter of $400 \text{ kWh}_{\text{el}}/(\text{unit} \cdot \text{a})$ has been stated. For main lines and remaining lines this results in energy consumption of $0.5 \text{ kWh}/(\text{m} \cdot \text{a})$ and $1.5 \text{ kWh}/(\text{m} \cdot \text{a})$, respectively. If we further assume that goods transportation predominately takes place on main lines, and use the allocation factor "gross tonne kilometre", we obtain an average point electricity consumption of $0.2 \text{ kWh}/(\text{m} \cdot \text{a})$ or $4.21\text{E-}05 \text{ kWh}/\text{tkm}$. The resulting figure, is significantly lower than the figure ($32 \text{ kWh}/\text{m} \cdot \text{a}$) stated in Frischknecht et al. (1996). The latter figure is obtained top down Manto (1994) and a lot of assumptions and allocation steps have been essential, resulting in a poor transparency and high uncertainty of the figure. In contrast, the figure for the point heating electricity consumption has been achieved with consistent assumption and thus is employed in this study. Figures for oil and natural gas combustion given in Frischknecht et al. (1996) and based on Manto (1994) are not included in this study for the same reason.

In addition, points need to be lubricated regularly. In the year 2000 the 69697 litre of lubricates have been consumed for point maintenance. The majority of the employed lubricates are biodegradable (SBB, 2002).

The use of herbicides in 2000 has been reported with 4.474 t SBB (2002). The concentration of the active agent is 360g Glyphosat/l. In spite of the fact that Glyphosat is considered as biodegradable Haas (2000) emissions to soil are accounted for in this project.

Operation and Maintenance of SBB Rail Tracks

Electricity consumption for points is directly taken from von Rozycki et al. (2003). The use of lubricates is estimated based on Swiss figures, assuming a use of $3.88 \text{ l}/\text{point} \cdot \text{a}$ and 2.1 point per km track. For the use and emissions of herbicides we use the figures derived from Swiss conditions.

Operation and Maintenance of Rail Tunnel

Data for operating expenditures are available from Maibach et al. (1999). The energy consumption of a rail tunnel has been estimated based on the electricity consumption of road tunnels. In contrast to road tunnels, lighting and ventilation are not installed in short rail tunnels, in Switzerland. In long, alp tunnels, however, the ventilation is required for thermal reason. In Maibach et al. (1999), the electricity consumption is estimated to be the half of a 12 h ventilation of a road tunnel and lighting has been neglected. In consequence the electricity consumption is calculated as $837 \text{ kWh}/(\text{m tunnel} \cdot \text{a})$.

7.4.4 Land Use

In this project we distinguish between land transformation and land occupation. Whilst transformation directly changes the ecological quality of land, occupation postpones changes of the ecological quality. Furthermore, we distinguish between the area actually used for rail tracks and embankments.

In Tab. 7.64 detailed information on the nature of the network and resulting occupied area are summarised for both, the entire Swiss network and the SBB rail network.

Tab. 7.64 Length and occupied area of the Swiss rail network and the SBB network.

Total Swiss Network		Width [m] ¹⁾	Length [km] ²⁾	Main area	Embankment ³⁾	Main area ⁴⁾	Embankment ⁴⁾
Unit		m	km	Km ²	Km ²	Km ²	Km ²
Normal gauge	One track	8.5	2418.0	20.6	18.7		
	Multi track	12	1268.3	15.2	13.8		
Siding		6	1400.0	8.4			
	Total		5086.3	44.2	32.5		
Narrow gauge	One Track	8	906.4	7.3	6.6		
	Multi track ⁶⁾	11.5	475.4	5.5	5.0		
	Total		1381.8	12.7	11.6		
Total CH			12936.1	56.9	44.1	50.2	46.87
SBB Network							
Normal gauge	One Track	8.5	1391.0	11.8	10.7		
	Multi track	12	1609.0	19.3	17.6		
Siding ⁵⁾		6	700	4.2	3.8		
	Total		3686.3	31.1	28.3	35.92	
Narrow gauge	One Track	8	74.0	0.6	0.5		
	Multi track	11.5		0.0	0.0		
	Total		74.0	0.6	0.5	0.45	
Total SBB			3060.3	35.9	32.7	36.4	

1: oral communication Müller Gleisbau

2: Bfs 2002

3: based on the ratio used in Hüsler: Embankment area = 0.9*Main area

4: as documented in Hüsler

5: the issue of Anschlussgleise is rather complicated. Currently very little is known how many kilometer of the Anschlussgleise are still in use. Thus, as an first assumption we merely account for 50% of the total length of Anschlussgleise.

6: Multi tracks are predominately two track routes. In 1993 the share of three track and four track routs on the total multi track network was about 4 %.

Land Occupation

Occupation refers to the time period for which the land is unavailable for other activities. The occupation intervention is expressed in terms of land use type, area and time. The unit is therefore square metres *year and the reference 1-metre rail track. In Tab. 7.65 the resulting specific land occupation for the SBB network is summarised.

Tab. 7.65 Specific land occupation for rail transport

	unit	rail network	rail embankments
Total traffic area occupation (SBB)	m ² /m*a	11.5	10.4
Total traffic area occupation (ICE track)	m ² /m*a	13.5	10.4 ¹⁾

1) in the absence of further information we employ the Swiss value for the German ICE track

Land Transformation

For transformation (the change of land use to be allocated to a certain user or user type) the situation is more difficult, mainly due to data variability. The yearly change in land use for a certain user may

change considerably, as land use change is not a continuous process, but very much determined by political decisions and economic constraints and opportunities. Thus, to assess transformation, the yearly trend has been determined based on a fairly long period of 30 years, from 1971 to 2000. In Tab. 7.66 specific land transformation figures for the SBB network are presented. For the ICE track we assume a lifespan of 100 years.

Tab. 7.66 Specific land transformation for rail transport

	unit	Rail network	Rail embankments
Total traffic area transformation (SBB)	m ² /m	2.66E-02	2.42E-02
Total traffic area transformation (ICE track)	m ² /m	1.3E-01	1.04E-02 ¹⁾

2) in the absence of further information we employ the Swiss value for the German ICE track

7.4.5 Disposal of Rail Infrastructure

The disposal of rail infrastructure reflects the SBB situation in 2000. Environmentally relevant components are contaminated gravel of the track bedding and contaminated gravel-sand of the substructures. Also, the disposal of creosoted timber railroad sleepers requires a special treatment. In contrast, for rails and concrete railroad sleepers a cut off allocation has been made, since the arising amount is completely recycled SBB (2002).

Currently, the SBB rebuild approximately 140-160 km/year of its rail network track bedding. In addition, for approximately 30 km of these tracks a complete renewal is performed including both, track bedding and substructure. The resulting figures (Tab. 7.68), correspondents with the figure presented in the SBB environmental report SBB (2002). The excavation waste is reported with 412'800 t/a, including both gravel and gravel-sand. Also it is claimed that approximately 80% of the used gravel are transported to specific washing facilities. The cleaned gravel then usually is used for other purposes. Thus, a cut off allocation is applied here. However, 20% of the total amount (82500 t/a) is deposited at specific landfills.

For the ballastless ICE track no disposal is accounted for.

Tab. 7.67 Yearly excavation waste of the SBB Oggier (2002)

Source		gravel & gravel-sand [m ³ /m]		Total gravel/ gravel sand [m ³ /a]		average [t/a] ¹⁾
	rebuilt/year [km/a]	min	max	min	max	
Track bedding rebuilding	120	0.6	1	72000	120000	163200
Substructure rebuilding	30	4	6	120000	180000	255000
Total SBB	150			192000	300000	418200

1: it has been assumed that 1 m³ gravel is equal to 1.7 kg gravel Oggier (2002).

Approximately 90% of wooden wastes are timber railroad sleepers. In 2000 and in particular 2001, the amount of wooden sleepers waste has increased significantly as demonstrated in Tab. 7.68. The underlying reason of this development is a change in legislation. Since autumn 2000, in Switzerland the sale of used wooden sleepers is prohibited. In 2001, the wooden sleepers are incinerated in thermal power plants. Thus, we merely include the transport of the wooden sleepers to thermal power plants. We assume an average distance of 200 km for the transport.

Tab. 7.68 Disposed railroad sleepers (modified from SBB (2002))

Waste type	unit	1996	1997	1998	1999	2000	2001
Timber sleeper	t	1090 ¹⁾	1180 ¹⁾	1040 ¹⁾	1.460 ¹⁾	2065 ²⁾	7330 ²⁾
Concrete sleeper	unit	n.a.	n.a.	n.a.	n.a.	23018	49296

1: the figures represent 90% of the total wood waste, as reported in SBB (2002)

2: since the significant increase of wood waste is due to an increase in waste from used wooden sleepers, not the 90% values has been taken but the 10% value of the reported wood waste quantity for 1999 has been subtracted from the values reported for 2000 and 2001 in SBB (2002)

7.4.6 Life Cycle Inventory Input Data

In this section the inventory input data for rail infrastructure processes are presented. For each component, a Swiss average track is assumed as well as a specific ICE rail track in Germany.

Tab. 7.69 Life cycle inventory input data for the construction of rail infrastructure in Switzerland

	Name	Location	Unit	railway track	UncertaintyType	Standard Deviation95%	GeneralComment	constr. railway track	constr. tunnel	constr. bridge
product	railway track	CH	m	1.00E+0						
technosphere	concrete, exacting, at plant	CH	m ³	4.31E-2	1	1.27	LCA case study; (3,3,1,3,3,4)	1.56E-02	2.61E-01	3.06E-01
	gravel, crushed, at mine	CH	kg	5.71E+2	1	1.27	LCA case study; (3,3,1,3,3,4)	5.30E+02	5.83E+02	
	steel, low-alloyed, at plant	RER	kg	5.60E-1	1	1.27	LCA case study; (3,3,1,3,3,4)	5.60E-01		
	reinforcing steel, at plant	RER	kg	1.35E+1	1	1.27	LCA case study; (3,3,1,3,3,4)	1.07E+01	2.45E+01	3.68E+01
	zinc for coating, at regional storage	RER	kg	6.00E-3	1	1.29	literature & oral communication; (3,3,3,3,3,4)	6.00E-03		
	copper, at regional storage	RER	kg	5.50E-2	1	1.29	literature & oral communication; (3,3,3,3,3,4)	5.50E-02		
	diesel, burned in building machine	GLO	MJ	9.50E+0	1	2.09	literature & oral communication; (3,3,3,3,3,4)	5.00E-02	1.35E+02	
	electricity, medium voltage, at grid	CH	kWh	3.99E+0	1	1.29	literature & oral communication; (3,3,3,3,3,4)	0.00E+00	5.69E+01	
	excavation, skid-steer loader	RER	m ³	1.20E+0	1	2.09	literature & oral communication; (3,3,3,3,3,4)	1.20E+00		
	transport, freight, rail	CH	tkm	8.48E+0	1	2.09	standard distance; (4,5,na,na,na,na)	6.79E+00	1.47E+01	2.21E+01
	transport, lorry 28t	CH	tkm	1.40E+1	1	2.09	standard distance; (4,5,na,na,na,na)	1.19E+01	2.44E+01	1.53E+01
emission to air	Heat, waste	-	MJ	1.44E+1	1	1.29	standard value; (3,3,3,3,3,4)			

Tab. 7.70 Life cycle inventory input data for operation and maintenance of rail infrastructure in Switzerland. Figures in this table represent ecoinvent data v1.1; the following elementary flows have been corrected for ecoinvent data v2.0: consumption of glyphosate (1.95E-3 kg/ma); emission of glyphosate (7.00E-4 kg/ma).

	Name	Location	Category	SubCategory	Unit	operation, maintenance, railway track	Uncertainty Type	Standard Deviation %	GeneralComment	operation railway track	operation tunnel
product technosphere	operation, maintenance, railway track	CH			ma	1.00E+0					
	electricity, medium voltage, at grid	CH			kWh	5.91E+1	1	1.20	(3,3,3,1,1,4); LCA case studies	4.82E-01	8.37E+02
	glyphosate, at regional storehouse	CH			kg	1.95E+0	1	1.05	(1,1,1,1,1,1); environmental report		
	lubricating oil, at plant	RER			kg	2.54E-2	1	1.05	(1,1,1,1,1,1); environmental report		
emission to air	Heat, waste	-	air	unspecified	MJ	2.13E+2	1	1.29	(3,3,3,3,3,4); standard value		
emissions to soil	Glyphosate	-	soil	agricultural	kg	7.00E-1	1	1.20	(1,1,1,1,1,1); environmental report and own calculations		
	Oils, biogenic	-	soil	unspecified	kg	2.54E-2	1	1.20	(1,1,1,1,1,1); environmental report and own calculations		
land use	Occupation, traffic area, rail network	-	resource	land	m2a	1.15E+1	1	1.52	(3,2,1,1,na,na); literature & own calculations		
	Occupation, traffic area, rail embankment	-	resource	land	m2a	1.04E+1	1	1.52	(3,2,1,1,na,na); literature & own calculations		
	Transformation, from unknown	-	resource	land	m2	5.08E-2	1	2.01	(3,2,1,1,na,na); literature & own calculations		
	Transformation, to traffic area, rail network	-	resource	land	m2	2.66E-2	1	2.01	(3,2,1,1,na,na); literature & own calculations		
	Transformation, to traffic area, rail embankment	-	resource	land	m2	2.42E-2	1	2.01	(3,2,1,1,na,na); literature & own calculations		

Tab. 7.71 Life cycle inventory input data for disposal of rail infrastructure in Switzerland. Figures in this table represent ecoinvent data v1.1; the following elementary flows have been corrected for ecoinvent data v2.0: The input of waste treatment service "disposal, pollutants from rail ballast, 0% water, to residual material landfill, CH, [kg]" is too high by a factor of 256. Correct value is 0.140 kg/ma. Additional waste management service input is added, namely: 38.65 kg/ma "disposal, inert waste, 5% water, to inert material landfill, CH, [kg]."

	3702	3703	3508	3706	3707
Explanations	Name	Location	Infrastructure- Process	Unit	disposal, railway track
	Location				CH
	InfrastructureProcess				1
	Unit				ma
Outputs	disposal, railway track	CH	1	ma	1.00E+0
Technosphere	transport, lorry 28t	CH	0	tkm	3.70E+1
	disposal, pollutants from rail ballast, 0% water, to residual material landfill	CH	0	kg	1.40E-1
	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	3.87E+1

Tab. 7.72 Life cycle inventory input data for the construction of ICE rail infrastructure (ballastless) in Germany

	Name	Location	Infrastr ucture Process s	Unit	railway track, ICE	UncertaintyTy pe	Standard Deviation95%	GeneralComment	constr. railway track	constr. tunnel	constr. bridge
	Location				DE				CH	CH	CH
	InfrastructureProcess				1				1	1	1
	Unit				ma				ma	ma	ma
product	railway track, ICE	DE	1	ma	1.00E+0						
technosphere	concrete, exacting, at plant	CH	0	m³	4.19E-2	1	1.27	literature studies; (3,3,1,3,3,4)	3.47E-02	9.44E-02	2.01E-02
	steel, low-alloyed, at plant	RER	0	kg	1.88E+0	1	1.27	literature studies; (3,3,1,3,3,4)	1.77E+00	1.35E+00	3.62E-01
	reinforcing steel, at plant	RER	0	kg	1.06E+1	1	1.27	literature studies; (3,3,1,3,3,4)	1.00E+01	7.68E+00	2.05E+00
	copper, at regional storage	RER	0	kg	4.24E-1	1	1.29	literature studies; (3,3,3,3,3,4)	4.24E-01		
	diesel, burned in building machine	GLO	0	MJ	5.00E-2	1	2.09	literature studies; (3,3,3,3,3,4)	5.00E-02		
	excavation, skid-steer loader	RER	0	m³	1.89E+0	1	2.09	literature studies; (3,3,3,3,3,4)	1.81E+00	1.04E+00	
	aluminium, production mix, at plant	RER	0	kg	1.25E-1	1	2.09	literature studies; (3,3,3,3,3,4)	1.25E-01		
	ceramic tiles, at regional storage	CH	0	kg	1.20E-1	1	2.09	literature studies; (3,3,3,3,3,4)	1.20E-01		
	pig iron, at plant	GLO	0	kg	7.40E-1	1	2.09	literature studies; (3,3,3,3,3,4)	7.40E-01		
	tin, at regional storage	RER	0	kg	5.00E-3	1	2.09	literature studies; (3,3,3,3,3,4)	5.00E-03		
	transport, freight, rail	RER	0	tkm	2.63E+0	1	2.09	standard distance; (4,5,na,na,na,na)	2.36E+03	3.61E+03	4.82E+02
	transport, lorry 32t	RER	0	tkm	4.28E+1	1	2.09	standard distance; (4,5,na,na,na,na)	4.00E+04	4.00E+04	8.83E+02
disposal	disposal, inert waste, 5%water, to inert material landfill	CH	0	kg	2.86E+0	1	1.27	own calculation based on the excavation figures; (3,3,3,1,1,5)	1.81E+00	1.04E+00	

Tab. 7.73 Life cycle inventory input data for operation and maintenance of ICE rail infrastructure in Germany. Figures in this table represent ecoinvent data v1.1; the following elementary flows have been corrected for ecoinvent data v2.0: consumption of glyphosate (1.95E-3 kg/ma); emission of glyphosate (7.00E-4 kg/ma).

Explanations	Name	Location	Infrastructure- Process	Unit	operation, maintenance, railway track, ICE
	Location				DE
	InfrastructureProcess				1
	Unit				ma
Outputs	operation, maintenance, railway track, ICE	DE	1	ma	1.00E+0
Technosphere	electricity, medium voltage, production DE, at grid	DE	0	kWh	6.46E+0
	glyphosate, at regional storehouse	CH	0	kg	7.00E-4
	lubricating oil, at plant	RER	0	kg	7.53E-3
	transport, freight, rail	RER	0	tkm	1.37E-3
resource, land	Occupation, traffic area, rail network			m2a	1.33E+1
	Occupation, traffic area, rail embankment			m2a	1.04E+1
	Transformation, from unknown			m2	2.37E-1
	Transformation, to traffic area, rail network			m2	1.33E-1
	Transformation, to traffic area, rail embankment			m2	1.04E-1
air, unspecified	Heat, waste			MJ	2.33E+1
soil, industrial	Glyphosate			kg	7.00E-4
soil, unspecified	Oils, biogenic			kg	7.53E-3

Data Quality and Uncertainties

The quality of the data for track expenditures is considered as good. The data for tunnel construction and bridges however is of lower quality, since it is based on set of assumptions, which are fairly uncertain. Furthermore the representativeness of the data for the total Swiss rail network is not known.

For land use data the representativeness and geographical correlation and hence the quality of the data is considered as good. The data is based on some simplification, such as that all multi lane tracks are accounted for as two way tracks, which leads to slight underestimation of the land use. Furthermore, the actual length of actually used siding is not available.

The representativeness for other countries or railway companies is unknown. Further uncertainties are at stake, when applying these data to the current transport performance and calculate a the specific land use per tkm.

7.5 Rail Transport

7.5.1 Method

In this section the demand factors of various transport components (as described in the previous sections) required to fulfill the functional unit of one tkm freight transport are presented. For instance, the module "transport, freight train" accesses the sub module "locomotive" with a demand factor $7.11\text{E-}10$; i.e. per functional unit $7.11\text{E-}10$ locomotives are required. The determination of the share of vehicles per transport performance unit [tkm] data is documented in Tab. 7.74 and Tab. 7.75.

In Tab. 7.76 and Tab. 7.78 the actual demand of vehicle- and road infrastructure per transport performance [tkm] are presented for Switzerland and Europe respectively.

Tab. 7.74 Transport performance of locomotives

	Unit	SBB 2000	SBB 1999
Locomotives total	unit	750	750
Locomotive goods trans.		307	307
Goods transport performance	tkm	$1.08\text{E}+10$	$9.80\text{E}+09$
Yearly transport performance per vehicle	tkm/(unit*a)	$3.52\text{E}+07$	$3.19\text{E}+07$
Life time transport performance (40a)	tkm/unit	$1.41\text{E}+09$	$1.28\text{E}+09$
Share of vehicle per performed tkm	unit/tkm	$7.11\text{E-}10$	$7.83\text{E-}10$

Tab. 7.75 Transport performance of goods wagons

	Unit	SBB 2000	SBB 1999
Goods wagons total	unit	19791	20119
Goods transport performance	tkm	$1.08\text{E}+10$	$9.80\text{E}+09$
Yearly transport performance per wagon	tkm/(unit*a)	$5.45\text{E}+05$	$4.87\text{E}+05$
Life time transport performance (40a)	tkm/unit	$2.18\text{E}+07$	$1.95\text{E}+07$
Share of wagon per performed tkm	unit/tkm	$4.59\text{E-}08$	$5.13\text{E-}08$

In order to account for the fact that the same rail tracks are used for both passenger and goods transportation we apply the allocation factors as documented in section 7.4.1.

The dataset transport of coal by train in China is based on "transport, freight, rail (RER)" but instead of "operation, freight train (RER)" three different inputs are defined to reflect the use of electric, diesel, and steam locomotives in the coal industry and in China in general:

For the USA, according to (Facanha & Horvath 2006) it can be assumed that US freight trains for coal transport are all operated using diesel engines. The new dataset "transport, freight, rail, diesel/US/tkm" (Tab. 7.80) is based on the existing European freight transport by train ("transport, freight, rail/RER/tkm"). Instead of the average European share of electric and diesel freight train transport, only diesel as fuel is used.

With the performance factors of Tab. 7.76 one may calculate the average weight of transported goods in the selected European countries (transport performance divided to the train performance) or the load factor (transport performance divided by traffic performance). The inverse of the load factor shows the load of the infrastructure due to the transport of one tkm. One calculated the demand of the track infrastructure with the Swiss value of $3.87\text{E-}05$ per Gtkm.

Tab. 7.76 Performance factors of European Railways {UIC, 2007 #91}

	av.weight of transported goods per train	average weight of train	Load factor	Gtkm/tkm	Demand of track infrastructure per tkm
	[t]	[t]	[%]		
Austria	427	951	45%	2.23	8.6E-05
Italy	407	661	62%	1.62	6.3E-05
France	445	1'023	43%	2.30	8.9E-05
Germany	449	999	45%	2.23	8.6E-05
Belgium	654	1'289	51%	1.97	7.6E-05
Switzerland	385	869	44%	2.26	8.7E-05

7.5.2 Life Cycle Inventory Input Data of freight transport

Tab. 7.77 Life cycle inventory input data for freight transport services in Switzerland

Explanations	Name	Location	Infrastructure- Process	Unit	transport, freight, rail	transport, freight, rail, diesel, with particle filter	uncertaintyType	StandardDeviation 95%	GeneralComment
	Location InfrastructureProcess Unit				CH 0 tkm	CH 0 tkm			
Technosphere	operation, freight train	CH	0	tkm	1.00E+0		1	1.30	own estimation
	operation, freight train, diesel, with particle filter	CH	0	tkm		1.00E+0	1	1.30	own estimation
	locomotive	RER	1	unit	7.11E-10	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	goods wagon	RER	1	unit	4.59E-8	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, goods wagon	RER	1	unit	4.59E-8	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, locomotive	RER	1	unit	7.11E-10	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	disposal, locomotive	RER	1	unit	7.11E-10	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	railway track	CH	1	ma	8.73E-5	8.73E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure legth -transport performance ratio and further geographical uncertainties due to extrapolation
	operation, maintenance, railway track	CH	1	ma	8.73E-5	8.73E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure legth -transport performance ratio and further geographical uncertainties due to extrapolation
	disposal, railway track	CH	1	ma	8.73E-5	8.73E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure legth -transport performance ratio and further geographical uncertainties due to extrapolation
products	transport, freight, rail	CH	0	tkm	1.00E+0				
	transport, freight, rail, diesel, with particle filter	CH	0	tkm		1.00E+0			

Tab. 7.78 Life cycle inventory input data for freight transport services in selected European countries

Explanations	Name	Location	Infrastructure-Process	Unit	transport, freight, rail	transport, freight, rail	transport, freight, rail	transport, freight, rail	transport, freight, rail	uncertaintyType	StandardDeviation 95%	GeneralComment
					AT 0 tkm	IT 0 tkm	FR 0 tkm	DE 0 tkm	BE 0 tkm			
Technosphere	operation, freight train	CH	0	tkm	1.00E+0					1	1.30	own estimation
	operation, freight train	AT	0	tkm		1.00E+0				1	1.30	own estimation
	operation, freight train	IT	0	tkm			1.00E+0			1	1.30	own estimation
	operation, freight train	FR	0	tkm				1.00E+0		1	1.30	own estimation
	operation, freight train	DE	0	tkm					1.00E+0	1	1.30	own estimation
	operation, freight train	BE	0	tkm						1	1.30	own estimation
	locomotive	RER	1	unit	7.11E-10	7.11E-10	7.11E-10	7.11E-10	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	goods wagon	RER	1	unit	4.59E-8	4.59E-8	4.59E-8	4.59E-8	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, goods wagon	RER	1	unit	4.59E-8	4.59E-8	4.59E-8	4.59E-8	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, locomotive	RER	1	unit	7.11E-10	7.11E-10	7.11E-10	7.11E-10	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	disposal, locomotive	RER	1	unit	7.11E-10	7.11E-10	7.11E-10	7.11E-10	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	railway track	CH	1	ma	8.63E-5	6.29E-5	8.91E-5	8.63E-5	7.63E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length-transport performance ratio and further geographical uncertainties due to extrapolation
	operation, maintenance, railway track	CH	1	ma	8.63E-5	6.29E-5	8.91E-5	8.63E-5	7.63E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length-transport performance ratio and further geographical uncertainties due to extrapolation
	disposal, railway track	CH	1	ma	8.63E-5	6.29E-5	8.91E-5	8.63E-5	7.63E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length-transport performance ratio and further geographical uncertainties due to extrapolation
products	transport, freight, rail	AT	0	tkm	1.00E+0							
	transport, freight, rail	IT	0	tkm		1.00E+0						
	transport, freight, rail	FR	0	tkm			1.00E+0					
	transport, freight, rail	DE	0	tkm				1.00E+0				
	transport, freight, rail	BE	0	tkm					1.00E+0			

Tab. 7.79 Life cycle inventory input data for European freight transport services

Explanations	Name	Location	Infrastructure-Process	Unit	transport, freight, rail	uncertaintyType	StandardDeviation 95%	GeneralComment
					RER 0 tkm			
Technosphere	operation, freight train	RER	0	tkm	1.00E+0	1	1.00	own estimation
	locomotive	RER	1	unit	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	goods wagon	RER	1	unit	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, goods wagon	RER	1	unit	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, locomotive	RER	1	unit	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	disposal, locomotive	RER	1	unit	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	railway track	CH	1	ma	9.30E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length-transport performance ratio and further geographical uncertainties due to extrapolation
	operation, maintenance, railway track	CH	1	ma	4.57E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length-transport performance ratio and further geographical uncertainties due to extrapolation
	disposal, railway track	CH	1	ma	9.30E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length-transport performance ratio and further geographical uncertainties due to extrapolation
product	transport, freight, rail	RER	0	tkm	1.00E+0			

Tab. 7.80 Inventory data for the CN Dataset "transport, coal freight, rail (CN)"

Explanations	Name	Location	Infrastructure- Process	Unit	transport, coal freight, rail	uncertaintyType	StandardDeviation 95%	GeneralComment
					CN 0 tkm			
Technosphere	operation, coal freight train, electricity	CN	0	tkm	2.50E-1	1	1.50	own assumption
	operation, coal freight train, diesel	CN	0	tkm	7.00E-1	1	1.50	own assumption
	operation, coal freight train, steam	CN	0	tkm	5.00E-2	1	1.50	own assumption
	disposal, railway track	CH	1	ma	9.30E-5	1	1.70	increased from 1.3 for corresponding item in "transport, freight, rail RER"
	goods wagon	RER	1	unit	4.59E-8	1	1.70	as for transport, freight, rail RER
	locomotive	RER	1	unit	7.11E-10	1	1.70	as for transport, freight, rail RER
	maintenance, goods wagon	RER	1	unit	4.59E-8	1	1.70	as for transport, freight, rail RER
	maintenance, locomotive	RER	1	unit	7.11E-10	1	1.70	as for transport, freight, rail RER
	operation, maintenance, railway track	CH	1	ma	4.57E-5	1	1.70	as for transport, freight, rail RER
	railway track	CH	1	ma	9.30E-5	1	1.70	as for transport, freight, rail RER
	disposal, locomotive	RER	1	unit	7.11E-10	1	1.70	as for transport, freight, rail RER
products	transport, coal freight, rail	CN	0	tkm	1.00E+0			

Tab. 7.81 Inventory data for freight transport by diesel train in the US.

Explanations	Name	Location	Infrastructure- Process	Unit	transport, freight, rail, diesel	uncertaintyType	StandardDeviation 95%	GeneralComment
					US 0 tkm			
Technosphere	disposal, locomotive	RER	1	unit	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation (3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length -transport performance ratio and further geographical uncertainties due to extrapolation
	disposal, railway track	CH	1	ma	9.30E-5	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	goods wagon	RER	1	unit	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	locomotive	RER	1	unit	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, goods wagon	RER	1	unit	4.59E-8	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	maintenance, locomotive	RER	1	unit	7.11E-10	1	1.30	own estimation based on comparison with the vehicle stock-transport performance ratio in 1999 and further geographical uncertainties due to extrapolation
	operation, freight train, diesel	RER	0	tkm	1.00E+0	1	1.00	own estimation
	operation, maintenance, railway track	CH	1	ma	4.57E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length -transport performance ratio and further geographical uncertainties due to extrapolation
	railway track	CH	1	ma	9.30E-5	1	1.30	(3,1,1,3,1,na); own estimation taken into account uncertainties in the infrastructure length -transport performance ratio and further geographical uncertainties due to extrapolation
products	transport, freight, rail, diesel	US	0	tkm	1.00E+0			

7.5.3 Life Cycle Inventory Input Data of Passenger transport

Tab. 7.82 Life cycle inventory input data for Swiss regional rail transport services

Explanations	Name	Location	Infrastructure- Process	Unit	transport, regional train, SBB mix	uncertaintyType	StandardDeviation	95%	GeneralComment
	Location InfrastructureProcess Unit								
Technosphere	operation, regional train, SBB mix	CH	0	pkm	1.00E+0	1	1.00		own estimation
	regional train	CH	1	unit	2.93E-9	1	1.20		own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	maintenance, regional train	CH	1	unit	2.93E-9	1	1.20		own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	disposal, regional train	CH	1	unit	2.93E-9	1	1.20		own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	railway track	CH	1	ma	1.31E-4	1	1.20		own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.
	operation, maintenance, railway track	CH	1	ma	1.31E-4	1	1.20		own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.
	disposal, railway track	CH	1	ma	1.31E-4	1	1.20		own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.
product	transport, regional train, SBB mix	CH	0	pkm	1.00E+0				

Tab. 7.83 Life cycle inventory input data for Swiss long distance rail transport services

Explanations	Name	Location	Infrastructure- Process	Unit	transport, long- distance train, SBB mix	uncertaintyType	StandardDeviation	95%	GeneralComment
	Location InfrastructureProcess Unit								
Technosphere	operation, long-distance train, SBB mix	CH	0	pkm	1.00E+0	1	1.00		own estimation
	long-distance train	CH	1	unit	2.48E-10	1	1.20		own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	maintenance, long-distance train	CH	1	unit	2.48E-10	1	1.20		own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	disposal, long-distance train	CH	1	unit	2.48E-10	1	1.20		own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	railway track	CH	1	ma	9.37E-5	1	1.20		own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.
	operation, maintenance, railway track	CH	1	ma	9.37E-5	1	1.20		own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.
	disposal, railway track	CH	1	ma	9.37E-5	1	1.20		own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.
product	transport, long-distance train, SBB mix	CH	0	pkm	1.00E+0				

Tab. 7.84 Life cycle inventory input data for Swiss metropolitan train ("S-Bahn")

Explanations	Name	Location	Infrastructure-Process	Unit	transport, metropolitan train, SBB mix			uncertaintyType	StandardDeviation	95%	GeneralComment
					CH	0	pkm				
Technosphere	operation, metropolitan train, SBB mix	CH	0	pkm	1.00E+0	1	1.00	own estimation			
	long-distance train	CH	1	unit	1.08E-9	1	1.20	own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.			
	maintenance, long-distance train	CH	1	unit	1.08E-9	1	1.20	own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.			
	disposal, long-distance train	CH	1	unit	1.08E-9	1	1.20	own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.			
	railway track	CH	1	ma	8.98E-5	1	1.20	own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.			
	operation, maintenance, railway track	CH	1	ma	8.98E-5	1	1.20	own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.			
	disposal, railway track	CH	1	ma	8.98E-5	1	1.20	own estimation taken into account uncertainties in the infrastructure length -transport performance ratio.			
product	transport, metropolitan train, SBB mix	CH	0	pkm	1.00E+0						

Tab. 7.85 Life cycle inventory input data for high-speed rail transport services in Germany, France and Italy

Explanations	Name	Location	Infrastructure-Process	Unit	transport, high speed train			uncertaintyType	StandardDeviation	95%	GeneralComment
					DE	FR	IT				
					0	0	0				
					pkm	pkm	pkm				
Technosphere	operation, high speed train	DE	0	pkm	1.00E+0			1	1.00	own estimation	
	operation, high speed train	FR	0	pkm		1.00E+0		1	1.00	own estimation	
	operation, high speed train	IT	0	pkm			1.00E+0	1	1.00	own estimation	
	ICE	DE	1	unit	3.24E-10	3.24E-10	3.24E-10	1	1.20	literature data. Uncertainties due to assumed kilometric performance, lifespan, and load factor	
	maintenance, ICE	DE	1	unit	3.24E-10	3.24E-10	3.24E-10	1	1.20	literature data. Uncertainties due to assumed kilometric performance, lifespan, and load factor	
	railway track, ICE	DE	1	ma	1.30E-4	1.30E-4	1.30E-4	1	1.20	literature data. Allocation principle (freight vs. Passenger): yearly kilometric performance.	
	operation, maintenance, railway track, ICE	DE	1	ma	1.30E-4	1.30E-4	1.30E-4	1	1.20	literature data. Allocation principle (freight vs. Passenger): yearly kilometric performance.	
	disposal, ICE	DE	1	unit	3.24E-10	3.24E-10	3.24E-10	1	1.20	literature data. Uncertainties due to assumed kilometric performance, lifespan, and load factor	
products	disposal, railway track	CH	1	ma	1.30E-4	1.30E-4	1.30E-4	1	1.20	literature data. Allocation principle (freight vs. Passenger): yearly kilometric performance.	
	transport, high speed train	DE	0	pkm	1.00E+0						
	transport, high speed train	FR	0	pkm		1.00E+0					
	transport, high speed train	IT	0	pkm			1.00E+0				

Tab. 7.86 Life cycle inventory input data for rail transport services (average) in selected European countries

Explanations	Name	Location	Infrastructure-Process	Unit	transport, average train, SBB mix	transport, average train	transport, average train	transport, average train	transport, average train	transport, average train	uncertainty Type	Standard Deviation 95%	GeneralComment
					CH 0 pkm	AT 0 pkm	IT 0 pkm	FR 0 pkm	DE 0 pkm	BE 0 pkm			
Technosphere	operation, average train, SBB mix	CH	0	pkm	1.00E+0						1	1.00	own estimation
	operation, average train	AT	0	pkm		1.00E+0					1	1.00	own estimation
	operation, average train	IT	0	pkm			1.00E+0				1	1.00	own estimation
	operation, average train	FR	0	pkm				1.00E+0			1	1.00	own estimation
	operation, average train	DE	0	pkm					1.00E+0		1	1.00	own estimation
	operation, average train	BE	0	pkm						1.00E+0	1	1.00	own estimation
	long-distance train	CH	1	unit	4.58E-10	5.35E-10	3.02E-10	2.42E-10	4.65E-10	4.06E-10	1	1.20	own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	maintenance, long-distance train	CH	1	unit	4.58E-10	5.35E-10	3.02E-10	2.42E-10	4.65E-10	4.06E-10	1	1.20	own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	disposal, long-distance train	CH	1	unit	4.58E-10	5.35E-10	3.02E-10	2.42E-10	4.65E-10	4.06E-10	1	1.20	own estimation based on comparison with the vehicle stock-transport performance ratio in 2000.
	railway track	CH	1	ma	9.28E-5	1.18E-4	6.19E-5	6.93E-5	8.43E-5	1.04E-4	1	1.20	own estimation taken into account uncertainties in the infrastructure length - transport performance ratio.
	operation, maintenance, railway track	CH	1	ma	9.28E-5	1.18E-4	6.19E-5	6.93E-5	8.43E-5	1.04E-4	1	1.20	own estimation taken into account uncertainties in the infrastructure length - transport performance ratio.
	disposal, railway track	CH	1	ma	9.28E-5	1.18E-4	6.19E-5	6.93E-5	8.43E-5	1.04E-4	1	1.20	own estimation taken into account uncertainties in the infrastructure length - transport performance ratio.
product	transport, average train, SBB mix	CH	0	pkm	1.00E+0								
	transport, average train	AT	0	pkm		1.00E+0							
	transport, average train	IT	0	pkm			1.00E+0						
	transport, average train	FR	0	pkm				1.00E+0					
	transport, average train	DE	0	pkm					1.00E+0				
	transport, average train	BE	0	pkm						1.00E+0			

7.6 References

- BFS (2000) BFS (2000) Schweizerische Verkehrstatistik 1996/2000. 11: Verkehr und Nachrichtenwesen. Bundesamt für Statistik (BfS), Bern.
- Borken et al. (1999) Borken J., Patyk A. and Reinhardt G. A. (1999) Basisdaten für ökologische Bilanzierungen: Einsatz von Nutzfahrzeugen in Transport, Landwirtschaft und Bergbau. Vieweg, Braunschweig/Wiesbaden
- Borken et al. (2003) Borken J., Helms H., Jungk N. and W. K. (2003) Energy consumptions and emissions of goods transports in Europe. Ifeu, Heidelberg
- BUWAL (2002) BUWAL (2002) PM10-Emissionen des Verkehrs. Statusbericht; Teil Schienenverkehr. 144. Bundesamt fuer Umwelt, Wald und Landschaft, Bern
- DB (2000) DB (2000) Die Bahn, Environmental Report. Deutsche Bahn AG, Berlin
- Elsener & Strub (1993) Elsener J. and Strub P. (1993) Material und Energiebilanz für den Güterverkehr - Teil Infrastruktur und Fahrzeugpark. ETH-Semesterarbeit. ETH Zürich, Zürich
- EU (2004) EU (2004), RICHTLINIE 2004/26/EG DES EUROPÄISCHEN PARLAMENTS UND DES RATES zur Änderung der Richtlinie 97/68/EG zur Angleichung der Rechtsvorschriften der Mitgliedstaaten über Maßnahmen zur Bekämpfung der Emission von gasförmigen Schadstoffen und luftverunreinigenden Partikeln aus Verbrennungsmotoren für mobile Maschinen und Geräte
- EUROSTAT (2000) EUROSTAT (2000) Glossary for transport statistics. Document prepared by the intersecretariat working group on transport statistics EUROSTAT, ECMT, UN/ECE, Luxembourg
- EUROSTAT (2001) EUROSTAT (2001) Panorama of transport Data 1970-1999. Theme 7 , Transport. European Commission, Luxembourg
- Frischknecht (2003) Frischknecht R. (2003) Strommix. In: *Sachbilanzen von Energiesystemen: Grundlagen für den ökologischen Vergleich von Energiesystemen und den Einbezug von Energiesystemen in Ökobilanzen für die Schweiz* (ed. Dones R.).

- Final report ecoinvent 2000 No. 6, Paul Scherrer Institut Villigen, Swiss Centre for Life Cycle Inventories, Duebendorf, CH, Online-Version under: www.ecoinvent.ch.
- Frischknecht et al. (1996) Frischknecht R., Bollens U., Bosshart S., Ciot M., Ciseri L., Doka G., Dones R., Gantner U. and Hirsch R. (1996) Ökoinventare von Energiesystemen: Grundlagen für den ökologischen Vergleich von Energiesystemen und den Einbezug von Energiesystemen in Ökobilanzen für die Schweiz. Auflage No. 3. Gruppe Energie - Stoffe - Umwelt (ESU), Eidgenössische Technische Hochschule Zuerich und Sektion Ganzheitliche Systemanalysen, Paul Scherrer Institut, Villigen, Bundesamt für Energie (Hrsg.), Bern
- INFRAS (1999) INFRAS (1999) Handbook emission factors for road transport. CD-Rom.
- Joergensen & Sorenson (1997) Joergensen M. W. and Sorenson S. C. (1997) Estimating Emissions from Railway Traffic. ET-EO-97-03. DTU Denmark, Lungby, Denmark
- Jungbluth (2003) Jungbluth N. (2003) Erdöl. In: *Sachbilanzen von Energiesystemen: Grundlagen für den ökologischen Vergleich von Energiesystemen und den Einbezug von Energiesystemen in Ökobilanzen für die Schweiz*. Final report ecoinvent 2000 No. 6, Paul Scherrer Institut Villigen, Swiss Centre for Life Cycle Inventories, Duebendorf, CH retrieved from: www.ecoinvent.ch.
- Knörr et al. (2000) Knörr W., Höpfner U., Lambrecht U. and Patyk A. (2000) Fortschreibung Daten- und Rechenmodell: Energieverbrauch und Schadstoffemissionen aus dem motorisierten Verkehr. Endbericht. Nr. 298 45 105. Institut für Energie- und Umweltforschung (ifeu), Heidelberg.
- Maibach et al. (1999) Maibach M., Peter D. and Seiler B. (1999) Ökoinventar Transporte: Grundlagen für den ökologischen Vergleich von Transportsystemen und den Einbezug von Transportsystemen in Ökobilanzen. Technischer Schlussbericht. In: *Schweizerischer Nationalfonds zur Förderung der wissenschaftlichen Forschung* (ed. SPP Umwelt M.). 2nd Edition. INFRAS, Zürich
- Manto (1994) Manto (1994) Consommation d'énergie totale actuelle de quatre domaines susceptibles d'être influence par de nouveaux moyens de télécommunication. 2.14. ETH Zürich, Zürich
- Oggier (2002) Oggier (2002) Entsorgung von Gleisaushub. pdf from http://www.sbb.ch/gs/pdf/umwelt_gleisaushub.pdf.
- Rozycki et al. (2003) von Rozycki C., Koeser H. and Schwarz H. (2003) Ecology Profile of the German High-speed Rail Passenger Transport System, ICE. In: *International Journal of Life Cycle Assessment*, 8(2), pp. 83-91.
- SBB (2002) SBB (2002) Umweltbericht 2000/2001. SBB, Bahnumweltzentrum, Bern, Online-Version under: www.sbb.ch/umwelt.
- SBB (2001a) SBB (2001a) Feinstaubemissionen von Gussklotzbremsen. Schweizerische Bundesbahnen, Bern
- SBB (2001b) SBB (2001b) Zahlen 2000. Kommunikation SBB, Bern
- Spielmann et al. (2004) Spielmann M., Kägi T., Stadler P. and Tietje O. (2004) Life cycle inventories of transport services. Final report ecoinvent 2000 No. 14. UNS, ETH-Zurich, Swiss Centre for Life Cycle Inventories, Duebendorf, CH, retrieved from: www.ecoinvent.ch.
- UIC (2007) UIC (2007) International Railway Statistics. UNION INTERNATIONALE DES CHEMINS DE FER (UIC), Paris, F.